

A Gentle Introduction to Intertemporal Choice

CHRISTOPHER D. CARROLL AND ALAN E. LUJAN SOLIS

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Chapter 1

Consumption Theory

Fisher's two-period problem opens the chapter. It exposes the logic of intertemporal choice in the simplest setting that contains anything interesting, one in which a budget constraint trades current against future consumption and a first-order condition equates the marginal rate of substitution between them to the interest factor.

Extending that problem in one direction gives the Diamond overlapping generations model, which lets us ask how public pensions and government debt shift resources across cohorts. Extending Fisher's problem in the other direction, toward many periods and uncertain income, gives the perfect foresight CRRA consumption function and then the random walk result. Admitting a precautionary motive yields buffer stock behavior, which reconciles the theory with observed age profiles of consumption and income (the hump the perfect foresight model cannot produce).

The chapter closes with the leading departures from that benchmark: habit formation, durable goods, time inconsistent discounting, and expectations that respond only sluggishly to news.

1.1 The Fisher Two-Period Optimal Consumption Problem

Irving Fisher [1930] first analyzed the optimization problem of a consumer who faces no uncertainty and lives for two periods.

1.1.1 Mathematical Analysis

In its most general form, the household's lifetime value function can be written

$$V(c_1, c_2) \quad (1.1.1)$$

where the first argument reflects consumption in “youth” while the second argument represents consumption in “old age” and we assume that the derivatives with respect to the first and second arguments are positive,

$$V_1, V_2 > 0, \quad (1.1.2)$$

while the second derivatives are negative,

$$V_{11}, V_{22} < 0. \quad (1.1.3)$$

The consumer begins the first period with resources of b_1 (think “bank balances”) and income of y_1 . Total resources are divided between consumption and end-of-period assets a_1 (“assets after all actions” in period 1):

$$\begin{aligned} b_1 + y_1 &= c_1 + a_1, \\ a_1 &= b_1 + y_1 - c_1. \end{aligned} \quad (1.1.4)$$

Balances at the beginning of period 2 are equal to end-of-first-period accumulated assets a_1 , rewarded by a gross interest factor $R = (1 + r)$:

$$b_2 = a_1 R. \quad (1.1.5)$$

This is the *dynamic budget constraint* or DBC for this problem. A DBC links two adjacent periods of time. A more comprehensive kind of constraint is the *intertemporal* budget constraint (IBC):

$$c_1 + c_2/R \leq y_1 + y_2/R + b_1, \quad (1.1.6)$$

which must be satisfied over an extended (multiperiod) span of time like a lifetime.

For various purposes, it is useful to keep track of *human wealth* h_t , defined as the present discounted value of future labor income (the operator $\mathbb{P}_t(\bullet)$ denotes the present discounted value of the variable $\bullet$ from the perspective of the beginning of period t through the end of the horizon),

$$\begin{aligned} h_t &= \mathbb{P}_t(y) \\ h_1 &= y_1 + y_2/R \end{aligned} \quad (1.1.7)$$

Because we have assumed (in (1.1.2)) that an additional unit of consumption always yields extra utility, we can reach our first conclusion (as opposed to assumption) in the model: Once the consumer has reached the last period of life, he will consume all available resources:

$$c_2 = b_2 + y_2. \quad (1.1.8)$$

This means that the IBC will hold with equality (if it did not, utility could be increased by consuming more in one or both periods). Thus, the IBC can be rewritten as

$$c_1 + c_2/R = h_1 + b_1. \quad (1.1.9)$$

The general form that the IBC will take is that the present discounted value of lifetime spending must equal the present discounted value of lifetime resources:

$$\mathbb{P}_t(c) = \mathbb{P}_t(y) + b_t. \quad (1.1.10)$$

Substituting in the definition of b_2 means that our problem can now be stated as:

$$\begin{aligned} \max_{\{c_1, c_2\}} \quad & V(c_1, c_2) \\ \text{s.t.} \quad & \\ & c_2 = (b_1 + y_1 - c_1)R + y_2. \end{aligned} \quad (1.1.11)$$

Now we can write the problem as a Lagrange multiplier problem, where the maximand is:

$$V(c_1, c_2) + (c_2 - (b_1 + y_1 - c_1)R - y_2) \lambda. \quad (1.1.12)$$

The first order conditions are:

$$\begin{aligned} V_1 + R\lambda &= 0 \\ V_2 &= -\lambda \end{aligned} \quad (1.1.13)$$

and substituting the second of these into the first we get

$$V_1 = RV_2. \quad (1.1.14)$$

This is the same condition you get when deciding between two commodities at a point in time, where we can now think of R as the intertemporal price: How much of good 2 (consumption in period 2) do I get in exchange for giving up a unit of good 1 (consumption in period 1).

Now suppose that the consumer's utility is *time-separable*, and the felicity function (felicity is the utility obtained in a single period of a multi-period problem) is the same in both periods of life, so that

$$V(c_1, c_2) = u(c_1) + \beta u(c_2) \quad (1.1.15)$$

where β is a *time preference factor* that specifies how the consumer trades off utility in period 1 against utility in period 2.¹

From our assumptions (1.1.2) and (1.1.3) we know that the felicity function must satisfy

$$\begin{aligned} u'(\bullet) &> 0 \\ u''(\bullet) &< 0, \end{aligned} \quad (1.1.16)$$

and since the felicity functions are the same in both periods we have that

$$\begin{aligned} V_1(c_1, c_2) &= u'(c_1) \\ V_2(c_1, c_2) &= \beta u'(c_2). \end{aligned} \quad (1.1.17)$$

Substituting these equations into (1.1.14) yields the *Euler equation* for consumption:

$$u'(c_1) = R\beta u'(c_2). \quad (1.1.18)$$

The Euler equation is a central result in intertemporal optimization theory, and will be used again and again as the course progresses. It is therefore worth studying carefully to be sure you understand it thoroughly.

To help obtain the intuition for why the Euler equation is necessary for optimality, consider the following thought experiment. Designate c_1^* and c_2^* as the optimal levels of consumption in this problem, the levels that solve the maximization problem under some set of circumstances. Thus, the highest attainable utility is

$$u(c_1^*) + \beta u(c_2^*). \quad (1.1.19)$$

¹Paul Samuelson [1937, 1958] introduced the discounting of future utility into the problem. See Frederick et al. [2002] for a comprehensive review of the still-controversial topic of time discounting.

Now consider reducing consumption by some small amount ϵ in period 1, investing that ϵ so that it grows to $R\epsilon$ in period 2, and then consuming it in period 2. What happens to utility?

Taking first-order Taylor expansions, the levels of first-period and second-period utility are now

$$\begin{aligned} u(c_1^* - \epsilon) &\approx u(c_1^*) - u'(c_1^*)\epsilon \\ u(c_2^* + R\epsilon) &\approx u(c_2^*) + u'(c_2^*)R\epsilon. \end{aligned} \quad (1.1.20)$$

Now the difference between the maximum possible utility and the new situation is given by

$$u(c_1^*) + \beta u(c_2^*) - [u(c_1^*) - u'(c_1^*)\epsilon + \beta (u(c_2^*) + u'(c_2^*)R\epsilon)] = u'(c_1^*)\epsilon - \beta u'(c_2^*)R\epsilon. \quad (1.1.21)$$

But it must be the case that (1.1.21) is approximately equal to zero. To see why, suppose it were a negative number. That would mean that moving from the original situation with $\{c_1, c_2\} = \{c_1^*, c_2^*\}$ to the new situation with $\{c_1, c_2\} = \{c_1^* - \epsilon, c_2^* + R\epsilon\}$ resulted in an *increase* in utility. But we assumed that c_1^*, c_2^* were already the utility-maximizing choices, which clearly could not be true if adjusting c_1^* downward by ϵ and c_2^* upward by $R\epsilon$ increased utility. Similarly, if the expression were positive, then utility could be increased by doing the opposite (i.e. increasing consumption in period 1 by ϵ and reducing it in period 2 by $R\epsilon$). Thus, in either case if the expression is not zero, we have a contradiction to the assumption that c_1^* and c_2^* are the utility-maximizing choices.

To make further progress, it is necessary to make more specific assumptions about the structure of the utility function. The most common assumption is that utility takes the Constant Relative Risk Aversion form,

$$u(c) = \frac{c^{1-\rho}}{1-\rho}, \quad (1.1.22)$$

with marginal utility

$$u'(c) = c^{-\rho}. \quad (1.1.23)$$

Consider equation (1.1.18) with CRRA utility,

$$\begin{aligned} c_1^{-\rho} &= R\beta c_2^{-\rho} \\ c_2/c_1 &= (R\beta)^{1/\rho} \\ c_2 &= (R\beta)^{1/\rho} c_1. \end{aligned} \quad (1.1.24)$$

Now note that this equation allows us to calculate the *intertemporal elasticity of substitution* as the change in the ratio of the log of c_2/c_1 to the log change in the intertemporal price R :

$$\begin{aligned} \left(\frac{d}{d \log R} \right) \log \left(\frac{c_2}{c_1} \right) &= \left(\frac{d}{d \log R} \right) \log (R\beta)^{1/\rho} \\ &= \rho^{-1}. \end{aligned} \quad (1.1.25)$$

Next note that from (1.1.24) we can calculate the PDV of lifetime consumption from the perspective of the first period of life as

$$\begin{aligned} \mathbb{P}_1(c) &= c_1 + R^{-1}c_2 \\ &= \left(1 + R^{-1}(R\beta)^{1/\rho} \right) c_1. \end{aligned} \quad (1.1.26)$$

Now we can use the intertemporal budget constraint:

$$\begin{aligned} \mathbb{P}_t(c) &= b_t + \mathbb{P}_t(y) \\ c_1 \left(1 + R^{-1}(R\beta)^{1/\rho} \right) &= b_1 + y_1 + R^{-1}y_2 \\ c_1 &= \frac{b_1 + h_1}{1 + R^{-1}(R\beta)^{1/\rho}}. \end{aligned} \quad (1.1.27)$$

Thus, we have solved the two-period life cycle saving problem for the *consumption function* c_1 relating the level of consumption to all of the parameters of the problem.

One of the surprising features of the solution goes by the name of “Fisherian Separation”: Notice that the *profile* of consumption growth over the lifetime is given by (1.1.24) *regardless of the shape of the income profile*. For two consumers with the same total amount of lifetime wealth (combined b_1 and h_1), the level and growth rates of consumption over the lifetime will be identical whether the consumer’s lifetime wealth is entirely concentrated in the first period (that is, $h_1 = 0$), entirely concentrated in the last period ($b_1 = 0$), split half-and-half, or organized any other way. Fisherian Separation is a pervasive feature of models that combine perfect foresight and a lack of liquidity constraints.

A common assumption (for simplicity, not realism) is that $\rho = 1$, which is equivalent to assuming that the utility function is logarithmic:²

$$\lim_{\rho \rightarrow 1} \frac{c^{1-\rho}}{1-\rho} = \log c. \quad (1.1.28)$$

In this case it turns out that we can simply substitute $\rho = 1$ into the solution for consumption, obtaining

$$c_1 = \frac{b_1 + h_1}{1 + \beta}. \quad (1.1.29)$$

1.1.2 Graphical Analysis

The classic graphical analysis of this problem is shown in Figure 1.1.

The top figure depicts a situation in which all of the consumer’s lifetime income is earned in the first period of life. The budget constraint in the initial situation, associated with a “Low R ”, yields an optimal consumption choice labeled as point A where the budget constraint is tangent to the indifference curve. When the interest factor is increased to the “High R ” situation, the optimal consumption choice moves to point C .

Note first that if all income is earned in the first period of life, an increase in the interest factor is unambiguously good for the consumer: the set of consumption possibilities is strictly larger.

Second, the movement from point A to point C can be decomposed into two parts: an income effect AB and a substitution effect BC .

Call the low and the high interest factors respectively $\underline{R}$ and $\bar{R}$.

The income effect is the answer to the question “Suppose we wanted to change lifetime value by the same amount as it is changed by going from $\underline{R}$ to $\bar{R}$, but we wanted to achieve this change in value at the initial interest factor $\underline{R}$. Supposing we gave the consumer enough extra initial resources to achieve the change in value, how would their consumption allocation change?”

In order to relate this back to the algebraic analysis above, it will be useful to rewrite lifetime value as a function simply of initial resources and the interest factor (taking $\{y_1, y_2\}$ and other parts of the problem as given):

$$V^*(b_1, R) = u(c_1^*(b_1, R)) + \beta u(c_2^*(b_1, R)) \quad (1.1.30)$$

Using this function, the income effect is obtained as the value of Δb_1 in the equation $V^*(b_1 + \Delta b_1, \underline{R}) = V^*(b_1, \bar{R})$.

The substitution effect is the answer to the question, “Staying on the new indifference curve, how much does the allocation of consumption change as a consequence of the difference in interest factors between $\underline{R}$ and $\bar{R}$?” This is captured in the movement from B to C .

Note that the income and the substitution effects on c_1 are opposite in sign. A higher interest factor gives consumers the incentive to substitute future for current consumption (c_1 is lower at point C than at B). But a higher interest factor also gives consumers the ability to consume more in both

²See the math fact showing that $\lim_{\rho \rightarrow 1} \frac{c^{1-\rho}}{1-\rho} = \log c$ in the [Math Facts section](#).

periods. Whether c_1 rises or falls in response to the increase in interest factors will depend on the relative magnitudes of the income and substitution effects.

The lower figure shows a similar experiment, with the sole difference that the consumer's lifetime resources are exclusively concentrated in period 2.

In this case, the period 1 consumer must borrow against future income in order to consume anything. An increase in interest factors is therefore unambiguously bad (the available set of consumption choices is strictly smaller).

The optimal choice again moves from point A to point C . However, we now decompose the movement into three parts. The first part is called the *human wealth effect*. It captures the fact that the present discounted value of lifetime resources $\mathbb{P}_1(y)$ is smaller when interest rates are higher; the magnitude of the change in human wealth is depicted on the horizontal axis of the figure. The *human wealth effect* is the consequence that an equivalent change in lifetime resources would have *in the absence of any change in interest factors*. So the human wealth effect takes the consumer from point A to point D .

Notice that once we have computed the human wealth effect, if we treat point D as the starting point of our analysis, the remaining analysis is identical to that for the upper figure: We can increase the interest factor from $\underline{R}$ to $\bar{R}$, which causes the equilibrium point to change from point D to point C , a movement that can be decomposed into an income effect DB (analogous to the income effect AB in the original analysis) and a substitution effect BC .

The terminology here is a modification (refinement) of the terminology often employed in micro textbooks, where the "income effect" is defined in a way that would incorporate both what I am calling the income effect and what I am calling the human wealth effect.

The reason to make this distinction is that it is important to distinguish between effects on behavior caused by the fact that the discounted value of future income is changed, and effects caused by the fact that the income that will be earned on savings is different. Summers [1981] vigorously made the point that in standard life cycle models, the quantitative magnitude of the human wealth effect dwarfs the size of either the income or the substitution effects, because for most people most of their lifetime income is in the future.

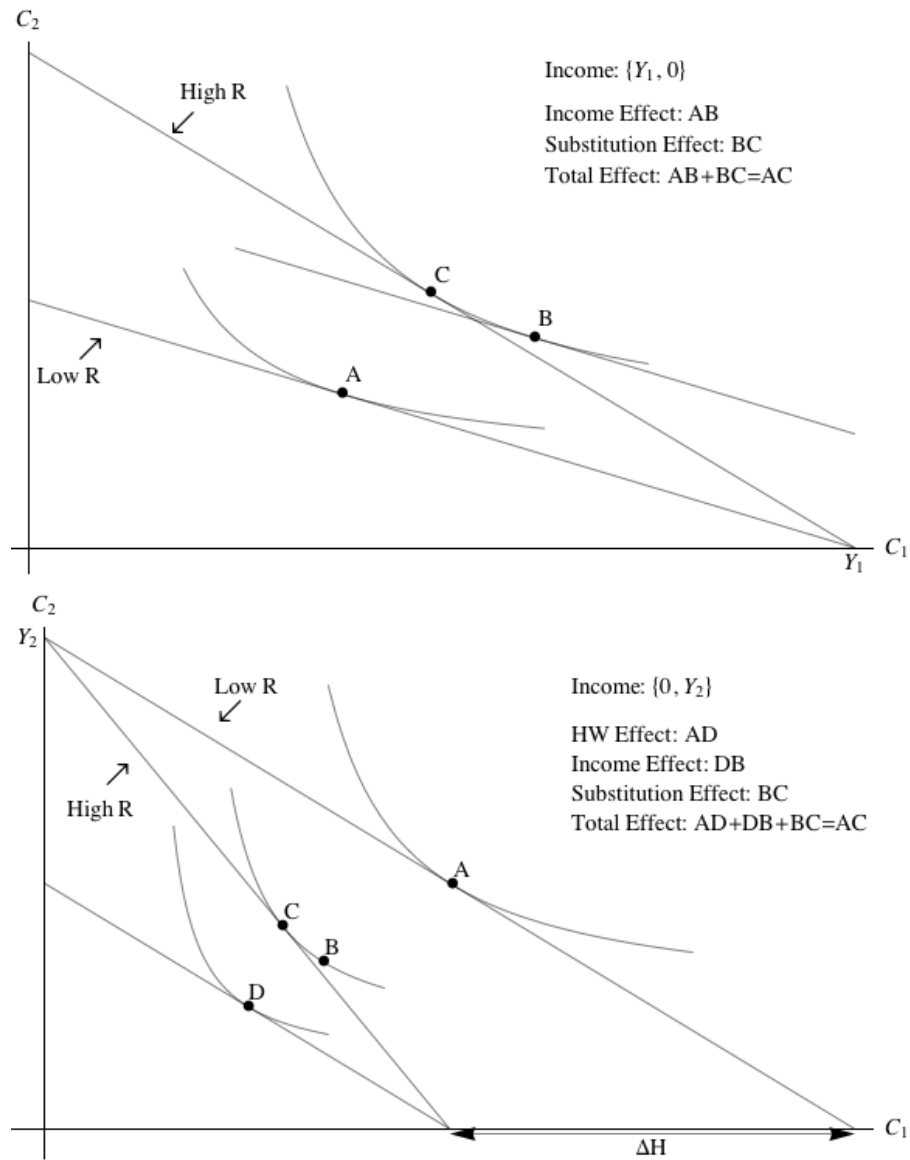


Figure 1.1: Fisher Figure Analysis

1.2 Consumption and Labor Supply

Consider a consumer who has a utility function

$$u(c_t, z_t) \quad (1.2.1)$$

where z_t is leisure (mnemonic: z for laZiness) and c_t is consumption. Normalize the maximum possible labor supply to 1; actual labor supply is ℓ_t , so that

$$\ell_t + z_t = 1. \quad (1.2.2)$$

The wage earned for working one unit of time is W_t , and labor income is the wage rate multiplied by the amount of labor supplied,

$$\begin{aligned} y_t &= W_t \ell_t \\ &= (1 - z_t)W_t. \end{aligned} \quad (1.2.3)$$

Suppose the consumer has a fixed amount x_t to spend in period t on consumption and leisure,

$$x_t = c_t + z_t W_t, \quad (1.2.4)$$

where x_t can differ from income y_t because this might be a single period in a multi-period problem.

The price of leisure is W_t (your income is lower by this amount for every extra unit of time you spend not working) and the price of consumption is 1, so the first order condition from the optimal choice of leisure says that the ratio of the marginal utility of leisure to the marginal utility of consumption should be

$$W_t = \left(\frac{u^z}{u^c} \right). \quad (1.2.5)$$

To see this formally, note that the consumer's goal is to

$$\max_{\{c_t, z_t\}} u(c_t, z_t) \quad (1.2.6)$$

subject to a budget constraint

$$c_t = x_t - W_t z_t \quad (1.2.7)$$

so (1.2.6) becomes

$$\max_{\{z_t\}} u(x_t - W_t z_t, z_t) \quad (1.2.8)$$

for which the FOC is

$$\begin{aligned} -u^c W_t + u^z &= 0 \\ W_t &= (u^z / u^c). \end{aligned} \quad (1.2.9)$$

This is just the classic condition that says that the ratio of prices of two goods should equal the ratio of their marginal utilities, which applies in any standard microeconomic problem. For a quantitative comparison of how this condition manifests itself in the U.S. and Europe, see Zweibel [2005].

Now, assume there is an “outer” utility function $f(\bullet)$ which depends on a Cobb-Douglas aggregate of consumption and leisure

$$u(c_t, z_t) = f\left(c_t^{1-\zeta} z_t^\zeta\right) \quad (1.2.10)$$

The inner function has the property that $z_t W_t = c_t \eta$ for $\eta = \zeta / (1 - \zeta)$, which implies utility can be written

$$f((W_t/\eta)^{-\zeta}c_t). \quad (1.2.11)$$

To see this, note that the maximization problem is

$$\max_{z_t} f\left((x_t - z_t W_t)^{1-\zeta} z_t^\zeta\right) \quad (1.2.12)$$

FOC:

$$\begin{aligned} (1-\zeta)W_t(x_t - z_t W_t)^{-\zeta} z_t^\zeta f' &= \zeta(x_t - z_t W_t)^{1-\zeta} z_t^{\zeta-1} f' \\ W_t z_t &= c_t \underbrace{\zeta/(1-\zeta)}_{\equiv \eta} \end{aligned} \quad (1.2.13)$$

so

$$\begin{aligned} f(c_t^{1-\zeta} z_t^\zeta) &= f(c_t^{1-\zeta} (\eta c_t / W_t)^\zeta) \\ &= f((W_t/\eta)^{-\zeta} c_t) \end{aligned} \quad (1.2.14)$$

Over long periods of time as wages have risen in the U.S., the proportion of time spent working has not changed very much (an old stylized fact recently reconfirmed by Ramey and Francis [2009]). Similarly, across countries with vastly different levels of per capita income, or across people with vastly different levels of wages, the amount of variation in z_t is small compared to the size of the difference in wages.

These facts motivate the choice of utility function; King et al. [1988] show that other choices of utility functions produce trends, but no such trends are evident in the data.

To see why the trends are produced, think about a model in which the lifetime lasts only a single period, with a lifetime budget constraint $W_t = c_t + z_t W_t$.

We can solve for the level of consumption over the lifetime as

$$\begin{aligned} W_t &= (1 + \eta)c_t \\ c_t &= W_t/(1 + \eta) \end{aligned} \quad (1.2.15)$$

implying that leisure is

$$\begin{aligned} z_t &= \eta c_t / W_t \\ &= \eta / (1 + \eta) \end{aligned} \quad (1.2.16)$$

which is a constant (i.e. the amount of leisure does not trend up or down with the level of wages). Obviously this is what motivates the choice of an “inner” utility function that is Cobb-Douglas: For such a function, people will choose to spend constant proportions of their resources on consumption and leisure as wages rise.

Now consider a two period lifetime version of the model in which each period of life is characterized by a utility function of the same form and the lifetime optimization problem is

$$\max u(c_1, z_1) + \beta u(c_2, z_2) \quad (1.2.17)$$

subject to a lifetime budget constraint

$$c_2 = (W_1(1 - z_1) - c_1)R + (1 - z_2)W_2 \quad (1.2.18)$$

If the “outer” utility function is of the CRRA form $f(\chi) = \chi^{1-\rho}/(1-\rho)$, the first order condition with respect to c_1 implies that

$$\begin{aligned} (W_1/\eta)^{-\zeta} f'_1 &= R\beta(W_2/\eta)^{-\zeta} f'_2 \\ W_1^{-\zeta} (c_1(W_1/\eta)^{-\zeta})^{-\rho} &= R\beta W_2^{-\zeta} (c_2(W_2/\eta)^{-\zeta})^{-\rho} \\ W_1^{-\zeta(1-\rho)} c_1^{-\rho} &= R\beta W_2^{-\zeta(1-\rho)} c_2^{-\rho} \\ c_2/c_1 &= (R\beta)^{1/\rho} (W_2/W_1)^{-\zeta(1-\rho)/\rho} \end{aligned} \quad (1.2.19)$$

so in general the growth rate of consumption depends on the growth rate of wages. From now on, assume that the “outer” utility function is logarithmic, $f(\chi) = \log \chi$, which is the $\rho \rightarrow 1$ limit of (1.2.19). The wage term then drops out and consumption growth depends only on the interest factor and the time preference factor,

$$c_2/c_1 = R\beta. \quad (1.2.20)$$

To obtain this result directly, start by noting that the lifetime budget constraint can be rewritten

$$\begin{aligned} c_2 &= (W_1 - \overbrace{z_1 W_1}^{=\eta c_1} - c_1)R + W_2 - \overbrace{W_2 z_2}^{\eta c_2} \\ 0 &= (W_1 - (1 + \eta)c_1)R + W_2 - (1 + \eta)c_2 \\ c_2 &= (RW_1 + W_2)/(1 + \eta) - Rc_1 \end{aligned} \quad (1.2.21)$$

so the lifetime optimization problem becomes

$$\max_{c_1} \{ \log c_1 - \zeta \log W_1 + \beta (\log c_2 - \zeta \log W_2) \} \quad (1.2.22)$$

with FOC

$$\begin{aligned} 1/c_1 &= R\beta/c_2 \\ c_2/c_1 &= R\beta. \end{aligned} \quad (1.2.23)$$

The *level* of consumption follows from the intertemporal budget constraint. Substituting (1.2.20), the present discounted value of consumption spending is

$$\begin{aligned} \mathbb{P}_1(c) &= c_1(1 + R^{-1}(R\beta)) \\ &= c_1(1 + \beta) \end{aligned} \quad (1.2.24)$$

while the present discounted value of labor income is

$$\begin{aligned} \mathbb{P}_1(y) &= W_1(1 - z_1) + R^{-1}(W_2(1 - z_2)) \\ &= W_1 + R^{-1}W_2 - \eta(c_1 + R^{-1}c_2) \end{aligned} \quad (1.2.25)$$

where the second line uses $W_t z_t = \eta c_t$ to replace spending on leisure. Equating the two and defining human wealth h_1 as the present discounted value of the wage stream,

$$\begin{aligned} \mathbb{P}_1(c) &= \mathbb{P}_1(y) \\ c_1(1 + \beta)(1 + \eta) &= W_1 + R^{-1}W_2 \equiv h_1 \\ c_1 &= h_1/((1 + \beta)(1 + \eta)) \end{aligned} \quad (1.2.26)$$

so consumption when young is a fixed fraction of human wealth, and the $(1 + \eta)$ term in the denominator appears because a constant share of resources goes to leisure rather than to consumption goods.

Now we want to compare this to the two period lifetime model with no labor supply decision. In that model, the profile of consumption was unrelated to the profile of labor income over the lifetime, so “Fisherian Separation” held. In this model, the Fisherian Separation proposition is that the profile of c is unrelated to the profile of wages W ; however, the lifetime profile of *leisure spending* is identical to the lifetime profile of consumption spending,

$$\begin{aligned} W_2 z_2 / W_1 z_1 &= \eta c_2 / \eta c_1 = R\beta \\ z_2 / z_1 &= R\beta W_1 / W_2 \\ (1 - \ell_2) / (1 - \ell_1) &= R\beta W_1 / W_2 \end{aligned} \quad (1.2.27)$$

so leisure moves in the opposite direction from wages, which means labor supply $\ell = 1 - z$ moves in the same direction as wages. This makes intuitive sense: You want to work harder when work pays better.

To make further progress, assume $R\beta = 1$ and define wage growth as $G = W_2/W_1 = (1 + g)$. Assume that young people tend to work about half of their waking hours $\ell_1 = (1/2)$ (remember vacations, weekends, etc!). Note that under these assumptions we can rewrite (1.2.27) as

$$\begin{aligned}(1 - \ell_2)G &= (1 - \ell_1) \\ g &= (1 + g)\ell_2 - \ell_1 \\ \ell_2 &= (g + \ell_1)/(1 + g) \\ &= (2g + 1)/2(1 + g)\end{aligned}\tag{1.2.28}$$

Empirically, wages in the U.S. tend to grow between youth and middle age by a factor of $G \approx 2 - 4$ (depending on occupation and education), so $g \approx 1 - 3$, but labor supply is about the same for 55 year olds as for 25 year olds, $\ell_2 \approx \ell_1$.

Suppose for analysis that $g = 2$. Then (1.2.28) becomes

$$\ell_2 = (5/6)\tag{1.2.29}$$

so the theory says middle aged people work more than young people by $(2/6)/(3/6) = 2/3$. This is of course absurd: it implies that middle aged people would barely have time to breathe because they were working so hard.

One objection to this analysis is that it assumed $R\beta = 1$, which implies that consumption when young equals consumption when middle aged. In fact, on average consumption grows by about the same amount as wages between youth and middle age. So perhaps the right assumption is $R\beta/G = 1$. Under this assumption, we obviously have $\ell_2 = \ell_1$, matching the empirical fact.

However, there is predictably different wage growth across occupations and education groups. Write $G_i = G\Gamma_i$, where Γ_i now will differ for people in different occupations indexed by i , and plausible values range from $\Gamma = 0.5$ (manual laborers) to $\Gamma = 1.5$ (doctors), leaving the average value of Γ across the two groups at $\Gamma = 1$. It is an empirical fact that the magnitude of variations in labor supply across these groups is rather small, both in youth and in middle age.

Assuming $R\beta/G = 1$, rewrite (1.2.27) for each occupation as

$$(1 - \ell_2)\Gamma_i = (1 - \ell_1)\tag{1.2.30}$$

For $\Gamma_i = 0.5$, if $\ell_1 = 1/2$ we have

$$(1 - \ell_2)0.5 = 1/2\tag{1.2.31}$$

implying $\ell_2 = 0$ - manual laborers would work zero hours. However, if $\Gamma = 1.5$ so that

$$\begin{aligned}(1 - \ell_2)(3/2) &= 1/2 \\ (1 - \ell_2)3 &= 1 \\ \ell_2 &= 2/3\end{aligned}\tag{1.2.32}$$

so doctors would be working much harder when middle aged than when young. Thus, the theory says that if labor supplies are equal when young (which is approximately true), they should differ drastically by middle age (which is not remotely true). That is, lifetime labor supply does not seem to respond very much to predictable variation in lifetime wages. This is described in the literature as a “small intertemporal elasticity of labor supply.”

1.3 The Diamond OLG Model

This section and the associated Jupyter notebook, DiamondOLG, present a canonical overlapping generations (OLG) model, like the one originally proposed by Diamond [1965], building on Samuelson [1958].³

1.3.1 Setup

The economy has the following features:

1. Two generations are alive at any point in time, the young (age 1) and old (age 2).
2. The size of the *young* generation in period t is given by $\mathbf{N}_t = \mathbf{N}_0 \Xi^t$ (note that $\mathbf{N}$ denotes only the young population, not the entire population, and this formula assumes constant population growth since time 0).
3. Households work only in the first period of life, earning income $Y_{1,t}$. They earn no income in the second period of life ($Y_{2,t+1} = 0$).
4. They consume part of their first-period income and save the rest to finance their consumption when old.
5. The assets of the young at the end of period t are the source of the capital used for aggregate production in period $t + 1$, $K_{t+1} = \mathbf{N}_t a_{1,t}$ where $a_{1,t}$ is the assets per young household *after* their consumption in period 1. (For convenience, we assume that there is no depreciation). Note that lower-case letters denote per-capita quantities, a convention used throughout.
6. The old in period t own the entire capital stock and (because they have no bequest motive) will consume it all, so dissaving by the old in period t will be $\mathbf{N}_{t-1} a_{1,t-1} = K_t$ (it is $\mathbf{N}_{t-1}$ because the old in t were young in $t - 1$). The old do receive interest on their capital, so their consumption will be K_t plus the interest income rK_t , but the rK_t component does not affect saving because it is part of both income and consumption.
7. Labor and capital markets are perfectly competitive and the aggregate production technology is CRS, $Y = F(K, L)$ (recall that this implies that $F(K, L) = F_L L + F_K K$, which is Eulers Theorem).

1.3.2 Analysis

Let's normalize everything by the period- t young population $\mathbf{N}_t$, writing normalized variables in lower case. Thus the per-young-capita aggregate production function becomes

$$f(k_t) \equiv F(K_t, \mathbf{N}_t) / \mathbf{N}_t = F(K_t / \mathbf{N}_t, 1). \quad (1.3.1)$$

The perfect competition assumption implies that wages and net interest rates are equal to the marginal products of labor and capital, respectively:

$$\begin{aligned} W_t &= f(k_t) - k_t f'(k_t), \\ r_t &= f'(k_t). \end{aligned} \quad (1.3.2)$$

To make further progress, we need to make specific assumptions about the utility function and the aggregate production function. Assume that utility is CRRA, $u(\bullet) = \bullet^{1-\rho} / (1 - \rho)$ and assume a Cobb-Douglas aggregate production function $F(K, L) = K^\varepsilon L^{1-\varepsilon} \Rightarrow f(k) = k^\varepsilon$.

In this case we can solve for wages and interest rates:

³For a remarkably clear statement of the questions addressed by OLG models, see Jefferson [1789]. For a review of the influence of Samuelson [1958]'s model, see Weil [2008].

$$\begin{aligned} W_t &= (1 - \varepsilon)k_t^\varepsilon \\ r_t &= \varepsilon k_t^{\varepsilon-1}. \end{aligned} \tag{1.3.3}$$

The individual's maximization problem yields the Euler equation (recall that $R = 1 + r$, an exception to the convention where lower-case does not signify per-capita):

$$u'(c_{1,t}) = \beta R_{t+1} u'(c_{2,t+1}). \tag{1.3.4}$$

Now let's assume that utility is logarithmic, $\rho = 1$. Beginning with consumption in levels and then moving to per-capita, this implies

$$\begin{aligned} c_{1,t} &= \frac{W_{1,t} + \overbrace{W_{2,t+1}}^{=0} / R_{t+1}}{1 + \beta} \\ c_{1,t} &= \frac{W_{1,t}}{1 + \beta} \\ a_{1,t} &= W_{1,t} - c_{1,t} \\ &= W_{1,t}(1 - 1/(1 + \beta)) \\ &= W_{1,t}(\beta/(1 + \beta)) \\ &= (1 - \varepsilon)k_t^\varepsilon \left(\frac{\beta}{1 + \beta} \right) \\ \overbrace{k_{t+1}}^{=a_{1,t}/\Xi} &= k_t^\varepsilon \left[\frac{(1 - \varepsilon)\beta}{\Xi(1 + \beta)} \right] \\ \frac{dk_{t+1}}{dk_t} &= k_t^{\varepsilon-1} \left[\frac{\varepsilon(1 - \varepsilon)\beta}{\Xi(1 + \beta)} \right]. \end{aligned} \tag{1.3.5}$$

The steady-state will be the point where $k_{t+1} = k_t$. For convenience, define a constant

$$\mathcal{Q} = \frac{(1 - \varepsilon)\beta}{\Xi(1 + \beta)}, \tag{1.3.6}$$

allowing us to rewrite (1.3.5) as

$$k_{t+1} = \mathcal{Q}k_t^\varepsilon. \tag{1.3.7}$$

Then the steady-state will be the point where $k_{t+1} = k_t = \bar{k}$

$$\begin{aligned} \bar{k} &= \mathcal{Q}\bar{k}^\varepsilon \\ \bar{k} &= \mathcal{Q}^{1/(1-\varepsilon)}. \end{aligned} \tag{1.3.8}$$

Dynamics of the model can be analyzed using a simple figure relating the capital stock per capita in period $t + 1$ to that in period t . The solid locus is a graph of equation (1.3.5). We depict the 45 degree line because it indicates the set of “steady-state” points where $k_{t+1} = k_t$ and thus any intersection of the 45 degree line with the $k_{t+1}(k_t)$ function indicates a steady-state of the model.

The experiment traced out in the figure is as follows. We start the economy in period $t = 0$ with capital per capita of k_0 , which, from (1.3.5), implies a certain value k_1 for capital in period $t + 1 = 1$. Now think about period 1 becoming t and period 2 becoming $t + 1$. To find the correct level of capital implied by the model in period $t + 2$ we need to find the point on the 45 degree line that corresponds to k_1 , then go vertically up from there to find $k_{t+1} = k_2$. When the same set of gyrations is repeated the result is that the level of capital converges to the steady-state level $\bar{k}$.

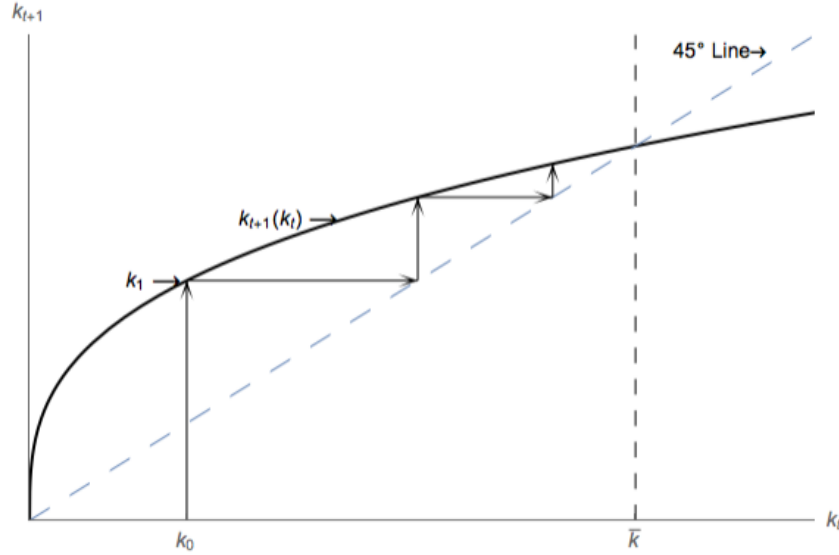


Figure 1.2: Convergence of OLG Economy to Steady State

1.3.3 Social Optimum

We have determined the outcome that will arise in a perfectly competitive economy in which households optimally choose their behavior given market prices with no government intervention.

Often in macroeconomic analysis this constellation of assumptions yields a conclusion that the steady state is optimal (and dynamics are also optimal) according to some plausible criteria. We now examine the optimality properties of the OLG model outcome.

As a preliminary, let's define the lifetime utility experienced by the young generation at time t as

$$v_t = u(c_{1,t}) + \beta u(c_{2,t+1}). \quad (1.3.9)$$

Suppose our definition of optimality reflects the choices that would be made by a benevolent social planner who maximizes a social welfare function of the form⁴

$$V_t = \beta u(c_{2,t}) + \sum_{n=0}^{\infty} \beta^n v_{t+n} \quad (1.3.10)$$

subject to the economy's aggregate resource constraint

$$\underbrace{K_t + F(K_t, N_t)}_{\text{Sources}} = \underbrace{K_{t+1} + N_t c_{1,t} + N_{t-1} c_{2,t}}_{\text{Uses}} \quad (1.3.11)$$

where the Hebrew letter $\beth$ reflects the social planner's discount factor and the planner must allocate the society's resources ("Sources") between consumption of the two generations alive at time t and the capital stock in period $t + 1$ ("Uses").

The idea is that the social planner cares about every generation's lifetime happiness, but discounts the happiness of future generations. (We will discuss why discounting is necessary later in the class).

It is possible to show (using methods not described in this section; see Blanchard and Fischer [1989] for details) that the socially optimal steady state is characterized by the equation

$$1 + f'(\bar{k}^*) = \beta \beth^{-1} \quad (1.3.12)$$

In the case of our Cobb-Douglas production function, this becomes

⁴The β multiplying the level of utility for the old generation at time t is necessary to prevent the social planner's problem from exhibiting time inconsistency.

$$\begin{aligned}
(\bar{k}^*)^{\varepsilon-1}\varepsilon &= \Xi\bar{\omega}^{-1} - 1 \\
\bar{k}^* &= \left(\frac{\Xi\bar{\omega}^{-1} - 1}{\varepsilon} \right)^{1/(\varepsilon-1)}.
\end{aligned} \tag{1.3.13}$$

Comparing this to the outcome that will actually arise,

$$\bar{k} = \left(\frac{\Xi(1+\beta)}{(1-\varepsilon)\beta} \right)^{1/(\varepsilon-1)}, \tag{1.3.14}$$

our point is that there is no particular relationship between the socially optimal outcome and the actual equilibrium outcome that will arise if the social planner is not involved. The actual outcome could have too little capital or too much, and there is no reason to expect it to be the “right” amount.

You might respond by saying that our definition of optimality here is too strong; we might hope that the economy would at least be able to avoid a Pareto inefficient outcome, even if we can’t expect perfect optimality according to the preferences of some mythical Godlike “social planner.”

It turns out, however, that even Pareto efficiency is not guaranteed. (In this context, Pareto efficiency must be defined across generations: The economy is Pareto efficient if there is no way to make one generation better off without making another generation worse off).

To examine Pareto efficiency, start by rewriting the aggregate DBC by dividing by the size of the labor force at time t :

$$k_t + f(k_t) = \Xi k_{t+1} + c_{1,t} + c_{2,t}/\Xi \tag{1.3.15}$$

Define an index of aggregate per capita consumption as

$$c_t = c_{1,t} + c_{2,t}/\Xi \tag{1.3.16}$$

In steady state, $k_{t+1} = k_t = \bar{k}$, so if $\bar{c}$ is the steady-state level of c_t then the accumulation equation implies

$$\begin{aligned}
\bar{k} + f(\bar{k}) &= \Xi\bar{k} + \bar{c} \\
f(\bar{k}) &= \underbrace{(\Xi - 1)}_{\xi} \bar{k} + \bar{c} \\
f(\bar{k}) - \xi\bar{k} &= \bar{c}
\end{aligned} \tag{1.3.17}$$

Now consider the effects of a change in $\bar{k}$ on $\bar{c}$:

$$\left(\frac{d\bar{c}}{d\bar{k}} \right) = f'(\bar{k}) - \xi. \tag{1.3.18}$$

There exists a $\bar{k}$ which maximizes per-capita steady-state consumption:

$$\max_{\bar{k}} f(\bar{k}) - \xi\bar{k} \tag{1.3.19}$$

whose solution is obtained from the FOC

$$\begin{aligned}
\varepsilon\bar{k}^{\varepsilon-1} &= \xi \\
\bar{k}^{**} &= (\xi/\varepsilon)^{1/(\varepsilon-1)},
\end{aligned} \tag{1.3.20}$$

and note that this means that for $\bar{k} \geq \bar{k}^{**}$ an extra bit of capital actually requires a *decline* in steady-state consumption. An economy in this circumstance of excessive saving is called “dynamically inefficient.”

Note further that there is actually a $\bar{k}$ so large that consumption would have to be zero:

$$\begin{aligned}
\bar{k}^\varepsilon &= \xi \bar{k} \\
\bar{k}^{\varepsilon-1} &= \xi \\
\bar{k} &= \xi^{1/(\varepsilon-1)}.
\end{aligned}
\tag{1.3.21}$$

These points are illustrated graphically in the remaining figure.

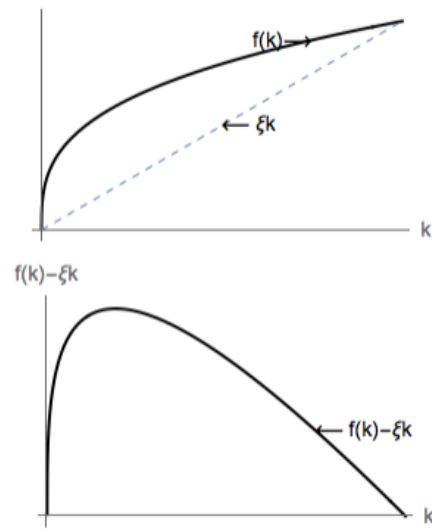


Figure 1.3: Gross and Net Per Capita Output as a Function of k

1.4 Generational Accounts and the Government

1.4.1 The Government Budget Constraint

Consider a government that raises taxes T_t , makes expenditures X_t , and has an outstanding stock of debt D_t at the beginning of period t , on which it must pay interest at rate r_t . The government can run a deficit only by raising funds via the issuing of new bonds (the alternative, printing money, leads to inflation and is the subject of monetary economics).

The government's Dynamic Budget Constraint (DBC) is given by

$$\begin{aligned} \overbrace{D_{t+1} - D_t}^{\text{Deficit}} &= \overbrace{(X_t + r_t D_t)}^{\text{Outlays}} - T_t \\ D_{t+1} &= X_t + R_t D_t - T_t \\ D_t &= \left(\frac{D_{t+1} + T_t - X_t}{R_t} \right). \end{aligned} \tag{1.4.1}$$

But we can obtain a similar formula for D_{t+1} in terms of D_{t+2} , and substitute it into (1.4.1). Continued substitution gives

$$\begin{aligned} D_t &= \overbrace{(T_t - X_t)}^{\equiv Z_t} / R_t + (T_{t+1} - X_{t+1}) / R_t R_{t+1} + \dots \\ &= Z_t / R_t + Z_{t+1} / R_t R_{t+1} + \dots \\ R_t D_t &= \mathbb{P}_t(Z) \\ &= \mathbb{P}(\text{Govt Primary Surpluses}) \end{aligned} \tag{1.4.2}$$

where $\mathbb{P}$ denotes the present discounted value. This equation says the government must plan to repay its debts, but places no constraint on what the government can do in any single period. Unlike households, the government has an infinite horizon and can obligate future generations to pay for the spending of current generations (Japan once offered 100-year mortgages, illustrating how intergenerational obligations can extend across multiple lifetimes). This can be rewritten

$$\mathbb{P}_t(X) = \mathbb{P}_t(T) - R_t D_t. \tag{1.4.3}$$

Equation (1.4.3) should look familiar: recall that in the consumption problem we had an Intertemporal Budget Constraint that said

$$\mathbb{P}_t(C) = \mathbb{P}_t(Y) + R_t K_t \tag{1.4.4}$$

where K_t is the beginning-of-period level of capital wealth (before interest has been earned), and $\mathbb{P}_t(Y) = H_t$ is human wealth.

In each case, the PDV of expenditures must be equal to the PDV of income plus current wealth. Thus, equations (1.4.2) and (1.4.3) are different ways to express the Government Intertemporal Budget Constraint (GIBC). The R_t multiplying debt reflects the timing convention that interest payments occur at the beginning of the period; for the consumer we were thinking about the situation after any interest income was received. The sign difference reflects the fact that D is debt while B represents asset balances.

Now let's suppose that the only kind of expenditures the government engages in are transfers, so that X_t simply reflects money handed out to some members of the population in period t . Then Z_t will be equal to total net transfers among the members of the population at period t (this is the primary surplus if we think of government activities more generally). Note that there is nothing that says that Z_t must be positive or negative in any particular period. The GIBC only places restrictions on the present discounted value of net transfers.

The fact that government only has to satisfy the GIBC means that the government can potentially treat different generations very differently from each other. It is therefore useful to have a mechanism to keep track of how different generations are treated. The standard way of doing this is to construct a set of 'generational accounts,' as initially proposed by Auerbach et al. [1991].

If we assume that consumers live two-period lives, the generational account for the generation born at time t is:

$$\begin{aligned}\bar{Z}_t &= Z_{1,t} + Z_{2,t+1}/R_{t+1} \\ &= \text{PDV of lifetime taxes net of transfer payments.}\end{aligned}\tag{1.4.5}$$

Note that either of these Z terms can be negative, which would signify net transfers to households rather than taxes collected.

In the US and most other countries, working-age people pay more in taxes than they receive in transfers, so $Z_{1,t}$ is positive, while old people receive more in transfers than they pay in taxes, so $Z_{2,t}$ is negative. The largest such transfers in the US are Social Security and Medicare.

Note now that the aggregate total of net transfers can be subdivided into the net transfers of the two age groups in the population,

$$Z_t = Z_{1,t} + Z_{2,t}.\tag{1.4.6}$$

Now write out the GIBC (1.4.2) explicitly:

$$\begin{aligned}\mathbb{P}_t(Z) &= Z_t + Z_{t+1}/R_{t+1} + \dots \\ &= Z_{1,t} + Z_{1,t+1}/R_{t+1} + Z_{1,t+2}/R_{t+1}R_{t+2} + \dots \\ &\quad + Z_{2,t} + Z_{2,t+1}/R_{t+1} + Z_{2,t+2}/R_{t+1}R_{t+2} + \dots \\ &= Z_{2,t} + [Z_{1,t} + Z_{2,t+1}/R_{t+1}] + [Z_{1,t+1} + Z_{2,t+2}/R_{t+2}]/R_{t+1} + \dots \\ &= Z_{2,t} + \bar{Z}_t + \bar{Z}_{t+1}/R_{t+1} + \bar{Z}_{t+2}/R_{t+1}R_{t+2} + \dots\end{aligned}\tag{1.4.7}$$

which again shows that the GIBC is consistent with any treatment of any particular generation; any pattern of generational accounts that satisfies the GIBC is feasible.

1.4.2 Social Security and Generational Accounts

Consider an economy that initially has no government so that $Z_{1,t} = Z_{2,t} = Z_{2,t-1} = 0$. Now consider introducing a Pay As You Go (PAYG) Social Security system at date s , which is to remain of constant size forever after introduction,

$$\begin{aligned}Z_{2,t} &= -Z_{1,t} \neq 0 \quad \forall t \geq s \\ Z_{1,t+1} &= Z_{1,t}.\end{aligned}\tag{1.4.8}$$

Consider the generation born at time $s-1$. It paid nothing into the Social Security system when young, yet gets $Z_{2,s}$ out when old. Its generational account is therefore

$$\begin{aligned}\bar{Z}_{s-1} &= Z_{1,s-1} + Z_{2,s}/R_s \\ &= 0 + Z_{2,s}/R_s\end{aligned}\tag{1.4.9}$$

so this generation benefits from the introduction of SS because it paid no taxes yet receives benefits. Since $Z_{2,s}$ is negative (a transfer to the old), this generation is better off than without Social Security.

The generational accounts for succeeding generations are

$$\begin{aligned}\bar{Z}_t &= Z_{1,t} + Z_{2,t+1}/R_{t+1} \\ &= Z_{1,t}(1 - 1/R_{t+1}) \\ &= r_{t+1}Z_{1,t}/R_{t+1}\end{aligned}\tag{1.4.10}$$

so future generations are worse off by this amount. Since taxes are positive, the effect on the lifetime budget constraint is the negative of the expression on the RHS of (1.4.10).

The reason the introduction of Social Security makes future generations worse off is that without SS they could have invested the amount $Z_{1,t}$ and earned interest on it of $r_{t+1}Z_{1,t}$ in period 2. Now the money is taken away from them when young and returned *without interest* when old. Thus, the loss is precisely the loss in interest income on $Z_{1,t}$ in period $t+1$, discounted back to the present.

Note that if there is zero population growth, the foregoing analysis all holds in per-capita terms as well, so that the per-capita change in generational accounts from introducing Social Security is

$$\bar{z}_t = r_{t+1}z_{1,t}/R_{t+1} \quad (1.4.11)$$

Effects of Population Growth

If there is perpetual population growth, it is possible to finance a positive rate of return on Social Security contributions. It will often be convenient to write generational accounts in per-capita terms rather than in aggregate terms, using lower-case letters for per-capita quantities. Define per capita contributions as (note that $z_{2,t+1}$ is divided by L_t rather than L_{t+1} to express the return from the perspective of the contributing generation):

$$z_{1,t} = Z_{1,t}/L_t \quad (1.4.12)$$

and assume there is constant population growth, $\Xi = L_{t+1}/L_t$. If we assume that Social Security taxes per capita are constant, then we can achieve a positive rate of return on Social Security contributions equal to the growth rate of population:

$$\begin{aligned} z_{2,t+1} &= Z_{2,t+1}/L_t \\ &= -Z_{1,t+1}/L_t \\ &= -\left(\frac{Z_{1,t+1}}{L_{t+1}}\right)\left(\frac{L_{t+1}}{L_t}\right) \\ &= -z_{1,t+1}\Xi = -z_{1,t}\Xi. \end{aligned} \quad (1.4.13)$$

Not only does this prove that it is *possible* for the Social Security system to pay a rate of return equal to the rate of population growth; it proves that the *only* rate of return that is consistent with constant per-capita taxes on the young is a rate of return of Ξ . In an economy with perpetual population growth, it is not only possible but *necessary* to pay a positive return.

Effects of Productivity Growth and Population Growth

Suppose there is wage growth G between t and $t+1$, and suppose that workers contribute a constant *percentage* of their incomes to the Social Security system, $z_{1,t} = \zeta W_{1,t}$. In this case it is possible to earn a rate of return on SS contributions equal to the product of the growth factor for wages and the growth factor for population:

$$\begin{aligned} z_{1,t} &= \zeta W_{1,t} \\ W_{1,t+1} &= GW_{1,t} \\ z_{2,t+1} &= -Z_{1,t+1}/L_t \\ &= -(Z_{1,t+1}/L_{t+1})(L_{t+1}/L_t) \\ &= -\zeta \underbrace{W_{1,t+1}}_{=GW_{1,t}} \Xi \\ &= -\zeta W_{1,t} G \Xi \\ &= -z_{1,t} G \Xi \end{aligned} \quad (1.4.14)$$

so viewed from the perspective of the young generation in period t , their Social Security contributions are returned to them larger by a factor of $G\Xi$ than what they paid in; the effective rate of return is therefore $G\Xi$.

Generational Accounts in a Growing Economy

Now consider the per-capita generational accounts in an economy with constant population growth and constant wage growth and a Social Security system that imposes a constant tax of ζ on the wages of the young:

$$\begin{aligned}
\bar{z}_t &= z_{1,t} + z_{2,t+1}/R_{t+1} \\
&= \zeta W_{1,t} - G\Xi \zeta W_{1,t}/R_{t+1} \\
&= \zeta W_{1,t} (1 - G\Xi/R_{t+1}) \\
&= \zeta W_{1,t} \left(\frac{R_{t+1} - G\Xi}{R_{t+1}} \right).
\end{aligned} \tag{1.4.15}$$

Note that this expression will be *negative* if $G\Xi > R_{t+1}$, meaning that the introduction of a Social Security system with a positive tax rate ζ actually *improves* the lifetime budget constraint! This is another way of seeing that an economy is *dynamically inefficient* if the return factor for capital R is less than the product of the population growth and productivity growth factors. (Or, using approximations, the rate of return is less than the sum of the population growth rate and the productivity growth rate).

1.5 Social Security and Capital Accumulation

Consider a household with a 2-period lifetime, whose optimization problem is written in the most general way possible, with the last line reflecting the assumption that no labor income is earned in period 2:

$$\begin{aligned}
 v(b_{1,t}) &= \max_{\{c_{1,t}\}} u(c_{1,t}) + \beta u(c_{2,t+1}) \\
 &\text{s.t.} \\
 y_{1,t} &= (W_{1,t} - z_{1,t}) \\
 a_{1,t} &= y_{1,t} - c_{1,t} \\
 y_{2,t+1} &= (W_{2,t+1} - z_{2,t+1}) = -z_{2,t+1} \\
 c_{2,t+1} &= R_{t+1}a_{1,t} + y_{2,t+1}.
 \end{aligned} \tag{1.5.1}$$

Under logarithmic utility, the [Fisher two-period model](#) section shows that the solution to this problem is

$$c_{1,t} = \left(\frac{y_{1,t} + y_{2,t+1}/R_{t+1}}{1 + \beta} \right). \tag{1.5.2}$$

The only role of government in this economy is to run a Social Security program. Suppose that initially this economy had no Social Security system and we are interested in the effects of introducing a Pay-As-You-Go Social Security system that is expected to remain a constant size from generation to generation from now on: $z_{2,t+1} = -z_{1,t+1}$ while $z_{1,t+1} = z_{1,t}$, so that taxes are greater than transfers when young and transfers are greater than taxes when old.

The effects of Social Security on first period consumption can be seen by writing out explicitly the value for $c_{1,t}$ from equation (1.5.2), substituting the definitions of $y_{1,t}$ and $y_{2,t+1}$:

$$\begin{aligned}
 c_{1,t} &= (W_{1,t} - z_{1,t} - z_{2,t+1}/R_{t+1}) / (1 + \beta) \\
 &= \left(W_{1,t} - \underbrace{r_{t+1}z_{1,t}/R_{t+1}}_{\text{consume less b/c poorer}} \right) / (1 + \beta)
 \end{aligned} \tag{1.5.3}$$

where the expression with the underbrace comes from the effect of introducing a constant-sized PAYG Social Security system in the [generational accounts](#) section. If taxes paid when young $z_{1,t}$ are positive (as they are after the introduction of the Social Security system) and the interest rate is positive, the expression with the underbrace is a positive number, and since it is being subtracted from $W_{1,t}$ it is clear that consumption in the first period of life will *decline* with the introduction of the Social Security system. The reason is that the household is poorer in a lifetime sense: the rate of return on Social Security contributions is lower than the market interest rate.

Does the decline in consumption mean the saving rate rises? No: saving is *after-tax* income minus consumption, and net taxes on the young have risen. For saving we have

$$\begin{aligned}
 a_{1,t} &= \overbrace{(W_{1,t} - z_{1,t})}^{\text{Less after-tax income}} - \overbrace{c_{1,t}}^{\text{Lower consumption b/c poorer}} \\
 &= W_{1,t}(1 - 1/(1 + \beta)) - z_{1,t} + [r_{t+1}z_{1,t}/R_{t+1}]/(1 + \beta) \\
 &= W_{1,t} \left(\frac{\beta}{1 + \beta} \right) - z_{1,t} \left(1 - \frac{r_{t+1}}{R_{t+1}(1 + \beta)} \right) \\
 &= W_{1,t} \left(\frac{\beta}{1 + \beta} \right) - z_{1,t} \left(\frac{R_{t+1}(1 + \beta) - r_{t+1}}{R_{t+1}(1 + \beta)} \right) \\
 &= W_{1,t} \left(\frac{\beta}{1 + \beta} \right) - z_{1,t} \left(\frac{1 + R_{t+1}\beta}{R_{t+1}(1 + \beta)} \right).
 \end{aligned} \tag{1.5.4}$$

So if $z_{1,t} > 0$ then saving is less than before the introduction of Social Security.

Now consider the implications in a Diamond [1965] OLG model where saving is the source of capital accumulation. Suppose there is no population growth so that

$$\begin{aligned}
 K_{t+1} &= a_{1,t} \\
 &= W_{1,t} \left(\frac{\beta}{1+\beta} \right) - z_{1,t} \left(\frac{1+R_{t+1}\beta}{R_{t+1}(1+\beta)} \right) \\
 &= (1-\varepsilon)K_t^\varepsilon \left(\frac{\beta}{1+\beta} \right) - z_{1,t} \left(\frac{1+R_{t+1}\beta}{R_{t+1}(1+\beta)} \right) \\
 &= \mathcal{Q}K_t^\varepsilon - z_{1,t} \left(\frac{1+R_{t+1}\beta}{R_{t+1}(1+\beta)} \right)
 \end{aligned} \tag{1.5.5}$$

where $\mathcal{Q} = (1-\varepsilon)\beta/(1+\beta)$ as before in the [Diamond OLG model](#) section.

Thus the capital accumulation curve is shifted down (the figure simplifies by assuming a constant downward shift, though strictly speaking R_{t+1} depends on k_{t+1}). The dynamics of the introduction of Social Security are captured in the figure, under the assumption that the economy was at its steady-state equilibrium level $\bar{k}$ before the Social Security system was introduced. The effect of introduction is an immediate increase in consumption, as the old generation spends everything it gets and the young generation doesn't need to do as much retirement saving as before. Over time the economy will converge to its new, lower level of capital $\bar{\bar{k}}$.

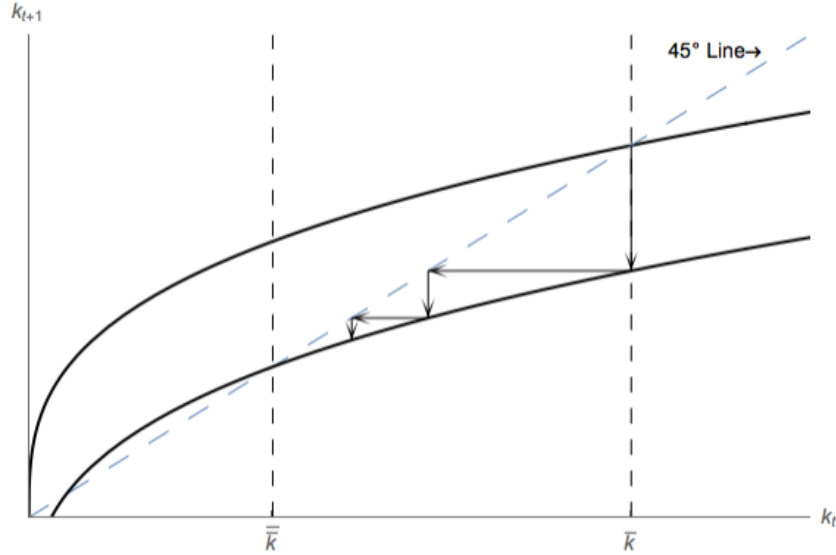


Figure 1.4: Convergence of OLG Economy After Intro of Social Security

1.6 The Envelope Theorem and the Euler Equation

This section shows how the Envelope theorem is used to derive the consumption Euler equation in a multiperiod optimization problem with geometric discounting and intertemporally separable utility.

The consumer's goal from the perspective of date t is to maximize the sum of discounted utilities, where geometric discounting means that utility n periods in the future is weighted by β^n :

$$\max \sum_{n=0}^{T-t} \beta^n u(c_{t+n}) \quad (1.6.1)$$

subject to the dynamic budget constraint

$$m_{t+1} = (m_t - c_t)R + y_{t+1}. \quad (1.6.2)$$

The problem can be written in Bellman equation form as

$$v_t(m_t) = \max_{\{c_t\}} u(c_t) + \beta v_{t+1}((m_t - c_t)R + y_{t+1}). \quad (1.6.3)$$

The first order condition for (1.6.3) can be written as

$$0 = u'(c_t) + \overbrace{\left(\frac{dm_{t+1}}{dc_t} \right)}^{=-R} \beta v'_{t+1}(m_{t+1}) \quad (1.6.4)$$

$$u'(c_t) = R\beta v'_{t+1}(m_{t+1}),$$

where the derivative $dm_{t+1}/dc_t = -R$ follows from (1.6.2). We can define a function $c_t(m)$ that returns the c_t that solves the max problem for any given m_t . That is, for $c_t = c_t(m_t)$ the first order condition (1.6.4) will hold so that

$$u'(c_t(m_t)) - R\beta v'_{t+1}((m_t - c_t(m_t))R + y_{t+1}) = 0. \quad (1.6.5)$$

Now define a function $\underline{v}_t$ (where the underscore indicates a weak lower bound on value, achieved only when c is chosen optimally):

$$\underline{v}_t(m_t, c_t) = u(c_t) + \beta v_{t+1}((m_t - c_t)R + y_{t+1}) \quad (1.6.6)$$

with partial derivatives

$$\underline{v}_t^c(m_t, c_t) \equiv \left(\frac{\partial \underline{v}_t}{\partial c_t} \right) = u'(c_t) - R\beta v_{t+1}^m((m_t - c_t)R + y_{t+1}) \quad (1.6.7)$$

$$\underline{v}_t^m(m_t, c_t) \equiv \left(\frac{\partial \underline{v}_t}{\partial m_t} \right) = R\beta v_{t+1}^m(m_{t+1}) \quad (1.6.8)$$

and note that by definition

$$v_t(m_t) = \underline{v}_t(m_t, c_t(m_t)). \quad (1.6.9)$$

The Chain Rule of differentiation tells us that

$$v'_t(m_t) \equiv v_t^m(m_t) \equiv \left(\frac{dv_t}{dm_t} \right) = \underline{v}_t^m(m_t, c_t(m_t)) + \left(\frac{\partial c_t(m_t)}{\partial m_t} \right) \underline{v}_t^c(m_t, c_t(m_t)). \quad (1.6.10)$$

Here's the key insight: The assumption that consumers are optimizing means that we will always be evaluating the value function and its derivatives at a c_t that satisfies the first-order optimality condition (1.6.5) (this reasoning would need modification if a liquidity constraint were binding). Thus we have from (1.6.7) that

$$\begin{aligned} \underline{v}_t^c(m_t, c_t(m_t)) &= u'(c_t(m_t)) - R\beta v'_{t+1}((m_t - c_t(m_t))R + y_{t+1}) \\ &= 0. \end{aligned} \quad (1.6.11)$$

This means that the second term in (1.6.10) is always equal to zero, so from (1.6.8) we obtain

$$v'_t(m_t) = R\beta v'_{t+1}(m_{t+1}). \quad (1.6.12)$$

Now notice that the RHS's of (1.6.4) and (1.6.12) are identical, so we can equate the left hand sides,

$$v'_t(m_t) = u'(c_t) \quad (1.6.13)$$

and since a corresponding equation will hold in period $t + 1$ we can rewrite (1.6.12) as

$$u'(c_t) = R\beta u'(c_{t+1}). \quad (1.6.14)$$

The general principle can be condensed into a rule of thumb by realizing that the Envelope theorem will always imply that the total derivative of a value function with respect to any choice variable must be equal to zero for optimizing consumers (because the first order condition holds). Thus we could have obtained the result immediately by treating c_t as though it were a constant (that is, treating the problem as though $c'_t(m_t) = 0$) and taking the derivative of Bellman's equation with respect to m_t directly. This leads immediately to the key result:

$$\begin{aligned} v_t(m_t) &= u(c(m_t)) + \beta v_{t+1}((m_t - c_t(m_t))R + y_{t+1}) \\ v'_t(m_t) &= \beta R v'_{t+1}(m_{t+1}). \end{aligned} \quad (1.6.15)$$

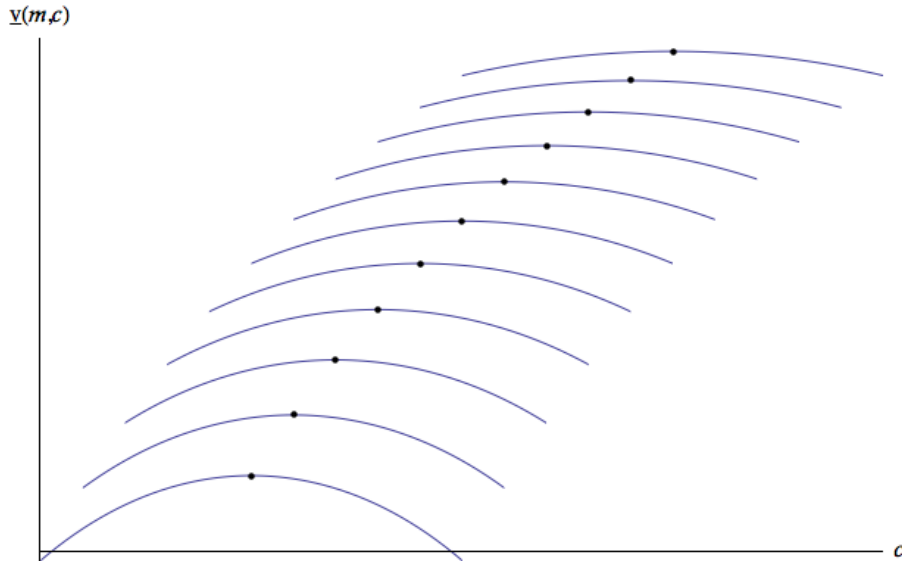


Figure 1.5: Illustration of the Envelope Theorem at Alternative Values of m

Each curve in the figure plots $\underline{v}(m, c)$ against c for a fixed value of m , with the dot marking the c that maximizes it; higher curves correspond to higher m , and the locus of dots traces out $v(m)$. The figure makes the Envelope theorem visible in the flatness of each curve at its peak. Consider the consequences of an increase in m : because the curve is flat where the dot sits, the consumer who spends the extra resources reaches almost exactly the same height as the consumer who saves all of them, so the rise in attainable utility is about the same whether the increment is consumed or not. The term involving $\partial c_t / \partial m_t$ can therefore be dropped.

1.7 Consumption Under Perfect Foresight and CRRA Utility

1.7.1 The Problem

This section solves the problem of a perfect foresight consumer with intertemporally separable CRRA utility $u(\bullet) = \bullet^{1-\rho}/(1-\rho)$ who discounts future utility geometrically by a factor β per period. The finite horizon solution, whose last period is T , extends to the infinite horizon case if intuitive “impatience” and “finite human wealth” conditions hold.

The consumer’s problem in period t is to

$$\max \sum_{n=0}^{T-t} \beta^n u(\mathbf{c}_{t+n}) \quad (1.7.1)$$

subject to the constraints below, in which we introduce $\mathbf{b}$ and $\mathbf{m}$ because it is useful to have a term for the consumer’s non-income resources at the beginning of the period,

$$\begin{aligned} \mathbf{a}_t &= \mathbf{m}_t - \mathbf{c}_t \\ \mathbf{b}_{t+1} &= \mathbf{a}_t R \\ \mathbf{m}_{t+1} &= \mathbf{b}_{t+1} + \mathbf{p}_{t+1} \end{aligned} \quad (1.7.2)$$

where $\mathbf{p}_{t+1}$ is “permanent labor income,” which always grows by a factor G :

$$\mathbf{p}_{t+1}/\mathbf{p}_t = G. \quad (1.7.3)$$

1.7.2 The Solution

It will be convenient to think of both market resources $\mathbf{m}_t$ and permanent noncapital (labor) income $\mathbf{p}_t$ as state variables in this problem. Bellman’s equation is⁵

$$v_t(\mathbf{m}_t, \mathbf{p}_t) = \max_{\mathbf{c}_t} \left\{ u(\mathbf{c}_t) + \beta v_{t+1} \left(\overbrace{(\mathbf{m}_t - \mathbf{c}_t)R + \mathbf{p}_{t+1}}^{=\mathbf{m}_{t+1}}, \mathbf{p}_{t+1} \right) \right\}. \quad (1.7.4)$$

The first order condition for this maximization is

$$u'(\mathbf{c}_t) = \beta \left(R v_{t+1}^{\mathbf{m}}(\mathbf{m}_{t+1}, \mathbf{p}_{t+1}) - \overbrace{\frac{d\mathbf{p}_{t+1}}{d\mathbf{c}_t}}^{=0} v_{t+1}^{\mathbf{p}}(\mathbf{m}_{t+1}, \mathbf{p}_{t+1}) \right), \quad (1.7.5)$$

and the Envelope theorem tells us that

$$v_t^{\mathbf{m}}(\mathbf{m}_t, \mathbf{p}_t) = R \beta v_{t+1}^{\mathbf{m}}(\mathbf{m}_{t+1}, \mathbf{p}_{t+1}). \quad (1.7.6)$$

But the right hand sides of (1.7.5) and (1.7.6) are identical, so

$$v_t^{\mathbf{m}}(\mathbf{m}_t, \mathbf{p}_t) = u'(\mathbf{c}_t) \quad (1.7.7)$$

and similar logic tells us that $v_{t+1}^{\mathbf{m}}(\mathbf{m}_{t+1}, \mathbf{p}_{t+1}) = u'(\mathbf{c}_{t+1})$, which (substituting u' for $v^{\mathbf{m}}$ in (1.7.6)) gives us the Euler equation for consumption:

$$\begin{aligned} u'(\mathbf{c}_t) &= R \beta u'(\mathbf{c}_{t+1}) \\ 1 &= R \beta \left(\frac{\mathbf{c}_{t+1}}{\mathbf{c}_t} \right)^{-\rho} \\ \left(\frac{\mathbf{c}_{t+1}}{\mathbf{c}_t} \right) &= (R \beta)^{1/\rho}. \end{aligned} \quad (1.7.8)$$

⁵Bellman’s equation relates v_t and v_{t+1} through the controls and the states, so it does not necessarily require writing out the budget constraint.

Thus, consumption grows in every period by a factor $\mathbf{P} \equiv (\mathbf{R}\beta)^{1/\rho}$, where we use the Old English letter $\mathbf{P}$ to measure what we will call the “absolute patience” factor. Specifically, if

$$\mathbf{P} < 1 \quad (1.7.9)$$

we will say that the consumer exhibits “absolute impatience” because this is the condition that guarantees that the level of consumption will be falling (and what better definition of absolute impatience could there be than deliberately spending so much that you will have to cut your spending in the future?). If $\mathbf{P} > 1$ the consumer exhibits “absolute patience” (the consumer wants to defer resources into the future in order to achieve consumption growth).

The Intertemporal Budget Constraint tells us⁶

$$\mathbb{P}_t^T(\mathbf{c}) = \mathbf{b}_t + \mathbb{P}_t^T(\mathbf{p}). \quad (1.7.10)$$

Fact FinSum can be used to show that the PDV of labor income (also called “human wealth” $\mathbf{h}_t$) is

$$\begin{aligned} \mathbf{h}_t &= \mathbb{P}_t^T(\mathbf{p}) = \sum_{n=0}^{T-t} \mathbf{R}^{-n} \mathbf{p}_{t+n} \\ &= \mathbf{p}_t \sum_{n=0}^{T-t} \mathbf{R}^{-n} \mathbf{G}^n = \mathbf{p}_t \sum_{n=0}^{T-t} (\mathbf{G}/\mathbf{R})^n \\ &= \mathbf{p}_t \left(\frac{1 - (\mathbf{G}/\mathbf{R})^{T-t+1}}{1 - (\mathbf{G}/\mathbf{R})} \right) \end{aligned} \quad (1.7.11)$$

while the PDV of consumption is

$$\begin{aligned} \mathbb{P}_t^T(\mathbf{c}) &= \sum_{n=0}^{T-t} \mathbf{R}^{-n} \mathbf{c}_{t+n} \\ &= \sum_{n=0}^{T-t} \mathbf{R}^{-n} \mathbf{c}_t ((\mathbf{R}\beta)^{1/\rho})^n \\ &= \mathbf{c}_t \sum_{n=0}^{T-t} [\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]^n \\ &= \mathbf{c}_t \left(\frac{1 - [\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]^{T-t+1}}{1 - [\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]} \right). \end{aligned} \quad (1.7.12)$$

We can solve the model by combining (1.7.12) and (1.7.11) using (1.7.10) to obtain:

$$\mathbf{c}_t = \underbrace{\left(\frac{1 - [\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]^{T-t+1}}{1 - [\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]} \right)}_{\equiv \kappa_t} \underbrace{\left[\mathbf{b}_t + \mathbf{p}_t \left(\frac{1 - (\mathbf{G}/\mathbf{R})^{T-t+1}}{1 - (\mathbf{G}/\mathbf{R})} \right) \right]}_{\equiv \mathbf{o}_t} \quad (1.7.13)$$

where κ_t is the marginal propensity to consume (MPC) out of overall (human plus nonhuman) wealth $\mathbf{o}_t$.

In order to apply InfSum to move to the infinite-horizon case ($T = \infty$), we need to impose the condition

$$\begin{aligned} \mathbf{G}/\mathbf{R} &< 1 \\ \mathbf{G} &< \mathbf{R}. \end{aligned} \quad (1.7.14)$$

⁶InfSum is the immediate implication of FinSum as $T \rightarrow \infty$. that the present discounted value of consumption must match the PDV of total resources:

Why? Because if income were expected to grow at a rate greater than the interest rate forever, then the PDV of future income would be infinite; with infinite human wealth, the problem has no well-defined solution. We henceforth call (1.7.14) the Finite Human Wealth Condition (FHWC).

Similarly, if consumption starts at a positive level and grows by the factor $\mathbf{P} = (R\beta)^{1/\rho}$, in order for the PDV of consumption to be finite we must impose:

$$\underbrace{\left(\frac{(R\beta)^{1/\rho}}{R} \right)}_{\mathbf{P}_R} < 1 \quad (1.7.15)$$

and we will henceforth call $\mathbf{P}_R$ the “return patience factor” whose log is the “return patience rate” $\mathbf{p}_r \equiv \log \mathbf{P}_R$ ($\mathbf{p}$ is the lower-case version of $\mathbf{P}$) and what (1.7.15) says is that the desired growth rate of consumption must be less than the interest rate in order for the model to have a well-defined solution. This condition therefore imposes a requirement that “impatience” be greater than some minimum amount. (For (much) more on the various definitions of impatience used in this section, their implications, and parallel conditions for models with uncertainty, see Carroll).

If both the RIC and the FHWC hold, then the model has a well-defined infinite horizon solution,⁷ as can be seen by realizing that

$$\begin{aligned} \lim_{T \rightarrow \infty} (G/R)^{T-t+1} &= 0 \\ \lim_{T \rightarrow \infty} (R^{-1}(R\beta)^{1/\rho})^{T-t+1} &= 0. \end{aligned} \quad (1.7.16)$$

Substituting these zeros into (1.7.13) yields

$$\begin{aligned} \mathbf{c}_t &= \left(1 - R^{-1}(R\beta)^{1/\rho} \right) \left[\mathbf{b}_t + \left(\frac{\mathbf{p}_t}{1 - (G/R)} \right) \right] \\ &= \left(1 - R^{-1}(R\beta)^{1/\rho} \right) (\mathbf{m}_t - \mathbf{p}_t + \mathbf{h}_t) \\ &= \underbrace{\left(\frac{R - (R\beta)^{1/\rho}}{R} \right)}_{\equiv \kappa} \mathbf{o}_t \end{aligned} \quad (1.7.17)$$

where $\mathbf{o}_t$ is the consumer’s “overall” or “total wealth,” the sum of human and nonhuman wealth, and κ is the infinite-horizon marginal propensity to consume.

Now consider the question “What is the level of $\mathbf{c}_t$ that will leave total wealth intact, allowing the same value of consumption in period $t + 1$ and forever after (that is, allowing $\mathbf{c}_{t+n} = \mathbf{c}_t \forall n > 0$)?”

The intuitive answer is that the wealth-preserving level of spending is exactly equal to the (properly conceived) interest earnings on one’s total wealth. We call this the “sustainable” level of consumption. The literature of the 1970s and 80s interpreted that quantity as “permanent income,” but we reserve that term here for $\mathbf{p}$ itself. Wealth in this framework behaves exactly like an asset that yields a dividend $\mathbf{p}$.

Because human wealth is exactly like any other kind of wealth in this perfect foresight framework, it is possible to work directly with the level of total wealth $\mathbf{o}$ to find the sustainable level of spending. Suppose we assume the consumer will spend fraction $\varkappa$ of total wealth in each period; the $\varkappa$ that leaves wealth intact will be given by $\varkappa$ in

⁷See Carroll for a discussion of the case where the conditions do not hold.

$$\begin{aligned}
\mathbf{o}_{t+1} &= (\mathbf{o}_t - \mathbf{c}_t)R \\
\bar{\mathbf{o}} &= (\bar{\mathbf{o}} - \varkappa \bar{\mathbf{o}})R \\
1 &= (1 - \varkappa)R \\
1/R &= (1 - \varkappa) \\
\varkappa &= 1 - 1/R \\
&= \left(\frac{R-1}{R} \right) \\
&= r/R.
\end{aligned} \tag{1.7.18}$$

Thus, the consumer can spend only the interest earnings r on wealth, divided by the return factor R . (The division occurs because the requirement is to be able to spend the same amount *next* period, so you need to account for the time cost of today's spending by dividing by R which connects today's spending to tomorrow's wealth.)

Note that the coefficient multiplying total wealth in (1.7.17) is also divided by R . Thus, whether the consumer is spending more than the sustainable amount, exactly the sustainable amount, or less than the sustainable amount depends upon whether the numerator in (1.7.17) is greater than, equal to, or less than r . As noted before, the consumer will be “absolutely impatient” if

$$\begin{aligned}
R - (R\beta)^{1/\rho} &> r \\
1 - (R\beta)^{1/\rho} &> 0 \\
1 &> (R\beta)^{1/\rho}.
\end{aligned} \tag{1.7.19}$$

Finally, if $R\beta = 1$ (which is to say, the interest rate exactly offsets the time preference rate), then $(R\beta)^{1/\rho} = 1$ regardless of the value of ρ so that the consumer is “poised” on the knife-edge between patience and impatience. We refer to such a consumer as “absolutely poised.” Similarly, we say that a consumer for whom $\mathbf{P}_R = 1$ is “return poised.”

The two conditions do different work. Return impatience guarantees a positive marginal propensity to consume, while absolute impatience guarantees a falling level of consumption. Whenever $r > 0$, return impatience holds even for a consumer who is poised with respect to absolute patience.

(The consumer will be impatient, spending more than his income, if $R\beta < 1$, and patient, spending less than his income, if $R\beta > 1$. Income here means income inclusive of the interest earnings on total wealth.)

Equation (1.7.17) can be simplified into something a bit easier to handle by making some approximations. If $\beta = 1/(1 + \vartheta)$, then we can use facts from the [MathFacts section](#) to discover that

$$\begin{aligned}
\log(R\beta)^{1/\rho}/R &= (1/\rho)(\log R + \log[1/(1 + \vartheta)]) - \log R \\
&= (1/\rho)(\log(1 + r) + \log 1 - \log(1 + \vartheta)) - \log R \\
&\approx \rho^{-1}(r - \vartheta) - r \\
(R\beta)^{1/\rho}/R &\approx 1 + (\rho^{-1}(r - \vartheta) - r).
\end{aligned} \tag{1.7.20}$$

Substituting this into (1.7.17) gives

$$\mathbf{c}_t \approx \left(r - \rho^{-1}(r - \vartheta) \right) \mathbf{o}_t. \tag{1.7.21}$$

From this we can see again that whether the consumer is return patient, return poised, or return impatient depends on the relationship between r and ϑ . Note also that if $\rho = \infty$ then the consumer is infinitely averse to changing the level of consumption, and so once again the consumer spends exactly the sustainable amount. (This consumer is “absolutely poised” but “return impatient”).

Now a brief digression on what “income” means in this model. Suppose for simplicity that the consumer had no capital assets (“bank balances” $\mathbf{b}_t = 0$), and suppose that income was expected to stay constant at level $\mathbf{p}_{t+n} = \mathbf{p} \forall n > 0$ forever. In this case human wealth would be:

$$\begin{aligned}
\mathbf{h}_t &= \mathbf{p} + \mathbf{p}/R + \mathbf{p}/R^2 + \dots \\
&= \mathbf{p}(1 + 1/R + 1/R^2 + \dots) \\
&= \mathbf{p} \left(\frac{1}{1 - 1/R} \right) \\
&= \mathbf{p} \left(\frac{R}{R - 1} \right) \\
&= \mathbf{p} \left(\frac{R}{r} \right).
\end{aligned} \tag{1.7.22}$$

We found in equation (1.7.18) that the level of consumption that leaves “wealth” $\mathbf{o}_t$ intact was

$$\begin{aligned}
\mathbf{c}_t &= \left(\frac{r}{R} \right) \mathbf{o}_t \\
&= \left(\frac{r}{R} \right) \underbrace{(\mathbf{b}_t)}_{=0} + \mathbf{h}_t \\
&= \left(\frac{r}{R} \right) \mathbf{p} \left(\frac{R}{r} \right) \\
&= \mathbf{p}.
\end{aligned} \tag{1.7.23}$$

So in this case, spending the “interest income on human wealth” corresponds to spending exactly your labor income. This seems less mysterious if you think of income $\mathbf{p}_t$ as the “return” on your human capital, which is an asset whose value is $\mathbf{h}_t$. If you “capitalize” your stream of income using the interest factor R and then spend the interest income on the capitalized stream, it stands to reason that you are spending the flow of income from that source.

With constant $\mathbf{p}$ we can rewrite (1.7.21) as

$$\mathbf{c}_t \approx \left(r - \rho^{-1}(r - \vartheta) \right) \left(\mathbf{b}_t + \mathbf{p} \left(\frac{R}{r} \right) \right). \tag{1.7.24}$$

r appears three times in this equation, which correspond (in order) to the income effect, the substitution effect, and the human wealth effect. To see this, note that an increase in the first r reflects an increase in the payout rate on total wealth (set $\mathbf{p} = 0$ and refer to our formula above for $\varkappa$, realizing that for small r , $r/R \approx r$.) That is, it simply reflects the consequence for consumption of an increase in interest income, so it captures the “income effect” of interest rates. The second term corresponds to the substitution effect, as can be seen from its dependence on the intertemporal elasticity of substitution ρ^{-1} . Finally, the $\mathbf{p}(R/r)$ term clearly corresponds to human wealth, and therefore the sensitivity of consumption to r coming through this term corresponds to the human wealth *effect*, which we introduced using the Fisher diagram in the [two-period model](#) section.

1.7.3 Normalizing By $\mathbf{p}$

The whole problem can be restated more simply by “dividing through” by the level of permanent income before solving. Hereafter, nonbold variables will be the normalized bold-letter equivalent, e.g. $c_t = \mathbf{c}_t/\mathbf{p}_t$, and note that if $\mathbf{p}_{t+1} = G\mathbf{p}_t \forall t$ then from the standpoint of date t ,

$$\begin{aligned}
u(\mathbf{c}_{t+n}) &= \frac{\mathbf{c}_{t+n}^{1-\rho}}{1-\rho} \\
&= \frac{(c_{t+n}\mathbf{p}_{t+n})^{1-\rho}}{1-\rho} \\
&= (\mathbf{p}_t G^n)^{1-\rho} \frac{c_{t+n}^{1-\rho}}{1-\rho}
\end{aligned} \tag{1.7.25}$$

which means that

$$\sum_{n=0}^{T-t} \beta^n \frac{\mathbf{c}_{t+n}^{1-\rho}}{1-\rho} = \mathbf{p}_t^{1-\rho} \sum_{n=0}^{T-t} (G^{1-\rho} \beta)^n \frac{c_{t+n}^{1-\rho}}{1-\rho}. \quad (1.7.26)$$

Furthermore, the accumulation equations can be rewritten by dividing both sides by $\mathbf{p}_{t+1}$:

$$\begin{aligned} \mathbf{b}_{t+1}/\mathbf{p}_{t+1} &= \frac{(\mathbf{m}_t - \mathbf{c}_t)\mathbf{R}}{\mathbf{p}_{t+1}} \\ b_{t+1} &= \left(\frac{(\mathbf{m}_t - \mathbf{c}_t)\mathbf{R}}{\mathbf{p}_t} \right) \left(\frac{\mathbf{p}_t}{\mathbf{p}_{t+1}} \right) \\ &= (m_t - c_t)(\mathbf{R}/G) \end{aligned} \quad (1.7.27)$$

$$\begin{aligned} \mathbf{m}_{t+1} &= \mathbf{b}_{t+1} + \mathbf{p}_{t+1} \\ m_{t+1} &= b_{t+1} + 1. \end{aligned} \quad (1.7.28)$$

Now if we define $\beth \equiv G^{1-\rho}\beta$ and $\mathcal{R}_G \equiv \mathbf{R}/G$, the original problem can be rewritten as:

$$\max \quad \mathbf{p}_t^{1-\rho} \sum_{n=0}^{T-t} \beth^n u(c_{t+n}) \quad (1.7.29)$$

subject to the constraints

$$\begin{aligned} a_t &= m_t - c_t \\ b_{t+1} &= a_t \mathcal{R}_G \\ m_{t+1} &= b_{t+1} + 1 \end{aligned} \quad (1.7.30)$$

and we can go through the same steps as above to find that the solution is

$$c_t = (1 - \mathcal{R}_G^{-1}(\mathcal{R}_G \beth)^{1/\rho}) \left[m_t - 1 + \overbrace{\left(\frac{1}{1 - 1/\mathcal{R}_G} \right)}^{\equiv h} \right] \quad (1.7.31)$$

subject to the “finite human wealth” condition

$$\begin{aligned} 1 &< \mathcal{R}_G \\ 1 &< \mathbf{R}/G \end{aligned} \quad (1.7.32)$$

which is the same condition (1.7.14) as above, and also subject to the “return impatience condition”

$$\begin{aligned} (\mathcal{R}_G \beth)^{1/\rho} &< \mathcal{R}_G \\ \left(\frac{\mathbf{R}}{G} \beta G^{1-\rho} \right)^{1/\rho} &< \mathbf{R}/G \\ (\mathbf{R}\beta)^{1/\rho} &< \mathbf{R} \end{aligned} \quad (1.7.33)$$

which is also the same as above in (1.7.15).

Now note that (1.7.31) can be rewritten

$$\begin{aligned} c_t &= \left(\frac{\mathcal{R}_G - (\mathcal{R}_G \beth)^{1/\rho}}{\mathcal{R}_G} \right) o_t \\ &= \underbrace{(1 - \mathbf{P}_R)}_{\equiv \kappa} o_t \end{aligned} \quad (1.7.34)$$

where o_t is the consumer’s total wealth-to-permanent-labor-income ratio, and κ is the “marginal propensity to consume” out of wealth.

As before, whether o is rising or falling depends upon the relationship between $\mathcal{R}_G - 1$ and $\mathcal{R}_G - (\mathcal{R}_G \beth)^{1/\rho}$. A consumer will be drawing down his wealth-to-income ratio if

$$\begin{aligned}
\mathcal{R}_G - (\mathcal{R}_G \beth)^{1/\rho} &> \mathcal{R}_G - 1 \\
1 - (\mathcal{R}_G \beth)^{1/\rho} &> 0 \\
1 &> (\mathcal{R}_G \beth)^{1/\rho}.
\end{aligned} \tag{1.7.35}$$

Now substituting the definitions of $\mathcal{R}_G$ and $\beth$ we see that whether o is rising or falling depends on whether

$$\begin{aligned}
1 &> \left(\frac{R}{G} \beta G^{1-\rho}\right)^{1/\rho} \\
1 &> (R \beta G^{-\rho})^{1/\rho} \\
1 &> \underbrace{\left(\frac{(R \beta)^{1/\rho}}{G}\right)}_{\mathbf{P}_G},
\end{aligned} \tag{1.7.36}$$

where $\mathbf{P}_G$ is the “growth patience factor.” We call (1.7.36) the “growth impatience condition” (GIC),⁸ and we say that the consumer is “growth impatient” if (1.7.36) holds.

Thus, whether the consumer is patient or impatient in the sense of building up or drawing down a wealth-to-income *ratio* depends on whether the growth rate of labor income is less than, equal to, or greater than the growth rate of consumption. Analogously to our earlier usages, a consumer for whom $\mathbf{P}_G = 1$ (equivalently, $\mathfrak{p}_g = 0$) would be “growth poised.”

To get the intuition for this, consider the case of a consumer with no nonhuman wealth, $b_t = 0$. This consumer’s absolute level of consumption will grow at $(R\beta)^{1/\rho}$ and absolute level of income grows at G , but the PDV of future consumption and future income must be equal. If income is growing faster than consumption but has the same PDV, consumption must be *starting out* at a level *higher* than income, which is the sense in which this consumer is impatient (spending more than his income). “Growth impatience” is therefore the condition that causes consumers with no assets to want to borrow.

1.7.4 Applications

How Large is the Human Wealth Effect?

We can now apply the model to answer our first useful question: How large does the model imply the “human wealth effect” is?

For simplicity, assume that $b_t = 0$. Then the original version of the approximate formula (1.7.21) tells us that the *level* of consumption will be given by:

$$\begin{aligned}
\mathbf{c}_t &\approx \left(r - \rho^{-1}(r - \vartheta)\right) \left(\frac{\mathbf{P}_t}{1 - G/R}\right) \\
&\approx \left(r - \rho^{-1}(r - \vartheta)\right) \left(\frac{\mathbf{P}_t}{r - g}\right).
\end{aligned} \tag{1.7.37}$$

We are interested only in calibrations of the model in which the consumer is “growth impatient” so that $g > \rho^{-1}(r - \vartheta)$ so if we define the rate of growth impatience as

$$\mathfrak{p}_g \equiv \rho^{-1}(r - \vartheta) - g \tag{1.7.38}$$

we can write this as

$$\begin{aligned}
\mathbf{c}_t &\approx \mathbf{P}_t \left(\frac{r - (g + \mathfrak{p}_g)}{r - g}\right) \\
&= \mathbf{P}_t \left(1 - \mathfrak{p}_g/(r - g)\right).
\end{aligned} \tag{1.7.39}$$

⁸Or, GIC-PF if we want to highlight that this is the condition for the perfect foresight model.

Remembering that imposition of the growth impatience condition is equivalent to assuming $b_g < 0$, while the FHWC requires $r > g$, it is clear that the expression $-b_g/(r - g)$ will be positive: The consumer will spend more than his permanent labor income.

Now suppose we choose plausible values for $(r, \vartheta, g, \rho) = (0.04, 0.04, 0.02, 2)$. Then (1.7.37) becomes:

$$\begin{aligned} c_t &\approx 0.04(p_t/0.02) \\ &= 2p_t. \end{aligned} \tag{1.7.40}$$

Now suppose the interest rate changes to $r = 0.03$, while all other parameters remain the same. Then (1.7.37) becomes:

$$\begin{aligned} c_t &\approx 0.035(p_t/0.01) \\ &= 3.5p_t. \end{aligned} \tag{1.7.41}$$

The point of this example is that for plausible parameter values, the human wealth effect is enormously stronger than the income and substitution effects, so that we should see large drops in consumption when interest rates rise and conversely strong gains when interest rates fall. This is a summary of the main point of the famous paper by Summers [1981]; Summers derives formulas for an economy with overlapping generations of finite-lifetime consumers, but those complications do not change the basic message.

How Does the Saving Rate Respond to Interest Rates?

The level of saving can be defined as total income minus total consumption:

$$s_t \approx ra_{t-1} + p_t - c_t \tag{1.7.42}$$

but substituting from (1.7.39) and (1.7.21),

$$c_t \approx p_t \left(1 - b_g/(r - g)\right) + (r - \rho^{-1}(r - \vartheta))b_t \tag{1.7.43}$$

this can be rewritten as

$$\begin{aligned} s_t &\approx ra_{t-1} + p_t - p_t \left(1 - b_g/(r - g)\right) - (r - \rho^{-1}(r - \vartheta))Ra_{t-1} \\ &= ra_{t-1} + p_t b_g/(r - g) - (r - \rho^{-1}(r - \vartheta))Ra_{t-1} \\ s_t &\approx ra_{t-1} + b_g/(r - g) - (r - \rho^{-1}(r - \vartheta))Ra_{t-1} \\ &\approx b_g/(r - g) + \rho^{-1}(r - \vartheta)a_{t-1} \end{aligned} \tag{1.7.44}$$

(where the last approximations come from the assumptions that $1/G \approx 1$) and that $r \times (r - \rho^{-1}(r - \vartheta))$ is “small.” The saving *rate* (for which we use the letter ς to distinguish it from s above) is the ratio of saving to *total* income (not just labor income):

$$\varsigma_t = \left(\frac{b_g/(r - g) + \rho^{-1}(r - \vartheta)a_{t-1}}{1 + ra_{t-1}} \right). \tag{1.7.45}$$

The first thing to notice about this expression is that as a_{t-1} approaches infinity, the saving rate asymptotes to

$$\varsigma_t \approx \left(\frac{\rho^{-1}(r - \vartheta)}{r} \right) \tag{1.7.46}$$

and whether the saving rate is positive or negative depends on whether the consumer is absolutely impatient, absolutely poised, or absolutely patient.⁹

Finally, if we rewrite this as

⁹In this partial equilibrium framework, we are assuming that the consumer’s wealth can go to infinity without any effect on the aggregate interest rate.

$$\varsigma \approx \rho^{-1}(1 - \vartheta r^{-1}) \quad (1.7.47)$$

then it is apparent that the response of the saving rate to the interest rate is

$$\left(\frac{d\varsigma}{dr}\right) = \rho^{-1}\vartheta r^{-2}. \quad (1.7.48)$$

If we consider almost any plausible configuration of parameter values, say $r = \vartheta = 0.05$ and $\rho = 2$, this translates to a very large response of the saving rate with respect to r (in the case of the parameter values mentioned above, $(1/2)(20) = 10$).

1.7.5 Appendix

The Limiting Solution to the Perfect Foresight Model if the FHCW Fails

When the RIC Holds Consider first a circumstance in which the RIC holds ($\mathbf{P}_R < 1$). In this case, the perfect foresight unconstrained model does not have a sensible solution because human wealth is infinite while the model implies that the optimal policy is to consume a positive proportion of human wealth. $c(m) = \infty \forall m$ is not a useful (or plausible!) solution.

When the RIC Fails The alternative case is when the RIC fails ($\mathbf{P}_R > 1$). Here, the only way to make sense of the model is to think about the limit of the finite horizon model as the horizon extends to infinity. This is because behavior reflects a competition between two pathologies that characterize the infinite horizon solution: It exhibits a limiting MPC of zero out of total wealth, which includes human wealth, which approaches infinity. A limiting solution of $c(m) = 0 \times \infty$ is even less useful than $c(m) = \infty$!

It turns out that the limiting solution is not ambiguous, however. The finite horizon solution implies that consumption out of human wealth when the end of life is n periods in the future is

$$\kappa_n h_n = \left(\frac{(\mathbf{R}^{-1}\mathbf{G})^{n+1} - 1}{[\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]^{n+1} - 1} \right) \quad (1.7.49)$$

whose limit is given by

$$\begin{aligned} \lim_{n \uparrow \infty} \kappa_n h_n &= \lim_{n \uparrow \infty} \left(\frac{(\mathbf{R}^{-1}\mathbf{G})^{n+1}}{[\mathbf{R}^{-1}(\mathbf{R}\beta)^{1/\rho}]^{n+1}} \right) \\ &= \lim_{n \uparrow \infty} \left(\frac{1}{\mathbf{P}_G^{(n+1)}} \right) \\ &= \infty \end{aligned} \quad (1.7.50)$$

since when the RIC fails and the FHCW also fails ($\mathbf{G} > \mathbf{R}$), both the numerator and the denominator diverge, so the limit is governed by the ratio of the dominant terms. That ratio is infinite as long as the growth patience factor satisfies $\mathbf{P}_G < 1$, in which case $\mathbf{P}_G^{n+1}$ approaches zero as $n \uparrow \infty$. (In the case treated above, where the RIC holds rather than fails, $\mathbf{P}_G < 1$ follows automatically, because $\mathbf{P} < \mathbf{R}$ and $\mathbf{R} < \mathbf{G}$ together imply the GIC $\mathbf{P} < \mathbf{G}$.)

Useful Analytical Results

Given the result from (1.7.8) that

$$c_{t+n} = \mathbf{P}^n c_t \quad (1.7.51)$$

we can rewrite the value function as

$$\begin{aligned}
v_t &= u(c_t) + \beta u(c_t \mathbf{P}) + \beta^2 u(c_t \mathbf{P}^2) + \dots \\
&= (1 - \rho)^{-1} \left(c_t^{1-\rho} + \beta (c_t \mathbf{P})^{1-\rho} + \beta^2 (c_t \mathbf{P}^2)^{1-\rho} + \dots \right) \\
&= (1 - \rho)^{-1} \left(c_t^{1-\rho} \left(1 + \beta \mathbf{P}^{1-\rho} + (\beta \mathbf{P}^{1-\rho})^2 + \dots \right) \right) \\
&= u(c_t) \left(1 + \beta \mathbf{P}^{1-\rho} + (\beta \mathbf{P}^{1-\rho})^2 + \dots \right)
\end{aligned} \tag{1.7.52}$$

but since $\beta \mathbf{P}^{1-\rho} = \mathbf{P}_R$,¹⁰ this reduces to

$$v_t = u(c_t) \overbrace{(1 + \mathbf{P}_R + \mathbf{P}_R^2 + \dots + \mathbf{P}_R^{T-t})}^{\equiv \mathbb{C}_t} \tag{1.7.54}$$

where $\mathbb{C}_t$ is the discounted value of future consumption growth (that is, the discounted value of the ratio of future consumption to today's consumption).

Carroll shows (in an appendix) that $\mathbb{C}_t = \kappa_t^{-1}$, which means that we can write value as

$$\begin{aligned}
v_t &= u(c_t) \kappa_t^{-1} \\
&= \left(\frac{(o_t \kappa_t)^{1-\rho}}{1 - \rho} \right) \kappa_t^{-1} \\
&= u(o_t) \kappa_t^{-\rho}
\end{aligned} \tag{1.7.55}$$

Additional Derivations

If consumption is simply a function of overall wealth bank balances o_t , we can derive a convenient recursive formula for the inverse of the MPC:

$$\begin{aligned}
u'(\kappa_t o_t) &= R \beta u'(\kappa_{t+1} \overbrace{o_t (1 - \kappa_t) R}^{o_{t+1}}) \\
\kappa_t o_t &= (R \beta)^{-1/\rho} \kappa_{t+1} o_t (1 - \kappa_t) R \\
\underbrace{R^{-1} (R \beta)^{1/\rho}}_{\mathbf{P}_R} \kappa_t &= \kappa_{t+1} (1 - \kappa_t) \\
(\mathbf{P}_R \kappa_t)^{-1} &= \kappa_{t+1}^{-1} (1 - \kappa_t)^{-1} \\
(1 - \kappa_t) \kappa_t^{-1} &= \mathbf{P}_R \kappa_{t+1}^{-1} \\
\kappa_t^{-1} - 1 &= \mathbf{P}_R \kappa_{t+1}^{-1} \\
\kappa_t^{-1} &= 1 + \mathbf{P}_R \kappa_{t+1}^{-1}
\end{aligned} \tag{1.7.56}$$

which implies that if the MPC in the last period T is $\kappa_T = 1$ then from any date $t \leq T$ we can write

$$\kappa_t^{-1} = 1 + \mathbf{P}_R + \mathbf{P}_R^2 + \dots + \mathbf{P}_R^{T-t}. \tag{1.7.57}$$

But the series on the RHS in (1.7.54) and (1.7.57) are identical! So $\kappa_t^{-1} = \mathbb{C}_t$, and we can equivalently write

$$v_t = u(c_t) \kappa_t^{-1}. \tag{1.7.58}$$

¹⁰

$$\begin{aligned}
\beta \mathbf{P}^{1-\rho} &= \beta (R \beta)^{\frac{1-\rho}{\rho}} \\
&= \beta (R \beta)^{1/\rho-1} \\
&= (R \beta)^{1/\rho} / R \\
&= \mathbf{P}_R
\end{aligned} \tag{1.7.53}$$

Now note that if we define a utility-inverse of the value function as $\Lambda \equiv ((1 - \rho)v)^{1/(1-\rho)}$, then consumption exceeds its minimum possible value at $\underline{m}$ (where consumption exceeds $c(\underline{m}) = 0$) by $c_t = \kappa_t(m - \underline{m})$:

$$\begin{aligned}\Lambda_t(m) &= \kappa_t(m_t - \underline{m}_t) \kappa_t^{-1/(1-\rho)} \\ &= (m - \underline{m}) \kappa^{((1-\rho)/(1-\rho) - 1/(1-\rho))} \\ &= (m - \underline{m}) \kappa^{-\rho/(1-\rho)}\end{aligned}\tag{1.7.59}$$

which is linear, and makes it very easy to compute

$$v_t(m) = u\left((m - \underline{m}) \kappa_t^{-\rho/(1-\rho)}\right)\tag{1.7.60}$$

1.8 The Random Walk Model of Consumption

This section derives the Hall [1978] random walk proposition for consumption.

The consumption Euler equation when future consumption is uncertain takes the form¹¹

$$u'(c_t) = \beta R \mathbb{E}_t[u'(c_{t+1})]. \quad (1.8.1)$$

Suppose the utility function is quadratic:

$$u(c) = -(1/2)(\bar{c} - c)^2 \quad (1.8.2)$$

where $\bar{c}$ is the “bliss point” level of consumption.¹² Marginal utility is

$$u'(c) = (\bar{c} - c) \quad (1.8.3)$$

and suppose further that $R\beta = 1$ so that (1.8.1) becomes

$$\begin{aligned} (\bar{c} - c_t) &= \mathbb{E}_t[(\bar{c} - c_{t+1})] \\ \mathbb{E}_t[c_{t+1}] &= c_t. \end{aligned} \quad (1.8.4)$$

Defining the innovation to consumption as

$$\begin{aligned} \epsilon_{t+1} &= c_{t+1} - c_t \\ &\equiv \Delta c_{t+1}, \end{aligned} \quad (1.8.5)$$

the random walk proposition is simply that the expectation of consumption changes is zero:

$$\mathbb{E}_t[\Delta c_{t+1}] = 0. \quad (1.8.6)$$

This means that no information known to the consumer when the consumption choice c_t was made can have any predictive power for how consumption will change between period t and $t + 1$ (or for any date beyond $t + 1$).

¹¹See the [Envelope theorem](#) section for the derivation of the Euler equation in the perfect foresight case; we will show later that the consequence of uncertainty is simply to insert the expectations operator.

¹²Assume that the consumer is sufficiently poor that it will be impossible for them ever to achieve consumption as large as $\bar{c}$.

1.9 Consumption Functions and the Permanent Income Hypothesis

This section derives the consumption function (the relation between consumption spending and the consumer's economic circumstances) for an optimizing consumer with Certainty Equivalent (CEQ) preferences like those assumed by Hall [1978] and with an income process that has a purely transitory and a purely permanent component.

The consumer wants to

$$\max \mathbb{E}_t \left[\sum_{n=0}^{\infty} \beta^n u(c_{t+n}) \right] \quad (1.9.1)$$

subject to the constraint

$$b_{t+1} = (b_t + y_t - c_t)R \quad (1.9.2)$$

where b_t is the consumer's beginning-of-period bank balances, y_t is current labor income, $R = (1+r)$ is the constant interest factor and β is the time preference factor. Suppose the consumer has quadratic utility $u(c) = -(1/2)(\bar{c} - c)^2$ where $\bar{c}$ is the “bliss point” level of consumption. Assume further that $\beta R = 1$.

Under these circumstances, the [random walk](#) section shows that consumption will follow a random walk,

$$\begin{aligned} \Delta c_{t+1} &= \epsilon_{t+1}, \\ \mathbb{E}_t[\epsilon_{t+n}] &= 0 \quad \forall n > 0. \end{aligned} \quad (1.9.3)$$

Hall [1978] tested this proposition by examining whether lagged variables had predictive power for consumption growth. Hall's approach largely supplanted a vast earlier literature that had attempted to estimate “the consumption function” which was interpreted as the relationship between observed economic variables like income, and household spending. The “Keynesian” consumption function, for example, was something along the lines of

$$\begin{aligned} c_t &= \alpha_0 + \alpha_1 y_t \\ \Delta c_t &= \alpha_1 \Delta y_t \end{aligned} \quad (1.9.4)$$

where y_t was disposable household income.

The reason Hall's approach was attractive is that Muth [1960] showed that the appropriate response of consumption to, say, a shock to current income depends on whether that income shock is transitory or permanent.

To see Muth's point clearly, suppose that the labor income process is

$$\begin{aligned} p_{t+1} &= p_t + \psi_{t+1} \\ y_{t+1} &= p_{t+1} + \theta_{t+1}, \end{aligned} \quad (1.9.5)$$

where θ is a white noise variable representing a transitory shock to labor income and ψ is a white noise variable representing a shock to permanent labor income, $\mathbb{E}_t[\theta_{t+n}] = \mathbb{E}_t[\psi_{t+n}] = 0 \quad \forall n > 0$.

We can solve for the level of consumption using the Intertemporal Budget Constraint, which says that the expected PDV of consumption must equal the expected PDV of total wealth, human and nonhuman:

$$\left[\sum_{n=0}^{\infty} R^n c_{t+n} \right] = b_t + \left[\sum_{n=0}^{\infty} R^n y_{t+n} \right] \quad (1.9.6)$$

Since the IBC must hold for any possible set of realizations of the stochastic shocks, it must hold in expectation, so

$$\begin{aligned}
\mathbb{E}_t \left[\sum_{s=t}^{\infty} R^{t-s} \right] c_t &= b_t + \theta_t + \mathbb{E}_t \left[\sum_{s=t}^{\infty} R^{t-s} p_s \right] \\
\left(\frac{1}{1 - R^{-1}} \right) c_t &= b_t + \theta_t + p_t \left(\frac{1}{1 - R^{-1}} \right) \\
c_t &= (R/R - 1/R) (b_t + \theta_t) + p_t \\
c_t &= \left(\frac{r}{R} \right) (b_t + \theta_t) + p_t.
\end{aligned} \tag{1.9.7}$$

From (1.9.7) we can see that if a given shock to income is perceived to be transitory, then the marginal propensity to consume will be $\alpha_1 = (r/R)$ which is a small number (say, 0.05) while if the shock is perceived to be permanent then α_1 will be 1.0. So there is no such thing as the “true” value of α_1 and the “consumption function” conceived as an estimated version of (1.9.4) is meaningless (though the consumption function conceived as (1.9.7) is perfectly sensible).

This problem with the existing literature explains why Hall’s innovation was so exciting: He showed a way to test the theory that did not depend on arbitrary and difficult-to-test assumptions about decisionmakers’ beliefs about the structure of the income process.

1.10 Sticky Expectations and Consumption Dynamics

Consider a consumer subject to the dynamic budget constraint

$$b_{t+1} = (b_t + y_t - c_t)R \quad (1.10.1)$$

where b_t is beginning-of-period bank balances, y_t is current labor income, and $R = (1 + r)$ is the constant interest factor. Actual labor income y is permanent labor income $\mathbf{p}$ modified by a transitory shock factor θ :

$$y_{t+1} = \mathbf{p}_{t+1}\theta_{t+1} \quad (1.10.2)$$

where $\mathbb{E}_t[\theta_{t+n}] = 1 \ \forall n > 0$. Permanent labor income grows by a predictable factor G from period to period, subject to a mean-one permanent shock ψ :

$$\mathbf{p}_{t+1} = G\mathbf{p}_t\psi_{t+1}, \quad (1.10.3)$$

so that the expected present discounted value of permanent labor income (“human wealth”) for an infinite-horizon consumer is

$$h_t = \left(\frac{\mathbf{p}_t}{1 - G/R} \right). \quad (1.10.4)$$

We carry permanent shocks throughout what follows. The simpler model without them is the restriction $\psi_{t+1} = 1 \ \forall t$, under which permanent income grows deterministically as $\mathbf{p}_{t+1} = G\mathbf{p}_t$; because $\hat{\Psi}_{t+1} \equiv \Psi_{t+1} - 1$ is then zero, every $\hat{\Psi}$ term drops out of the expressions derived below. Both cases are therefore covered by the general expressions.

We will assume that the consumer behaves according to the consumption rule

$$c_t = \underbrace{(b_t + (\theta_t - 1)\mathbf{p}_t + h_t)}_{\equiv \mathbf{o}_t} \underbrace{(r/R)}_{\equiv \kappa}, \quad (1.10.5)$$

where κ is the “marginal propensity to consume” out of total wealth $\mathbf{o}$.¹³

Under these circumstances, the [random walk](#) section shows that consumption will follow a random walk,

$$\Delta c_{t+1} = \epsilon_{t+1}, \quad \mathbb{E}_t[\epsilon_{t+n}] = 0 \ \forall n > 0. \quad (1.10.6)$$

Now assume that the economy is populated by a set of measure one of consumers indexed by a superscript i distributed uniformly along the unit interval. Per capita values of all variables, designated by the upper case, are the integral over all individuals in the economy, as in the [Aggregation](#) appendix, so that

$$C_t = \int_0^1 c_t^i di = \int_0^1 (r/R) \mathbf{o}_t^i di = \mathbf{O}_t \kappa. \quad (1.10.7)$$

This equation implies that an aggregate version of equation (1.10.6) holds,¹⁴

$$\Delta C_{t+1} = \epsilon_{t+1}. \quad (1.10.8)$$

In principle, we could allow each individual in this economy to experience a different transitory and permanent shock from every other individual in each period. However, for our purposes it is

¹³This is the optimal consumption function for a utility-maximizing consumer with $R\beta = 1$ if that consumer has quadratic utility (Hall [1978]) or if the consumer has CRRA utility and perfect foresight and anticipates $\theta_{t+n} = \psi_{t+n} = 1 \ \forall n > 0$. See the [consumption function](#) section for a derivation of this consumption function under quadratic utility, and the [perfect foresight CRRA](#) section for the derivation in the perfect foresight CRRA case. Deaton [1992] argues that the “Permanent Income Hypothesis” should be **defined** as the hypothesis that consumption is determined according to (1.10.5); but this differs sharply from Friedman [1957]’s definition, and has not become universally accepted.

¹⁴The crucial feature of the model that allows us to aggregate analytically is the linearity of the consumption rule in p , b , and Θ .

useful to assume that everyone experiences the same shocks in a given period; that is $\theta_t^i = \Theta_t \forall t$ and $\psi_t^i = \Psi_t \forall t$.

Assuming (here and henceforth) that the growth factor for permanent income is $G = 1$, the figures below show the path of consumption and bank balances (the solid dots) for an economy populated by omniscient consumers who in periods $t - n$ for $n > 0$ had experienced $\Theta_{t-n} = \Psi_{t-n} = 1$; that is, this economy has had no shocks to income in the past. (For convenience, the consumer is assumed to have arrived in period t with $B_t = 0$). In period t the consumer draws $\Theta_t = 2$ and $\Psi_t = 1$; thereafter $\Psi_{t+n} = \Theta_{t+n} = 1$. The figures show $C_{t-2}, C_{t-1}, C_t, \mathbb{E}_t[C_{t+1}], \mathbb{E}_t[C_{t+2}], \dots$ and the corresponding values for B .

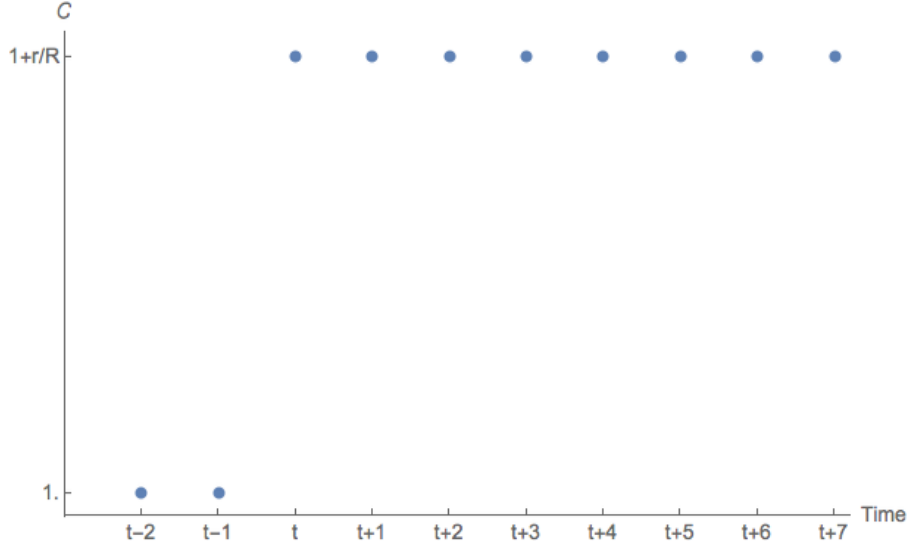


Figure 1.6: Path of C after a shock $\Theta_t = 2$, Omniscient Consumers

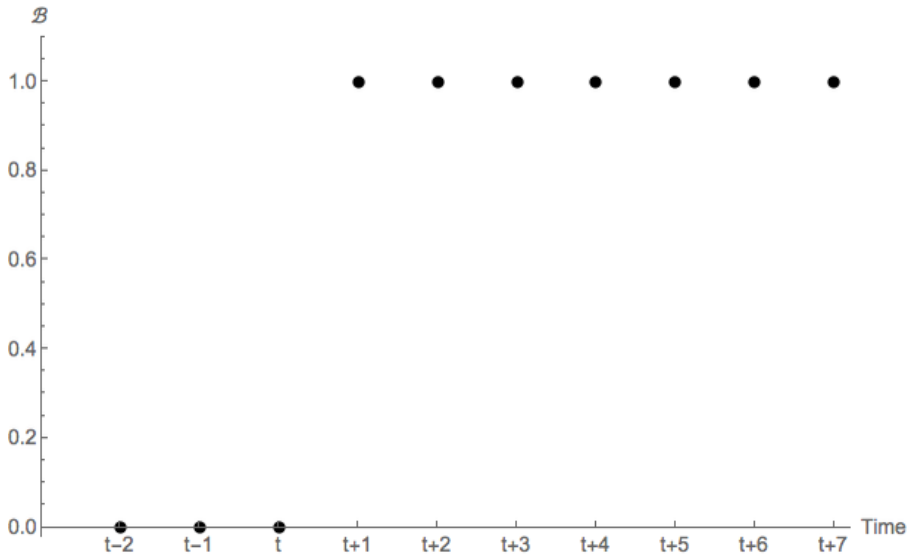


Figure 1.7: Path of B after a shock $\Theta_t = 2$, Omniscient Consumers

Now suppose that not every consumer updates expectations in every period. Instead, expectations are “sticky”: each consumer updates with probability Π in each period. Whether the consumer at location i updates in period t is determined by the realization of the dichotomous random variable

$$\pi_t^i = \begin{cases} 1 & \text{if consumer } i \text{ updates in period } t \\ 0 & \text{if consumer } i \text{ does not update in period } t, \end{cases} \quad (1.10.9)$$

and each period's updaters are chosen randomly such that a constant proportion Π update in each period:

$$\mathbb{E}_t^i[\pi_{t+1}^i] = \Pi \forall t, \quad \int_0^1 \pi_\tau^i di = \Pi \forall \tau. \quad (1.10.10)$$

It will also be convenient to define the date of consumer i 's most recent update; we call this object τ_t^i . We designate the value of a variable $\bullet$ at date t for a consumer who last updated his expectations at date $t-n$ by $\bullet_{t|t=n}$, where the part of the subscript following $|$ indicates criteria that must be matched by the consumer's (or group's) τ . To illustrate, consider a consumer who updated his expectations most recently in period $t-3$ so that $n(i) = 3$. This consumer's actual consumption in period t will be $c_t^i = c_{t-3|t=3}$, the level of consumption that was chosen in period $t-3$ given beliefs at that time.

We are assuming that the probability of adjusting one's expectations is independent of the level of income or wealth; therefore, the average levels of wealth and income among those consumers who adjust their expectations this period will be the same as the average levels of wealth and income in the economy as a whole. This independence assumption is crucial for aggregation: it ensures that the subset of consumers who update in any period is representative of the population.

We need a notation to represent sets of consumers defined by the period of their most recent update. We denote such a set by the condition on τ_t^i ; for example, the set of consumers whose most recent update, as of date t , was prior to period $t-1$ would be $\mathcal{T} = \{\tau_t^i < t-1\}$. We denote the per-capita value of a variable $\bullet$, among consumers in a set $\mathcal{T}$ as of date t , by $\bullet_{t|\mathcal{T}}$. Dropping the i superscripts to reduce clutter, per-capita consumption among households who have updated in period t is therefore

$$C_{t|\tau_t=t} = \Pi^{-1} \int_0^1 \pi_t c_t di. \quad (1.10.11)$$

Dropping the i superscript for notational simplicity, the level of consumption per capita that would prevail if all consumers were to update in period t is $C_{t|t} = \int_0^1 c_{t|t} di$. But since the set of consumers who updated was randomly selected from the population, the average level of consumption-per-capita for the updaters must equal the average level of consumption-per-capita that would characterize the population as a whole if everyone in the economy were to update. This is the key step that allows aggregation to work analytically.

In periods when expectations are not updated, the consumer continues to spend the same amount as in the most recent period when his expectations **were** updated.¹⁵ If the economy is large the proportion of consumers who update their expectations every period will be Π .¹⁶ Average consumption among those who are not updating in the current period (for whom $1 - \pi_t = 1$) is then

$$C_{t|\tau_t < t} = (1 - \Pi)^{-1} \int_0^1 (1 - \pi_t) c_t di = C_{t-1} \quad (1.10.12)$$

because consumption per capita among those who are not updating in the current period is (by assumption) identical to their consumption per capita in the prior period, which must match aggregate consumption per capita in the prior period because the set who do **not** update today is randomly selected from the $t-1$ population.

Now note that

$$C_{t+1} = \Pi C_{t+1|\tau_{t+1}=t+1} + (1 - \Pi) \overbrace{C_{t+1|\tau_{t+1} < t+1}}^{=C_t}, \quad \Delta C_{t+1} = \Pi(C_{t+1|\tau_{t+1}=t+1} - C_{t|\tau_t=t}) + (1 - \Pi)\Delta C_t \quad (1.10.13)$$

and

¹⁵This makes sense because under the Hall [1978] assumptions the expected change in consumption is zero under our assumption that $R\beta = 1$.

¹⁶For consumers who are not updating at date t , $c_t > 0$ but $\pi_t = 0$, so the integral produces the sum of consumption among only those i who are updating in period t .

$$C_t = \Pi C_{t|\tau_t=t} + (1-\Pi)C_{t-1}, \quad \Delta C_t = \Pi(C_{t|\tau_t=t} - C_{t-1}) = \Pi(C_{t|\tau_t=t} - C_t + C_t - C_{t-1}), \quad (1-\Pi)\Delta C_t = \Pi(C_{t|\tau_t=t} - C_t) \quad (1.10.14)$$

while, defining $\hat{\Theta} = \Theta - 1$ and $\hat{\Psi} = \Psi - 1$,

$$\begin{aligned} C_{t+1|\tau_{t+1}=t+1} &= \left(\overbrace{(B_t + \hat{\Theta}_t - C_t)R + \hat{\Theta}_{t+1}}^{=B_{t+1}} \right) \kappa + 1 + \hat{\Psi}_{t+1} \\ C_{t+1|\tau_{t+1}=t+1} &= \left((B_t + \hat{\Theta}_t - C_t + C_{t|\tau_t=t} - C_{t|\tau_t=t})R + \hat{\Theta}_{t+1} \right) \kappa + 1 + \hat{\Psi}_{t+1} \\ &= \underbrace{\left((B_t + \hat{\Theta}_t - C_{t|\tau_t=t})R + \hat{\Theta}_{t+1} \right) \kappa + 1 + \hat{\Psi}_{t+1}}_{=\mathbb{E}_t[C_{t+1|\tau_t=t}] + \hat{\Theta}_{t+1}\kappa + \hat{\Psi}_{t+1}} + (C_{t|\tau_t=t} - C_t)R\kappa \\ &= C_{t|\tau_t=t} + \hat{\Theta}_{t+1}\kappa + \hat{\Psi}_{t+1} + (C_{t|\tau_t=t} - C_t)\kappa R \end{aligned} \quad (1.10.15)$$

where (1.10.15) follows its predecessor since, among consumers who have updated in period t , the random walk proposition says that $\mathbb{E}_t[C_{t+1|\tau_t=t}] = C_{t,\tau_t=t}$. Subtracting $C_{t|\tau_t=t}$ from both sides of (1.10.15) and substituting the result into (1.10.13), and using (1.10.14) to substitute for $\Pi(C_{t|\tau_t=t} - C_t)$, yields

$$\begin{aligned} \Delta C_{t+1} &= (1-\Pi)\Delta C_t + \underbrace{\Pi(C_{t|\tau_t=t} - C_t)}_{=(1-\Pi)\Delta C_t} \underbrace{R\kappa}_{=r} + \underbrace{\Pi(\hat{\Theta}_{t+1}\kappa + \hat{\Psi}_{t+1})}_{\equiv \xi_{t+1}} \\ &= (1-\Pi)\Delta C_t + (1-\Pi)r\Delta C_t + \xi_{t+1} \\ &= (1-\Pi)R\Delta C_t + \xi_{t+1} \end{aligned} \quad (1.10.16)$$

where ξ is a white noise variable ($\mathbb{E}_t[\xi_{t+n}] = 0 \forall n > 0$).

We are finally in position to show how aggregate consumption and wealth would respond in this economy to a transitory positive shock to aggregate labor income like the one considered above for the omniscient model.

Consider the case of a positive shock of size $\hat{\Theta}_t = 1$, as before. In the first period consumption rises only by $\Pi\kappa$, rather than the full amount corresponding to the permanent income associated with the new level of wealth. Therefore aggregate wealth in period $t+1$ will be greater than it would have been in the omniscient model. Similarly for all subsequent periods. Thus, in contrast with the omniscient model, the sluggish adjustment of consumption to the shock means that the shock has a permanent effect on the level of aggregate wealth, and therefore on the level of aggregate consumption. (The figures below depict the results.)

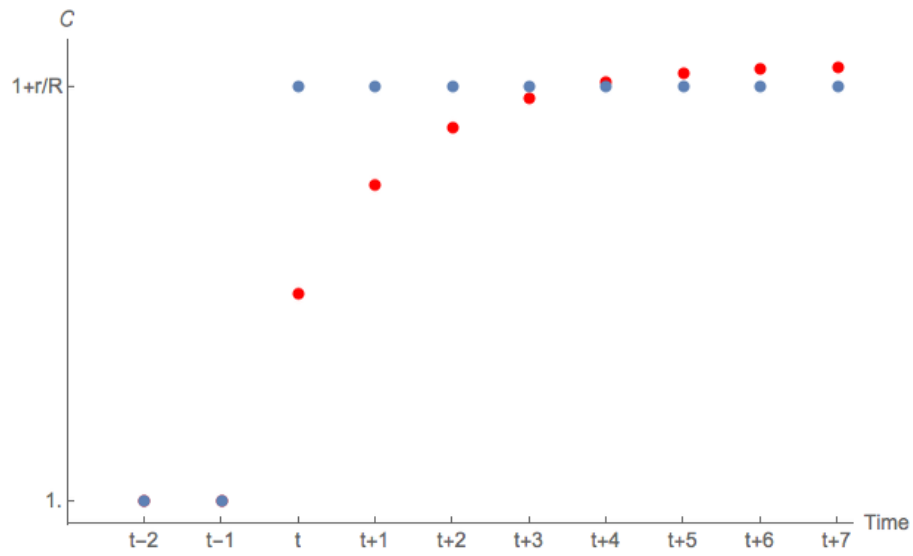
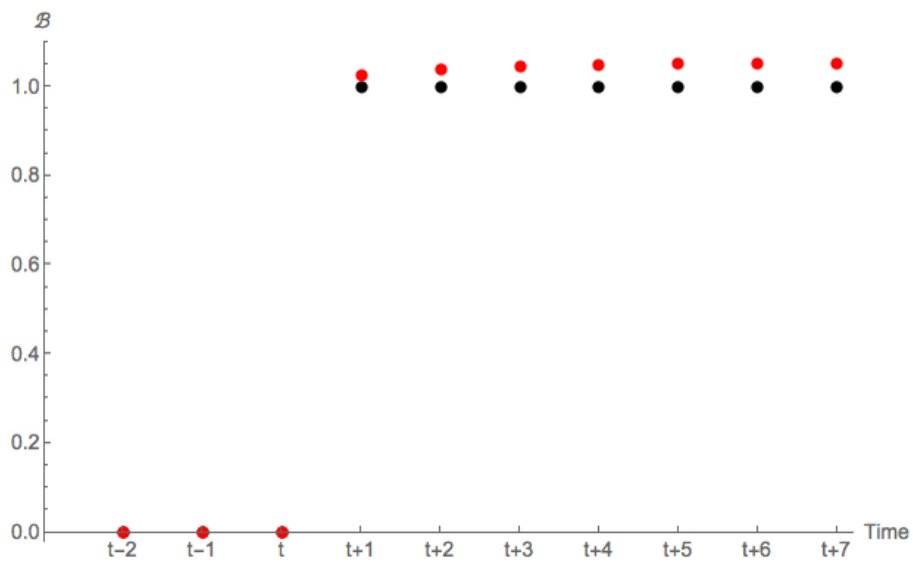
The sticky expectations model says that consumption growth today can be statistically related to any variable that is related to lagged consumption growth. In particular, if lagged consumption growth is related to lagged income growth (as it certainly will be), then there **should** be a statistically significant effect of lagged income growth on current consumption growth if expectations are sticky.

If the model derived here could be taken literally, it would suggest estimating an equation of the form

$$\Delta C_{t+1} = \alpha_0 + \alpha_1 \Delta C_t + \varepsilon_{t+1} \quad (1.10.17)$$

and interpreting the coefficient α_1 as a measure of $R(1-\Pi)$.

However, if there is potential measurement error in C_t the coefficient obtained from estimating (1.10.17) would be biased toward zero for standard errors-in-variables reasons (just as regressing consumption on actual income yields a downward-biased estimate of the response of consumption to permanent income), which means that the estimate of Π would be biased toward 1 (i.e. the omniscient model in which everyone adjusts all the time). Under these circumstances, direct estimation of (1.10.17) would not be a reliable way to estimate Π .

Figure 1.8: Path of C after a shock $\Theta_t = 2$; Sticky Expectations in Red/GrayFigure 1.9: Path of B after a shock $\Theta_t = 2$; Sticky Expectations in Red/Gray

For estimation methods that get around this problem see Sommer [2007], Carroll et al. [2011b], Carroll et al. [2011a]. Those papers consistently find that the proportion of updaters is about $\Pi = 0.25$ per quarter, so that the serial correlation of “true” consumption growth is about $(1 - \Pi) = 0.75$ per quarter.

1.10.1 Appendix

This appendix provides additional derivations and notation useful for simulating the model. One way of interpreting consumers’ behavior in this model is to attribute to them the beliefs that would rationalize their actions. Define $\mathbf{O}$ as the level of wealth (human and nonhuman) that the consumer perceives. Then the Deaton definition of the permanent income hypothesis is that

$$\mathbf{C}_t = \mathbf{O}_t \kappa \quad (1.10.18)$$

and the reason consumption follows a random walk is that $\kappa = (r/R)$ is precisely the amount that ensures that $\mathbb{E}_t[\mathbf{O}_{t+1}] = \mathbf{O}_t$.

Writing the “believed” level of wealth as $\bar{\mathbf{O}}$, we could then interpret the failure of the sticky expectations consumer to change his consumption during the period of nonupdating as reflecting his optimal forecast that, in the absence of further information, $\bar{\mathbf{O}}_{t+1} = \bar{\mathbf{O}}_{t+2} + \dots = \mathbf{O}_t$.

To pursue this interpretation, it is useful to write the budget constraint more explicitly, as before; start with the constraint in levels, then decompose variables into ratios to permanent income (nonbold variables) and the level of permanent income:

$$\begin{aligned} \mathbf{B}_{t+1} &= \mathbf{A}_t \mathbf{R}_{t+1} \\ B_{t+1} P_{t+1} &= A_t P_t R_{t+1} \\ B_{t+1} &= A_t \underbrace{(R_{t+1}/G_{t+1})}_{\equiv \mathcal{R}_{t+1}} \end{aligned} \quad (1.10.19)$$

where we permit a time subscript on R and G because we want to allow for the possibility that beliefs about the interest rate or growth rate might change over time.

Consider an economy that comes into existence in period 0 with a population of consumers who are identical in every respect, including their beliefs about current and future values of the economy’s variables.

First we examine the case where neither R nor G can change after date 0. In that case, we can track the dynamics of believed and actual variables as follows.

$$\begin{aligned} \bar{\mathbf{B}}_{t+1} &= \Pi \mathbf{B}_{t+1} + (1 - \Pi) \bar{\mathbf{A}}_t \mathbf{R} \\ \bar{B}_{t+1} \bar{P}_{t+1} &= \Pi B_{t+1} P_{t+1} + (1 - \Pi) \bar{A}_t \bar{P}_t R \\ \bar{B}_{t+1} &= \Pi B_{t+1} \underbrace{(P_{t+1}/\bar{P}_{t+1})}_{\equiv Q_{t+1}} + (1 - \Pi) \bar{A}_t \underbrace{(R/G \bar{\Psi}_{t+1})}_{\equiv \bar{\mathcal{R}}_{t+1}} \end{aligned} \quad (1.10.20)$$

where capturing the dynamics of the ratio of true permanent income to believed permanent income Q_{t+1} requires us to compute

$$\begin{aligned} \bar{P}_{t+1} &= (\Pi \Psi_{t+1} P_t + (1 - \Pi) \bar{P}_t) G \\ \bar{\Psi}_{t+1} &\equiv \bar{P}_{t+1}/(\bar{P}_t G) \\ Q_{t+1} &= (\Psi_{t+1}/\bar{\Psi}_{t+1}) Q_t \end{aligned} \quad (1.10.21)$$

and so

$$\begin{aligned} (\bar{\Theta}_{t+1} - 1) \bar{P}_{t+1} &= \Pi (\Theta_{t+1} - 1) P_{t+1} + (1 - \Pi) (1 - 1) \bar{P}_t G \\ \bar{\Theta}_{t+1} - 1 &= \Pi (\Theta_{t+1} - 1) Q_{t+1} \end{aligned} \quad (1.10.22)$$

with the crucially useful fact that since by assumption neither R nor G is changing, normalized human wealth does not change from

$$h = \left(\frac{1}{1 - G/R} \right) \quad (1.10.23)$$

so that

$$\begin{aligned} \bar{H}_t &= h\bar{P}_t \\ H_t &= hP_t \\ \bar{H}_{t+1} &= \left(\Pi Q_{t+1} + (1 - \Pi)/\bar{\Psi}_{t+1} \right) h\bar{P}_{t+1} \end{aligned} \quad (1.10.24)$$

so that perceived wealth and consumption will be

$$\begin{aligned} \bar{O}_{t+1} &= \bar{B}_{t+1} + (\bar{\Theta}_{t+1} - 1) + \bar{H}_{t+1} \\ \bar{C}_{t+1} &= \bar{O}_{t+1}\kappa \\ \bar{A}_{t+1} &= \bar{B}_{t+1} + \bar{\Theta}_{t+1} - \bar{C}_{t+1}. \end{aligned} \quad (1.10.25)$$

Matters are more complex if expectations about R and G are allowed to change over time.

Suppose again that we begin our economy in period 0 with population with homogeneous views: Everyone believes $R = R^0$ and $G = G^0$; so long as these views are universally held in the population, aggregate dynamics are captured by the foregoing analysis.

Suppose, however, that in some period $n > 0$ the economy's "true" values of R or G change. Updating consumers see this change immediately. But nonupdaters will not discover the changed nature of the economy's dynamics until they update again.

We capture this modification to the model by keeping track of the aggregate values of the variables for the set of consumers who adhere to each differing opinion, along with the population mass associated with the different opinions. Specifically, suppose there are J different opinions in the population, each of whom constitutes population mass $\mathcal{L}_t^j$ such that $\sum_j \mathcal{L}_t^j = 1 \forall t$. Then for each such population, it will be necessary to keep track of their average beliefs about macroeconomic variables.

Suppose, for example, that through period x there have been only $j - 1$ different opinion groups in the population. In period $x + 1$ either G or R changes. We need then to define group j by $R^j = R_x$ and $G^j = G_x$ and to define $\bar{A}_x^j \equiv A_x$, $\bar{P}_x^j = P_x$, and so on. We will henceforth need to keep track of dynamics of the consumers who remain in belief group j by, e.g.,

$$\begin{aligned} \bar{B}_{t+1}^j &= \mathbf{A}_t^j R^j \\ \bar{B}_{t+1}^j \bar{P}_{t+1}^j &= \bar{A}_t^j \bar{P}_t^j R^j \\ \bar{B}_{t+1}^j &= \bar{A}_t^j \underbrace{(R^j / G^j \bar{\Psi}_{t+1}^j)}_{\equiv \bar{R}^j} \\ \bar{P}_{t+1}^j &= \bar{P}_t^j G^j \end{aligned} \quad (1.10.26)$$

while we need to keep track of the populations of the differing groups by, e.g.,

$$\begin{aligned} \mathcal{L}_t^j &= \Pi \\ \mathcal{L}_{t+1}^j &= \Pi(1 - \Pi) \\ \mathcal{L}_{t+2}^j &= \Pi(1 - \Pi)^2 \end{aligned} \quad (1.10.27)$$

and so on. These population dynamics continue forever, but the population of households continuing to hold any specific belief configuration dwindles toward zero as time progresses.

Aggregate variables for the population as a whole can be constructed as the population-weighted sums across all the differing belief groups, weighted by their masses:

$$\bar{P}_t = \Pi P_t + (1 - \Pi) \sum_j \mathcal{L}_t^j \bar{P}_t^j \quad (1.10.28)$$

and note that if beliefs change back to a configuration that has been seen before it is possible to add the population mass and aggregate values of the variables associated with the new population

with that belief configuration to the corresponding figures for the old population that holds the same beliefs. This reduces the number of groups that the simulations must track in the case where beliefs switch between a limited number of distinct values.

1.11 Consumption Models with Habit Formation

1.11.1 The Problem

Consider a consumer whose goal at date t is to solve the problem (this section is a simplified version of Carroll [2000a]):

$$\max \sum_{n=0}^{T-t} \beta^n u(c_{t+n}, h_{t+n}) \quad (1.11.1)$$

where h_{t+n} is the habit stock, and all other variables are as usually defined. Note that here “habit formation” refers to something different from the colloquial usage of the term: it is not about developing behavioral routines or rules of thumb for saving. Rather, it refers to preferences in which past consumption affects current utility. Utility is decreasing in the habit stock ($u^h < 0$): higher past consumption raises the reference point against which current consumption is compared, making any given level of current consumption less satisfying. This utility function is not “time separable” in the original sense, but once habits are included as a state variable it becomes time-separable. The DBC is

$$m_{t+1} = (m_t - c_t)R + y_{t+1}. \quad (1.11.2)$$

However, when habits affect utility we must also specify a process that describes how habits evolve over time. Our assumption will be:

$$h_{t+1} = h_t + \lambda(c_t - h_t), \quad (1.11.3)$$

where $\lambda \in (0, 1]$ governs how fast the habit stock adjusts toward current consumption. The derivations below adopt the limiting case $\lambda = 1$, in which the habit stock is simply last period’s consumption:

$$h_{t+1} = c_t. \quad (1.11.4)$$

Bellman’s equation for this problem is therefore

$$v_t(m_t, h_t) = \max_{\{c_t\}} u(c_t, h_t) + \beta v_{t+1}((m_t - c_t)R + y_{t+1}, c_t). \quad (1.11.5)$$

To clarify the workings of the Envelope theorem in the case with two state variables, let’s define a function $\underline{v}_t$ that represents value for any choice of c_t , not just the optimal one:

$$\underline{v}_t(m_t, h_t, c_t) = u(c_t, h_t) + \beta v_{t+1}((m_t - c_t)R + y_{t+1}, c_t) \quad (1.11.6)$$

Viewed as a function of m_t , this $\underline{v}_t$ lies below v_t everywhere except at the optimal c_t , where the two touch. The Envelope theorem exploits that tangency.

Now define the function $\mathbf{c}_t(m_t, h_t)$ as the choice of c_t that solves the maximization (1.11.5), so that we have

$$v_t(m_t, h_t) = \underline{v}_t(m_t, h_t, \mathbf{c}_t(m_t, h_t)). \quad (1.11.7)$$

1.11.2 Optimality Conditions

The First Order Condition

The first order condition for (1.11.5) with respect to c_t is (dropping arguments for brevity and denoting the derivative of f with respect to x at time t as f_t^x):

$$0 = u_t^c + \beta \left(v_{t+1}^h - R v_{t+1}^m \right) \quad (1.11.8)$$

or, equivalently,

$$u_t^c = \beta \left(Rv_{t+1}^m - v_{t+1}^h \right). \quad (1.11.9)$$

The intuition is as follows. Note first that if utility is not affected by habits, then $v_{t+1}^h = 0$ and equation (1.11.9) reduces to the usual first order condition for consumption, which tells us that increasing consumption by ϵ today and reducing it by $R\epsilon$ in the next period must not change expected discounted utility. With habits, an increase in consumption today has a consequence beyond its effect on tomorrow's resources m_{t+1} : tomorrow's habit stock will be changed as well. An increase in consumption today of size ϵ increases the size of the habit stock which tomorrow's consumption is compared to, and therefore reduces tomorrow's utility by an amount corresponding to the marginal utility of higher habits tomorrow v_{t+1}^h . Since v_{t+1}^h is negative (higher habits make utility lower), this tells us that the RHS of equation (1.11.9) will be a larger positive number than it would be without habits. This means that the level of c_t that satisfies the first order condition will be a lower number (higher marginal utility) than before. Hence, habits increase the willingness to delay spending, and increase the saving rate.

Note that the first order condition also implies that

$$\frac{d\underline{v}_t}{dc_t} = 0 \quad (1.11.10)$$

when evaluated at $c_t = \mathbf{c}_t(m_t, h_t)$.

Envelope Conditions

Now consider the total derivative of $\underline{v}_t(m_t, h_t, \mathbf{c}_t(m_t, h_t))$ with respect to m_t . (To reduce clutter, I will write $d\mathbf{c}_t(m_t, h_t)/dm_t$ as $d\mathbf{c}_t/dm_t$). The chain rule tells us that

$$\begin{aligned} \frac{d\underline{v}_t}{dm_t} &= \frac{d\mathbf{c}_t}{dm_t} u_t^c + \overbrace{\frac{dh_t}{dm_t}}^{=0} u_t^h + \beta \left(\left(\frac{d\mathbf{c}_t}{dm_t} \right) (v_{t+1}^h - Rv_{t+1}^m) + Rv_{t+1}^m \right) \\ &= \left(\frac{d\mathbf{c}_t}{dm_t} \right) \underbrace{\left(u_t^c + \beta (v_{t+1}^h - Rv_{t+1}^m) \right)}_{=0} + \beta Rv_{t+1}^m \end{aligned} \quad (1.11.11)$$

where the underbraced term vanishes at $c_t = \mathbf{c}_t(m_t, h_t)$ by the first-order condition (1.11.8). Thus we have that

$$\begin{aligned} v_t^m &= \frac{d\underline{v}_t}{dm_t} \Big|_{c_t=\mathbf{c}_t(m_t, h_t)} \\ &= \beta Rv_{t+1}^m. \end{aligned} \quad (1.11.12)$$

The Envelope theorem is the shortcut way to obtain this conclusion. The clearest way to use the theorem is by taking the partial derivatives of the $\underline{v}_t$ function with respect to each of its three arguments, using the Chain Rule to take into account the possible dependency of h_t and c_t on m_t :

$$\begin{aligned} v_t(m_t, h_t) &= \underline{v}_t(m_t, h_t, \mathbf{c}_t(m_t, h_t)) \\ v_t^m &= \frac{\partial \underline{v}_t}{\partial m_t} + \frac{\partial \underline{v}_t}{\partial h_t} \underbrace{\frac{\partial h_t}{\partial m_t}}_{=0} + \underbrace{\frac{\partial \underline{v}_t}{\partial c_t}}_{=0} \frac{\partial c_t}{\partial m_t} + \frac{\partial \underline{v}_t}{\partial c_t} \frac{\partial c_t}{\partial h_t} \underbrace{\frac{\partial h_t}{\partial m_t}}_{=0} \end{aligned} \quad (1.11.13)$$

where the Envelope theorem is what tells you that the $\partial \underline{v}_t / \partial c_t$ term is equal to zero because you are evaluating the function at $c_t = \mathbf{c}_t(m_t, h_t)$ (and $\partial h_t / \partial m_t$ is zero by the assumed structure of the problem in which h_t is predetermined).

Now writing out $\partial \underline{v}_t / \partial m_t$, (1.11.13) becomes

$$v_t^m = \frac{\partial}{\partial m_t} [\beta v_{t+1}((m_t - c_t)R + y_{t+1}, h_{t+1})] \quad (1.11.14)$$

which the envelope theorem says is equivalent to

$$\mathbf{v}_t^m = \beta R \mathbf{v}_{t+1}^m. \quad (1.11.15)$$

There is a potentially confusing thing about doing it this way, however: when you reach an expression like (1.11.14) it is tempting to think to yourself as follows: “ c_t is a function of m_t , and $h_{t+1} = c_t$ is also indirectly a function of m_t , so the chain rule tells me that when I take the derivative in (1.11.14) I need to keep track of these.” In fact, you must treat $\partial c_t / \partial m_t$ and $\partial h_{t+1} / \partial m_t$ as zero here. The reason is that this is a *partial* derivative with respect to m_t . The dependence of c_t (and indirectly h_{t+1}) on m_t has already been taken care of in the two terms in (1.11.13) that were equal to zero. The confusion here is caused largely by the fact that partial differentiation is an area where standard mathematical notation is basically confusing and poorly chosen.¹⁷

The shortest way to obtain the end result is, as in the single variable problem, to start with Bellman’s equation and take the partial derivative with respect to m_t directly (treating the problem as though c_t were a constant):

$$\begin{aligned} \mathbf{v}_t(m_t, h_t) &= u(c_t, h_t) + \beta \mathbf{v}_{t+1}((m_t - c_t)R + y_{t+1}, h_{t+1}) \\ \mathbf{v}_t^m(m_t, h_t) &= \beta R \mathbf{v}_{t+1}^m(m_{t+1}, h_{t+1}). \end{aligned} \quad (1.11.16)$$

Whichever way you do it, substituting (1.11.15) into the FOC equation (1.11.9) gives

$$\mathbf{v}_t^m = u_t^c + \beta \mathbf{v}_{t+1}^h. \quad (1.11.17)$$

The intuition for this is as follows. The marginal value of wealth must be equal to the marginal value associated with a tiny bit more consumption. In the presence of habits, the extra consumption yields extra utility today u_t^c but affects value next period by $\mathbf{v}_{t+1}^h$ (which is a negative number), the discounted consequence of which from today’s perspective is the $\beta \mathbf{v}_{t+1}^h$ term.

Envelope Theorem for h_t

In a problem with two state variables, the Envelope theorem can be applied to each state (and indeed in general must be applied in order to solve the model).

Again let’s start the brute force way by working through the total derivative of $\mathbf{v}_t$. For this problem, the total derivative (again denoting $d\mathbf{c}_t(m_t, h_t)/dh_t$ as $d\mathbf{c}_t/dh_t$) is:

$$\begin{aligned} \frac{d\mathbf{v}_t}{dh_t} &= \frac{d\mathbf{c}_t}{dh_t} u_t^c + u_t^h + \beta \left(\frac{dh_{t+1}}{dh_t} \mathbf{v}_{t+1}^h + \frac{dm_{t+1}}{dh_t} \mathbf{v}_{t+1}^m \right) \\ &= \frac{d\mathbf{c}_t}{dh_t} u_t^c + u_t^h + \beta \left(\frac{dh_{t+1}}{d\mathbf{c}_t} \frac{d\mathbf{c}_t}{dh_t} \mathbf{v}_{t+1}^h + \frac{dm_{t+1}}{d\mathbf{c}_t} \frac{d\mathbf{c}_t}{dh_t} \mathbf{v}_{t+1}^m \right) \\ &= u_t^h + \frac{d\mathbf{c}_t}{dh_t} \underbrace{\left(u_t^c + \beta (\mathbf{v}_{t+1}^h - R \mathbf{v}_{t+1}^m) \right)}_{=0} \end{aligned} \quad (1.11.18)$$

where again the underbraced term vanishes at $c_t = \mathbf{c}_t(m_t, h_t)$ by (1.11.8). Thus we have

$$\begin{aligned} \mathbf{v}_t^h &= \frac{d\mathbf{v}_t}{dh_t} \Big|_{c_t = \mathbf{c}_t(m_t, h_t)} \\ &= u_t^h. \end{aligned} \quad (1.11.19)$$

Turning now to more direct use of the envelope theorem, the Chain Rule tells us

$$\mathbf{v}_t^h = \frac{\partial \mathbf{v}_t}{\partial m_t} \overbrace{\frac{\partial m_t}{\partial h_t}}^{=0} + \frac{\partial \mathbf{v}_t}{\partial h_t} + \frac{\partial \mathbf{v}_t}{\partial c_t} \frac{\partial c_t}{\partial h_t} \quad (1.11.20)$$

while the Envelope theorem once again says $\partial \mathbf{v}_t / \partial c_t = 0$ at $c_t = \mathbf{c}_t(m_t, h_t)$ so we obtain

¹⁷Google the string “partial differentiation confusing OCW” to find a fuller description of the problems of standard notation on partial differentiation.

$$\begin{aligned} v_t^h &= \frac{\partial \underline{v}_t}{\partial h_t} \\ &= u_t^h \end{aligned} \quad (1.11.21)$$

since h_t appears directly only in the $u(c_t, h_t)$ part of $\underline{v}_t(m_t, h_t, c_t)$. And once again, the shortest way to the answer is to treat c_t as though it were a constant in the value function, which yields

$$\begin{aligned} v_t(m_t, h_t) &= u(c_t, h_t) + \beta v_{t+1}((m_t - c_t)R + y_{t+1}, c_t) \\ v_t^h(m_t, h_t) &= u_t^h. \end{aligned} \quad (1.11.22)$$

From (1.11.17) this implies that

$$v_t^m = u_t^c + \beta u_{t+1}^h. \quad (1.11.23)$$

Roll this equation forward one period and substitute into equation (1.11.15) to obtain:

$$u_t^c + \beta u_{t+1}^h = R\beta [u_{t+1}^c + \beta u_{t+2}^h] \quad (1.11.24)$$

Note that if $u_{t+1}^h = u_{t+2}^h = 0$ so that habits have no effect on utility, (1.11.24) again is solved by the standard time-separable Euler equation.

Now assume that the utility function takes the specific form

$$u(c, h) = \mathbf{f}(c - \alpha h) \quad (1.11.25)$$

which implies derivatives of

$$\begin{aligned} u^c &= \mathbf{f}' \\ u^h &= -\alpha \mathbf{f}'. \end{aligned} \quad (1.11.26)$$

Substituting these into equation (1.11.24) we obtain,

$$\mathbf{f}'_t - \alpha \beta \mathbf{f}'_{t+1} = R\beta [\mathbf{f}'_{t+1} - \alpha \beta \mathbf{f}'_{t+2}] \quad (1.11.27)$$

Now assume that there is a solution in which marginal utility of consumption grows at a constant rate over time, $\mathbf{f}'_t = k \mathbf{f}'_{t+1}$ and substitute into (1.11.27)

$$\begin{aligned} \mathbf{f}'_{t+1}(k - \alpha\beta) &= R\beta [\mathbf{f}'_{t+2}(k - \alpha\beta)] \\ k \mathbf{f}'_{t+2}(k - \alpha\beta) &= R\beta [\mathbf{f}'_{t+2}(k - \alpha\beta)] \\ k &= R\beta \end{aligned} \quad (1.11.28)$$

so marginal utility grows at rate $1/R\beta$. Note that if we assume $\alpha = 0$ so that habits do not matter, we again obtain the standard result that $u'(c_t) = R\beta u'(c_{t+1})$.

Now make the final assumption that $\mathbf{f}(z) = z^{1-\rho}/(1-\rho)$, implying of course that $\mathbf{f}'(z) = z^{-\rho}$. Equation (1.11.28) can be rewritten

$$1 = R\beta (z_{t+1}/z_t)^{-\rho} \quad (1.11.29)$$

Now expand z_{t+1}/z_t

$$\frac{c_{t+1} - \alpha c_t}{c_t - \alpha c_{t-1}} = \frac{c_{t+1}/c_t - \alpha}{1 - \alpha c_{t-1}/c_t} \quad (1.11.30)$$

$$\approx \frac{1 + \Delta \log c_{t+1} - \alpha}{1 - \alpha + \alpha \Delta \log c_t} \quad (1.11.31)$$

$$= \frac{1 - \alpha + \Delta \log c_{t+1}}{1 - \alpha + \alpha \Delta \log c_t} \quad (1.11.32)$$

$$= \frac{1 + (\Delta \log c_{t+1})/(1 - \alpha)}{1 + (\alpha/(1 - \alpha))\Delta \log c_t} \quad (1.11.33)$$

$$\approx 1 + \left(\frac{1}{1 - \alpha} \right) (\Delta \log c_{t+1} - \alpha \Delta \log c_t) \quad (1.11.34)$$

where (1.11.31) follows from (1.11.30) because $c_{t+1}/c_t = 1 + (c_{t+1} - c_t)/c_t \approx 1 + \Delta \log c_{t+1}$ and $c_{t-1}/c_t = (c_t - (c_t - c_{t-1}))/c_t \approx 1 - \Delta \log c_t$, and (1.11.34) follows from (1.11.33) because for small η and ϵ , $(1 + \eta)/(1 + \epsilon) \approx 1 + \eta - \epsilon$.

Substituting (1.11.34) into (1.11.29) gives

$$\begin{aligned} 1 &\approx R\beta \left(1 + \left(\frac{1}{1 - \alpha} \right) (\Delta \log c_{t+1} - \alpha \Delta \log c_t) \right)^{-\rho} \\ 0 &\approx \log[R\beta] - \rho \left(\frac{1}{1 - \alpha} \right) (\Delta \log c_{t+1} - \alpha \Delta \log c_t) \\ \Delta \log c_{t+1} &\approx (1 - \alpha)\rho^{-1}(r - \vartheta) + \alpha \Delta \log c_t. \end{aligned} \quad (1.11.35)$$

Thus, this formulation of habit formation implies that the growth rate of consumption is serially correlated.

1.12 Durables

A durable good is one that provides utility over multiple periods rather than being consumed immediately. The consumer here derives utility from nondurable consumption c_t as well as from a stock of durable goods d_t .¹⁸ The consumer's goal is to

$$\max \sum_{s=t}^T \beta^{s-t} u(c_s, d_s) \quad (1.12.1)$$

where d_s is the stock of the durable good, and all other variables are as usually defined. We will assume that the stock of the durable good evolves over time according to

$$d_{t+1} = (1 - \delta)d_t + x_{t+1}, \quad (1.12.2)$$

where x_t is period- t expenditure on the durable good and δ is the durable good's depreciation rate (a good with a lower value of δ is said to be "more durable"). This geometric depreciation assumption contrasts with alternatives like "one-hoss-shay" depreciation, where each period there is some probability that the durable completely fails.

For simplicity, we abstract from asymmetries between buying and selling the durable good (there is no restriction requiring $x_t > 0$).

The dynamic budget constraint subtracts expenditures on durables from available resources:

$$m_{t+1} = (m_t - c_t - x_t)R + y_{t+1}. \quad (1.12.3)$$

Bellman's equation is

$$v_t(m_t, d_{t-1}) = \max_{\{c_t, x_t\}} [u(c_t, d_t) + \beta v_{t+1}(m_{t+1}, d_t)], \quad (1.12.4)$$

where d_{t-1} is the state variable because the level of durables in period t is not determined until you choose spending on durables in that period. Equivalently, treating d_t directly as the control variable:

$$v_t(m_t, d_{t-1}) = \max_{\{c_t, d_t\}} [u(c_t, d_t) + \beta v_{t+1}(m_{t+1}, d_t)], \quad (1.12.5)$$

subject to

$$m_{t+1} = \left(m_t - c_t - \overbrace{(d_t - (1 - \delta)d_{t-1})}^{=x_t} \right) R + y_{t+1} \quad (1.12.6)$$

or (substituting this into (1.12.5)),

$$v_t(m_t, d_{t-1}) = \max_{\{c_t, d_t\}} \{u(c_t, d_t) + \beta v_{t+1}((m_t - c_t - (d_t - (1 - \delta)d_{t-1}))R + y_{t+1}, d_t)\}. \quad (1.12.7)$$

Since this equation has two control variables, c_t and d_t , there are two first order conditions:

wrt c_t :

$$\begin{aligned} u_t^c - R\beta v_{t+1}^m &= 0 \\ u_t^c &= R\beta v_{t+1}^m \end{aligned} \quad (1.12.8)$$

wrt d_t :

$$u_t^d = \beta(Rv_{t+1}^m - v_{t+1}^d) = R\beta v_{t+1}^m - \beta v_{t+1}^d. \quad (1.12.9)$$

¹⁸The basic ideas in this section are derived from Mankiw [1982]. See Carroll and Dunn [1997] for further discussion of the frictionless model and empirical estimates, as well as a model that incorporates transactions costs.

Note that when taking the derivative with respect to c_t you assume that $\partial d_t / \partial c_t = 0$ and vice versa. Although the first order conditions will define a relationship between the *optimal* values of c_t and d_t , there is no *mechanical* link that applies at this point.

Now we want to apply the Envelope theorem. Basically, the Envelope theorem says that at the optimal levels of the control variables the partial derivative of the entire value function with respect to each control variable is zero. This means that when taking the derivative with respect to a *state* variable you can simply ignore all terms that involve $\partial c_t / \partial m_t$, $\partial d_t / \partial m_t$, $\partial c_t / \partial d_{t-1}$, and $\partial d_t / \partial d_{t-1}$. So, for example, the full expression for the derivative of the value function with respect to m_t is:

$$\begin{aligned} v_t^m = & \frac{\partial u(c_t, d_t)}{\partial c_t} \frac{\partial c_t}{\partial m_t} + \frac{\partial u(c_t, d_t)}{\partial d_t} \frac{\partial d_t}{\partial m_t} \\ & + \left[\frac{\partial m_{t+1}}{\partial m_t} + \frac{\partial m_{t+1}}{\partial c_t} \frac{\partial c_t}{\partial m_t} + \frac{\partial m_{t+1}}{\partial d_t} \frac{\partial d_t}{\partial m_t} \right] \beta v_{t+1}^m + \beta v_{t+1}^d \frac{\partial d_t}{\partial m_t} \end{aligned} \quad (1.12.10)$$

but the Envelope theorem tells us to ignore all the terms that involve $\partial c_t / \partial m_t$ or $\partial d_t / \partial m_t$; then because the only term in that whole mess above that does *not* involve either $\partial c_t / \partial m_t$ or $\partial d_t / \partial m_t$ is $\partial m_{t+1} / \partial m_t = R$ we have:

$$v_t^m = R \beta v_{t+1}^m \quad (1.12.11)$$

Applying the same Envelope theorem logic for d_{t-1} yields:¹⁹

$$\begin{aligned} v_t^d &= R(1 - \delta) \beta v_{t+1}^m \\ &= (1 - \delta) R \beta v_{t+1}^m \\ &= (1 - \delta) v_t^m \end{aligned} \quad (1.12.12)$$

Think now about the case where depreciation is 100 percent ($\delta = 1$); from (1.12.12) it is clear that in this case $v_t^d = 0$. This makes sense because in this case the “durable” good is really a totally nondurable good. $v_t^{d_{t-1}} = 0$ because the amount that you consumed of a nondurable good last period has no direct effect on your current utility (we have assumed that utility is time separable).

The $\delta = 0$ case is more interesting. In this case the marginal utility of having an extra unit of durable good last period is equal to the marginal utility of having an extra unit of wealth this period. Why? Because if $\delta = 0$ the durable good doesn’t depreciate at all. How much would it cost to buy another unit of durable good today? One unit of wealth. Because the durable does not depreciate from period to period and can be transformed into and out of wealth at a one-to-one price, it is exactly as valuable as a unit of wealth.

Now we want to try to derive a relationship between the contemporaneous marginal utilities of d and c . From (1.12.9) we have:

$$u_t^d = R \beta v_{t+1}^m - \beta v_{t+1}^d. \quad (1.12.13)$$

and $R \beta v_{t+1}^m = u_t^c$ and from (1.12.12) $v_{t+1}^d = (1 - \delta) v_{t+1}^m$. Substituting these into (1.12.13):

$$\begin{aligned} u_t^d &= u_t^c - \beta(1 - \delta) v_{t+1}^m \\ &= u_t^c - \frac{(1 - \delta)}{R} R \beta v_{t+1}^m \\ &= \left[1 - \frac{(1 - \delta)}{R} \right] u_t^c \\ &= \left[\frac{r + \delta}{R} \right] u_t^c \end{aligned} \quad (1.12.14)$$

Assuming $\delta < 1$, this equation tells us that the marginal utility *in the current period* of a unit of spending on the durable good is lower than the marginal utility of spending on the nondurable. Why? Because the durable good will yield utility in the future as well as in the present. What should be equated to the marginal utility of nondurables consumption is the total discounted lifetime utility

¹⁹Both here and in (1.12.10), the derivative v_t^d should be understood as a differentiation with respect to d_{t-1} .

from an extra unit of the durable good, not simply the marginal utility it yields right now. You don't buy a car because it is worth \$20,000 to you on the day you buy it; you buy a car because its expected discounted value over its lifetime is \$20,000 or more.

Now assume the utility function is of the Cobb-Douglas form: $u(c, d) = \frac{(c^{1-\alpha}d^\alpha)^{1-\rho}}{1-\rho}$. This implies that the instantaneous marginal utilities with respect to c and d are:

$$\begin{aligned} u^c &= (c^{1-\alpha}d^\alpha)^{-\rho}(1-\alpha)c^{-\alpha}d^\alpha \\ &= (c^{1-\alpha}d^\alpha)^{-\rho}(1-\alpha)(d/c)^\alpha \\ u^d &= (c^{1-\alpha}d^\alpha)^{-\rho}\alpha c^{1-\alpha}d^{\alpha-1} \\ &= (c^{1-\alpha}d^\alpha)^{-\rho}\alpha(d/c)^{\alpha-1} \end{aligned} \tag{1.12.15}$$

Substituting these definitions into (1.12.14) gives:

$$\begin{aligned} (c^{1-\alpha}d^\alpha)^{-\rho}\alpha(d/c)^{\alpha-1} &= (c^{1-\alpha}d^\alpha)^{-\rho}(1-\alpha)(d/c)^\alpha \left(\frac{r+\delta}{R} \right) \\ \frac{\alpha}{1-\alpha} &= (d/c) \left(\frac{r+\delta}{R} \right) \\ d/c &= \left(\frac{\alpha}{1-\alpha} \right) \left(\frac{R}{r+\delta} \right) \equiv \gamma \end{aligned} \tag{1.12.16}$$

What this implies is that whenever the level of nondurables consumption changes, the level of the *stock* of durables should change by the same proportion. Because *expenditures* on durable goods are equal to the *change* in the stock plus depreciation, a change in c implies spending on durables large enough to immediately adjust the stock to the new target level. (Recall that d_t was the *stock* of durable good owned in period t , while *spending* on the durable good was defined as $x_t = d_t - (1-\delta)d_{t-1}$.)

Define $\gamma = d_t/c_t$ as in (1.12.16). Now consider a consumer who had consumed the same amount of the nondurable good for periods $c_{t-2} = c_{t-1}$ but who between period $t-1$ and period t learns some good news about permanent income; she adjusts her nondurables consumption up so that $c_t/c_{t-1} = (1+\epsilon_t)$. This implies that the level of *spending* in period t is:

$$\begin{aligned} x_t &= d_t - (1-\delta)d_{t-1} = \gamma[c_t - (1-\delta)c_{t-1}] \\ x_{t-1} &= \gamma[c_{t-1} - (1-\delta)c_{t-2}] \\ &= \gamma\delta c_{t-1} \\ x_t/x_{t-1} &= \gamma[c_{t-1}(1+\epsilon_t) - (1-\delta)c_{t-1}]/\gamma\delta c_{t-1} \\ &= \frac{\epsilon_t + \delta}{\delta} \end{aligned} \tag{1.12.17}$$

Assuming $\delta < 1$, this equation implies that *spending on durable goods should be more variable than spending on nondurable goods*. For goods with a low depreciation rate, spending should be much more variable. This is true because the ratio of the stock of durables to income is much larger than the ratio of the average level of spending on durables to income. Consider housing: a house might be worth 3 times permanent income, so fluctuations in permanent income mean perpetually buying new “chunks” of house (when P rises) or selling off portions (when P falls). This model abstracts from transactions costs, which are important in reality; durables spending is in fact empirically much more volatile than nondurables spending.

A further implication of this model is that the degree of correlation between nondurables spending growth and durables spending growth depends on the frequency under consideration. For a given quarterly depreciation rate (say, 5 percent per quarter), the durable good will have almost completely depreciated over the course of 10 years = 40 quarters because $0.95^{40} = 0.12$. According to the model, over an interval long enough for the durable to have completely depreciated, the rate of growth of spending on the durable should match the rate of growth of spending of the nondurable, because over such a long interval they are really both nondurable.

Some evidence on this proposition is provided in the Jupyter notebook available [here](#).

1.13 Risk Premia and Precautionary Premia

A consumer who faces a risk to future consumption suffers two distinct consequences, and it is easy to conflate them. The first is that the risk makes the consumer worse off; we would like to know how much the consumer would pay to be rid of it. The second is that the risk changes the consumer's behavior, inducing extra saving; we would like to know how much. The two questions have different answers, and each is governed by a different property of the utility function. This section defines the measure that answers each one.

Suppose the consumer would consume $\bar{c}$ in the absence of risk, and now faces an additive mean-zero shock ϵ that takes the value $-\alpha$ or $+\alpha$, each with probability one half. The parameter $\alpha > 0$ indexes the size of the risk.

1.13.1 The Pratt-Arrow Risk Premium

The measure of how much the consumer dislikes the risk was introduced by Pratt [1964]. The risk premium π^* is the amount by which certain consumption would have to be reduced to make the consumer exactly as badly off as the risk makes them:

$$u(\bar{c} - \pi^*) = \mathbb{E}[u(\bar{c} + \epsilon)]. \quad (1.13.1)$$

In words, π^* measures willingness to pay, in units of certain consumption, to have the risk removed.

Figure 1.10 constructs π^* geometrically. The two possible outcomes $\bar{c} - \alpha$ and $\bar{c} + \alpha$ are marked on the horizontal axis, and the dashed straight line joining the corresponding points on the utility function is the chord between them. Because each outcome occurs with probability one half, expected utility $\mathbb{E}[u(\bar{c} + \epsilon)]$ is the height of the midpoint of that chord, which sits directly above $\bar{c}$. The figure shows this height lying strictly below $u(\bar{c})$, the height of the curve itself at $\bar{c}$; this is Jensen's inequality, and it holds for any mean-zero risk whenever u is concave, $u'' < 0$. Tracing that height horizontally back to the utility function locates the certainty-equivalent consumption level $\bar{c} - \pi^*$, and the horizontal distance from there to $\bar{c}$ is the risk premium. Concavity of u is therefore exactly what makes π^* positive: risk aversion is what makes the risk unwelcome.

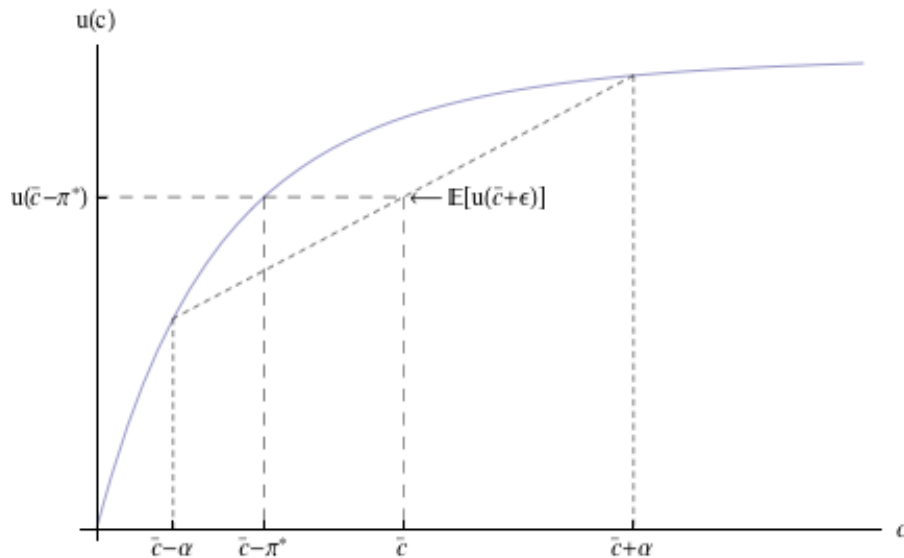


Figure 1.10: The Pratt-Arrow risk premium π^* . Expected utility is the midpoint of the chord, which lies below the utility function because u is concave.

1.13.2 The Kimball Precautionary Premium

Willingness to pay, however, is not what determines saving. Saving is chosen by comparing marginal utilities, so the object that matters for behavior is the effect of the risk on expected *marginal* utility.

The corresponding measure comes from Kimball [1990], who defines the precautionary premium μ^* as the reduction in certain consumption that would raise marginal utility by as much as the risk does:

$$u'(\bar{c} - \mu^*) = \mathbb{E}[u'(\bar{c} + \epsilon)]. \quad (1.13.2)$$

The definition is (1.13.1) with u' written everywhere in place of u . Its behavioral content is that a consumer facing the risk acts, at the margin, like a consumer with no risk whose consumption has been cut by μ^* ; the extra saving that produces the cut is precautionary saving.

Figure 1.11 repeats the geometry of the previous figure one derivative up. The chord now joins $u'(\bar{c} - \alpha)$ to $u'(\bar{c} + \alpha)$, and its midpoint gives $\mathbb{E}[u'(\bar{c} + \epsilon)]$. The direction of the inequality reverses: because u' is a *convex* function, the chord lies *above* the curve, so the risk *raises* expected marginal utility. Convexity of u' means $u''' > 0$, the property Kimball named prudence. Since u' is downward sloping, reading the elevated marginal utility back to the curve locates a consumption level *below* $\bar{c}$, namely $\bar{c} - \mu^*$, so $\mu^* > 0$. Prudence, not risk aversion, is what generates precautionary saving.

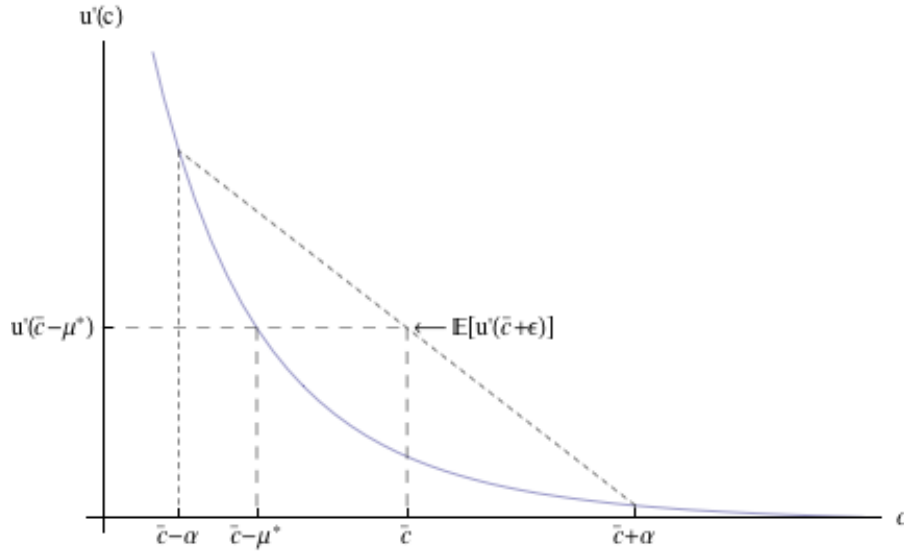


Figure 1.11: The Kimball precautionary premium μ^* . Expected marginal utility is the midpoint of the chord, which lies above u' because u' is convex.

1.13.3 Absolute Risk Aversion and Absolute Prudence

The parallel between the two definitions carries over to the parallel between the two measures of curvature that govern them. Writing $\sigma^2 = \mathbb{E}[\epsilon^2] = \alpha^2$ for the variance of the risk, a second-order expansion of each side of (1.13.1) around $\bar{c}$ gives

$$\pi^* \approx \left(\frac{\sigma^2}{2} \right) \left(\frac{-u''(\bar{c})}{u'(\bar{c})} \right), \quad (1.13.3)$$

in which the second factor is the coefficient of absolute risk aversion. The identical calculation applied to (1.13.2), with u' in place of u , gives

$$\mu^* \approx \left(\frac{\sigma^2}{2} \right) \left(\frac{-u'''(\bar{c})}{u''(\bar{c})} \right), \quad (1.13.4)$$

in which the second factor is what Kimball calls the coefficient of absolute prudence. Absolute prudence therefore stands to the precautionary premium exactly as absolute risk aversion stands to the risk premium.²⁰

²⁰Multiplying each coefficient by $\bar{c}$ yields the relative measures, relative risk aversion $-\bar{c}u''/u'$ and relative prudence $-\bar{c}u'''/u''$.

For the CRRA utility function used throughout these notes, absolute risk aversion is $\rho/\bar{c}$ while absolute prudence is $(\rho + 1)/\bar{c}$. Two things follow. First, prudence is implied by risk aversion in this family, so any CRRA consumer who dislikes risk also saves against it. Second, prudence strictly exceeds risk aversion, so $\mu^* > \pi^*$: the risk moves behavior by more than it moves welfare. That the two coefficients differ by exactly one unit of relative curvature is a special feature of CRRA, not a general result.

Logarithmic utility makes the gap concrete. Setting $\rho = 1$ gives relative risk aversion of one but relative prudence of two, so the log consumer's precautionary saving is governed by a curvature twice as large as the curvature that governs their distaste for the risk. Quoting a coefficient of relative risk aversion therefore does not by itself pin down how much precautionary saving a risk will induce.

1.13.4 A Numerical Illustration

Take relative risk aversion $\rho = 6$, baseline consumption $\bar{c} = 3$, and a risk of $\alpha = 0.75$, so consumption is either 2.25 or 3.75 with equal probability. Solving (1.13.1) and (1.13.2) exactly gives $\pi^* \approx 0.454$ and $\mu^* \approx 0.494$, confirming $\mu^* > \pi^*$.

The approximations are less impressive. Equations (1.13.3) and (1.13.4) return 0.563 and 0.656, overstating both premia by more than twenty percent. The discrepancy is what a local approximation should be expected to produce when the risk is a shock of ± 25 percent of consumption. The approximations are reliable guides to how the premia depend on preferences and on the scale of the risk, and unreliable guides to their level when the risk is large.

Prudence is the engine of the buffer stock saving behavior taken up in the sections that follow, and Carroll and Kimball [2007] survey how precautionary motives translate into accumulated wealth.

1.14 Age Consumption/Income Profiles In Data and Theory

This section reproduces figures from Carroll [1997] showing the profiles of income and consumption by age across different occupation groups from the 1960-61 *Consumer Expenditure Survey*. The unconstrained perfect foresight/certainty equivalent life cycle model implies that there should be no relationship between the pattern of the consumption profile and the income profile, since the shape of the consumption profile is determined by tastes independently of the shape of the income profile.

In practice, there seems to be a strong connection between the income and consumption profiles. The second set of figures shows the implications of a “buffer stock” model of optimizing life cycle behavior in which consumers are modestly impatient, have a standard CRRA utility function with a precautionary saving motive, and face uncertain income. Consumers in this model optimally engage in buffer stock saving behavior until 10-15 years before retirement, as a result of their combination of impatience and prudence. This model is thus capable of matching the broad pattern of the data.

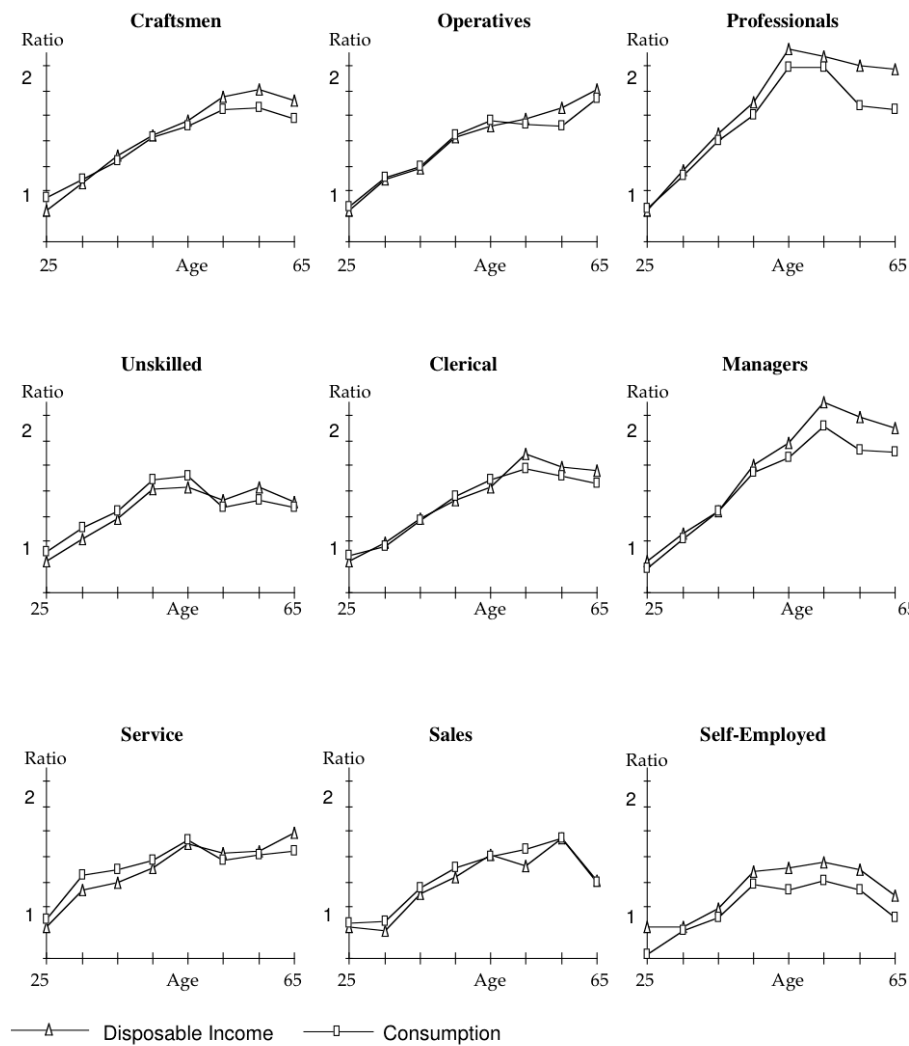


Figure IV
Lifetime Consumption and Income Profiles

Figure 1.12: Age Profiles of Income and Consumption in Data

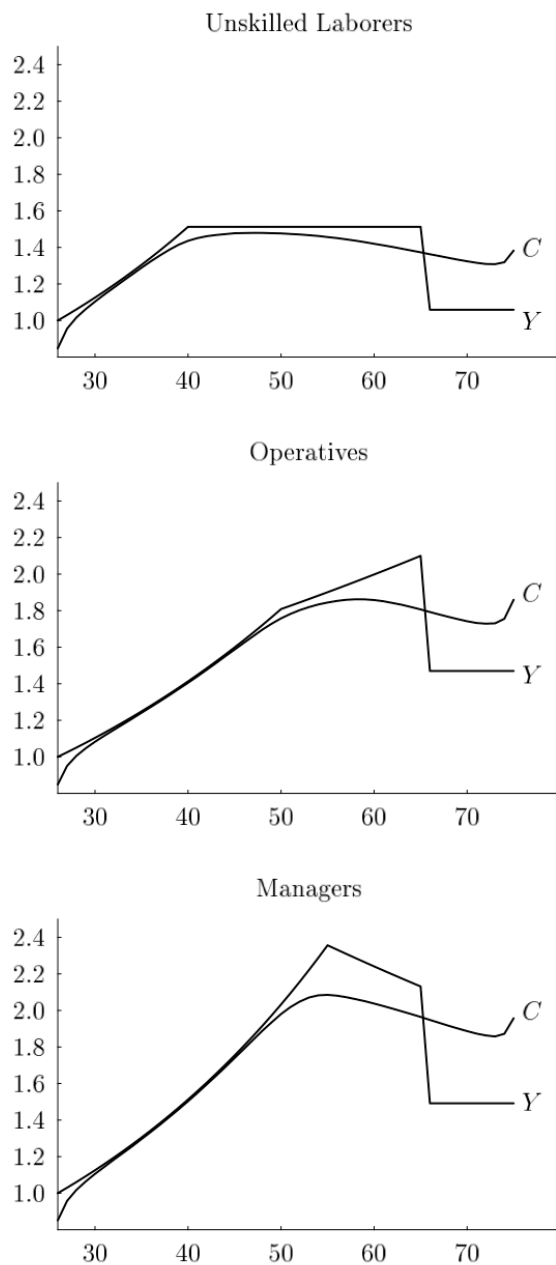


Figure V
Age Profiles of Consumption Predicted by the Model
for Three Profiles of Lifetime Income

Figure 1.13: Age Profiles of Income and Consumption in Theory (Buffer Stock Model)

1.15 The Laibson Model of Time Inconsistency

This section provides a simple example of a discrete-time solution to the problem of a consumer with a self-control problem *à la* Laibson [1997].²¹

Suppose a value function $v_{t+1}(m_{t+1})$ exists for period $t + 1$. Then for any period- t consumption function χ_t we can define

$$\begin{aligned} v_t(m_t; \chi_t) &= u(\chi_t(m_t)) + \beta \mathbb{E}_t[v_{t+1}((m_t - \chi_t(m_t))R + y_{t+1})] \\ \mathbf{v}_t(m_t; \chi_t) &= u(\chi_t(m_t)) + \delta \beta \mathbb{E}_t[v_{t+1}((m_t - \chi_t(m_t))R + y_{t+1})] \end{aligned} \quad (1.15.1)$$

Notice that these functions are well defined for any consumption function $\chi_t(m_t)$ that is feasible; they are *not* Bellman equations because they do not assume that the consumption function χ_t is optimal. For example, these functions would be well defined for $\chi_t(m_t) = m_t$, or for $\chi_t(m_t) = 1$, or for many other potential consumption rules.

What these functions capture is the value of behaving according to the rule χ_t in the current period, under two possible assumptions about discounting of the future: Either next period's value is discounted by the factor β (for v_t) or by $\delta\beta$ (for $\mathbf{v}_t$).

Now consider two possible candidates for χ_t :

$$\begin{aligned} \mathbf{c}_t(m_t) &= \arg \max_c u(c) + \beta \mathbb{E}_t[v_{t+1}((m_t - c)R + y_{t+1})] \\ \mathbf{c}_t(m_t) &= \arg \max_c u(c) + \delta \beta \mathbb{E}_t[v_{t+1}((m_t - c)R + y_{t+1})] \end{aligned} \quad (1.15.2)$$

If we solve the problem recursively using $\chi = \mathbf{c}$ in every period, we obtain the standard time consistent solution. (Think about why).

The Laibson alternative is to suppose that there is something special about “now”: Next period's value is discounted not only by the standard geometric discount factor β , but also by an extra factor δ (Laibson argues that at an annual frequency the appropriate value of δ is about 0.7). This may reflect the fact that certain areas of the brain associated with emotional rewards are activated only by instant gratification, and are not activated by thoughts of future gratification (see, e.g., Cohen et al. [2004]).

It is clear from comparing the equations in (1.15.2) that the consumer with Laibson preferences will consume more in the current period, because he values future rewards less.

More insight about the solution can be obtained from the modified Euler equation that can be derived for the Laibson problem. This is derived as follows.

Note first that if $\chi_t = \mathbf{c}_t$ the [Envelope theorem](#) implies that

$$\mathbf{v}_t^m(m_t) = u'(c_t) \quad (1.15.3)$$

while the first order condition from the maximization problem implies that

$$u'(c_t) = \delta R \beta \mathbb{E}_t[v_{t+1}^m(m_{t+1})] \quad (1.15.4)$$

Now note that for $\chi_t = \mathbf{c}_t$ there is a simple identity linking v and $\mathbf{v}$:

$$\delta v_t = \mathbf{v}_t - (1 - \delta)u(\mathbf{c}_t(m_t)) \quad (1.15.5)$$

(to see this, multiply the first equation in (1.15.1) by δ and note that the difference between the result and the second equation is $(1 - \delta)u(c_t)$). Now differentiate (1.15.5)

$$\delta v_t^m = \mathbf{v}_t^m - (1 - \delta)u'(c_t)\mathbf{c}_t^m(m_t) \quad (1.15.6)$$

and substitute the envelope relation $\mathbf{v}_t^m(m_t) = u'(c_t)$ to express the marginal long-run value of wealth in terms of the marginal utility of consumption,

²¹Laibson's own lecture notes are available on his website, and are very good; as of this writing, they are in lecture 6 of his course 2010c, handouts, lecture06.

$$\delta v_t^m(m_t) = (1 - (1 - \delta)\mathfrak{c}_t^m(m_t)) u'(c_t). \quad (1.15.7)$$

Advancing (1.15.7) by one period and substituting it into (1.15.4) yields the modified Euler equation,

$$u'(c_t) = R\beta\mathbb{E}_t[(1 - (1 - \delta)\mathfrak{c}_{t+1}^m(m_{t+1})) u'(c_{t+1})]. \quad (1.15.8)$$

If $\delta = 1$, this collapses to the usual consumption Euler equation $u'(c_t) = R\beta\mathbb{E}_t[u'(c_{t+1})]$. However, if $\delta < 1$ (the Laibson case), the equation says several interesting things. First, note that since $(1 - \delta)$ and $u'(c_{t+1})$ and $\mathfrak{c}_{t+1}^m$ are all positive, the contribution of the “Laibson” term in (1.15.8) is to reduce the RHS of the equation. In order to match a lower RHS, the LHS must be smaller. But a smaller marginal utility of consumption implies a higher level of consumption, so the Laibson consumer spends more.

Second, notice that the magnitude of the “present-bias” effect depends on the size of next period’s marginal propensity to consume $\mathfrak{c}_{t+1}^m$. If the MPC is small, the size of the Laibson bias will be small. The reason is transparent if the bracketed term in (1.15.8) is rewritten as

$$1 - (1 - \delta)\mathfrak{c}_{t+1}^m = \delta\mathfrak{c}_{t+1}^m + (1 - \mathfrak{c}_{t+1}^m), \quad (1.15.9)$$

which shows that the effective discount factor applied to next period’s marginal utility is a weighted average of the short-horizon factor $\delta\beta$ and the long-horizon factor β . The weight on the short-horizon factor is next period’s marginal propensity to consume $\mathfrak{c}_{t+1}^m$ and the weight on the long-horizon factor is next period’s marginal propensity to save $1 - \mathfrak{c}_{t+1}^m$. Resources that the next-period self will spend immediately get discounted at the present-biased rate, while resources that the next-period self will pass along to still-later selves get discounted geometrically. A consumer whose next-period MPC is near zero therefore behaves almost like an exponential discounter.

Finally, notice that this model nicely captures the commonplace psychological tension in which the cost of deviating from the optimal plan in a single period may be trivially small (“eating dessert this one time will not make me fat”), but the consequences of perpetual deviation could be quite large (“but if I give in to temptation this time, maybe that means I will always give in.”)

1.16 Consumption out of Risky Assets

Consider a consumer with CRRA utility whose only available financial asset has a risky return factor $\acute{\mathbf{R}}$ which is lognormally distributed, $\log \acute{\mathbf{R}}_{t+1} \sim \mathcal{N}(\mathbf{r} - \sigma_{\mathbf{r}}^2/2, \sigma_{\mathbf{r}}^2)$.

With market assets m , the dynamic budget constraint is:

$$m_{t+1} = (m_t - c_t)\acute{\mathbf{R}}_{t+1}. \quad (1.16.1)$$

Start with the standard Euler equation for consumption under CRRA utility:

$$1 = \beta \mathbb{E}_t \left[\acute{\mathbf{R}}_{t+1} \left(\frac{c_{t+1}}{c_t} \right)^{-\rho} \right] \quad (1.16.2)$$

and postulate a solution of the form $c_t = \kappa m_t$. The guess-and-verify method works here because market resources m_t appear in both numerator and denominator, allowing them to cancel; if labor income appeared in the numerator, this approach would fail.

$$\begin{aligned} 1 &= \beta \mathbb{E}_t \left[\acute{\mathbf{R}}_{t+1} \left(\frac{\kappa m_{t+1}}{\kappa m_t} \right)^{-\rho} \right] \\ &= \beta \mathbb{E}_t \left[\acute{\mathbf{R}}_{t+1} \left(\frac{(m_t - c_t)\acute{\mathbf{R}}_{t+1}}{m_t} \right)^{-\rho} \right] \\ &= \beta \mathbb{E}_t \left[\acute{\mathbf{R}}_{t+1} \left(\frac{(1 - \kappa)m_t \acute{\mathbf{R}}_{t+1}}{m_t} \right)^{-\rho} \right] \\ &= \beta \mathbb{E}_t \left[\acute{\mathbf{R}}_{t+1} \left((1 - \kappa)\acute{\mathbf{R}}_{t+1} \right)^{-\rho} \right] \\ &= \beta (1 - \kappa)^{-\rho} \mathbb{E}_t [\acute{\mathbf{R}}_{t+1}^{1-\rho}] \\ (1 - \kappa)^\rho &= \beta \mathbb{E}_t [\acute{\mathbf{R}}_{t+1}^{1-\rho}] \\ (1 - \kappa) &= \left(\beta \mathbb{E}_t [\acute{\mathbf{R}}_{t+1}^{1-\rho}] \right)^{1/\rho} \\ \kappa &= 1 - \left(\beta \mathbb{E}_t [\acute{\mathbf{R}}_{t+1}^{1-\rho}] \right)^{1/\rho} \end{aligned} \quad (1.16.3)$$

which (finally) yields an exact formula for κ :

$$\kappa = 1 - \left(\beta \mathbb{E}_t [\acute{\mathbf{R}}_{t+1}^{1-\rho}] \right)^{1/\rho}. \quad (1.16.4)$$

Since $\log \acute{\mathbf{R}}_{t+1}^{1-\rho} = (1-\rho) \log \acute{\mathbf{R}}_{t+1}$, fact ELogNormTimes implies that (using the definition $\exp(\bullet) \equiv e^\bullet$),

$$\begin{aligned} \mathbb{E}_t [\acute{\mathbf{R}}_{t+1}^{1-\rho}] &= \exp[(1-\rho)(\mathbf{r} - \sigma_{\mathbf{r}}^2/2) + (1-\rho)^2 \sigma_{\mathbf{r}}^2/2] \\ &= \exp[(1-\rho)\mathbf{r} - (1-\rho)(\sigma_{\mathbf{r}}^2/2) + (1-\rho)(\sigma_{\mathbf{r}}^2/2) - \rho(1-\rho)\sigma_{\mathbf{r}}^2/2] \\ &= \exp[(1-\rho)\mathbf{r} - \rho(1-\rho)\sigma_{\mathbf{r}}^2/2]. \end{aligned} \quad (1.16.5)$$

Substituting in (1.16.4):

$$\begin{aligned} \kappa &= 1 - \beta^{1/\rho} \exp \left[\rho \left((1/\rho - 1)\mathbf{r} - (1-\rho)\sigma_{\mathbf{r}}^2/2 \right) \right]^{1/\rho} \\ &= 1 - \beta^{1/\rho} \exp \left[(1/\rho - 1)\mathbf{r} - (1-\rho)\sigma_{\mathbf{r}}^2/2 \right]. \end{aligned} \quad (1.16.6)$$

Now use OverPlus and TaylorOne,

$$\begin{aligned} \beta^{1/\rho} &= \left(\frac{1}{1 + \vartheta} \right)^{1/\rho} \\ &\approx 1 - \rho^{-1} \vartheta \\ &\approx \exp(-\rho^{-1} \vartheta) \end{aligned} \quad (1.16.7)$$

which hold if $\rho^{-1}\vartheta$ is close to zero. Substituting into (1.16.6) and using ExpPlus and LogEps gives

$$\begin{aligned}\kappa &\approx 1 - (1 + \rho^{-1}(\dot{\mathbf{r}} - \vartheta) - \dot{\mathbf{r}} + (\rho - 1)\sigma_{\mathbf{r}}^2/2) \\ &= \dot{\mathbf{r}} - \rho^{-1}(\dot{\mathbf{r}} - \vartheta) - (\rho - 1)(\sigma_{\mathbf{r}}^2/2)\end{aligned}\tag{1.16.8}$$

which, when $\sigma_{\mathbf{r}}^2 = 0$, reduces to the usual perfect foresight formula $\kappa = \dot{\mathbf{r}} - \rho^{-1}(\dot{\mathbf{r}} - \vartheta)$.

This equation implies the plausible result that as unavoidable uncertainty in the financial return goes up ($\sigma_{\mathbf{r}}^2$ rises) the level of consumption falls (because $\rho > 1$, so $-(\rho - 1)$ which multiplies $\sigma_{\mathbf{r}}^2$ is negative). The reduction in consumption as risk increases reflects the precautionary saving motive.²²

The top figure plots the marginal propensity to consume as a function of the coefficient of relative risk aversion (for both the true MPC and the approximation derived above), under parameter values such that $\vartheta - \dot{\mathbf{r}} \approx 0$ so that a change in ρ does not affect the MPC through the intertemporal elasticity of substitution channel. As intuition would suggest, as consumers become more risk averse, they save more (the MPC is lower; that is, the plotted loci are downward-sloping).

The other way to see the precautionary effect is to examine the effect on the MPC of a change in risk. For a consumer with relative risk aversion of 3, the bottom figure shows that as the size of the risk increases, the MPC κ falls.

1.16.1 Relation Between MPC and Parameters

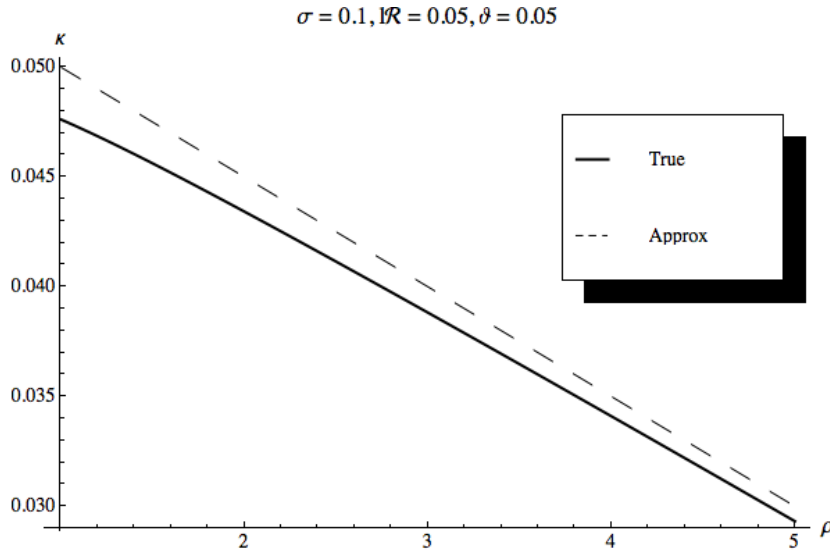
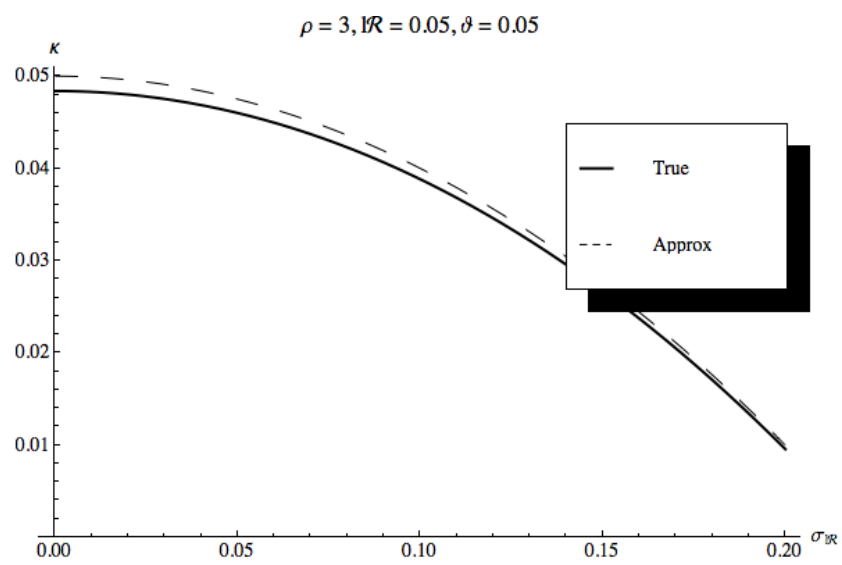


Figure 1.14: Marginal Propensity to Consume Falls as Relative Risk Aversion ρ Rises

²²It is surprising to note that for a consumer with logarithmic utility, a mean-preserving spread in risk has no effect on the level of consumption (this can be seen by substituting $\rho = 1$ into (1.16.8), which causes the term involving risk $\sigma_{\mathbf{r}}^2$ to disappear from the equation). The reason this is surprising is that intuition suggests that if the consumer's consumption (and therefore current saving) are unchanged, the increase in uncertainty must constitute a mean-preserving spread in future consumption, which by Jensen's inequality should yield higher expected marginal utility. The place where this argument goes wrong is that it forgets that the expectation in the Euler equation $u'(c_t) = \beta \mathbb{E}_t[\dot{\mathbf{R}}_{t+1} u'(c_{t+1})]$ is also affected by a covariance between $\dot{\mathbf{R}}_{t+1}$ and $u'(c_{t+1})$; the case of log utility is the special case where this boils down to a constant times $\mathbb{E}_t[\dot{\mathbf{R}}_{t+1}/\dot{\mathbf{R}}_{t+1}] = 1$, which is why the expected marginal utility is unaffected by the unavoidable increase in risk. This is yet another reason (if any more were needed) to conclude that logarithmic utility does not exhibit sufficient curvature to plausibly represent attitudes toward risk. ($\rho \geq 2$ seems a plausible lower bound).

Figure 1.15: Marginal Propensity to Consume Falls as Risk σ Rises

1.17 Consumption with Constant Absolute Risk Aversion (CARA) Utility

Consider the optimization problem of a consumer with a constant absolute risk aversion instantaneous utility function $u(C) = -(1/\alpha)e^{-\alpha C}$ implying $u'(C) = e^{-\alpha C}$ facing an interest rate that is constant at $r = R - 1$.²³ The consumer's optimization problem is

$$\max_{\{C\}_t^T} \mathbb{E}_t \left\{ \sum_{s=t}^T \beta^{s-t} u(C_s) \right\} \quad (1.17.1)$$

subject to the constraints

$$\begin{aligned} B_{t+1} &= (M_t - C_t)R \\ M_{t+1} &= B_{t+1} + Y_{t+1} \end{aligned} \quad (1.17.2)$$

Here M_t denotes “market resources,” the consumer's total resources available for consumption at the beginning of period t , combining beginning-of-period bank balances B_t with current income. The consumer's idiosyncratic income Y_{t+1} exhibits a random-walk deviation from an exogenously-growing trend:

$$\begin{aligned} \bar{P}_{t+1} &= \Gamma \bar{P}_t \\ Y_{t+1} &= \bar{P}_{t+1} + P_{t+1} \\ P_{t+1} &= P_t + \Psi_{t+1}. \end{aligned} \quad (1.17.3)$$

Bellman's equation relates the value function at time t to the value function at $t + 1$ through the choice variables and state variables:

$$V_t(M_t, \bar{P}_t, P_t) = \max_{\{C\}_t^T} u(C_t) + \mathbb{E}_t[\beta V_{t+1}(M_{t+1}, \bar{P}_{t+1}, P_{t+1})]. \quad (1.17.4)$$

The first order condition (FOC) for the CARA utility problem is

$$u'(C_t) = R\beta \mathbb{E}_t[V_{t+1}^M] \quad (1.17.5)$$

and the [Envelope theorem](#) tells us that

$$V_t^M = R\beta \mathbb{E}_t[V_{t+1}^M]. \quad (1.17.6)$$

In the perfect foresight version of the model in which $\Psi_t = 0 \forall t$, the Euler equation will be

$$\begin{aligned} u'(C_t) &= R\beta u'(C_{t+1}) \\ \exp[-\alpha C_t] &= R\beta \exp[-\alpha C_{t+1}] \\ 1 &= R\beta \exp[-\alpha(C_{t+1} - C_t)] \\ \exp[\alpha(C_{t+1} - C_t)] &= R\beta \\ \alpha(C_{t+1} - C_t) &= \log R\beta \\ C_{t+1} &= C_t + \log(R\beta)^{1/\alpha}. \end{aligned} \quad (1.17.7)$$

The $\log(R\beta)^{1/\alpha}$ term reflects the intertemporal substitution factor in consumption. Notice that intertemporal substitution takes the form of *additive* changes in the *level* of consumption in the CARA utility model, rather than multiplicative changes that affect the growth rate of consumption, as in the CRRA model.

Now suppose we are interested in the case where permanent income shocks are distributed normally, $\Psi_t \sim \mathcal{N}(0, \sigma_\Psi^2)$. Then it turns out that the process

$$C_{t+1} = C_t + \log(R\beta)^{1/\alpha} + \alpha\sigma_\Psi^2/2 + \Psi_{t+1} \quad (1.17.8)$$

²³A problem like this was considered in a well-known paper by Caballero [1990].

satisfies the FOC under uncertainty:

$$\begin{aligned}
1 &= R\beta\mathbb{E}_t[\exp[-\alpha(c_{t+1} - c_t)]] \\
1 &= R\beta\mathbb{E}_t[\exp[-\alpha(\alpha\sigma_\Psi^2/2 + \Psi_{t+1} + (1/\alpha)\log(R\beta) + c_t - c_t)]] \\
1 &= R\beta\exp[-\alpha^2\sigma_\Psi^2/2]\mathbb{E}_t\{\exp[-\alpha\Psi_{t+1}]\}\exp[-\alpha(1/\alpha)\log R\beta] \\
1 &= R\beta\exp[-\alpha^2(\sigma_\Psi^2/2)]\exp[\alpha^2(\sigma_\Psi^2/2)]\exp[\log(R\beta)^{-1}] \\
1 &= R\beta(R\beta)^{-1} \\
1 &= 1.
\end{aligned} \tag{1.17.9}$$

Define $\kappa = \log(R\beta)^{1/\alpha} + \alpha\sigma_\Psi^2/2$, so that (1.17.8) becomes:

$$C_{t+1} = C_t + \Psi_{t+1} + \kappa. \tag{1.17.10}$$

The expected present discounted value of consumption is²⁴

$$\begin{aligned}
\mathbb{P}_t(C) &= C_t + (C_t + \Psi_{t+1} + \kappa)/R + (C_t + \Psi_{t+1} + \kappa + \Psi_{t+2} + \kappa)/R^2 + \dots \\
\mathbb{E}_t[\mathbb{P}_t(C)] &= C_t + C_t/R + C_t/R^2 + \dots + \kappa/R + 2\kappa/R^2 + 3\kappa/R^3 + \dots \\
&= C_t(1 + R^{-1} + R^{-2} + \dots) + \kappa \sum_{i=1}^{\infty} i/R^i.
\end{aligned} \tag{1.17.11}$$

Now we need InfSumMult: if $R > 1$, then $\sum_{i=0}^{\infty} i/R^i = \left(\frac{R}{(R-1)^2}\right)$.

Thus, the expectation of the infinite horizon PDV of consumption is:

$$\mathbb{E}_t[\mathbb{P}_t(C)] = C_t \left(\frac{1}{1 - 1/R}\right) + \left(\frac{\kappa R}{(1 - R)^2}\right). \tag{1.17.12}$$

Given the process for income described above, we have

$$\begin{aligned}
\mathbb{P}_t(Y) &= Y_t + Y_{t+1}/R + Y_{t+2}/R^2 + \dots \\
&= \bar{P}_t + P_t + (\Gamma\bar{P}_t + P_{t+1})/R + (\Gamma^2\bar{P}_t + P_{t+2})/R^2 + \dots \\
&= \bar{P}_t \left(1 + \Gamma/R + (\Gamma/R)^2 + \dots\right) + \\
&\quad P_t + (P_t + \Psi_{t+1})/R + (P_t + \Psi_{t+1} + \Psi_{t+2})/R^2 + \dots \\
\mathbb{E}_t[\mathbb{P}_t(Y)] &= \left(\frac{\bar{P}_t}{1 - \Gamma/R}\right) + P_t \sum_{s=0}^{\infty} R^{-s} \\
&= \left(\frac{\bar{P}_t}{1 - \Gamma/R}\right) + \left(\frac{P_t}{1 - 1/R}\right)
\end{aligned} \tag{1.17.13}$$

The IBC says

$$\mathbb{P}_t(C) = B_t + \mathbb{P}_t(Y), \tag{1.17.14}$$

Because the intertemporal budget constraint must hold in every state of the world, the expectation of the PDV of consumption must equal current wealth plus the expectation of the PDV of income. Thus,

²⁴The Ψ_{t+n} terms disappear when expectations are taken.

$$\begin{aligned}
\mathbb{E}_t[\mathbb{P}_t(C)] &= B_t + \mathbb{E}_t[\mathbb{P}_t(Y)] \\
C_t \left(\frac{1}{1 - 1/R} \right) &= B_t + \left(\frac{\bar{P}_t}{1 - \Gamma/R} \right) + \left(\frac{P_t}{1 - 1/R} \right) - \left(\frac{\kappa R}{(1 - R)^2} \right) \\
C_t &= P_t + \left(\frac{r}{R} \right) \left[B_t + \left(\frac{\bar{P}_t}{1 - \Gamma/R} \right) - \left(\frac{\kappa R}{(1 - R)^2} \right) \right] \\
&= P_t + \left(\frac{r}{R} \right) \left[B_t + \left(\frac{\bar{P}_t}{1 - \Gamma/R} \right) \right] - r \left(\frac{\log(R\beta)^{1/\alpha} + \alpha\sigma_\Psi^2/2}{(1 - R)^2} \right)
\end{aligned} \tag{1.17.15}$$

The P_t term reflects the consumer's idiosyncratic level of permanent income, which has no systematic growth (or decline). The next term reflects the MPC out of total “certain” wealth, human and nonhuman. The final term reflects the combination of the intertemporal substitution motive (in the $\log(R\beta)^{1/\alpha}$ term) and the precautionary motive in the $\alpha\sigma_\Psi^2$ term, as is evident from the fact that it equals zero if either there is no precautionary motive ($\alpha = 0$) or there is no uncertainty $\sigma_\Psi^2 = 0$.

Note some peculiar aspects of this solution. First, observe that, marginally, the consumer spends exactly the interest income on capital, $dC_t/dB_t = r/R$. The reason this is peculiar is that the MPC out of capital *does not depend on how impatient the consumer is*. Impatience is reflected in the change in consumption over time, but not in the level of consumption except as that is affected by the budget constraint.

Second, notice that the effect of income uncertainty on saving is the same in absolute dollars regardless of the level of resources or permanent income.

The finite-horizon version of the problem, together with the “impatience” and “finite human wealth” conditions under which the horizon can be extended to infinity, is worked out for CRRA utility in the [perfect foresight CRRA](#) section. The CARA counterpart of that apparatus replaces the multiplicative consumption growth factor $(R\beta)^{1/\rho}$ with the additive drift $\log(R\beta)^{1/\alpha}$ derived in (1.17.7).

1.18 Dynamics of Consumption with Time Varying R

The intertemporal budget constraint for an infinite-horizon representative agent can be written as

$$\begin{aligned}\mathbb{P}_t(C) &= B_t + \overbrace{\mathbb{P}_t(Y)}^{\equiv H_t} \\ &= O_t\end{aligned}\tag{1.18.1}$$

where B_t is the consumer's beginning-of-period stock of physical assets, H_t is human wealth, and O_t is total wealth, human and nonhuman.

R_{t+1} is the riskless, but time-varying, return factor at $t + 1$, and so we can define the dynamic budget constraint for total wealth as

$$O_{t+1} = (O_t - C_t)R_{t+1}.\tag{1.18.2}$$

Campbell and Mankiw [1989] show that the dynamic budget constraint can be manipulated to generate an expression relating the current levels of wealth and consumption to future interest rates. First, divide both sides of (1.18.2) by O_t to obtain

$$\begin{aligned}\left(\frac{O_{t+1}}{O_t}\right) &= \left(1 - \frac{C_t}{O_t}\right) R_{t+1} \\ \Delta o_{t+1} &\approx r_{t+1} + \log(1 - \exp(c_t - o_t)),\end{aligned}\tag{1.18.3}$$

where the lower-case variables represent the logarithms of their upper-case equivalents. Define $x_t \equiv c_t - o_t$ and assume that any variations in interest rates over time are stationary, $\lim_{n \rightarrow \infty} \mathbb{E}_t[R_{t+n}] = R$. In this case, the ratio of consumption to total wealth x_t will be a stationary variable. It seems reasonable, therefore, to consider a Taylor expansion of the DBC around the steady-state value for $c_t - o_t$, which we will designate as x

$$\begin{aligned}\Delta o_{t+1} &\approx r_{t+1} + \log(1 - \exp(x_t)) \\ &\approx r_{t+1} + \log(1 - \exp(x)) + \left(\frac{d}{dx} \log(1 - \exp(x))\right) (x_t - x) \\ &= r_{t+1} + \log(1 - \exp(x)) - \left(\frac{\exp(x)}{1 - \exp(x)}\right) (x_t - x),\end{aligned}\tag{1.18.4}$$

or, for simplicity defining a constant $\xi = 1 - \exp(x)$ (which will be a number slightly less than one)²⁵

$$\begin{aligned}\Delta o_{t+1} &\approx r_{t+1} + \log \xi + \left(\frac{1 - \exp(x) - 1}{\xi}\right) (x_t - x) \\ &= r_{t+1} + \log \xi + \left(\frac{\xi - 1}{\xi}\right) (x_t - x) \\ &= r_{t+1} + \log \xi + \left(1 - \frac{1}{\xi}\right) (x_t - x) \\ &= \log \xi - (1 - 1/\xi)x + r_{t+1} + \left(1 - \frac{1}{\xi}\right) (c_t - o_t) \\ &= \underbrace{\log \xi - (1 - 1/\xi) \log \xi}_{\equiv k} + r_{t+1} + \left(1 - \frac{1}{\xi}\right) (c_t - o_t) \\ &= k + r_{t+1} + \left(1 - \frac{1}{\xi}\right) (c_t - o_t).\end{aligned}\tag{1.18.5}$$

But the definition of the change in wealth is

²⁵ ξ slightly less than one because $\exp(x) = C/W$ is small.

$$\begin{aligned}
\Delta o_t &= o_{t+1} - o_t \\
&= o_{t+1} - c_{t+1} + c_{t+1} - c_t + c_t - o_t \\
&= \Delta c_{t+1} + (c_t - o_t) - (c_{t+1} - o_{t+1})
\end{aligned} \tag{1.18.6}$$

Now set (1.18.6) equal to (1.18.5) and solve for $c_t - o_t$ to get

$$\begin{aligned}
\Delta c_{t+1} + (c_t - o_t) - (c_{t+1} - o_{t+1}) &= k + r_{t+1} + \left(1 - \frac{1}{\xi}\right)(c_t - o_t) \\
(c_t - o_t) \left[1 - \left(1 - \frac{1}{\xi}\right)\right] &= k + r_{t+1} + (c_{t+1} - o_{t+1}) \\
c_t - o_t &= \xi(r_{t+1} - \Delta c_{t+1}) + \xi(c_{t+1} - o_{t+1}) + \xi k.
\end{aligned} \tag{1.18.7}$$

Of course, an equivalent expression can be derived for $c_{t+1} - o_{t+1}$; repeated substitution leads to

$$\begin{aligned}
c_t - o_t &= \xi(r_{t+1} - \Delta c_{t+1}) + \xi(\xi(r_{t+2} - \Delta c_{t+2}) + \xi(c_{t+2} - o_{t+2}) + \xi k) + \xi k \\
&= \sum_{j=1}^{\infty} \xi^j (r_{t+j} - \Delta c_{t+j}) + \xi k / (1 - \xi).
\end{aligned} \tag{1.18.8}$$

This equation is interesting: It says that the ratio of consumption to wealth today (that is, the log difference) must equal the discounted value of the rate of return on wealth minus the growth rate of consumption, plus a constant term. Note that this result derives purely from the dynamic budget constraint, with no behavioral assumptions yet. Thus, holding consumption growth and current wealth constant, higher future interest rates must correspond to higher current consumption. This is just the income effect: If interest rates are higher and future consumption growth the same, you will have more lifetime resources and therefore must spend more today if all resources are to be exhausted (as the IBC requires). Alternatively, if you will have fast consumption growth in the future, you need to have either low consumption today or higher interest rates in the future to earn the income required to finance that fast consumption growth.

This equation is purely the result of the dynamic budget constraint; so far we have said nothing about how consumption is chosen. Now consider a perfect-foresight CRRA utility $u(c) = c^{1-\rho}/(1-\rho)$ model with risk aversion ρ , which implies the Euler equation

$$\begin{aligned}
\left(\frac{C_{t+1}}{C_t}\right) &= (R_{t+1}\beta)^{1/\rho} \\
\Delta c_{t+1} &= \underbrace{\rho^{-1} \log \beta}_{\equiv \mu} + \rho^{-1} r_{t+1}
\end{aligned} \tag{1.18.9}$$

where ρ^{-1} is the intertemporal elasticity of substitution. This equation for consumption growth can be substituted into (1.18.8), to generate

$$c_t - o_t = (1 - \rho^{-1}) \sum_{j=1}^{\infty} \xi^j r_{t+j} + \xi(k - \mu)/(1 - \xi). \tag{1.18.10}$$

All of these results were derived under the assumption of perfect foresight: Interest rates vary over time, but the consumer knows in advance what the pattern of variation will be. If we wish to allow for truly stochastic interest rates, things get somewhat more complicated. Recall that if interest rates are fixed at R and income grows by factor G from period to period, human wealth is

$$\begin{aligned}
H_t &= \left(\frac{Y_t}{1 - G/R}\right) \\
&\approx \left(\frac{Y_t}{r - g}\right).
\end{aligned} \tag{1.18.11}$$

Summers [1981] showed that a permanent change in interest rates has an enormous effect on the value of human wealth. In a model with stochastic interest rates, there is still a large human wealth effect even if interest rates eventually return to some “natural” rate following a shock. Thus, a proper

analysis of the effect of changes in interest rates must take account of the effect of that change not only on the expectations of future interest rates on the RHS of (1.18.10) but also on the level of total wealth o_t on the LHS of that equation.

Chapter 2

Asset Pricing

Once we understand consumption and saving, the next question is how the saving gets allocated across assets, and what the resulting demand implies about prices. This chapter runs the argument in both directions.

It begins with the portfolio problem taken in isolation. Under CARA utility the solution is a dollar amount held in the risky asset, independent of wealth; under CRRA utility the Merton-Samuelson result instead makes the risky *share* independent of wealth. The logic then reverses. Lucas takes consumption as exogenous and solves for the price that makes households willing to hold the market, which delivers the consumption CAPM and, applied to data, the equity premium puzzle, since no plausible degree of risk aversion explains the historical excess return on equity (matching it takes a coefficient of relative risk aversion around 20, while the riskless rate over the same period wants one below one).

Two further questions close the chapter. What prices are consistent with the absence of arbitrage even when they diverge from fundamentals, and what changes when the household chooses consumption and portfolio share jointly rather than in sequence.

2.1 Portfolio Choice With CARA Utility

Consider a consumer with Constant Absolute Risk Aversion utility $u(c) = -\alpha^{-1}e^{-\alpha c}$, with assets a_{T-1} who is deciding how much to invest in a risky security that will earn a normally distributed stochastic return $\mathbf{R}_T \sim \mathcal{N}(\mathbf{R}, \sigma)$ versus a safe asset that will earn return $\mathbf{R} < \mathbf{R}$ (the risky asset gets a bold font because you must be a bold person to invest in a risky asset!).¹²

Consumption in the last period of life will be the entire amount of resources. If the consumer invests an absolute amount of money $\$S$ in the risky asset, then

$$\begin{aligned} c_T &= S\mathbf{R}_T + (a_{T-1} - S)\mathbf{R} \\ &= a_{T-1}\mathbf{R} + \underbrace{(\mathbf{R}_T - \mathbf{R})S}_{\equiv \boldsymbol{\varphi}_T} \end{aligned} \quad (2.1.1)$$

where $\boldsymbol{\varphi}_T$ is the “excess return” realized in period T . Note that $\boldsymbol{\varphi}_T$ could be negative: if the risky asset performs badly enough, shareholders lose not just the excess return they hoped to earn, but part of what they would have earned from the safe asset. Given S and defining the expected risk premium as the expected value of the expected return $\boldsymbol{\varphi} = \mathbb{E}_{T-1}[\mathbf{R}_T - \mathbf{R}]$, the expectation as of time $T - 1$ is:

$$\begin{aligned} \mathbb{E}_{T-1}[u(c_T)] &= \mathbb{E}_{T-1}[-\alpha^{-1}e^{-\alpha(a_{T-1}\mathbf{R} + (\mathbf{R}_T - \mathbf{R})S)}] \\ &= -\alpha^{-1}e^{-\alpha(a_{T-1}\mathbf{R})}\mathbb{E}_{T-1}[e^{-\alpha(\mathbf{R}_T - \mathbf{R})S}] \\ &= -\alpha^{-1}e^{-\alpha(a_{T-1}\mathbf{R})}e^{-\alpha S\boldsymbol{\varphi} + (\alpha S)^2\sigma^2/2} \\ &= -\alpha^{-1}e^{-\alpha(a_{T-1}\mathbf{R})}e^{-\alpha(S\boldsymbol{\varphi} - \alpha S^2\sigma^2/2)} \end{aligned} \quad (2.1.2)$$

and the third line follows from the second because if $z \sim \mathcal{N}(\boldsymbol{\varphi}_z, \sigma_z^2)$ then $\mathbb{E}[e^z] = e^{\boldsymbol{\varphi}_z + \sigma_z^2/2}$. (See ELogNorm).

Because (2.1.2) is negative, the optimal S will be the one that yields the largest negative exponent on e , which occurs at the value of S given by

$$\max_S \left\{ S\boldsymbol{\varphi} - \frac{\alpha S^2\sigma^2}{2} \right\} \quad (2.1.3)$$

with FOC

$$\begin{aligned} \boldsymbol{\varphi} &= \alpha S\sigma^2 \\ S &= \frac{\boldsymbol{\varphi}}{\alpha\sigma^2}. \end{aligned} \quad (2.1.4)$$

This yields the intuitive result that the greater is risk aversion or the greater is the risk, the less the consumer wants to invest in the risky asset, while the greater is the expected excess return, the more the consumer wants to invest. Note, however, that the model implausibly says that the dollar amount invested in the risky asset does not depend on the total dollar amount of resources a_{T-1} . So, Warren Buffett and Homer Simpson should have the exact same dollar holdings of the risky asset! If Buffett is richer than Simpson, Buffett’s excess wealth is held in the safe form. Not very plausible. (That is why models with CARA utility are increasingly unfashionable in the economics and finance literatures).

¹The seminal paper examining this problem (in continuous time) was by Merton [1969]; that paper also examines the case with CRRA utility and lognormal returns.

²The assumption that returns are normally distributed is highly implausible. This means that with some positive probability, $\mathbf{R}_T < 0$. So, owning a \$1 of the risky asset in period $T - 1$ could result in *negative* wealth in period T . You can lose *more than everything*, which is a violation of the legal principle of limited liability. (For a detailed history of limited liability, see Mickelthwait and Wooldridge [2002].) Lognormally distributed returns are therefore much more plausible.

2.2 Portfolio Choice with CRRA Utility (Merton-Samuelson)

Merton [1969] and Samuelson [1969] study the optimal portfolio choice of a consumer with constant relative risk aversion ρ .³ This consumer has assets at the end of period t equal to a_t and is deciding how much to invest in a risky asset⁴ with a lognormally distributed return factor $\mathbf{R}_{t+1}$ whose log can be written in either of two ways:

$$\begin{aligned} \mathbf{r}_{1,t+1} &= \overbrace{\mathbf{r}_1 + 0.5\sigma_{\mathbf{r}}^2}^{\equiv \hat{\mathbf{r}}_1} + \theta_{1,t+1} \\ &= \underbrace{\hat{\mathbf{r}}_1 - 0.5\sigma_{\mathbf{r}}^2}_{\equiv \mathbf{r}_1} + \underbrace{0.5\sigma_{\mathbf{r}}^2 + \theta_{1,t+1}}_{\equiv \theta_{1,t+1}} \end{aligned} \quad (2.2.1)$$

where $\theta_{1,t+1} \sim \mathcal{N}(-0.5\sigma_{\mathbf{r}}^2, \sigma_{\mathbf{r}}^2)$; the notation for θ is motivated by the fact that the inclusion of the extra term $0.5\sigma_{\mathbf{r}}^2$ “cancels” the nonzero mean of θ , so that $\theta_{1,t+1} \sim \mathcal{N}(0, \sigma_{\mathbf{r}}^2)$.

The alternative to the risky asset is a riskfree asset that earns return factor $\mathbf{R} = e^r$.⁵ Importantly, the consumer is assumed to have no labor income and to face no risk except from the investment in the risky asset.⁶⁷

Both papers consider a multiperiod optimization problem, but here we examine a consumer for whom period t is the second-to-last period of life (the insights, and even the formulas, carry over to the multiperiod case).⁸

If the period- t consumer invests proportion ς in the risky asset, spending all available resources in the last period of life $t + 1$ will yield:

$$\begin{aligned} c_{t+1} &= (\mathbf{R}(1 - \varsigma) + \mathbf{R}_{t+1}\varsigma) a_t \\ &= \underbrace{(\mathbf{R} + (\mathbf{R}_{t+1} - \mathbf{R})\varsigma)}_{\equiv \tilde{\mathbf{R}}_{t+1}} a_t \end{aligned} \quad (2.2.2)$$

where $\tilde{\mathbf{R}}_{t+1}$ is the realized arithmetic⁹ return factor for the portfolio.

The optimal portfolio share will be the one that maximizes expected utility:

$$\varsigma = \arg \max_{\varsigma} \mathbb{E}_t[u(c_{t+1})] \quad (2.2.3)$$

and can be calculated numerically for any arbitrary distribution of rates of return.

Campbell and Viceira [2002] point out that if we define

$$\varphi_{t+1} = \mathbf{r}_{t+1} - \mathbf{r} + (1/2)\sigma_{\theta}^2 \quad (2.2.4)$$

then for many distributions a good approximation to the rate of return (the log of the return factor) is obtained by¹⁰

³ $u(c) = (1 - \rho)^{-1} c^{1-\rho}$.

⁴Both papers present the solution in the case with multiple risky assets; for the two-asset case, see the [two risky assets](#) section.

⁵Fact ELogNorm tells us that a variable with this lognormal distribution has an expected return factor of $\mathbb{E}_t[e^{\mathbf{r}_{t+1}}] = e^{\mathbf{r}} = \mathbf{R}$ (where upper-case variables like $\mathbf{R}$ without a subscript are the time-invariant mean).

⁶A common interpretation is that this is the problem of a retired investor who expects to receive no further labor income. Note however that all risks other than the returns from financial investments have been ruled out; for example, health expense risk is not possible in this model, though recent research has argued such risk is important (maybe even dominant) later in life (cf. Ameriks et al. [2011]).

⁷Riskless labor income can trivially be added to the problem, because its risklessness means that (in the absence of liquidity constraints) it is indistinguishable from a lump sum of extra current wealth with a value equal to the present discounted value (using the riskless rate) of the (riskless) future labor income. Of course, in practice, labor income is not riskless, but when labor income is risky the problem no longer has the tidy analytical solution described here and must be solved numerically. See Carroll [2023] for an introduction to numerical solution methods.

⁸Samuelson [1979].

⁹Google “arithmetic geometric mean wiki” for a refresher on the difference between arithmetic and geometric means. If the portfolio return is instead geometric $\mathbf{R}^{1-\varsigma}\mathbf{R}^{\varsigma}$ then the approximate formulas below become exact.

¹⁰See the appendix for further details.

$$\tilde{r}_{t+1} = r + \varsigma\varphi_{t+1} + \varsigma\sigma_r^2/2 - \varsigma^2\sigma_r^2/2 \quad (2.2.5)$$

Using this approximation, the expectation as of date t of utility at date $t + 1$ is:

$$\begin{aligned} \mathbb{E}_t[u(c_{t+1})] &\approx (1 - \rho)^{-1} \mathbb{E}_t \left[\left(a_t e^r e^{\varsigma\varphi_{t+1} + \varsigma(1-\varsigma)\sigma_r^2/2} \right)^{1-\rho} \right] \\ &\approx (1 - \rho)^{-1} \mathbb{E}_t \left[(a_t R)^{1-\rho} \left(e^{\varsigma\varphi_{t+1} + \varsigma(1-\varsigma)\sigma_r^2/2} \right)^{1-\rho} \right] \\ &\approx (1 - \rho)^{-1} (a_t R)^{1-\rho} \mathbb{E}_t \left[e^{(\varsigma\varphi_{t+1} + \varsigma(1-\varsigma)\sigma_r^2/2)(1-\rho)} \right] \\ &\approx \underbrace{(1 - \rho)^{-1} (a_t R)^{1-\rho}}_{\text{constant} < 0} \underbrace{e^{(1-\rho)\varsigma(1-\varsigma)\sigma_r^2/2} \mathbb{E}_t \left[e^{\varsigma\varphi_{t+1}(1-\rho)} \right]}_{\text{excess return utility factor}} \end{aligned} \quad (2.2.6)$$

where the first term is a negative constant under the usual assumption that relative risk aversion $\rho > 1$.

For the special (but reasonable) case of a lognormally distributed return, we can make substantial further progress, by obtaining an analytical approximation to the numerical optimum. In this case $\varsigma(1 - \rho)\varphi_{t+1} \sim \mathcal{N}(\varsigma(1 - \rho)(\varphi - \sigma_r^2/2), (\varsigma(1 - \rho))^2\sigma_r^2)$ (again using LogELogNormTimes). With a couple of extra lines of derivation we can show that the log of the expectation in (2.2.6) is

$$\begin{aligned} \log \mathbb{E}_t \left[e^{\varsigma\varphi_{t+1}(1-\rho)} \right] &= (1 - \rho)\varsigma\varphi - (1 - \rho)\varsigma\sigma_r^2/2 + ((1 - \rho)\varsigma)^2\sigma_r^2/2 \\ &= (1 - \rho)\varsigma\varphi - (1 - \rho)\varsigma(1 - \varsigma)\sigma_r^2/2 \\ &= (1 - \rho)\varsigma\varphi - (1 - \rho)\varsigma(1 - \varsigma)\sigma_r^2/2 - \rho(1 - \rho)\varsigma^2\sigma_r^2/2 \end{aligned} \quad (2.2.7)$$

Substitute from (2.2.7) for the log of the expectation in (2.2.6) and note that the resulting expression simplifies because it contains $(1 - \rho)\varsigma\sigma_r^2/2 - (1 - \rho)\varsigma\sigma_r^2/2 = 0$; thus the log of the “excess return utility factor” in (2.2.6) is

$$-(\rho - 1)\varsigma\varphi - (\rho - 1)(-\rho\varsigma^2\sigma_r^2/2) \quad (2.2.8)$$

and the ς that minimizes the log will also minimize the level; minimizing this when $\rho > 1$ is equivalent to maximizing the terms multiplied by $-(\rho - 1)$, so our problem reduces to

$$\max_{\varsigma} \quad \varsigma\varphi - \rho\varsigma^2\sigma_r^2/2 \quad (2.2.9)$$

with FOC

$$\begin{aligned} \varphi - \rho\sigma_r^2\varsigma &= 0 \\ \varsigma &= \left(\frac{\varphi}{\rho\sigma_r^2} \right) \end{aligned} \quad (2.2.10)$$

Equation (2.2.10) says¹¹ that the consumer allocates a higher proportion of his net worth to the high-risk, high-return asset when

1. the amount φ by which the risky asset’s return exceeds the riskless return is greater
2. the consumer is less risk averse (ρ is lower)
3. riskiness σ_r^2 is less

¹¹This expression differs slightly from that derived by Campbell and Viceira [2002], because we adjust the mean logarithmic return of the risky investment for its variance in order to keep the mean return factor constant for different values of the variance (cf. (2.2.4)), which makes comparisons of alternative levels of risk more transparent.

If there is no excess return, nothing will be put in the risky asset. Similarly, if risk aversion or the variance of the risk is infinity, again nothing will be put in the risky asset.¹²

This formula hints at the existence of an “equity premium puzzle” (Mehra and Prescott [1985]). Interpreting the risky asset as the aggregate stock market, the annual standard deviation of the log of U.S. stock returns has historically been about $\sigma_r = 0.2$ yielding $\sigma_r^2 = 0.04$. The equity premium over historical periods has been something like $\varphi = 0.08$ (eight percent). With risk aversion of $\rho = 2$ this formula implies that the share of risky assets in your portfolio should be $0.08/0.08$ or 100 percent! The fact that most people have less than 100 percent of their wealth invested in stocks is the “stockholding puzzle,” the microeconomic manifestation of the equity premium puzzle (Haliassos and Bertaut [1995]).

To avoid the problems caused by a prediction of a risky portfolio share greater than one, we can calibrate the model with more modest expectations for the equity premium. Some researchers have argued that when evidence for other countries and longer time periods is taken into account, a plausible average value of the premium might be as low as three percent. The figures show the relationship between the portfolio share and relative risk aversion for a calibration that assumes a modest premium of 3 percent and a large standard deviation of $\sigma = 0.2$. Even when risks are this high and the premium is this low, if relative risk aversion is close to logarithmic ($\rho = 1$) the investor wants to put well over half of the portfolio in the risky asset. Only for values of risk aversion greater than 2 does the predicted portfolio share reach plausible small values.

But remember that these calculations are all assuming that the consumer’s *entire* consumption spending is financed by asset income. If the consumer has other income (for example, labor or transfer income) that is not perfectly correlated with returns on the risky asset, they should be willing to take more risk. Since, for most consumers, most of their future consumption will be financed from labor or transfer income, it is not surprising to learn that models calibrated to actual data on capital and noncapital income dynamics imply that people should be investing most of their non-human wealth in the risky asset (with reasonable values of ρ).

A final interesting question is what the expected rate of return on the consumer’s portfolio will be once the portfolio share in risky assets has been chosen optimally. Note first that (2.2.16) implies that

$$\log \mathbb{E}_t[e^{\tilde{r}_{t+1}-r}] = \varsigma\varphi \quad (2.2.11)$$

while the variance of the log of the excess return factor for the portfolio is $\sigma_{\tilde{r}}^2 = \varsigma^2\sigma_r^2$. Substituting the solution (2.2.10) for ς into (2.2.11), we have

$$\begin{aligned} \varsigma\varphi &= \left(\frac{\varphi^2}{\rho\sigma_r^2} \right) \\ &= (\varphi/\sigma_r)^2/\rho \end{aligned} \quad (2.2.12)$$

which is an interesting formula for the excess return of the optimally chosen portfolio because the object φ/σ_r (the excess return divided by the standard deviation) is a well-known tool in finance for evaluating the tradeoff between risk and return (the “Sharpe ratio”). Equation (2.2.12) says that the consumer will choose a portfolio that earns an excess return that is directly related to the (square of the) Sharpe ratio and inversely related to the risk aversion coefficient. Higher reward (per unit of risk) convinces the consumer to take the risk necessary to earn higher returns; but higher risk aversion convinces the investor to sacrifice (risky) return for safety.

Finally, we can ask what effect an exogenous increase in the risk of the risky asset has on the endogenous riskiness of the portfolio once the consumer has chosen optimally. The answer is surprising: The variance of the optimally-chosen portfolio is

$$\begin{aligned} \varsigma^2\sigma_r^2 &= \left(\frac{\varphi}{\rho\sigma_r^2} \right)^2 \sigma_r^2 \\ &= \left(\frac{(\varphi/\rho)^2}{\sigma_r^2} \right) \end{aligned} \quad (2.2.13)$$

¹²Figure 2.2 below shows the quality of the approximation.

which is actually *smaller* when σ_r^2 is larger. Upon reflection, maybe this makes sense. Imagine that the consumer had adjusted his portfolio share in the risky asset downward just enough to restore the portfolio's riskiness to its original level before the increase in risk. The consumer would now be bearing the same degree of risk but for a lower (mean) rate of return (because of his reduction in exposure to the risky asset). It makes intuitive sense that the consumer will not be satisfied with this "same riskiness, lower return" outcome and therefore that the undesirableness of the risky asset must have increased enough to make him want to hold even less than the amount that would return his portfolio's riskiness to its original value.

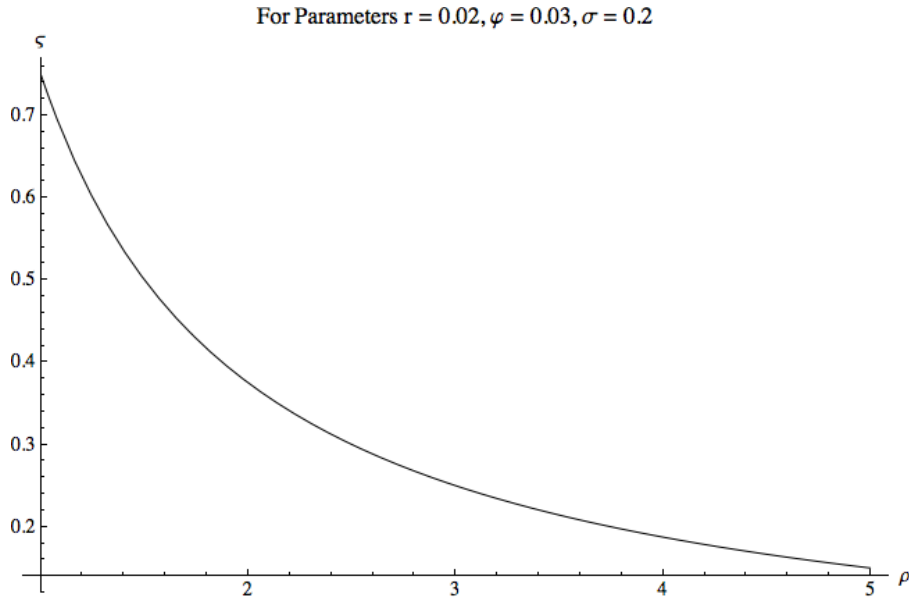


Figure 2.1: The Approximate Risky Portfolio Share ς Declines as Relative Risk Aversion ρ Increases

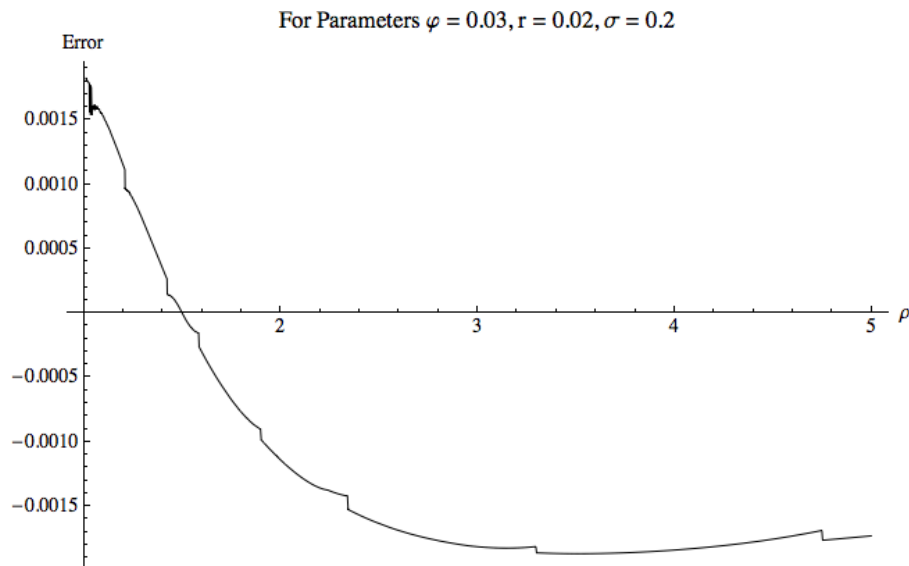


Figure 2.2: The Approximation Error for the Portfolio Share in Risky Assets ς Is Small
 Note: The approximation error is computed by solving for the exactly optimal portfolio share numerically. See the `doA11.nb` Mathematica notebook for details.

2.2.1 Appendix: The Campbell and Viceira [2002] Approximation

For mathematical analysis (especially under the assumption of CRRA utility) it would be convenient if we could approximate the realized arithmetic portfolio return factor by the log of the realized

geometric return factor $\tilde{R}_{t+1} = \mathbf{R}^{1-\varsigma} \mathbf{R}_{t+1}^\varsigma$, because then the logarithm of the return factor would be $\tilde{r}_{t+1} = r(1-\varsigma) + \mathbf{r}_{t+1}\varsigma = r + (\mathbf{r}_{t+1} - r)\varsigma = r + \varphi_{t+1}\varsigma$ and the realized “portfolio excess return” would be simply $\tilde{r}_{t+1} - r = \varsigma\varphi_{t+1}$. Unfortunately, for ς values well away from 0 and 1 (that is, for any *interesting* values of portfolio shares), the log of the geometric mean is a badly biased approximation to the log of the arithmetic mean when the variance of the risky asset is substantial.

Campbell and Viceira [2002] propose instead

$$\tilde{r}_{t+1} \approx r + \varsigma\varphi_{t+1} + \varsigma\sigma_{\mathbf{r}}^2/2 - \varsigma^2\sigma_{\mathbf{r}}^2/2 \quad (2.2.14)$$

To see one virtue of this approximation,¹³ note (using NormTimes and SumNormsIsNorm) that since the mean and variance of $\varphi_{t+1}\varsigma$ are respectively $\varsigma(\mathbf{r} - \sigma_{\mathbf{r}}^2/2 - r)$ and $\varsigma^2\sigma_{\mathbf{r}}^2$, fact LogELogNormTimes implies that

$$\log \mathbb{E}_t[e^{\varphi_{t+1}\varsigma}] = \varsigma(\mathbf{r} - \sigma_{\mathbf{r}}^2/2 - r) + \varsigma^2\sigma_{\mathbf{r}}^2/2 \quad (2.2.15)$$

which means that exponentiating then taking the expectation then taking the logarithm of (2.2.5) gives

$$\begin{aligned} \log \mathbb{E}_t[e^{\tilde{r}_{t+1}}] &= \log e^r + \log \mathbb{E}_t[e^{\varsigma\varphi_{t+1}}] + \log e^{\varsigma\sigma_{\mathbf{r}}^2/2 - \varsigma^2\sigma_{\mathbf{r}}^2/2} \\ &= r + \varsigma(\mathbf{r} - \sigma_{\mathbf{r}}^2/2 - r) + \varsigma^2\sigma_{\mathbf{r}}^2/2 + \varsigma\sigma_{\mathbf{r}}^2/2 - \varsigma^2\sigma_{\mathbf{r}}^2/2 \\ \log \mathbb{E}_t[e^{\tilde{r}_{t+1}}] - r &= \varsigma(\mathbf{r} - r) \end{aligned} \quad (2.2.16)$$

or, in words: The expected excess portfolio return is equal to the proportion invested in the risky asset times the expected return of the risky asset.¹⁴

¹³The approximation is motivated by the continuous-time solution, which is obtained using Ito’s lemma.

¹⁴We use the word “return” always to mean the logarithm of the corresponding “factor”; and when not explicitly specified, we always take the expectation before taking the log; if we wanted to refer to $\mathbb{E}_t[\tilde{r}_{t+1}]$ we would call it the expected log portfolio return (to distinguish it from the expected portfolio return, $\log \mathbb{E}_t[e^{\tilde{r}_{t+1}}]$).

2.3 The Lucas Asset Pricing Model

2.3.1 Introduction/Setup

Lucas [1978] considers an economy populated by infinitely many¹⁵ identical individual consumers, in which the only assets are a set of identical infinitely-lived trees. Aggregate output is the fruit that falls from the trees, and cannot be stored (it would rot!); because $u'(c) > 0 \forall c$, the fruit is all eaten:

$$c_t L_t = d_t K_t \quad (2.3.1)$$

where c_t is consumption of fruit per person, L_t is the population, K_t measures the stock of trees, and d_t is the exogenous output of fruit that drops from each tree. A crucial assumption is that the stock of trees is exogenous: you cannot consume a little less fruit and have more trees next period.

(In a given year, each tree produces exactly the same amount of fruit as every other tree, but d_t varies from year to year depending on the weather.) An economy like this, in which output arrives without any deliberate actions on the part of residents, is called an “endowment” economy (or, sometimes, an “exchange” economy).¹⁶

2.3.2 The Market for Trees

We now consider a market in which individual consumers can buy and sell trees. This requires some mental gymnastics: with infinitely many identical consumers, we must find a price at which every one of them is simultaneously satisfied. If there is a perfect capital market for trees, the price of trees P_t must be such that, each period, each (identical) consumer does not want either to increase or to decrease their holding of trees.¹⁷

If a tree is sold, the sale is assumed to occur after the existing owner receives that period’s fruit (P_t is the “ex-dividend” price). Total resources available to consumer i in period t are the sum of the fruit received from the trees owned, $d_t k_t^i$, plus the potential proceeds if the consumer were to sell all his stock of trees, $P_t k_t^i$. Total resources are divided into two uses: Current consumption c_t^i and the purchase of trees for next period k_{t+1}^i at price P_t ,

$$\overbrace{k_{t+1}^i P_t + c_t^i}^{\text{Uses of resources}} = \overbrace{d_t k_t^i + P_t k_t^i}^{\text{Total resources}} \quad (2.3.2)$$

$$k_{t+1}^i = (1 + d_t/P_t)k_t^i - c_t^i/P_t. \quad (2.3.3)$$

2.3.3 The Problem of an Individual Consumer

Consumer i maximizes

$$v(m_t^i) = \max \mathbb{E}_t^i \left[\sum_{n=0}^{\infty} \beta^n u(c_{t+n}^i) \right] \quad (2.3.4)$$

subject to

$$k_{t+1}^i = (1 + d_t/P_t)k_t^i - c_t^i/P_t \quad (2.3.5)$$

$$m_{t+1}^i = (P_{t+1} + d_{t+1})k_{t+1}^i. \quad (2.3.6)$$

Rewriting in the form of Bellman’s equation,

¹⁵As in the [Aggregation](#) section.

¹⁶The alternative is a “production” economy, in which factors of production (labor, capital, maybe land, maybe knowledge) combine somehow to generate the output.

¹⁷If, at a hypothesized equilibrium price, every identical consumer wanted (say) to increase their holdings, that price could not be an equilibrium price, because with a fixed supply of trees everyone cannot increase their holding of trees at once!

$$v(m_t^i) = \max_{\{c_t^i\}} u(c_t^i) + \beta \mathbb{E}_t^i [v(m_{t+1}^i)], \quad (2.3.7)$$

the first order condition tells us that

$$0 = u'(c_t^i) + \beta \mathbb{E}_t^i \left[v'(m_{t+1}^i) \frac{\partial}{\partial c_t^i} \left(\overbrace{(P_{t+1} + d_{t+1}) \left((1 + d_t/P_t) k_t^i - c_t^i/P_t \right)}^{m_{t+1}^i} \right) \right] \quad (2.3.8)$$

Spending one more unit of consumption today buys $1/P_t$ fewer trees, each of which would have paid off $P_{t+1} + d_{t+1}$ tomorrow, so the derivative inside the expectation is

$$\frac{\partial m_{t+1}^i}{\partial c_t^i} = -\frac{P_{t+1} + d_{t+1}}{P_t} \equiv -\mathbf{R}_{t+1}. \quad (2.3.9)$$

Substituting this and moving the resulting negative term to the left-hand side,

$$u'(c_t^i) = \beta \mathbb{E}_t^i \left[v'(m_{t+1}^i) \left(\underbrace{\frac{P_{t+1} + d_{t+1}}{P_t}}_{\equiv \mathbf{R}_{t+1}} \right) \right] = \beta \mathbb{E}_t^i [\mathbf{R}_{t+1} v'(m_{t+1}^i)] \quad (2.3.10)$$

where $\mathbf{R}_{t+1}$ is the return factor that measures the resources in period $t+1$ that are the reward for owning a unit of trees at the end of t .

The [Envelope theorem](#) tells us that $v'(m_{t+1}^i) = u'(c_{t+1}^i)$, so (2.3.10) becomes

$$u'(c_t^i) = \beta \mathbb{E}_t^i \left[u'(c_{t+1}^i) \left(\frac{P_{t+1} + d_{t+1}}{P_t} \right) \right] \quad (2.3.11)$$

$$P_t = \beta \mathbb{E}_t^i \left[\left(\frac{u'(c_{t+1}^i)}{u'(c_t^i)} \right) (P_{t+1} + d_{t+1}) \right]. \quad (2.3.12)$$

2.3.4 Aggregation

Since all consumers are identical, $c_t^i = c_t^j \forall i, j$, so henceforth we just call consumption per capita c_t . Since aggregate consumption must equal aggregate production because fruit cannot be stored, normalizing the population to $L_t = 1 \forall t$ and stock of trees to $K_t = 1 \forall t$, equation (2.3.1) becomes:

$$c_t = d_t. \quad (2.3.13)$$

Substituting c_t and c_{t+1} for c_t^i and c_{t+1}^i in (2.3.11) and then substituting d_t for c_t we get

$$P_t = \beta \mathbb{E}_t \left[\left(\frac{u'(d_{t+1})}{u'(d_t)} \right) (P_{t+1} + d_{t+1}) \right]. \quad (2.3.14)$$

We can rewrite this more simply if we define

$$M_{t,t+n} = \beta^n \left(\frac{u'(d_{t+n})}{u'(d_t)} \right) \quad (2.3.15)$$

which is called the “stochastic discount factor” because (a) it is stochastic (thanks to the shocks between t and $t+n$ that determine the value of d_{t+n}); and (b) it measures the rate at which all agents in this economy in period t will discount a unit of value received in a future period, e.g. $t+1$:

$$P_t = \mathbb{E}_t [M_{t,t+1} (P_{t+1} + d_{t+1})]. \quad (2.3.16)$$

A corresponding equation will hold in period $t+1$ (and in period $t+2$ and beyond):

$$P_{t+1} = \mathbb{E}_{t+1} [M_{t+1,t+2} (P_{t+2} + d_{t+2})] \quad (2.3.17)$$

so we can use repeated substitution, e.g. of (2.3.17) into (2.3.16), and so on to get

$$P_t = \mathbb{E}_t [M_{t,t+1}d_{t+1}] + \mathbb{E}_t [M_{t,t+1}\mathbb{E}_{t+1}[M_{t+1,t+2}d_{t+2}]] + \dots \quad (2.3.18)$$

The “law of iterated expectations” says that $\mathbb{E}_t[\mathbb{E}_{t+1}[P_{t+2}]] = \mathbb{E}_t[P_{t+2}]$; given this, and noting that $M_{t,t+2} = M_{t,t+1}M_{t+1,t+2}$, (2.3.18) becomes:

$$P_t = \mathbb{E}_t [M_{t,t+1}d_{t+1} + M_{t,t+2}d_{t+2} + M_{t,t+3}d_{t+3} + \dots]. \quad (2.3.19)$$

So, the price of the asset is the present discounted value of the stream of future “dividends,” where the stochastic factor by which (potentially stochastic) dividends received in $t+n$ are discounted back to t is $M_{t,t+n}$.

2.3.5 Specializing the Model

This is as far as we can go without making explicit assumptions about the structure of utility. If utility is CRRA, $u(c) = (1-\rho)^{-1}c^{1-\rho}$, substituting $u'(d) = d^{-\rho}$ into (2.3.14) yields

$$P_t = \beta d_t^\rho \mathbb{E}_t [d_{t+1}^{-\rho} (P_{t+1} + d_{t+1})] \quad (2.3.20)$$

and the particularly special case of logarithmic utility (which Lucas emphasizes) corresponds to $\rho = 1$, which (again using the law of iterated expectations) allows us to simplify (2.3.20) to

$$\begin{aligned} \frac{P_t}{d_t} &= \beta \mathbb{E}_t [d_{t+1}^{-1} (P_{t+1} + d_{t+1})] \\ &= \beta \left(1 + \mathbb{E}_t \left[\frac{P_{t+1}}{d_{t+1}} \right] \right) \\ &= \beta \left(1 + \beta \left(1 + \mathbb{E}_t \left[\frac{P_{t+2}}{d_{t+2}} \right] \right) \right) \\ &= \beta \left(1 + \beta + \beta^2 + \dots + \mathbb{E}_t \left[\lim_{n \rightarrow \infty} \beta^{n-1} \left[\frac{P_{t+n}}{d_{t+n}} \right] \right] \right) \\ &= \frac{\beta}{1-\beta} + \beta \mathbb{E}_t \left\{ \lim_{n \rightarrow \infty} \beta^{n-1} \left[\frac{P_{t+n}}{d_{t+n}} \right] \right\}. \end{aligned} \quad (2.3.21)$$

If the price is bounded (it cannot ever go, for example, to a value such that it would cost more than the economy’s entire output to buy a single tree), it is possible to show that the limit term in this equation goes to zero. Using the usual definition of the time preference factor as $\beta = 1/(1+\vartheta)$ where ϑ is the time preference rate, the equilibrium price is:

$$P_t = d_t \left(\frac{\beta}{1-\beta} \right) = d_t \left(\frac{1}{1/\beta - 1} \right) = d_t \left(\frac{1}{1+\vartheta - 1} \right) = \frac{d_t}{\vartheta} \quad (2.3.22)$$

or, equivalently, the “dividend-price ratio” is always $d_t/P_t = \vartheta$.¹⁸

It may surprise you that the equilibrium price of trees today does not depend on the expected level of fruit output in the future. If the weather was bad this year, but is expected to return to normal next year (and, by definition, is expected to be equal to normal in subsequent years), you might think that the price today would mostly reflect the “normal” value of fruit production that the trees produce, not the (temporarily low) value that happens to obtain today.

The above derivation says that intuition is wrong: Today’s price depends only on today’s output.

Nevertheless, the logic (higher future output is a reason for higher current prices) is not wrong; but it is (exactly) counterbalanced by another, and subtler, fact: Since future consumption will equal future fruit output, higher expected fruit output means lower marginal utility of consumption in that future period of (more) abundant fruit (basically, people are starving today, which reduces the attractiveness of cutting their consumption to buy trees that will produce more in a period when they expect *not* to be starving). These two forces are the manifestation of the (pure) income effect and

¹⁸A derivation parallel to the one above shows that in the CRRA utility case the solution is $d_t^\rho/P_t = \vartheta$.

substitution effect in this model (there is no human wealth, and therefore no human wealth effect). In the special case of logarithmic utility considered here, income and substitution effects are of the same size and opposite sign so the two forces exactly offset.

2.3.6 The Interest Rate and the Rate of Return in a Lucas Model

We can decompose the return factor attributable to ownership of a share of capital (cf. (2.3.10)) by adding and subtracting P_t in the numerator:

$$\mathbf{R}_{t+1} = \left(\frac{P_{t+1} + P_t - P_t + d_{t+1}}{P_t} \right) = \left(1 + \frac{\Delta P_{t+1}}{P_t} + \frac{d_{t+1}}{P_t} \right) \quad (2.3.23)$$

so the “rate of return” is

$$\mathbf{r}_{t+1} = \frac{\Delta P_{t+1}}{P_t} + \frac{d_{t+1}}{P_t} \quad (2.3.24)$$

which is a useful decomposition because the two components have natural interpretations: The first is a “capital gain” (or loss), and the second can plausibly be identified as “the interest rate” paid by the asset (because it corresponds to income received regardless of whether the asset is liquidated).

In models that do not explicitly discuss asset pricing, the implicit assumption is usually that the price of capital is constant (which might be plausible if capital consists mostly of reproducible items like machines,¹⁹ rather than Lucas trees). In this case

$$\mathbf{R}_{t+1} = \left(1 + \frac{d_{t+1}}{P_t} \right) \quad (2.3.25)$$

says that the only risk in the rate of return is attributable to unpredictable variation in the size of dividend/interest payments. Indeed, if additional assumptions are made (e.g., perfect capital markets) that yield the conclusion that the interest rate matches the marginal product of capital, then such models generally imply that variation in returns (at least at high frequencies) is very small, because aggregate capital typically is very stable from one period to the next in such models, and, if the aggregate production function is stable, this implies great stability in the marginal product of capital.

2.3.7 Aggregate Returns Versus Individual Returns

One of the subtler entries in Aristotle’s catalog of common human reasoning errors was the “fallacy of composition,” in which the reasoner supposes that if a proposition is true of each element of a whole, then it must be true of the whole.

The Lucas model provides a counterexample. From the standpoint of any individual (atomistic) agent, it is quite true that a decision to save one more unit will yield greater future resources, in the amount $\mathbf{R}_{t+1}$. But from the standpoint of the society as a whole, if everyone decided to do the same thing (save one more unit), there would be no effect on aggregate resources in period $t + 1$. Put another way, for any individual agent, it appears that the “marginal product of capital” is $\mathbf{R}_{t+1}$, but for the society as a whole the marginal product of capital is zero.

The proposition that the return for society as a whole must be the same as the return that is available to individuals is an error because it implicitly assumes that there are no general equilibrium effects of a generalized desire to save more (or, more broadly, that there is no interaction between the decisions one person makes and the outcomes for another person). The Lucas model provides a counterexample in which, if everyone’s preferences change (e.g., ϑ goes down for everyone), the price of the future asset is affected, indeed, it is affected in a way that is sufficient to exactly counteract the increased desire for ownership of future dividends (since there is a fixed supply of assets to be owned, the demand must be reconciled with that preexisting supply).

Aristotle was a smart guy!

¹⁹The key insights below remain true even if there is a gradual trend in the real price of capital goods, as has in fact been true.

2.3.8 Analytical and Numerical Solutions

The appendices derive various results about the solution to the model under different assumptions. But, unfortunately, the model has analytical solutions (like, $P = d/\vartheta$) or approximate analytical solutions only in very special circumstances. The accompanying DemARK notebook shows how to solve the model numerically for a simple case where there is no such analytical solution (the case where dividends follow an AR(1) process), and also shows how the numerical solution compares with the approximate analytical solution in the CRRA utility case.

2.3.9 Appendix: Analytical Solutions in CRRA Utility Case

When Dividends are IID

Suppose d_{t+n} is identically individually distributed in every future period, so that its expectation as of t is the same for any date $n > 0$:

$$\dot{d} \equiv \mathbb{E}_t[d_{t+n}^{1-\rho}]. \quad (2.3.26)$$

Now note that (2.3.20) can be rewritten as

$$\frac{P_t}{d_t^\rho} = \beta \left(\dot{d} + \mathbb{E}_t \left[\frac{P_{t+1}}{d_{t+1}^\rho} \right] \right) \quad (2.3.27)$$

$$= \beta \dot{d} \left(1 + \beta + \beta \mathbb{E}_t \left[\frac{P_{t+2}}{d_{t+2}^\rho} \right] \right) \quad (2.3.28)$$

$$= \beta \dot{d} \left(1 + \beta + \beta^2 + \dots + \underbrace{\mathbb{E}_t \left[\lim_{n \rightarrow \infty} \beta^{n-1} \left[\frac{P_{t+n}}{d_{t+n}^\rho} \right] \right]}_{\text{assume goes to zero}} \right) \quad (2.3.29)$$

$$= \left(\frac{\beta \dot{d}}{1 - \beta} \right) = \left(\frac{\dot{d}}{\beta^{-1} - 1} \right) \quad (2.3.30)$$

To make further progress, suppose that the iid process for dividends is a mean-one lognormal: $\log d_{t+n} \sim \mathcal{N}(-\sigma^2/2, \sigma^2) \forall n$ so that $\mathbb{E}_t[d_{t+n}] = 1 \forall n$ (see ELogNormMeanOne), in which case ELogNormTimes can be used to show that

$$\dot{d} = e^{\rho(\rho-1)(1/2)\sigma^2} \quad (2.3.31)$$

and if we approximate $\beta \approx e^{-\vartheta}$ then $\beta^{-1} \approx 1 + \vartheta$ and so (2.3.30) becomes

$$\frac{P_t}{d_t^\rho} \approx \left(\frac{e^{\rho(\rho-1)(1/2)\sigma^2}}{\vartheta} \right) \quad (2.3.32)$$

$$P_t \approx \left(\frac{d_t^\rho e^{\rho(\rho-1)(1/2)\sigma^2}}{\vartheta} \right) \quad (2.3.33)$$

So the log is

$$\log P_t \approx \rho \log d_t + \rho(\rho-1)(1/2)\sigma^2 - \log \vartheta \quad (2.3.34)$$

and thus the variances obey

$$\text{var}(\log P) = \rho^2 \text{var}(\log d). \quad (2.3.35)$$

Given that $\rho > 1$, this derivation yields some interesting insights:

1. (the log of) asset prices will be more volatile than (the log of) dividends

2. An increase in risk aversion ρ increases the price P_t (because $\rho(\rho - 1)\sigma^2/2 > 0$ and an increase in ρ increases its size)

The second point is surprising, so let me say it again: an *increase* in risk aversion *increases* the price of the risky asset. In a sense, this is an implication of the proposition that risk aversion increases the volatility of asset prices (when they are high, they must be *very* high; when low, *very* low). But, it does not correspond very well to the common narrative in which market analysts often attribute a decline in asset prices to “increased risk aversion.”

When Dividends Follow a Random Walk

The polar alternative to IID shocks would be for dividends to follow a random walk: $\log(d_{t+1}/d_t) \sim \mathcal{N}(-\sigma^2/2, \sigma^2)$.

Now divide both sides of (2.3.20) by d_t , and rewrite the object inside the expectations operator by multiplying the first term by d_{t+1} and dividing the second term by d_{t+1} , yielding

$$\left(\frac{P_t}{d_t}\right) = \beta d_t^{-(1-\rho)} \mathbb{E}_t \left[d_{t+1}^{1-\rho} \left(\frac{P_{t+1}}{d_{t+1}} + 1 \right) \right] = \beta \mathbb{E}_t \left[\left(\frac{d_{t+1}}{d_t} \right)^{1-\rho} \left(\frac{P_{t+1}}{d_{t+1}} + 1 \right) \right]. \quad (2.3.36)$$

Now (1) note that our assumption here about the distribution of d_{t+1}/d_t is identical to the assumption about d_{t+1} above, so the expectation will be the same $\dot{d}$; and (2) hypothesize that there will be a solution under which the price-dividend ratio is a constant; call it $\mathbf{r}^{-1}$:

$$\mathbf{r}^{-1} = \beta \left[\dot{d}(\mathbf{r}^{-1} + 1) \right] \quad (2.3.37)$$

$$1 = \beta \dot{d}(1 + \mathbf{r}) \quad (2.3.38)$$

$$\left(\frac{1 - \beta \dot{d}}{\beta \dot{d}} \right) = \mathbf{r} \quad (2.3.39)$$

$$\left(\frac{1}{(\beta \dot{d})^{-1} - 1} \right) = \mathbf{r}^{-1} \quad (2.3.40)$$

so that we obtain a formula for $\mathbf{r}^{-1} = P_t/d_t$

$$\log P_t \approx \log d_t - \log(\vartheta - (1/2)\rho(\rho - 1)\sigma^2) \quad (2.3.41)$$

The difference with (2.3.34) is only the absence of the ρ multiplying $\log d_t$. The main substantive difference is therefore that the variance of (log) prices and the variance of (log) dividends is now the same. The surprising result that the price-dividend ratio increases when risk aversion increases continues to hold.

2.3.10 When Dividends Follow an AR(1) Process

The two cases just examined are the polar ones, and the interesting territory lies between them. Dividends that are IID have no memory at all, while dividends that follow a random walk never forget. Actual dividends fall somewhere in the middle, high this year if they were high last year, though not permanently so.

We must impose the persistence on the *log* of dividends rather than on their level, and it is worth being clear about why, because the tempting alternative fails in two separate ways. An AR(1) in levels, $d_{t+1} = \alpha d_t + z_{t+1}$ with normal z , would put positive probability on a negative dividend, which is not a thing a tree can produce. It would also leave d_t sitting inside the expectation in (2.3.36) in a form admitting no constant price-dividend ratio, so the analytical thread would be lost at the first step.

So write $x_t \equiv \log d_t$ and assume

$$x_{t+1} = \alpha x_t + z_{t+1}, \quad z_{t+1} \sim \mathcal{N}(-\sigma^2/2, \sigma^2), \quad (2.3.42)$$

which contains both earlier cases: $\alpha = 0$ is the IID assumption and $\alpha = 1$ is the random walk.

Rather than guess at a price-dividend ratio, we price each future dividend separately and add up (the guess-and-verify route used above needs a constant ratio, and there is not one to find here). The Lucas price is the present discounted value of the dividend stream evaluated at the consumer's own marginal utility, so with $c = d$ and $u'(c) = c^{-\rho}$,

$$\left(\frac{P_t}{d_t}\right) = \sum_{n=1}^{\infty} \beta^n \mathbb{E}_t \left[\left(\frac{d_{t+n}}{d_t}\right)^{1-\rho} \right]. \quad (2.3.43)$$

Iterating (2.3.42) gives $x_{t+n} - x_t = (\alpha^n - 1)x_t + \sum_{j=1}^n \alpha^{n-j} z_{t+j}$, and because the z are independent and lognormal each strip can be evaluated with ELogNorm. Writing $S_n = \sum_{k=0}^{n-1} \alpha^k$ and $Q_n = \sum_{k=0}^{n-1} \alpha^{2k}$,

$$\mathbb{E}_t \left[\left(\frac{d_{t+n}}{d_t}\right)^{1-\rho} \right] = \exp \left((1-\rho)(\alpha^n - 1)x_t - \frac{\sigma^2}{2}(1-\rho)S_n + \frac{\sigma^2}{2}(1-\rho)^2 Q_n \right). \quad (2.3.44)$$

Substituting (2.3.44) into (2.3.43) gives the price-dividend ratio for any α . We know of no closed form for $0 < \alpha < 1$, since α^n appears in an exponent and resists the geometric sum that rescues the polar cases, though the series itself converges geometrically and evaluating it costs a single line of code.

The two polar cases fall out. Setting $\alpha = 1$ makes $S_n = Q_n = n$ and kills the x_t term, so every strip is d^n and the sum collapses to the constant ratio of (2.3.41). Setting $\alpha = 0$ makes $S_n = Q_n = 1$ for every $n \geq 1$, leaving $d_t^{\rho-1} d \beta / (1 - \beta)$, which is (2.3.34).

Between the poles the price-dividend ratio is no longer a constant. It depends on d_t through the $(\alpha^n - 1)x_t$ term, and it is constant only in the random walk limit, where the level of dividends carries no information about their growth. Raising α raises the ratio, steeply as α nears one: at $\rho = 3$, $\sigma = 0.1$, $\vartheta = 0.04$ and $d_t = 1$, the ratio runs 25.8, 26.4, 29.5, 47.6 and 108 for $\alpha = 0, 0.6, 0.9, 0.99$ and 1. Mean reversion is what keeps the ratio finite and modest. Remove it and the price of the tree rises sharply, because a high dividend today then forecasts high dividends forever (which is why the random walk deserves to be called the extreme case, and why the ratio there sits so far above the values that surround it).

2.4 The Equity Premium Puzzle and the Riskfree Rate

This section derives the equity premium puzzle (Mehra and Prescott [1985]). Consider a representative agent solving the joint consumption and portfolio allocation problem:

$$\begin{aligned} v(m_t) &= \max_{\{c_t, \varsigma_t\}} u(c_t) + \mathbb{E}_t \left[\sum_{n=1}^{\infty} \beta^n u(c_{t+n}) \right] \\ \text{s.t.} \\ m_{t+1} &= (m_t - c_t) \tilde{R}_{t+1} + y_{t+1} \\ \tilde{R}_{t+1} &= \varsigma_t \mathbf{R}_{t+1} + (1 - \varsigma_t) R \end{aligned} \quad (2.4.1)$$

where R denotes the return on a perfectly riskless asset and $\mathbf{R}_{t+1}$ denotes the return on equities (the risky asset) held between periods t and $t+1$, ς_t is the share of end-of-period savings invested in the risky asset, $\tilde{R}_{t+1}$ is the portfolio-weighted rate of return, and y_{t+1} is noncapital income in period $t+1$.

As usual, the objective can be rewritten in recursive form:

$$v(m_t) = \max_{\{c_t, \varsigma_t\}} u(c_t) + \beta \mathbb{E}_t \left[v \left(\underbrace{[\varsigma_t \mathbf{R}_{t+1} + (1 - \varsigma_t) R]}_{\tilde{R}_{t+1}} (m_t - c_t) + y_{t+1} \right) \right] \quad (2.4.2)$$

The first order condition with respect to c_t is

$$u'(c_t) = \beta \mathbb{E}_t [\tilde{R}_{t+1} v'(m_{t+1})] \quad (2.4.3)$$

and the FOC with respect to ς_t is

$$\begin{aligned} \mathbb{E}_t [(\mathbf{R}_{t+1} - R) v'(m_{t+1}) (m_t - c_t)] &= 0 \\ \mathbb{E}_t [(\mathbf{R}_{t+1} - R) v'(m_{t+1})] &= 0 \end{aligned} \quad (2.4.4)$$

But the usual logic of the [Envelope theorem](#) tells us that

$$u'(c_{t+1}) = v'(m_{t+1}) \quad (2.4.5)$$

so, substituting (2.4.5) into (2.4.3) and (2.4.4) we have

$$\begin{aligned} u'(c_t) &= \mathbb{E}_t [\beta \tilde{R}_{t+1} u'(c_{t+1})] \\ \mathbb{E}_t [(\mathbf{R}_{t+1} - R) u'(c_{t+1})] &= 0 \end{aligned} \quad (2.4.6)$$

Now assume CRRA utility, $u(c) = c^{1-\rho}/(1-\rho)$ and divide both sides by $c_t^{-\rho}$ to get

$$\mathbb{E}_t [(c_{t+1}/c_t)^{-\rho} (\mathbf{R}_{t+1} - R)] = 0 \quad (2.4.7)$$

Now recall the following two facts:

Fact 1: If $\Delta c_{t+1}/c_t$ is small, $c_{t+1}/c_t \approx 1 + \Delta \log c_{t+1}$.

Fact 2: If z is small, $(1+z)^\lambda \approx 1 + \lambda z$.

Using these two facts, equation (2.4.7) can be approximated by

$$\mathbb{E}_t [(1 - \rho \Delta \log c_{t+1}) (\mathbf{R}_{t+1} - R)] \approx 0 \quad (2.4.8)$$

Using one more fact,

Table 2.1: Certainty equivalent X for different coefficients of relative risk aversion

ρ	X
1	70,711
3	63,246
5	58,565
10	53,991
20	51,858
30	51,209
∞	50,000

Fact 3: $\mathbb{E}[xy] = \mathbb{E}[x]\mathbb{E}[y] + \text{cov}(x, y)$

we get

$$(1 - \rho \mathbb{E}_t[\Delta \log c_{t+1}])(R - \mathbb{E}_t[\mathbf{R}_{t+1}]) + \text{cov}_t(-\rho \Delta \log c_{t+1}, -\mathbf{R}_{t+1}) \approx 0 \quad (2.4.9)$$

or

$$\begin{aligned} \mathbb{E}_t[\mathbf{R}_{t+1}] - R &\approx \frac{\rho \text{cov}_t(\Delta \log c_{t+1}, \mathbf{R}_{t+1})}{1 - \rho \mathbb{E}_t[\Delta \log c_{t+1}]} \\ &\approx \rho \text{cov}_t(\Delta \log c_{t+1}, \mathbf{R}_{t+1}) \end{aligned} \quad (2.4.10)$$

where the last approximation holds because $\mathbb{E}_t[\Delta \log c_{t+1}]$ is small.

2.4.1 The Equity Premium Puzzle

Because this expression must hold at all t , we can check it empirically by calculating empirical estimates of the two components and assuming that the sample averages correspond to the representative agent's expectations. That is, if we have data for periods $1 \dots n$, we assume that the unconditional expectations correspond to the sample means, $\mathbb{E}[\mathbf{R}] = (1/n) \sum_{s=1}^n \mathbf{R}_s$; $\mathbb{E}[\Delta \log c] = (1/n) \sum_{s=1}^n \Delta \log c_s$; and $\text{cov}(\Delta \log c, \mathbf{R}) = (1/n) \sum_{s=1}^n (\Delta \log c_s - \mathbb{E}[\Delta \log c])(\mathbf{R}_s - \mathbb{E}[\mathbf{R}])$.

The equity premium puzzle is essentially that $\text{cov}(\Delta \log c, \mathbf{R})$ is very small (about 0.004) but $\mathbb{E}[\mathbf{R}] - R$ is about 0.08 (stocks have earned real returns of about 8 percent more than riskless assets over the historical period), which means that the only way equation (2.4.10) can hold is if ρ is implausibly large (these values imply a value of $\rho = 20$).

How do we know what plausible values of ρ are? Consider the following. You must choose between a gamble in which you consume \$50,000 for the rest of your life with probability 0.5 and \$100,000 with probability 0.5, or consuming some amount X with certainty. The coefficient of relative risk aversion determines the X which would make you indifferent between consuming X or being exposed to the gamble. For example, if $\rho = 0$, then you have no risk aversion at all and you will be indifferent between \$75,000 with certainty and the 50/50 gamble with expected value of \$75,000. Here are the values of X associated with different values of ρ (table taken from Mankiw and Zeldes [1991]).

2.4.2 The Riskfree Rate Puzzle

Rewrite the consumption Euler equation (2.4.6) as

$$u'(c_t) = \mathbb{E}_t [\beta(R + \varsigma_t[\mathbf{R}_{t+1} - R])u'(c_{t+1})] \quad (2.4.11)$$

and note that from (2.4.7) we know that $\mathbb{E}_t[\beta \varsigma_t(\mathbf{R}_{t+1} - R)u'(c_{t+1})] = 0$ so that (2.4.11) reduces to the ordinary Euler equation

$$\begin{aligned} u'(c_t) &= \mathbb{E}_t[\beta R u'(c_{t+1})] \\ 1 &= \beta R \mathbb{E}_t[(c_{t+1}/c_t)^{-\rho}] \end{aligned} \quad (2.4.12)$$

Using the same “facts” and approximations as above, we get the standard approximation to the Euler equation,

$$\Delta \log c_{t+1} \approx (1/\rho)(r - \vartheta) \quad (2.4.13)$$

The “riskfree rate puzzle” is that average consumption growth per capita has been about 1.5 percent (in the US in the postwar period) while real riskfree interest rates have been at most 1 percent. Even if we assume a time preference rate of $\vartheta = 0$ (no impatience at all, e.g. $\beta = 1$), the only way this equation can hold is if ρ is a very small number (maybe even less than one). Of course, this is precisely the opposite of the conclusion of the equity premium puzzle, which implies the ρ must be very large.

The riskfree rate puzzle might in principle be explicable by overlapping generations, though in practice it is hard to make that work well. A precautionary saving motive also cuts against the puzzle, by adding a variance term to consumption growth.

2.5 The Consumption Capital Asset Pricing Model (C-CAPM)

Consider a representative agent solving the joint consumption and portfolio allocation problem:

$$\begin{aligned}
 v(m_t) &= \max u(c_t) + \mathbb{E}_t \left[\sum_{n=1}^{\infty} \beta^n u(c_{t+n}) \right] \\
 \text{s.t.} \\
 m_{t+1} &= (m_t - c_t) \tilde{R}_{t+1} + y_{t+1} \\
 \tilde{R}_{t+1} &= \sum_{i=1}^m \omega_{t,i} \mathbf{R}_{t+1,i} + \left(1 - \sum_{i=1}^m \omega_{t,i} \right) R
 \end{aligned} \tag{2.5.1}$$

where R denotes the return on a perfectly riskless asset and $\mathbf{R}_{t+1,i}$ denotes the return on asset i between periods t and $t+1$, $\omega_{t,i}$ is the share of end-of-period savings invested in asset i , and $\tilde{R}_{t+1}$ is the portfolio-weighted rate of return, and y_{t+1} is noncapital income in period $t+1$.

As usual, the objective can be rewritten in recursive form:

$$v(m_t) = \max_{\{c_t, \omega_{1,t}, \omega_{2,t}, \dots\}} u(c_t) + \beta \mathbb{E}_t \left[v(\tilde{R}_{t+1}(m_t - c_t) + y_{t+1}) \right] \tag{2.5.2}$$

The first order condition with respect to c_t is

$$u'(c_t) = \beta \mathbb{E}_t [\tilde{R}_{t+1} v'(m_{t+1})]. \tag{2.5.3}$$

and the FOC with respect to $\omega_{t,i}$ is

$$\mathbb{E}_t [(\mathbf{R}_{t+1,i} - R) v'(m_{t+1})] = 0 \tag{2.5.4}$$

But the usual logic of the Envelope theorem tells us that

$$u'(c_{t+1}) = v'(m_{t+1}), \tag{2.5.5}$$

so, substituting (2.5.5) into (2.5.3) and (2.5.4) we have

$$u'(c_t) = \mathbb{E}_t [\beta \tilde{R}_{t+1} u'(c_{t+1})] \tag{2.5.6}$$

$$\mathbb{E}_t [(\mathbf{R}_{t+1,i} - R) u'(c_{t+1})] = 0. \tag{2.5.7}$$

Now assume CRRA utility, $u(c) \equiv c^{1-\rho}/(1-\rho)$ and divide both sides of (2.5.7) by $c_t^{-\rho}$ to get

$$\mathbb{E}_t [(c_{t+1}/c_t)^{-\rho} (\mathbf{R}_{t+1,i} - R)] = 0 \tag{2.5.8}$$

We can now follow the same steps as in the [Equity Premium Puzzle](#) section to obtain the relation that *for every asset i*

$$\begin{aligned}
 \mathbb{E}_t [\mathbf{R}_{t+1,i}] - R &\approx \frac{\rho \text{cov}_t(\Delta \log c_{t+1}, \mathbf{R}_{t+1,i})}{1 - \rho \mathbb{E}_t [\Delta \log c_{t+1}]} \\
 &\approx \rho \text{cov}_t(\Delta \log c_{t+1}, \mathbf{R}_{t+1,i})
 \end{aligned} \tag{2.5.9}$$

What does this imply about asset pricing?

Consider an asset for which the return covaries positively with consumption, $\text{cov}(\mathbf{R}_{t+1,i}, \Delta \log c_{t+1}) > 0$. For such an asset, the *marginal utility* will negatively covary with the return. Thus the expected return must be higher for an asset that “does well” when consumption is high. But for a given average stream of dividends or payouts, if the average return is high, the average price must be low. Thus this indicates that prices should be low for assets whose payoffs are procyclical, and high for assets whose payoffs are countercyclical.

2.6 Canonical Asset Pricing and Rational Bubbles

2.6.1 Prices as the PDV of Dividends

Denoting dividends as d_t for a stock with a market price of P_t per share, consider an investor who owns K_t shares at the beginning of period t yielding total wealth $M_t = P_t K_t + d_t K_t$. Assuming the investor has no other source of income (no labor income, for instance) the investor's budget constraint will be

$$\begin{aligned} M_t &= \underbrace{d_t K_t + P_t K_t}_{\text{Total resources}} \\ &= \underbrace{K_{t+1} P_t + C_t}_{\text{Uses of resources}} \end{aligned} \quad (2.6.1)$$

which can be rearranged to indicate how many shares the investor will own next period, as a function of this period's wealth and consumption:

$$\begin{aligned} K_{t+1} &= \frac{M_t - C_t}{P_t} \\ M_{t+1} &= K_{t+1}(P_{t+1} + d_{t+1}) \\ &= (P_{t+1} + d_{t+1}) \left(\frac{M_t - C_t}{P_t} \right). \end{aligned} \quad (2.6.2)$$

If this investor's only goal is to maximize the present discounted utility of consumption and the investor uses a discount factor of $\mathbf{R}^{-1}$ then we have

$$V_t(M_t) = \max_{\{C_t\}} u(C_t) + \mathbf{R}^{-1} \mathbb{E}_t \left[V_{t+1} \left((P_{t+1} + d_{t+1}) \left(\frac{M_t - C_t}{P_t} \right) \right) \right] \quad (2.6.3)$$

with FOC

$$\begin{aligned} u'(C_t) &= \mathbf{R}^{-1} \mathbb{E}_t \left[\left(\frac{P_{t+1} + d_{t+1}}{P_t} \right) V'_{t+1}(M_{t+1}) \right] \\ u'(C_t) &= \mathbf{R}^{-1} \mathbb{E}_t \left[\left(\frac{P_{t+1} + d_{t+1}}{P_t} \right) u'_{t+1}(C_{t+1}) \right]. \end{aligned} \quad (2.6.4)$$

Now suppose that the investor is risk neutral ($u(C) = C$) so that $u'(C_{t+1}) = u'(C_t) = 1$; (2.6.4) becomes

$$\begin{aligned} \mathbf{R} &= \mathbb{E}_t \left[\frac{P_{t+1} + d_{t+1}}{P_t} \right] \\ P_t &= \mathbb{E}_t \left[\frac{P_{t+1} + d_{t+1}}{\mathbf{R}} \right] \end{aligned} \quad (2.6.5)$$

Of course, similar logic can be employed to show that

$$P_{t+1} = \mathbb{E}_{t+1}[(P_{t+2} + d_{t+2})/\mathbf{R}] \quad (2.6.6)$$

and we can use the law of iterated expectations to substitute repeatedly, obtaining

$$P_t = \mathbb{E}_t \left[\sum_{s=t+1}^{T+1} \mathbf{R}^{t-s} d_s \right] + \mathbf{R}^{-(T+1-t)} \mathbb{E}_t[P_{T+1}] \quad (2.6.7)$$

We usually assume the “no-bubbles” condition that says that $\lim_{T \rightarrow \infty} \mathbb{E}_t[\mathbf{R}^{-(T+1-t)} P_{T+1}] = 0$. In this case it is clear that the equilibrium price must equal the present discounted value of dividends:

$$P_t^* = \mathbb{E}_t \left[\sum_{s=t+1}^{\infty} \mathbf{R}^{t-s} d_s \right] \quad (2.6.8)$$

This result is very similar to what emerges from the [Lucas Asset Pricing model](#), though there the stochastic discount factor plays the role that $\mathbf{R}$ plays here.

Suppose now that dividends are expected to grow by a constant factor Γ henceforth. In that case we have

$$\begin{aligned}
 P_t^* &= \mathbb{E}_t \left[\sum_{s=t+1}^{\infty} \mathbf{R}^{t-s} \Gamma^{s-t} d_t \right] \\
 &= \mathbb{E}_t \left[\sum_{s=t+1}^{\infty} (\Gamma/\mathbf{R})^{s-t} d_t \right] \\
 &= d_t \left(\frac{\Gamma/\mathbf{R}}{1 - \Gamma/\mathbf{R}} \right) \\
 &\approx d_t \left(\frac{1}{\mathbf{r} - \gamma} \right),
 \end{aligned} \tag{2.6.9}$$

where the final approximation uses log rates: $\mathbf{r} \equiv \log \mathbf{R}$ and $\gamma \equiv \log \Gamma$. This is known as the “Gordon formula.” The tricky thing in applying the formula is to know what to assume for $\mathbf{R}$ and Γ . The interest rate $\mathbf{R}$ should be the interest rate “appropriate” for discounting risky quantities. The usual assumption is that $\mathbf{R} = R + \varphi$ where R is the rate of return on perfectly safe (riskfree) assets and φ is the rate-of-return premium that people demand as compensation for the risk inherent in future dividends.

2.6.2 The Random Walk of Asset Pricing

The Law of Iterated Expectations

Suppose that a security price at time t , P_t , can be written as the rational expectation of some “fundamental value” V^* conditional on information available at time t (the usual example of the “fundamental value” in question is the present discounted value of dividends). Then we have

$$P_t = \mathbb{E}_t[V^*]. \tag{2.6.10}$$

The same formula holds in period $t + 1$:

$$P_{t+1} = \mathbb{E}_{t+1}[V^*]. \tag{2.6.11}$$

Then the expectation of the change in the price over the next period is

$$\begin{aligned}
 \mathbb{E}_t[P_{t+1} - P_t] &= \mathbb{E}_t[\mathbb{E}_{t+1}[V^*] - \mathbb{E}_t[V^*]] \\
 &= \mathbb{E}_t[V^*] - \mathbb{E}_t[V^*] \\
 &= 0
 \end{aligned} \tag{2.6.12}$$

because any information known at time t must be known at time $t + 1$ and so the only thing that should cause a *change* in prices should be the arrival of new information that was not known at time t .

2.6.3 Deterministic Bubbles

Note, however, that we simply *assumed* the no-bubbles condition; we did not justify it with economic logic. Consider the following candidate process for P_t :

$$\begin{aligned}
 P_t &= P_t^* + B_t \\
 B_{t+1} &= R B_t.
 \end{aligned} \tag{2.6.13}$$

That is, price is equal to the fundamental price plus a “bubble” term B_t which grows nonstochastically at rate R from period to period. (Here we will assume that the risk premium φ is zero, which

would be true in equilibrium for risk-neutral consumers). The question at hand is whether this equation satisfies the first order condition (2.6.5). We can show that it does by starting with the formula for P_{t+1} and working backwards:

$$\begin{aligned}
P_{t+1} &= P_{t+1}^* + B_{t+1} \\
\mathbb{E}_t[P_{t+1}] &= \mathbb{E}_t[P_{t+1}^*] + B_t R \\
\mathbb{E}_t[P_{t+1}] &= \mathbb{E}_t \left[\sum_{s=t+2}^{\infty} R^{t+1-s} d_s \right] + B_t R \\
\mathbb{E}_t[P_{t+1}]/R &= \mathbb{E}_t \left[\sum_{s=t+2}^{\infty} R^{t-s} d_s \right] + B_t \\
\underbrace{\mathbb{E}_t[P_{t+1} + d_{t+1}]/R}_{=P_t} &= \mathbb{E}_t[d_{t+1}/R] + \mathbb{E}_t \left[\sum_{s=t+2}^{\infty} R^{t-s} d_s \right] + B_t \\
P_t &= \mathbb{E}_t \left[\sum_{s=t+1}^{\infty} R^{t-s} d_s \right] + B_t \\
P_t &= P_t^* + B_t
\end{aligned} \tag{2.6.14}$$

where the underbraced equality follows from (2.6.5).

In words, this says that the first order condition has an infinite number of solutions of the form $P_t = P_t^* + B_t$. Thus, nothing about the logic of the problem thus far rules out a *rational deterministic bubble*, which is a bubble whose size grows at the rate of interest forever. Thus, in principle any level of the stock price is possible at period t ; all that the theory implies is that *if a bubble exists*, its value must rise by a factor R in every period.

2.6.4 Stochastically Bursting Bubbles

Blanchard [1979] considers another possible candidate process for P :

$$P_t = P_t^* + q_t \tag{2.6.15}$$

$$q_{t+1} = \begin{cases} (R/\alpha)q_t & \text{with probability } \alpha \\ 0 & \text{with probability } 1 - \alpha \end{cases} \tag{2.6.16}$$

Roll equation (2.6.15) forward one period, and take its expectation as of time t :

$$\begin{aligned}
P_{t+1} &= P_{t+1}^* + q_{t+1} \\
\mathbb{E}_t[P_{t+1}] &= \mathbb{E}_t[P_{t+1}^*] + q_t(R/\alpha)\alpha + (0)(1 - \alpha) \\
\mathbb{E}_t[P_{t+1}] &= \mathbb{E}_t \left[\sum_{s=t+2}^{\infty} R^{t+1-s} d_s \right] + q_t R \\
\mathbb{E}_t[P_{t+1}]/R &= \mathbb{E}_t \left[\sum_{s=t+2}^{\infty} R^{t-s} d_s \right] + q_t \\
\mathbb{E}_t[P_{t+1} + d_{t+1}]/R &= \mathbb{E}_t[d_{t+1}/R] + \mathbb{E}_t \left[\sum_{s=t+2}^{\infty} R^{t-s} d_s \right] + q_t \\
P_t &= \mathbb{E}_t \left[\sum_{s=t+1}^{\infty} R^{t-s} d_s \right] + q_t
\end{aligned} \tag{2.6.17}$$

Thus, the model allows stochastic bubbles which, during the period of their inflation, rise at a rate that is enough faster than the gross interest rate to exactly compensate shareholders (in expected value terms) for their expected capital loss when the bubble collapses.

Note that the probability that the bubble has burst by time $t + s$ is the probability that it bursts in $t + 1$ plus the probability that it bursts in $t + 2$ given that it did not burst in $t + 1$, and so on:

$$\text{Prob}(\text{burst by } t+s) = (1 - \alpha)(1 + \alpha + \alpha^2 + \dots + \alpha^{s-1}). \quad (2.6.18)$$

Summing the geometric series collapses the right-hand side of (2.6.18) to $1 - \alpha^s$. Since $\alpha < 1$, letting $s \rightarrow \infty$ drives α^s to zero, so the probability that the bubble has burst by time $t + s$ approaches one: a stochastic bubble bursts eventually with probability one.

2.6.5 Arguments for Why Bubbles Cannot Exist

Note that the bubble term in equation (2.6.13) rises without bound. It turns out that this fact rules out negative bubbles. To see why, note that if B_t is negative and if the fundamental price P_t^* is bounded, then eventually B_t grows large enough so that the predicted price of a share is negative. But if share prices were negative, people could make themselves better off by simply throwing away their stock certificates. Thus, the restriction that prices must be positive rules out negative bubbles.

Bubbles can also be ruled out if there is a maximum possible price that the asset can have. Consider, for example, the question of whether there can be a bubble on the price of diamonds. Suppose that there is a fixed supply of natural diamonds in existence, but suppose that new artificial diamonds can be made at some price $\bar{P} > 4P_t$, that is diamonds can be made at a cost 4 times higher than the current market price. But if there is a bubble, then it will imply that eventually the market price would exceed $\bar{P}$. At that point, nobody will be willing to pay $P_t > \bar{P}$ for natural diamonds, so the bubble's price cannot keep rising beyond $\bar{P}$. Thus, rational bubbles are ruled out for assets which are reproducible.

We assumed, in deriving these results, that the investor's utility function was linear. It is more difficult to justify rational bubbles in an economy with risk averse investors.

There are also some general equilibrium arguments against bubbles, which basically boil down to the observation that if the value of the bubble is growing forever, its size will eventually exceed the size of the entire capital stock, and in that case productive capital will have been driven to zero because everybody owns the bubble instead of capital, which cannot make sense.

2.7 Consumption with Optimal Portfolio Choice

The [consumption out of risky assets](#) section shows that for a Merton [1969]-Samuelson [1969] consumer facing return $\log \hat{\mathbf{R}}_{t+1} \sim \mathcal{N}(\hat{\mathbf{r}} - \sigma_{\hat{\mathbf{r}}}^2/2, \sigma_{\hat{\mathbf{r}}}^2)$ on the only financial asset available, the optimal marginal propensity to consume is approximately

$$\kappa \approx \hat{\mathbf{r}} - \rho^{-1}(\hat{\mathbf{r}} - \vartheta) - (\rho - 1) \left(\sigma_{\hat{\mathbf{r}}}^2/2 \right) \quad (2.7.1)$$

where the precautionary effect of financial risk on the MPC is captured by the $-(\rho - 1) (\sigma_{\hat{\mathbf{r}}}^2/2)$ term. Since $\rho > 1$ by assumption, this equation yields the plausible conclusion that an increase in *unavoidable* financial risk $\sigma_{\hat{\mathbf{r}}}^2$ reduces the level of consumption.

We are interested here in understanding how the results change when the consumer can choose how much to invest in the risky asset, so that financial risk can be avoided by reducing the share of the portfolio allocated to the risky asset. The [CRRRA portfolio choice](#) section derives the portfolio share ς that an optimizing consumer will invest in a risky asset earning return $\log \hat{\mathbf{R}}_{t+1} \sim \mathcal{N}(\hat{\mathbf{r}} - \sigma_{\hat{\mathbf{r}}}^2/2, \sigma_{\hat{\mathbf{r}}}^2)$, so that $\log \hat{\mathbf{R}} = \hat{\mathbf{r}}$ (where the subscriptless version of a variable denotes its expectation unless otherwise noted, e.g. $\hat{\mathbf{R}} \equiv \mathbb{E}_t[\hat{\mathbf{R}}_{t+1}]$).²⁰

The remaining proportion $(1 - \varsigma)$ of the portfolio earns a riskless return $r = \log \mathbf{R}$,²¹ and we write the log expected return premium factor as $\varphi \equiv \hat{\mathbf{R}}/\mathbf{R}$ with log expected return premium $\varphi \equiv \log \varphi = \hat{\mathbf{r}} - r = \log \hat{\mathbf{R}}/\mathbf{R}$; optimal choice of ς yields a portfolio whose realized return factor is written $\tilde{R}$ and the log of whose realization is well approximated by²²

$$\begin{aligned} \tilde{r}_{t+1} - r &\approx \varsigma \overbrace{(\hat{\mathbf{r}}_{t+1} - r)}^{\equiv \varphi_{t+1}} + \varsigma(1 - \varsigma)\sigma_{\hat{\mathbf{r}}}^2/2 \\ &= \varsigma(\hat{\mathbf{r}}_{t+1} - r + \sigma_{\hat{\mathbf{r}}}^2/2) - \varsigma^2\sigma_{\hat{\mathbf{r}}}^2/2, \end{aligned} \quad (2.7.2)$$

whose variance (using facts SumNormsIsNorm and NormTimes) is

$$\sigma_{\tilde{r}}^2 = \varsigma^2\sigma_{\hat{\mathbf{r}}}^2, \quad (2.7.3)$$

and since $\mathbb{E}_t[\hat{\mathbf{r}}_{t+1}] = \hat{\mathbf{r}} - \sigma_{\hat{\mathbf{r}}}^2/2$ the expectation of (2.7.2) will be

$$\begin{aligned} \mathbb{E}_t[\tilde{r}_{t+1} - r] &\approx (\hat{\mathbf{r}} - \sigma_{\hat{\mathbf{r}}}^2/2 - r + \sigma_{\hat{\mathbf{r}}}^2/2)\varsigma - \varsigma^2\sigma_{\hat{\mathbf{r}}}^2/2 \\ &= \varphi\varsigma - \varsigma^2\sigma_{\hat{\mathbf{r}}}^2/2 \end{aligned} \quad (2.7.4)$$

so that, using fact SumNormsIsNorm,

$$\tilde{r}_{t+1} - r \sim \mathcal{N}(\varphi\varsigma - \varsigma^2\sigma_{\hat{\mathbf{r}}}^2/2, \varsigma^2\sigma_{\hat{\mathbf{r}}}^2) \quad (2.7.5)$$

yielding, using fact LogELogNormTimes,

$$\log \tilde{R} \equiv \log \mathbb{E}_t[\tilde{R}_{t+1}] = \varsigma\varphi + r \quad (2.7.6)$$

and the [CRRRA portfolio choice](#) section shows that under these circumstances the optimal risky portfolio share is well approximated by

$$\varsigma \approx \left(\frac{\varphi}{\rho\sigma_{\hat{\mathbf{r}}}^2} \right) \quad (2.7.7)$$

so that substituting from (2.7.7) and (2.7.3), the precautionary effect after taking account of optimal portfolio adjustment is

²⁰(See fact LogMeanMPS, and note that $\hat{\mathbf{R}} = \mathbf{R}$ under the given assumptions; that is, an increase in the degree of risk $\sigma_{\hat{\mathbf{r}}}^2$ does not change the expected return factor in levels). Note that the subscriptless version of the log of the risky return, $\hat{\mathbf{r}}_{t+1}$, is *not* equal to its mean: $\mathbb{E}_t[\hat{\mathbf{r}}_{t+1}] = \hat{\mathbf{r}} - \sigma_{\hat{\mathbf{r}}}^2/2$.

²¹The bold font is used for the risky return and the narrow font for the safe return.

²²See the [CRRRA portfolio choice](#) section for a figure examining the quality of the approximation.

$$\begin{aligned}
-(\rho - 1) \left(\sigma_{\tilde{r}}^2 / 2 \right) &= -(\rho - 1) \left(\frac{\varsigma^2 \sigma_{\mathbf{r}}^2}{2} \right) \\
&= -(\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right)
\end{aligned} \tag{2.7.8}$$

which says that (for $\rho > 1$) the absolute size of the precautionary term *shrinks* as the risk grows larger. To put it another way, when $\rho > 1$ an *increase* in the riskiness of the risky asset causes consumption to *rise*. At first, this seems bizarre: Intuition suggests that in reality people *cut back* on consumption in the face of greater risk. The way to understand the result is to break the response to the increase in riskiness into two components. The first is the direct precautionary effect examined in the [consumption out of risky assets](#) section, which works in the way intuition suggests, since more risk implies lower consumption. The second is the “portfolio rebalancing” effect examined in the [CRRA portfolio choice](#) section, because an increase in riskiness makes the consumer choose to invest less in the risky asset. The resolution is that the consumer’s “flight from risk” is so effective in reducing riskiness (because the portfolio share ς enters as a *squared* term in (2.7.8)) that the riskiness of the *portfolio* actually *declines* as the riskiness of the risky asset increases, a result also derived in the [CRRA portfolio choice](#) section. The overall reduction in consumption coming through the precautionary term then makes sense: precautionary saving is less than before *because the consumer optimally chooses less exposure to risk* than before.

The fact that precautionary saving diminishes when risk is larger does not, by itself, tell us whether consumption increases in response to an increase in risk; in addition to the precautionary channel, the MPC is also affected by the decline in the portfolio’s expected rate of return that results from the consumer’s choice to invest less in the high-expected-return risky asset and more in the low-return safe asset. Substituting (2.7.7) into (2.7.6) indicates that the log of the expected portfolio excess return factor becomes

$$\log \tilde{R}/R \equiv \log \mathbb{E}_t[\tilde{R}_{t+1}/R] = \frac{\varphi^2}{\rho\sigma_{\mathbf{r}}^2} \tag{2.7.9}$$

so an increase in $\sigma_{\mathbf{r}}^2$ can have quite a powerful effect in reducing the consumer’s expected portfolio return.

As with the change in the riskfree rate analyzed in the [perfect foresight CRRA](#) section, this change in the portfolio return has two consequences: An income and a substitution effect, captured respectively by the first and second terms of (2.7.1). Substituting $\tilde{r} = \varphi^2/\rho\sigma_{\mathbf{r}}^2 + r$ and $\sigma_{\tilde{r}}^2 = (\varphi/\rho)^2/\sigma_{\mathbf{r}}^2$ into (2.7.1) yields

$$\begin{aligned}
\kappa &\approx \tilde{r} - \rho^{-1}(\tilde{r} - \vartheta) - (\rho - 1) \left(\sigma_{\tilde{r}}^2 / 2 \right) \\
&= \rho^{-1}\vartheta + \tilde{r}(1 - \rho^{-1}) - (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \\
&= \rho^{-1}\vartheta + (r + \varphi^2/\rho\sigma_{\mathbf{r}}^2)(1 - \rho^{-1}) - (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \\
&= r - \rho^{-1}(r - \vartheta) + \underbrace{(\rho - 1)(\varphi^2/\rho^2\sigma_{\mathbf{r}}^2)}_{\text{net inc and sub effect}} - (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right).
\end{aligned} \tag{2.7.10}$$

Thus, for $\rho > 1$ the income effect outweighs the substitution effect so that when an increase in financial risk causes consumers to shy away from the risky asset the reduction in consumption from the drop in income (“the income effect”) is larger than the increase in consumption from the lowering of the incentive to delay consumption (“the substitution effect”).

Note, finally, that the net of the income and substitution effects is (remarkably) of exactly the same functional form as the precautionary effect, but of opposite sign and twice as large. So the combination of all three effects (income, substitution, precautionary) yields:

$$\kappa \approx r - \rho^{-1}(r - \vartheta) + (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \quad (2.7.11)$$

and we can see that the overall effect of the increase in financial risk is to reduce the marginal propensity to consume because the net of the income and substitution effects channels outweighs the precautionary effect (see the [perfect foresight CRRA](#) section for a refresher on income and substitution effects, as well as human wealth effects (which are absent here but could easily be added by giving the consumer an expected stream of future noncapital income)).

Figure 2.3 confirms, both using the approximate formulas derived above and using a numerically exact solution, that the MPC declines as risk increases.

Finally, in addition to the effects on consumption, we are interested in the effects on saving. Since saving is income minus consumption, this means we need to know the effects on income. But income for our consumer is entirely from his portfolio investments, so the income effect is captured simply by $\tilde{r}$. The “saving effect” is therefore captured by

$$\begin{aligned} \tilde{r} - \kappa &\approx \rho^{-1}(\tilde{r} - \vartheta) + (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \\ &= \rho^{-1} \left(\frac{\varphi^2}{\rho\sigma_{\mathbf{r}}^2} + r - \vartheta \right) + (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \\ &= \rho^{-1}(r - \vartheta) + \frac{\varphi^2}{\rho^2\sigma_{\mathbf{r}}^2} + (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \\ &= \rho^{-1}(r - \vartheta) + \frac{\varphi^2/\rho^2}{\sigma_{\mathbf{r}}^2} + (\rho - 1) \left(\frac{(\varphi/\rho)^2}{2\sigma_{\mathbf{r}}^2} \right) \\ &= \rho^{-1}(r - \vartheta) + \left(\frac{\rho + 1}{2} \right) \left(\frac{\varphi^2}{\sigma_{\mathbf{r}}^2\rho^2} \right) \end{aligned} \quad (2.7.12)$$

which indicates that an increase in the riskiness of the financial asset will reduce net saving; although the increase in risk diminishes consumption, the decline in expected income from the decline in the consumer’s investment in the high-return asset is greater, and so expected saving declines despite the decline in consumption. It is worth emphasizing again, though, that this decline in saving is *not* properly called a precautionary saving effect; it is a consequence of the portfolio rebalancing which actually reduces the size of the precautionary effect. It would be confusing to refer to this simply as a precautionary effect, when in fact the ultimate outcome (reduced saving) depends also on portfolio rebalancing, income, and substitution effects. This can be seen most clearly in the case of logarithmic utility, where the precautionary effect is precisely zero (see the discussion in the [consumption out of risky assets](#) section), yet the portfolio and other effects exist and still generate the conclusion that the increase in risk reduces saving.

This analysis has been entirely in partial equilibrium. General equilibrium considerations (which arise, for example, in attempting to use models of this kind to understand global imbalances) complicate the picture further. In particular, consumers’ efforts to flee the risky asset result in a lower equilibrium price for that asset (and a correspondingly higher price for the riskless asset, and therefore a lower riskfree rate). This induces yet a *third* round of responses to the increase in risk.

A final important observation is that the assumption that the consumer has no labor income means that perhaps the largest “classical” effect of interest rates on consumption and saving is entirely omitted from the analysis here: The human wealth effect. (See Summers [1981] for a statement of the argument that the human wealth effect of interest rates is likely in practice to be much larger than the income and substitution effects.) The general equilibrium decline in riskfree rates attendant upon an increase in riskiness of the risky asset should boost human wealth, increase consumption, and reduce saving. For careful and insightful treatments of the general equilibrium problem, see Angeletos and Panousi [2011] and Corneli [2011].

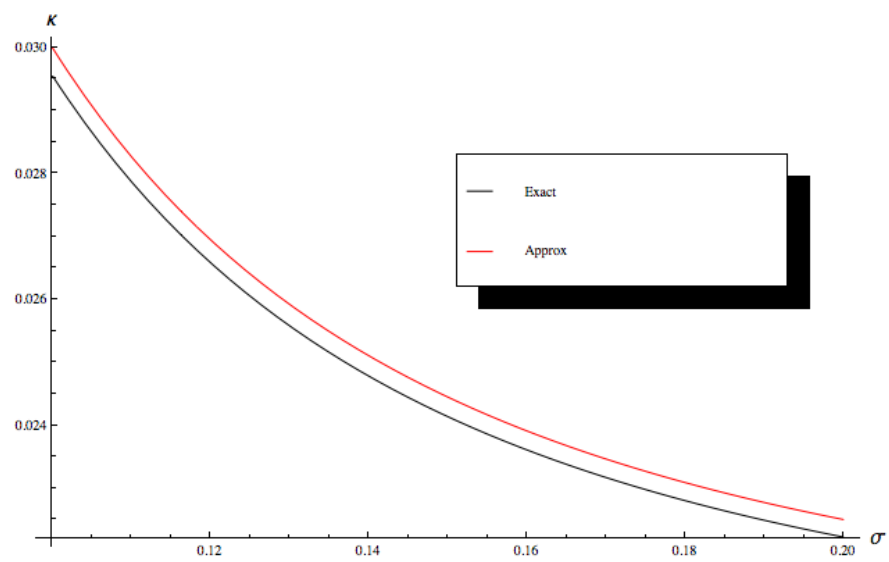


Figure 2.3: MPC as a Function of σ Calculated Using (Numerically) Exact and Approximate Methods

2.8 CRRA Portfolio Choice with Two Risky Assets

Merton [1969] and Samuelson [1969] study optimal portfolio allocation for a consumer with Constant Relative Risk Aversion utility $u(c) = (1 - \rho)^{-1} c^{1-\rho}$ who can choose among many risky investment options.

Using their framework, here we study a consumer who has wealth a_t at the end of period t , and is deciding how much to invest in two risky assets with lognormally distributed return factors $\mathbf{R}_{t+1} = (\mathbf{R}_{1,t+1}, \mathbf{R}_{2,t+1})'$, $\log \mathbf{R}_{t+1} = \mathbf{r}_{t+1} = (\mathbf{r}_{1,t+1}, \mathbf{r}_{2,t+1})' \sim (\mathcal{N}(\mathbf{r}_1, \sigma_1^2), \mathcal{N}(\mathbf{r}_2, \sigma_2^2))'$, with covariance matrix

$$\begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}. \quad (2.8.1)$$

If the period- t consumer invests proportion ς_i of a_t in risky asset i , $i = 1, 2$ (so that $\varsigma_1 = (1 - \varsigma_2)$ and vice-versa), spending all available resources in the last period of life²³ $t + 1$ will yield:

$$c_{t+1} = \underbrace{(\varsigma \cdot \mathbf{R}_{t+1})}_{\equiv \tilde{R}_{t+1}} a_t \quad (2.8.2)$$

where $\tilde{R}_{t+1}$ is the portfolio-weighted return factor.

Campbell and Viceira [2002] point out that a good approximation to the portfolio rate of return is obtained by

$$\tilde{r}_{t+1} = \mathbf{r}_{1,t+1} + \varsigma_2(\mathbf{r}_{2,t+1} - \mathbf{r}_{1,t+1}) + \varsigma_2(1 - \varsigma_2)\eta/2 \quad (2.8.3)$$

where

$$\eta = (\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}). \quad (2.8.4)$$

Using this approximation, the expectation as of date t of utility at date $t + 1$ is:

$$\begin{aligned} \mathbb{E}_t[u(c_{t+1})] &\approx (1 - \rho)^{-1} \mathbb{E}_t \left[\left(a_t e^{\mathbf{r}_{1,t+1}} e^{\varsigma_2(\mathbf{r}_{2,t+1} - \mathbf{r}_{1,t+1}) + \varsigma_2(1 - \varsigma_2)\eta/2} \right)^{1-\rho} \right] \\ &\approx \underbrace{(1 - \rho)^{-1} a_t^{1-\rho}}_{\text{constant} < 0} \underbrace{e^{(1-\rho)\varsigma_2(1-\varsigma_2)\eta/2} \mathbb{E}_t \left[e^{(\mathbf{r}_{1,t+1} + \varsigma_2(\mathbf{r}_{2,t+1} - \mathbf{r}_{1,t+1}))(1-\rho)} \right]}_{\text{excess return utility factor}} \end{aligned} \quad (2.8.5)$$

where the first term is a negative constant under the usual assumption that relative risk aversion $\rho > 1$.

Our foregoing assumptions imply that

$$(1 - \rho)(\varsigma_1 \mathbf{r}_{1,t+1} + \varsigma_2 \mathbf{r}_{2,t+1}) \sim \mathcal{N}((1 - \rho)(\varsigma_1 \mathbf{r}_1 + \varsigma_2 \mathbf{r}_2), (1 - \rho)^2(\varsigma_1^2 \sigma_1^2 + \varsigma_2^2 \sigma_2^2 + 2\varsigma_1 \varsigma_2 \sigma_{12})) \quad (2.8.6)$$

(using LogELogNormTimes). With a couple of extra lines of derivation we can show that the log of the expectation in (2.8.5) is

$$\log \mathbb{E}_t \left[e^{(\mathbf{r}_{1,t+1} + \varsigma_2(\mathbf{r}_{2,t+1} - \mathbf{r}_{1,t+1}))(1-\rho)} \right] = (1 - \rho)(\varsigma_1 \mathbf{r}_1 + \varsigma_2 \mathbf{r}_2) + (1 - \rho)^2(\varsigma_1^2 \sigma_1^2 + \varsigma_2^2 \sigma_2^2 + 2\varsigma_1 \varsigma_2 \sigma_{12})/2 \quad (2.8.7)$$

Substituting from (2.8.7) for the log of the expectation in (2.8.5), the log of the “excess return utility factor” in (2.8.5) is

$$(1 - \rho)\varsigma_2(1 - \varsigma_2)\eta/2 + (1 - \rho)(\mathbf{r}_1 + \varsigma_2(\mathbf{r}_2 - \mathbf{r}_1)) + (\rho - 1)^2(\sigma_1^2 + \varsigma_2^2\eta + 2\varsigma_2(\sigma_{12} - \sigma_1^2))/2. \quad (2.8.8)$$

²³The portfolio allocation solution obtained below induces back to earlier periods of life, as Samuelson [1963] and Samuelson [1989] famously emphasized.

The ς that minimizes this log will also minimize the level; the FOC for minimizing this expression is

$$\begin{aligned} (1 - 2\varsigma_2)\eta/2 + \mathbf{r}_2 - \mathbf{r}_1 + (1 - \rho)(\varsigma_2\eta + (\sigma_{12} - \sigma_1^2)) &= 0 \\ (\mathbf{r}_2 - \mathbf{r}_1 + \frac{\eta}{2}) + (1 - \rho)(\sigma_{12} - \sigma_1^2) &= \rho\eta\varsigma_2. \end{aligned} \quad (2.8.9)$$

So

$$\varsigma_2 = \left(\frac{\mathbf{r}_2 - \mathbf{r}_1 + \eta/2 + (1 - \rho)(\sigma_{12} - \sigma_1^2)}{\rho\eta} \right) \quad (2.8.10)$$

and note that if the first asset is riskfree so that $\sigma_1 = \sigma_{12} = 0$ then this reduces to

$$\varsigma_2 = \left(\frac{\mathbf{r}_2 - \mathbf{r}_1 + \sigma_2^2/2}{\rho\sigma_2^2} \right) \quad (2.8.11)$$

but the log of the expected return premium (in levels) on the risky over the safe asset in this case is $\varphi \equiv \log(\mathbf{R}_2/\mathbf{R}_1) = \mathbf{r}_2 - \mathbf{r}_1 + \sigma_2^2/2$ (recalling that we have assumed $\sigma_{12} = \sigma_1^2 = 0$), so (2.8.11) becomes

$$\varsigma_2 = \left(\frac{\varphi}{\rho\sigma_2^2} \right) \quad (2.8.12)$$

which corresponds to the solution obtained for the case of a single risky asset in the [CRRA portfolio choice](#) section.

Chapter 3

Investment Theory

Investment is the firm's version of the consumption problem, and we exploit the parallel throughout. It is close enough that the Hall-Jorgenson user cost of capital reads directly off an intertemporal first-order condition (the same Euler equation, with the marginal product of capital where the interest factor stood). The trouble with that frictionless benchmark is what it leaves out. It pins down the desired capital *stock* but says nothing about the *rate* at which a firm approaches it, so whenever user cost falls below the marginal product of capital the implied investment rate is unbounded.

Abel and Hayashi cure the indeterminacy by making adjustment costly. The shadow value of installed capital, q , then becomes the state variable governing investment. The chapter closes with imperfect capital markets, which break the separation between a firm's investment decision and its financial position, and so explain why cash flow helps predict investment even after controlling for q .

3.1 Investment Notes

3.1.1 Broad Comments

Why Study Investment?

1. Although investment is much smaller as a fraction of GDP than consumption, investment is much more *variable* than consumption. So *fluctuations* in investment spending account for a large proportion of business-cycle frequency fluctuations in GDP. (See the figure below for one way of measuring this).
2. In the long run, the average magnitude of investment spending determines the size of the capital stock. If we think accounting for capital is an important component of explaining macroeconomic outcomes, then we need an understanding of investment.

History of Thought

- Keynes: I determined by “animal spirits” which fluctuated strongly
- Samuelson [1939]/Hansen/Harrod: Keynesian “Multiplier/Accelerator” model of investment
- Hall and Jorgenson [1967] neoclassical model with no costs of adjustment
- Tobin [1969]’s q model
- Abel [1981]-Hayashi [1982] “marginal q ” model with smooth convex costs of adjustment
- Models with nonconvex costs of adjustment
- Models with capital market imperfections

3.1.2 The Samuelson Multiplier-Accelerator Model

From today’s perspective, the distinctive feature of Samuelson’s model of investment is its emphasis on the proposition that levels of investment were determined by impulses from outside the market for investment. While the model is highly sophisticated in many ways, the essence of it for our purposes can be captured by a very simple equation:

$$I_t = \alpha Y_t + \gamma r_t \quad (3.1.1)$$

When paired with equations determining consumption and income, and with some lags added, the model generated the classic Keynesian multiplier-accelerator framework. But the key feature of the investment component of the model was that it assumed that the desired level of investment depended in a simple linear way on the autonomous evolution of other variables: Output Y and interest rates r . These variables were in turn affected by I , and the model was capable of producing rich dynamics as a result. But for the purpose of explaining investment itself, the model’s key implication could be boiled down to (3.1.1).

In purely statistical terms, an equation like (3.1.1) performs remarkably well in “explaining” investment spending. But this is not very satisfactory because income, investment, and interest rates are presumably all determined by deeper underlying features of the environment, and so it is highly problematic to interpret this equation as instructive about the fundamental determinants of investment.

3.1.3 Hall and Jorgenson

Hall and Jorgenson [1967] made an important step forward by constructing a model in which an optimizing firm chooses the level of its capital stock with reference to fundamental features of the economic environment like its production function, depreciation rates, tax considerations, and a comparison of interest rates to the productivity of the firm’s available uses of capital.

The [Hall-Jorgenson](#) section presents a stripped-down version of this model.

An important objection to this framework, however, was that it assumes that firms have the ability to instantly and costlessly move their capital stocks to the level that would be justified by the

prevailing economic environment. If something happens to make the firm wish to be larger, it can, say, quadruple its capital stock overnight, paying no cost of adjustment for such a radical change.

3.1.4 q Models

A seminal paper by Tobin [1969] provided the intuition for how to move forward. Tobin defined a variable

$$Q = \left(\frac{\text{Stock Mkt Val of Firm}}{\text{Replacement Cost of K}} \right) \quad (3.1.2)$$

and proposed that investment should obey the rule

$$\begin{cases} i_t > 0 & \text{if } Q_t > 1 \\ i_t < 0 & \text{if } Q_t < 1 \end{cases} \quad (3.1.3)$$

That is, firms which have a value greater than what it would cost to reproduce their capital should be growing, while firms which are not worth what it would cost to reproduce them should be shrinking.

Tobin's Q is now one of the basic tools of financial market analysis; it is used regularly as one of several tools designed to try to assess firms' prospects.

Tobin's paper inspired a large amount of empirical and theoretical work, culminating in two classic papers, by Abel [1981] and Hayashi [1982], which put the theory on a rigorous mathematical basis. The canonical model that emerged from this literature is summarized in the [marginal \$q\$ model](#) section.

One unsatisfactory feature of the q model is that the motivations of the firm's owners are not clearly delineated: shareholders are assumed to be risk-neutral return maximizers, so the model has no real implications for the firm's dividend policy or holdings of cash. The [entrepreneur's problem](#) section takes up the alternative of a firm wholly owned by a utility-maximizing entrepreneur whose consumption is the firm's dividend payout.

3.1.5 Imperfect Capital Markets

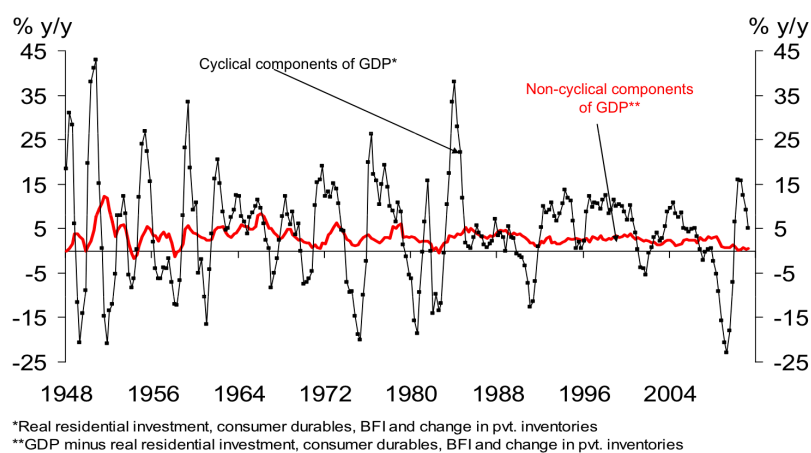
One of the foundations of the q model as developed by Abel and Hayashi is its reliance on an assumption that capital markets are perfect, in the sense that in these models investment is pursued right up to the point where the marginal value of an additional unit of investment has been driven down to the exact value of the riskless interest rate (after accounting appropriately for tax features, depreciation, etc), which reflects the optimal quantity of investment.

The key reason that the assumption of perfect capital markets is so useful is that it implies that it is not necessary to keep track of the firm's cash management or financial activities.

Useful though it may be, the assumption of perfect capital markets is clearly very far from reality. A vast literature has explored what happens when a wide variety of imperfections are introduced. The handout [DynanImperfCapMkts.pdf](#) summarizes one particular model of capital market imperfections that produces implications that are similar to those that emerge from many other models.

The implication of capital market imperfections that has been most vigorously explored is that the amount of investment may be affected by a firm's financial position. The crux of that literature is summarized in the [investment and cash flow](#) section.

The business cycle is driven by cyclical components of GDP, but cyclical components only 20%-25% of GDP



*Real residential investment, consumer durables, BFI and change in pvt. inventories
 **GDP minus real residential investment, consumer durables, BFI and change in pvt. inventories

Source: BEA, DB Global Markets Research

Deutsche Bank

Torsten Slok, torsten.slok@db.com +1 212 250-2155 October 2011

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Figure 3.1: Investment Drives the Cycle. Source: Torsten Sløk, Deutsche Bank, October 2011.

3.2 The Hall-Jorgenson Model of Investment

Hall and Jorgenson [1967] consider the problem of a firm that produces output using capital k as its only input,

$$y = f(k) \quad (3.2.1)$$

and which obtains its capital k from a market in which a unit of capital can be rented for a unit of time at rate $\varkappa_t$.

In period t , the firm maximizes profit (implicitly normalizing the price of output to 1),

$$\max_{k_t} k_t^\alpha - \varkappa_t k_t \quad (3.2.2)$$

yielding first order conditions

$$\begin{aligned} f'(k_t) &= \varkappa_t \\ \alpha k_t^{\alpha-1} &= \varkappa_t \\ (y_t/k_t)\alpha &= \varkappa_t \\ k_t &= (y_t/\varkappa_t)\alpha. \end{aligned} \quad (3.2.3)$$

This equation says the level of capital is always instantly adjusted to y_t and $\varkappa_t$, with no costs of adjustment.

What determines the cost of capital? In the simple case with no taxes and no capital market frictions of any kind, an investor must be indifferent between putting his money in the bank and earning interest at rate r , and buying a unit of capital, renting it out at rate $\varkappa_t$, and then reselling it the next period.

The purchase price at which capital goods can be bought at date t (distinct from the rental rate, and typically much larger) is:

$$P_t - \text{purchase price of one unit of capital,} \quad (3.2.4)$$

and in continuous time, the rate of change of P_t is $\dot{P}_t$. Assume that capital depreciates geometrically at rate δ . The net profit from the *continuous time* purchase-and-rent strategy is

$$\begin{aligned} \varkappa_t - \delta P_t + \dot{P}_t - & \text{Income from renting, minus loss from depreciation} \\ & \text{plus capital gain from the change in price of capital.} \end{aligned} \quad (3.2.5)$$

Thus, the no-arbitrage condition is

$$\begin{aligned} rP_t &= \varkappa_t - \delta P_t + \dot{P}_t \\ (r + \delta)P_t &= \varkappa_t + \dot{P}_t. \end{aligned} \quad (3.2.6)$$

where the left-hand side is the return from putting money in the bank at rate r , which we assume is perfectly certain and time-invariant.

Now to simplify our lives we will assume constant capital goods prices, $\dot{P}_t = 0$. Thus, substituting the value for $\varkappa_t$ from (3.2.6) into (3.2.3) we have:

$$\begin{aligned} k_t &= \alpha y_t / \varkappa_t \\ &= \frac{\alpha y_t}{(r + \delta)P_t}. \end{aligned} \quad (3.2.7)$$

Now let's introduce taxes, defined as follows:

$$\begin{aligned} \tau - & \text{corporate tax rate } (\approx 0.34 \text{ in US}) \\ \zeta - & \text{investment tax credit (sometimes 10 percent, sometimes 0)} \end{aligned} \quad (3.2.8)$$

The net, discounted, after-tax price of capital to the firm is¹

$$\hat{P}_t = (1 - \zeta)P_t. \quad (3.2.9)$$

Now let's rewrite the arbitrage equation (3.2.6) taking account of taxes. The rental income $\varkappa_t$ must be multiplied by $(1 - \tau)$ because the capital-rental business must pay taxes too:

$$(r + \delta)\hat{P}_t = (1 - \tau)\varkappa_t + \dot{\hat{P}}_t. \quad (3.2.10)$$

If we simplify again by assuming that $\dot{\hat{P}}_t = 0$, we have

$$\varkappa_t = (r + \delta)P_t(1 - \zeta)/(1 - \tau). \quad (3.2.11)$$

Note that so far we have not derived a formula for investment; we have derived a formula for the *level* of the capital stock. But net investment is just the difference between the capital stock in periods t and $t - 1$. Thus, the Hall-Jorgenson model of gross investment is

$$\begin{aligned} i_{t-1} &= k_t - k_{t-1} + \delta k_{t-1} \\ &= \left(\Delta \frac{y_t}{\varkappa_t} \right) \alpha + \delta k_{t-1} \end{aligned} \quad (3.2.12)$$

(where we neglect some minor complications having to do with the distinction between continuous and discrete time).

¹Assume that if the firm sells the capital, it must repay the ITC on a pro-rata basis; this prevents tax arbitrage opportunities.

3.3 The Abel (1981)-Hayashi (1982) Marginal q Model

This section presents a discrete-time version of the Abel-Hayashi marginal q model of investment ([Abel, 1981]; [Hayashi, 1982]).

A corresponding Jupyter Notebook implements numerical solutions to the model using HARK and dolo.

3.3.1 Definitions

To simplify some algebra, we assume that a unit of investment purchased in period t does not become productive until $t + 1$; the cost at t reflects the present discounted value of the period- $t + 1$ price of capital.²

Adjustment costs are priced the same way.³ We repeatedly make approximations motivated by results from the continuous-time model; the key approximation will be the usual calculus result that if ϵ and γ are “small” then $\epsilon\gamma$ can be approximated by zero. (We will call this fact `SmallSmallZero` in the `MathFactsList`.)⁴

Symbol		Definition
k_t	-	Firm’s capital stock at the beginning of period t
$f(k)$	-	Gross output excluding investment and adjustment costs
τ	-	Tax rate on corporate earnings
$\not\tau$	-	$1 - \tau =$ Portion of earnings remaining after corporate tax
$\pi_t = f(k_t) \not\tau$	-	After tax revenues
i_t	-	Investment in period t (affects capital stock in period $t + 1$)
$j_t = j(i_t, k_t)$	-	AdJustment costs incurred in period t ; smooth and convex
$\beta = R^{-1}$	-	Discount factor for future profits (inverse of interest factor)
ζ	-	Investment tax credit (ITC)
$\not\zeta$	-	$= 1 - \zeta =$ Cost of investment after ITC

The price of one unit of investment is P_t , so $\mathcal{P}_t = \not\zeta P_t$ is the effective after-tax price of one unit of investment. (In the special case of a constant price $P_t = 1$, this collapses to $\not\zeta$.)

²This assumption simplifies many of the expressions that arise in the discrete-time framework; in continuous time the model presentation is simpler, but is hard to map the continuous-time theory into a transparent computational solution like the one that accompanies these notes.

³To avoid arbitrage opportunities, assume that if you claimed an investment tax credit on a unit of investment in period t , then if you resell the portion that remains after depreciation in a future period, you must repay the ITC corresponding to the remaining capital. Actual tax treatment of depreciation is too complicated to be worth incorporating into the model; these assumptions capture the core of it.

⁴We neglect tax depreciation because, as shown in the [Hall-Jorgenson](#) section, it matters only insofar as it affects the cost of capital; since the investment tax credit has a more transparent and direct effect on the cost of capital, including tax depreciation would add complication without adding any fundamental insights to the analysis. In the perfect foresight framework analyzed here, any attempt to analyze the effects of changes in tax depreciation should be translatable into an equivalent modification to the ITC. See House and Shapiro [2008] for elaboration.

Symbol		Definition
$\xi_t = (i_t + j_t)\mathcal{P}_{t+1}\beta$	-	Total after-tax period- t spending on investment
δ	-	Depreciation rate
$\bar{\gamma}$	-	Depreciation factor = $(1 - \delta)$
ω	-	Adjustment cost parameter

3.3.2 The Problem

The q model assumes that firms maximize the net profits payable to shareholders, definable as the present discounted value of after-tax revenues after subtracting off costs of investment:

$$e_t(k_t) = \max_{\{i\}_t^\infty} \mathbb{E}_t \left[\sum_{n=0}^{\infty} \beta^n (\pi_{t+n} - \xi_{t+n}) \right]. \quad (3.3.1)$$

Next period's capital is what remains of this period's capital after depreciation, plus current investment:⁵

$$k_{t+1} = k_t \bar{\gamma} + i_t. \quad (3.3.2)$$

If capital markets are efficient, $e_t(k_t)$ will also be the stock market value (“equity” is the mnemonic) of the profit-maximizing firm because it is precisely the amount that a rational investor will be willing to pay if they care only about discounted after-tax income derived from owning (a share of) the firm.

We can simplify by thinking about the firm's shareholders as the suppliers of physical capital, not just financial capital. In this interpretation, k represents not just the value of the physical machinery owned by the firm, but also the number of shares of stock outstanding in the firm. We can think of the firm in this way if we suppose that every time the firm purchases new physical capital, it does so by issuing new shares at a price equal to the marginal valuation of the firm's capital stock, purchasing the unit of capital at the price given by the after-tax cost of that capital (and after paying any associated adjustment costs).⁶

The Bellman equation for the firm's value can be derived from

$$\begin{aligned} e_t(k_t) &= \max_{i_t} \pi_t - \xi_t + \beta \mathbb{E}_t \left[\max_{\{i\}_{t+1}^\infty} \sum_{n=0}^{\infty} \beta^n (\pi_{t+1+n} - \xi_{t+1+n}) \right] \\ &= \max_{i_t} \pi_t - \xi_t + \beta \mathbb{E}_t [e_{t+1}(k_t \bar{\gamma} + i_t)] \end{aligned} \quad (3.3.3)$$

which is equivalent to

$$e_t(k_t) = \max_{k_{t+1}} \left\{ \pi_t - \overbrace{(k_{t+1} - k_t \bar{\gamma})}^{=i_t} + j(k_{t+1} - k_t \bar{\gamma}, k_t) \mathcal{P}_{t+1} \beta + \beta \mathbb{E}_t [e_{t+1}(k_{t+1})] \right\} \quad (3.3.4)$$

and defining $j_t^i = j^i(i_t, k_t)$ as the derivative of adjustment costs with respect to the level of investment,⁷ the first order condition for optimization with respect to k_{t+1} (or, equivalently, i_t) is

$$(1 + j_t^i) \mathcal{P}_{t+1} \beta = \beta \mathbb{E}_t [e_{t+1}^k(k_{t+1})]. \quad (3.3.5)$$

⁵The software archive that produces the figures for this section makes a slightly different assumption about the timing of depreciation: $k_{t+1} = (i_t + k_t) \bar{\gamma}$. The timing choice makes no qualitative and very little quantitative difference, but the alternative specification is slightly better for computational reasons.

⁶This leads to some subtlety in thinking about what happens to the firm's share price when an ITC is introduced; we discuss this briefly below.

⁷We implicitly assume in the following derivations that the structure of the j function is appropriate to our needs; later we will define a specific j that will always work, but here we want to leave the structure of the function general.

Thus: The PDV of the marginal cost (after tax, including adjustment costs) of an additional unit of investment should match the discounted expected marginal value of the resulting extra capital. (Recall that investment performed today i_t is paid for tomorrow at price $\mathcal{P}_{t+1}$ so today's cost is $\mathcal{P}_{t+1}\beta$.)

Recalling that $\pi_t = \mathcal{F}(k_t)$, the [Envelope theorem](#) for this problem can be used on either (3.3.3) or (3.3.4):

$$\begin{aligned} e_t^k(k_t) &= \mathcal{F}^k(k_t) - j_t^k \mathcal{P}_{t+1} \beta + \mathcal{T} \beta \mathbb{E}_t[e_{t+1}^k(k_{t+1})] \\ e_t^k(k_t) &= \mathcal{F}^k(k_t) + ((1 + j_t^i) \mathcal{T} - j_t^k) \mathcal{P}_{t+1} \beta \end{aligned} \quad (3.3.6)$$

and equivalently for period $t + 1$ so that (3.3.5) can be rewritten as the Euler equation for investment,

$$\begin{aligned} (1 + j_t^i) \mathcal{P}_{t+1} &= \mathbb{E}_t[\mathcal{F}^k(k_{t+1}) + (\mathcal{T} + \mathcal{T} j_{t+1}^i - j_{t+1}^k) \mathcal{P}_{t+2} \beta] \\ &= \mathbb{E}_t[\mathcal{F}^k(k_{t+1}) + (\mathcal{T} + j_{t+1}^i - \delta j_{t+1}^i - j_{t+1}^k) \mathcal{P}_{t+2} \beta]. \end{aligned} \quad (3.3.7)$$

It will be useful to define the *net investment ratio* as the Greek letter ι (the absence of a dot distinguishes ι from the level of investment i),

$$\iota_t = (i_t/k_t - \delta) \quad (3.3.8)$$

which measures how much investment differs from the proportion δ necessary to maintain the capital stock unchanged. It has derivatives

$$\begin{aligned} \iota_t^i &= (1/k_t) \\ \iota_t^k &= -(i_t/k_t)/k_t \\ &= -(\iota_t + \delta)/k_t. \end{aligned} \quad (3.3.9)$$

We now specify a convex (quadratic) adjustment cost function centered around the depreciation rate δ . The rationale for this centering is that when $i_t = \delta k_t$ (investment exactly replaces depreciated capital), the firm is simply maintaining its existing capital stock and incurs no adjustment costs. Adjustment costs arise only when the firm deviates from this “replacement investment” benchmark:

$$\begin{aligned} j(i_t, k_t) &= (k_t/2) \left(\frac{i_t - \delta k_t}{k_t} \right)^2 \omega \\ &= (k_t/2) \iota_t^2 \omega \end{aligned} \quad (3.3.10)$$

with derivatives (using (3.3.9) to simplify)

$$\begin{aligned} j^i &= k \iota \omega \iota^i \\ &= \iota \omega \\ j^k &= (\iota^2/2 - k \iota (\iota + \delta)/k) \omega \\ &= (\iota^2/2 - \iota (\iota + \delta)) \omega \\ &= -(\iota^2/2 + \iota \delta) \omega \\ j^i - \delta j^i - j^k &= (\iota - \delta \iota + (\iota^2/2 + \iota \delta)) \omega \\ &= (\iota + \iota^2/2) \omega \\ &= j^i + (\omega/2) \iota^2 \end{aligned} \quad (3.3.11)$$

so the Euler equation for investment (3.3.7) can be written

$$(1 + j_t^i) \mathcal{P}_{t+1} = \mathbb{E}_t[\mathcal{F}^k(k_{t+1}) + (\mathcal{T} + j_{t+1}^i + (\omega/2) \iota_{t+1}^2) \mathcal{P}_{t+2} \beta]. \quad (3.3.12)$$

To begin interpreting this equation, consider first the case where the costs of adjustment are zero, $\omega = 0$. In this case $j^i = j^k = 0$ and the Euler equation reduces to

$$\mathcal{P}_{t+1} = \mathbb{E}_t[\not{f}f^k(k_{t+1}) + \mathbb{T}\mathcal{P}_{t+2}\beta]. \quad (3.3.13)$$

Simplifying further, suppose that capital prices are constant at $\mathcal{P}_t = 1$ and the ITC is unchanging so that the after-tax price of capital is constant at $\mathcal{P}$. Then since $1 + r + \delta \approx 1/\beta\mathbb{T}$, the equation becomes

$$\begin{aligned} \mathcal{P} &= \mathbb{E}_t[\not{f}f^k(k_{t+1})] + \mathcal{P}\beta\mathbb{T} \\ (r + \delta)\mathcal{P} &\approx \mathbb{E}_t[\not{f}f^k(k_{t+1})]. \end{aligned} \quad (3.3.14)$$

This says that the cost of buying one unit of capital, $\mathcal{P}$, is equal to the opportunity cost in lost interest plus the value lost to depreciation, $(r + \delta)$, which must match the (after-tax) payoff from ownership of that capital. This corresponds exactly to the formula for the equilibrium cost of capital in the [Hall-Jorgenson](#) model: In the presence of an investment tax credit at rate ζ , the after-tax price of capital is $\mathcal{P} = \zeta$, and the firm will adjust its holdings of capital to the point where

$$\zeta(r + \delta)/\not{f} = f^k(k). \quad (3.3.15)$$

Now define $\lambda_t \equiv e_t^k$ as the marginal value to the firm of ownership of one more unit of capital at the beginning of period t ; using this definition, the envelope condition can be written

$$\begin{aligned} \lambda_t &= \not{f}f^k(k_t) - j_t^k\mathcal{P}_{t+1}\beta + \beta\mathbb{T}\mathbb{E}_t[\lambda_{t+1}] \\ &\approx \not{f}f^k(k_t) - j_t^k\mathcal{P}_{t+1}\beta + (1 - \delta - r)\mathbb{E}_t[\lambda_t + \lambda_{t+1} - \lambda_t] \\ &= \not{f}f^k(k_t) - j_t^k\mathcal{P}_{t+1}\beta + (1 - \delta - r)(\lambda_t + \mathbb{E}_t[\Delta\lambda_{t+1}]) \\ (r + \delta)\lambda_t &\approx \not{f}f^k(k_t) - j_t^k\mathcal{P}_{t+1}\beta + \mathbb{E}_t[\Delta\lambda_{t+1}] \end{aligned} \quad (3.3.16)$$

where the last approximation uses `SmallSmallZero` in the form $(r + \delta)\mathbb{E}_t[\Delta\lambda_{t+1}] \approx 0$. Equation (3.3.16) can be rearranged as

$$\mathbb{E}_t[\Delta\lambda_{t+1}] \approx r\lambda_t - [\not{f}f^k(k_t) - j_t^k\mathcal{P}_{t+1}\beta - \delta\lambda_t]. \quad (3.3.17)$$

This equation can best be understood as an arbitrage equation for the share price of the company if capital markets are efficient.⁸ The first term on the RHS $r\lambda_t$ is the flow of income that would be obtained from putting the value of an extra unit of capital in the bank. The term in brackets $[\]$ is the flow value of having an extra unit of capital inside the firm: Extra after-tax revenues are measured by the first term, the second term accounts for the effect of the extra capital on costs of adjustment, and the final term reflects the cost to the firm of the extra depreciation that results from having more capital.

Think first about the $\mathbb{E}_t[\Delta\lambda_{t+1}] = 0$ case, in which the firm's value, share price, and size will be unchanging because the marginal value of capital inside the firm is equal to the opportunity cost of employing that capital outside the firm (leaving it in the bank). If these two options yield equivalent returns, it is because the firm is already the "right" size and should be neither growing nor shrinking.

Now consider the case where $\mathbb{E}_t[\Delta\lambda_{t+1}] < 0$, because

$$r\lambda_t < [\not{f}f^k(k_t) - \mathcal{P}_{t+1}\beta j_t^k - \delta\lambda_t]. \quad (3.3.18)$$

This says that an extra unit of capital is more valuable inside the firm than outside it, which means that 1) λ_t is above its steady-state value; 2) the firm will have positive net investment; and 3) the firm's share value will be falling over time (because the *level* of its share value today is high, reflecting the fact that the high marginal valuation of the firm's future investment has already been incorporated into λ_t).⁹

⁸This interpretation requires the assumption (made above) that the number of shares outstanding for the firm is equal to the number of units of capital the firm has; this is justifiable by the assumption that in an efficient capital market shares can always be issued or repurchased at an implicit interest rate corresponding to the riskless rate.

⁹The case with rising share prices is symmetric.

Now define “marginal q ” as the value of an additional unit of capital inside the firm divided by the after-tax purchase price of an additional unit of capital,¹⁰

$$q_t = \lambda_t / \mathcal{P}_t. \quad (3.3.19)$$

The investment first order condition (3.3.5) implies

$$\begin{aligned} (1 + j^i) \mathcal{P}_{t+1} \beta &= \beta \lambda_{t+1} \\ 1 + \iota_t \omega &= q_{t+1} \end{aligned} \quad (3.3.20)$$

which constitutes the implicit definition of a function

$$\begin{aligned} \iota(q_{t+1}) &\equiv (q_{t+1} - 1) / \omega \\ i_t &= (\iota(q_{t+1}) + \delta) k_t \end{aligned} \quad (3.3.21)$$

and notice that this implies

- At a value of $q_{t+1} = 1$, investment takes place at a rate exactly equal to the depreciation rate ($\iota(1) = 0$)
- The investment ratio ι_t is monotonically increasing in q_{t+1} ($\iota'(q_{t+1}) > 0$)
- The strength with which ι_t is related to q_{t+1} depends on the magnitude of adjustment costs ($d|\iota_t|/d\omega < 0$)

3.3.3 Phase Diagrams

Dynamics of k

The capital accumulation equation can be rewritten as

$$\begin{aligned} k_{t+1} &= (1 - \delta) k_t + i_t \\ \Delta k_{t+1} &= i_t - \delta k_t \\ &= \iota(q_{t+1}) k_t. \end{aligned} \quad (3.3.22)$$

Dynamics of q

To construct a phase diagram involving q , we need to transform our equation (3.3.16) for the dynamics of λ into an equation for the dynamics of q . As a preliminary, define the proportional change in the after-tax price of capital as

$$\nabla \mathcal{P}_{t+1} \equiv \Delta \mathcal{P}_{t+1} / \mathcal{P}_t. \quad (3.3.23)$$

Recalling that $\mathcal{P}_{t+1} = \Delta \mathcal{P}_{t+1} + \mathcal{P}_t$, dividing both sides of (3.3.16) by $\mathcal{P}_t$ yields

$$\begin{aligned} (r + \delta) q_t &= \mathcal{F}^k(k_t) / \mathcal{P}_t - j_t^k (\Delta \mathcal{P}_{t+1} + \mathcal{P}_t) \beta / \mathcal{P}_t + \mathbb{E}_t \left[\left(\frac{\lambda_{t+1}}{\mathcal{P}_t} \right) - \left(\frac{\lambda_t}{\mathcal{P}_t} \right) \right] \\ &\approx \mathcal{F}^k(k_t) / \mathcal{P}_t - (1 + \nabla \mathcal{P}_{t+1}) j_t^k \beta + \mathbb{E}_t \left[\left(\frac{\lambda_{t+1}}{\mathcal{P}_{t+1}} (1 + \nabla \mathcal{P}_{t+1}) \right) - q_t \right] \\ &= \mathcal{F}^k(k_t) / \mathcal{P}_t - (1 + \nabla \mathcal{P}_{t+1}) j_t^k \beta + \mathbb{E}_t [(1 + \nabla \mathcal{P}_{t+1}) q_{t+1} - q_t]. \end{aligned} \quad (3.3.24)$$

Now assuming that Δq_{t+1} , $\nabla \mathcal{P}_{t+1}$, r , and j_t^k are all “small” so that their interactions are approximately 0, we have

$$\mathbb{E}_t [\Delta q_{t+1}] \approx (r + \delta - \nabla \mathcal{P}_{t+1}) q_t - [\mathcal{F}^k(k_t) / \mathcal{P}_t - j_t^k \beta]. \quad (3.3.25)$$

Simplifying further, if the ITC is unchanging, and the pretax price of capital is unchanging at $\mathcal{P}_{t+1} = \mathcal{P}_t = 1$, and $\delta = 0$, equation (3.3.25) becomes

¹⁰The variable naming conventions here differ subtly from those in Romer [2011]: what we call λ Romer calls q . Romer also makes different assumptions about the production function.

$$\mathbb{E}_t[\Delta q_{t+1}] \approx r q_t - f^k(k_t)/\mathcal{T}_t + j_t^k \beta \quad (3.3.26)$$

where

$$\mathcal{T}_t = \zeta_t / \not\! \zeta_t \quad (3.3.27)$$

combines the effects of the corporate tax and the investment tax credit into a single tax term. Its inverse, $\mathcal{T}_t^{-1} = \not\! \zeta_t / \zeta_t$, appears in subsequent analysis where it is convenient to express the tax-adjusted marginal product of capital as $\mathcal{T}^{-1} f^k$.

Results

Figure 3.2 and Figure 3.3 present two phase diagrams, one for k and λ and one for k and q .

For most purposes, q diagram is simpler, because our facts about the ι function imply that the $\Delta k_{t+1} = 0$ locus is always a horizontal line at $q = 1$. This is because $q = 1$ always corresponds to the circumstance in which the value of a unit of capital inside the firm, λ , matches the after-tax cost of a unit of capital, $\mathcal{P}$; $q = 1$ is the only value of q at which the firm does not wish to change size ($\Delta k_{t+1} = 0$).

The slope of the $\mathbb{E}_t[\Delta q_{t+1}] = 0$ locus is easiest to think about near the steady state value of k where we can approximate $j_t^k \approx 0$.

Pick a point on the $\mathbb{E}_t[\Delta q_{t+1}] = 0$ locus. Now consider a value of q that is slightly larger. From (3.3.26), at the initial value of k we would have $\mathbb{E}_t[\Delta q_{t+1}] > 0$. Thus, the value of k corresponding to $\mathbb{E}_t[\Delta q_{t+1}] = 0$ must be one that balances the higher q by a higher value of f^k , which is to say a lower value of k . This means that higher q will be associated with lower k so that the locus is downward-sloping.

For appropriate choices of parameter values the problem satisfies the usual conditions for stability and will therefore have a saddle path solution, as depicted in the diagram.

The λ diagram is virtually indistinguishable from the q diagram; the only difference is that the Δk_{t+1} locus is located at the point $\lambda = \mathcal{P}$ (i.e. the marginal value of investment is equal to the price of a unit of investment). The distinction between the diagrams reflects the fact that an increase in the investment tax credit will result in a rise in the steady-state value of k which implies a fall in the pretax marginal product of capital.

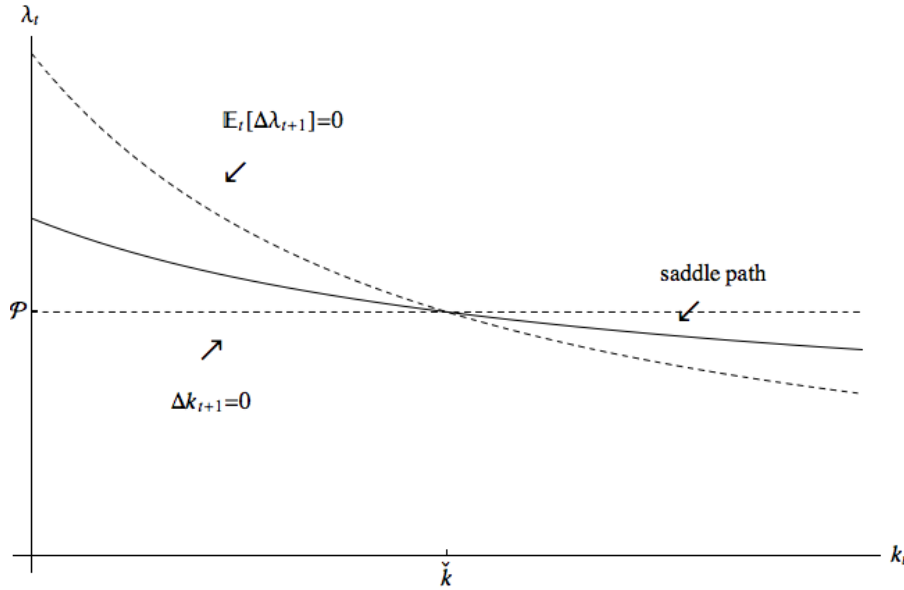
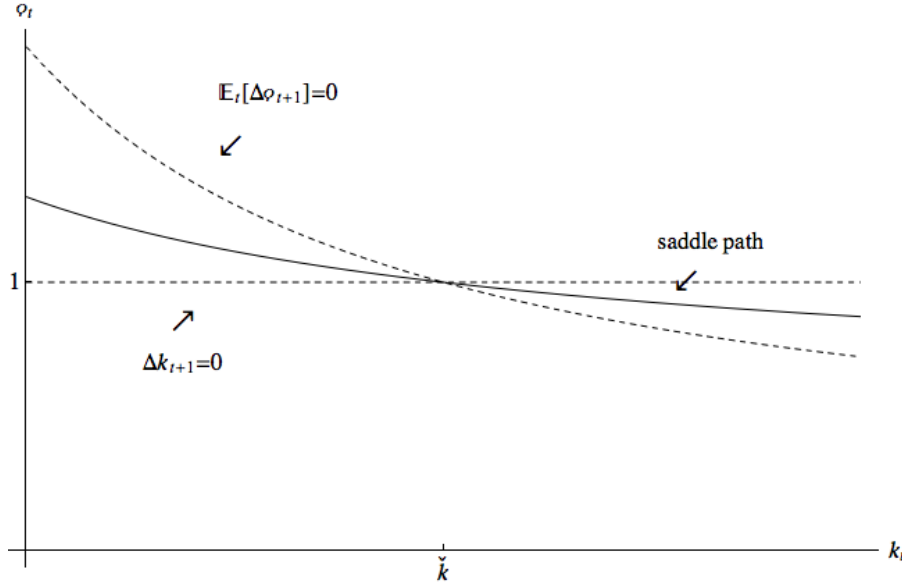


Figure 3.2: Phase diagram for λ

Figure 3.3: Phase diagram for q

3.3.4 Dynamics

Steady State

The key to understanding the model's dynamics (as, really, with all infinite horizon models) is to figure out the steady state toward which it is heading, then to work out how it gets there. The key to the steady state, in turn, is that the capital stock will eventually reach a point where $j^k = j^i = 0$.

A Positive Shock to Productivity

Suppose that the production function for the firm suddenly, permanently, and unexpectedly improves; specifically, leading up to period t the firm was in steady state, but in periods $t + 1$ and beyond the production function will be $f_{\geq}(k) = \Psi f_{<}(k)$ for some $\Psi > 1$ where $f_{<}$ and $f_{\geq}$ indicate the production functions before and after the increase in productivity.

Note first that none of the tax terms has changed, and in the long run there is nothing to prevent the firm from adjusting its capital stock to the point consistent with the new level of productivity and then leaving it fixed there so that $j^k = j^i = 0$. Thus (3.3.26) implies that at the new steady state $\check{k}_{\geq}$ we will have $r\check{q} = \mathcal{T}^{-1}f_{\geq}^k(\check{k}_{\geq}) = \Psi \mathcal{T}^{-1}f_{<}^k(\check{k}_{\geq})$ which implies $\check{k}_{\geq} > \check{k}_{<}$, since the steady state value of q never changes: $\check{q}_{\geq} = \check{q}_{<} = 1$. That is, with higher productivity, the equilibrium capital stock is larger, but the equilibrium *tax adjusted marginal product* of capital is the same.

Obviously in order to get from an initial capital stock of $\check{k}_{<}$ to a larger equilibrium capital stock of $\check{k}_{\geq}$ the firm will need to engage in investment in excess of the depreciation rate, incurring costs of adjustment. In the absence of a change in the environment, *expected* costs of adjustment will always be declining toward zero, because the firm's capital stock will always be moving toward its equilibrium value in which those costs are zero.

So we can tell the story as follows. Suppose that leading up to period t the firm was in its steady-state. When the productivity shock occurs, f^k jumps up. j_t^k had been zero (because the firm was at steady state), but now the firm wishes it had more capital because extra capital would reduce future adjustment costs (the firm knows that its old steady-state capital stock is now too small, so it will have to be engaging in $\iota > 0$ for a while), so j_t^k becomes negative (that is, the firm knows that having more capital will reduce the adjustment costs associated with the higher investment that it will be undertaking). The combination $\mathcal{T}^{-1}f_t^k - j_t^k\beta$ therefore becomes a larger positive number, so at the initial level of q the RHS of (3.3.26) would imply $\mathbb{E}_t[\Delta q_{t+1}]$ less than zero, so the new $\mathbb{E}_t[\Delta q_{t+1}] = 0$ locus must be higher (because the equilibrating value of q is higher for any k). The saddle path is therefore also higher. So q , and therefore ι , jump up instantly when the new higher

level of productivity is revealed, corresponding also to an immediate increase in the firm's share price (the marginal valuation of an additional unit of capital), since $\mathcal{T}$ has not changed.

The phase diagrams with the saddle paths before and after the productivity increase together with the impulse response functions are plotted in Figure 3.4.

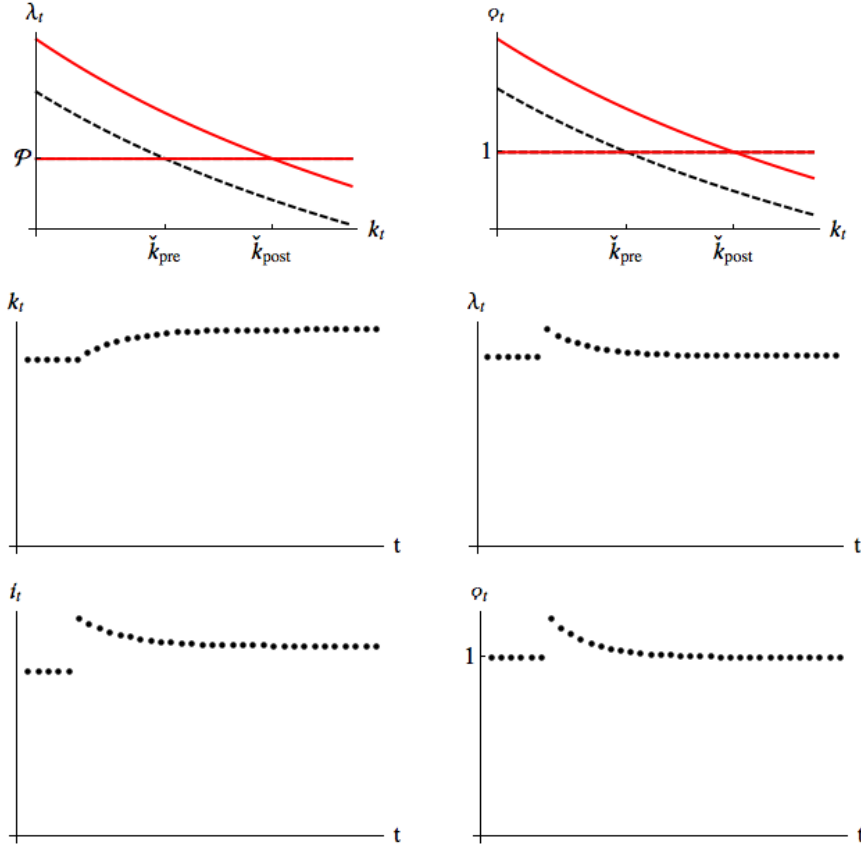


Figure 3.4: Increase in productivity. Phase diagrams with saddle paths (dashed black and continuous red lines respectively pre and post the productivity increase) and impulse response functions.

A Permanent Tax Cut

Again starting from the steady state equilibrium, suppose $\mathcal{T}$ unexpectedly and permanently decreases, which could happen because of a cut in corporate taxes or an increase in the ITC. Equation (3.3.26) implies that in steady state

$$\begin{aligned} r \overbrace{q}^{=1} &= f^k / \mathcal{T} \\ f^k &= r\mathcal{T}. \end{aligned} \quad (3.3.28)$$

Dynamically, the story is as follows. Equation (3.3.26) implies that following the tax change the $\mathbb{E}_t[\Delta q_{t+1}] = 0$ locus must be higher because at any given q the $-f^k / \mathcal{T}$ term is a larger negative number, while at the initial k the j_t^k term is also now negative; so the $\mathbb{E}_t[\Delta q_{t+1}] = 0$ locus shifts up.

In contrast to the case with a productivity shock, the equilibrium marginal product of capital will be lower than before. Arbitrage equalizes the *after-tax* marginal product of capital with the interest rate, but with a lower tax rate, that equilibration will occur at a higher level of capital.

Notice that the qualitative story is the same whether the change in $\mathcal{T}$ is due to a permanent reduction in the corporate tax rate (increase in $\not\mathcal{T}$) or a permanent increase in the investment tax credit (reduction in ζ). In either case, q and investment jump upward at time t and then gradually decline back downward (though the equilibrium level of investment is higher than before the change).

There is, however, one interesting distinction between a decrease in $\mathcal{T}$ due to a reduction in corporate taxes and a decrease caused by an increase in ζ . Since $\lambda = \zeta q$, an increase in ζ reduces

ζ and therefore reduces the equilibrium value of λ , while a change in $\not\lambda$ has no effect on equilibrium λ . This reflects a subtle distinction. λ is the after-tax marginal value of extra capital, and the equilibrium in this model will occur at the point where that marginal value is equal to the marginal cost. Changing ζ changes that marginal cost, so it changes the equilibrium after-tax marginal value. Changing $\not\lambda$ does not change the marginal cost of capital, so the equilibrium *after-tax* marginal value of capital is unchanged. The marginal *product* of capital is lower after a tax cut (equilibrium f^k is smaller), but that is exactly counterbalanced by the larger value of $\not\lambda$ so that $\not\lambda f^k$ is unchanged in the long run by the change in $\not\lambda$.

The phase diagrams with the saddle paths before and after the corporate tax reduction and the ITC increase, together with the impulse response functions, are plotted respectively in Figure 3.5 and Figure 3.6. Note that the λ saddle path actually jumps *downward* after the ITC increase. This is not an error; rather, recall that λ reflects marginal value of a unit of capital inside the firm, and recall that the *price* of purchasing that capital has gone down. Remembering that we are assuming that capital can move in and out of the firm, this has the surprising consequence that, for the original owners of the firm, the ITC is *bad* news because it means that the capital they own has a lower value (its value is ultimately tied to the price of capital, which has gone down). For a potential new shareholder, the investment tax credit means that you can obtain ownership of a share of the firm's capital by buying the capital at the ITC-discounted price, paying the adjustment costs, then giving the capital to the firm. Thus, the ITC has the effect of increasing the absolute value of a dollar of money relative to the value of a unit of capital inside the firm. So in this special case, you should think of the ITC as something that provides a discount to purchasing shares or capital λ . While the new saddle path for λ is lower than the old one, that does not reflect the adjustment for the fact that the new capital is being purchased at a cheaper price. The dynamics of q , in this case, are more intuitive than those of λ : q unambiguously increases, reflecting the fact that the value of capital to the firm exceeds its new (cheaper) cost.

In sum: In terms of effects on capital, the outcome from a corporate tax cut and an ITC tax cut are similar, but the analytics of λ are different, because the former affects the after-tax interest rate while the latter affects the after-tax cost of capital.

A Future Shock to Productivity

Now consider a circumstance where the firm knows that at some date in the future, $t + n$, the level of productivity will increase so that $f_{\geq t+n} = \Psi f_{< t+n}$ for $\Psi > 1$.

The long run steady state is of course the same as in the example where the increase in productivity is immediately effective.

To determine the short run dynamics, notice several things. First, there can be no *anticipated* big jumps in the share price of the firm (the marginal productivity of capital inside the firm). Thus, if the productivity jump occurs in period $t + n$ and the time periods are short enough, we must have

$$\mathbb{E}_{t+n-1}[\lambda_{t+n} - \lambda_{t+n-1}] \approx 0. \quad (3.3.29)$$

But because the equilibrium capital stock is larger, we know that $j_{\geq t+n}^k < 0$ and will stay negative thereafter (asymptoting to zero from below). This reflects the fact that if you know you will need higher capital in the future, the most efficient way to minimize the cost of obtaining that capital is to gradually start building some of it even before you need it, rather than trying to do it all at once. Note further that before period $t + n$ the model behaves according to the equations of motion defined by the problem under the $<$ parameter values,¹¹ while at $t + n$ and after it behaves according to the new $\geq$ equations of motion.

¹¹Strictly speaking, this is not true, because j_t^k will now differ from the value associated with the initial problem. For purposes of analyzing problems of this kind (announced future changes in parameters) we will neglect the effects of the path of j_t^k on the equations of motion. Except under extreme circumstances, this should not change the qualitative results of the analysis, and doing anything else would require a very intricate analysis. This treatment is admittedly a bit inconsistent, since in the case under consideration it is precisely the change in j_t^k that motivates the firm to start adjusting its capital stock before the productivity change comes into effect; effectively, we are taking into account the effect of j_t^k on the level of λ before period $t + n$ while neglecting its effects on λ 's dynamics during this interval. C'est la vie.

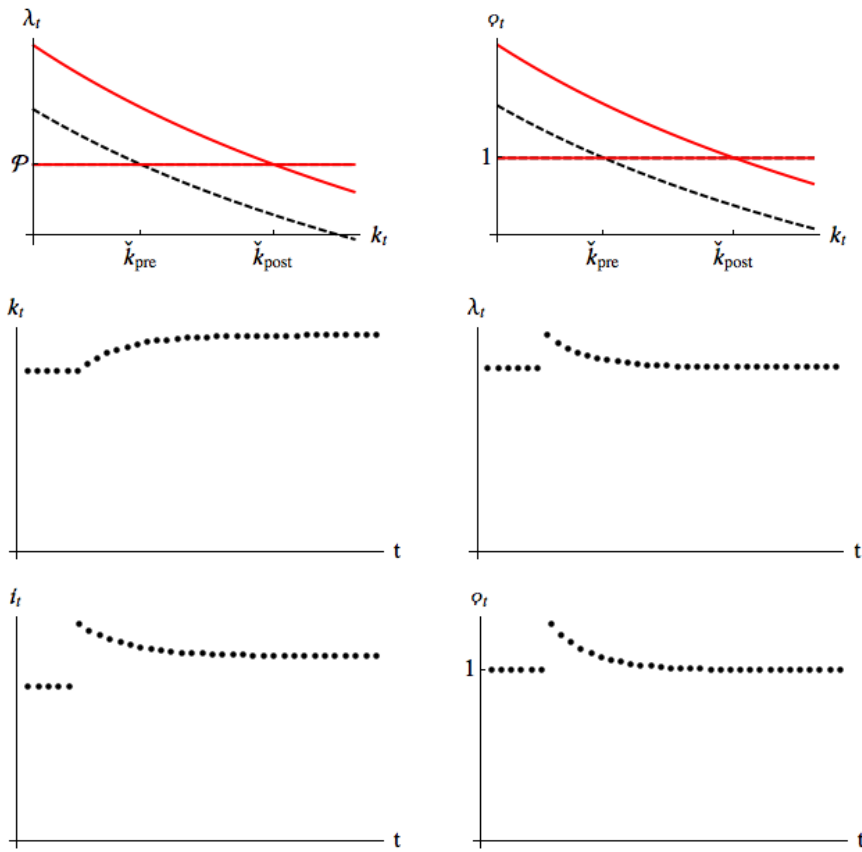


Figure 3.5: Corporate tax reduction. Phase diagrams with saddle paths (dashed black and continuous red lines respectively pre and post the corporate tax reduction) and impulse response functions.

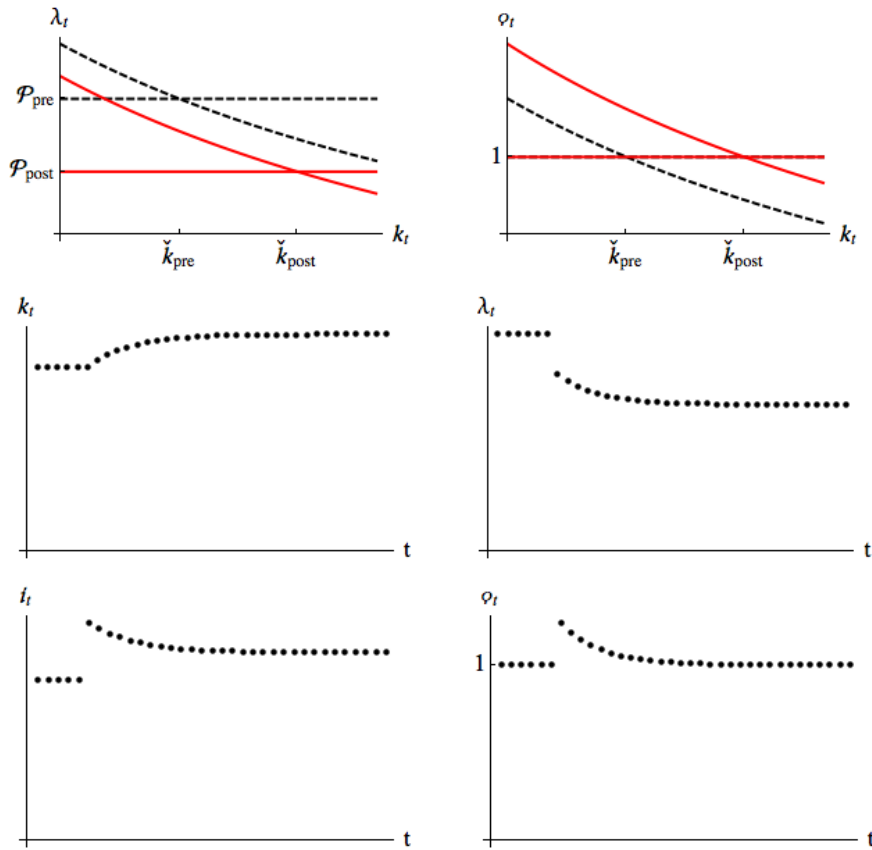


Figure 3.6: ITC increase. Phase diagrams with saddle paths (dashed black and continuous red lines respectively pre and post the ITC increase) and impulse response functions.

Putting all this together, the story is as follows. Upon announcement of the productivity increase, λ jumps to the level such that, evolving exactly according to its $<$ equations of motion, it will arrive in period $t + n$ at a point exactly on the saddle path of the model corresponding to the $\geq$ equations of motion. Thereafter it will evolve toward the steady state, which will be at a higher level of capital than before, $\check{k}_{\geq} > \check{k}_{<}$, because the greater productivity justifies a higher equilibrium capital stock.

Thus, λ jumps up at time t , evolves to the northeast until time $t + n$, and thereafter asymptotes downward toward the same equilibrium value it had originally before the productivity change. Since $\mathcal{T}$ has not changed, the dynamics of q and ι are the same as those of λ .

A Future Increase in the ITC

Consider now the consequences if a surprise increase in the investment tax credit is passed at date t that will become effective at date $t + n > t$.

Inspection of (3.3.26) might suggest that the effects of a future tax cut would be identical to the effects of a future increase in f^k , since the terms enter multiplicatively via $\mathcal{T}^{-1}f^k$. And indeed, with respect to the dynamics of λ the two experiments are basically the same. And of course the steady-state value of q is always equal to one.

During the transition, however, q has interesting dynamics. From periods t to $t + n - 1$, the ITC does not change, leaving $\mathcal{T}^{-1}$ and the after-tax marginal product of capital unchanged, and so the dynamics of q are basically the same as those of λ . But between $t + n - 1$ and $t + n$, λ cannot jump but $\mathcal{T}^{-1}$ does jump, which implies that q must jump (so there is a predictable change in q).

Dynamics of investment are determined by dynamics of q , so the path of ι is: At t , a discrete jump up; between t and $t + n - 1$, a gently rising path; between $t + n - 1$ and $t + n$, an upward jump; and after $t + n$, a path that asymptotes downward toward the steady state level of investment.

The steady-state effects on λ are of course determined by the same considerations as apply to the unanticipated tax cut, so they depend on whether the tax change is a drop in τ or an increase in ζ .

3.3.5 More Figures

Figures for a variety of other experiments, including changes in the interest rate, can be constructed using the notebook.

3.4 An Entrepreneur's Problem Under Perfect Foresight

Consider a firm characterized by the following:

Symbol	Description
k_t	Firm's capital stock at the beginning of period t
$f(k)$	The firm's total output depends only on k
i_t	Investment in period t
$j(i, k)$	Adjustment costs associated with investment i given capital k
$\xi_t = i_t + j_t$	Expenditures (purchases plus adjustment costs) on investment
$\beta = 1/R$	Discount factor for future profits (inverse of interest factor)

Taxes are left out here to keep the algebra transparent; the [q model](#) section works through the same problem with corporate taxes and an investment tax credit.

Suppose that the firm's goal is to pick the sequence i_t that solves:

$$e(k_t) = \max_{\{i\}_t^\infty} \sum_{n=0}^{\infty} \beta^n (f_{t+n} - i_{t+n} - j_{t+n}) \quad (3.4.1)$$

subject to the transition equation for capital,

$$k_{t+1} = (k_t + i_t)\Upsilon \quad (3.4.2)$$

where $\Upsilon = (1 - \delta)$ is the amount of capital left after one period of depreciation at rate δ .¹² e_t is the value of the profit-maximizing firm: If capital markets are efficient this is the equity value that the firm would command if somebody wanted to buy it.

The firm's Bellman equation can be written:

$$\begin{aligned} e_t(k_t) &= \max_{\{i\}_t^\infty} \sum_{n=0}^{\infty} \beta^n (f_{t+n} - i_{t+n} - j_{t+n}) \\ &= \max_{\{i_t\}} f_t - i_t - j(i_t, k_t) + \beta \left[\max_{\{i\}_{t+1}^\infty} \sum_{n=0}^{\infty} \beta^n (f_{t+1+n} - i_{t+1+n} - j_{t+1+n}) \right] \\ &= \max_{\{i_t\}} f_t - i_t - j(i_t, k_t) + \beta e_{t+1}((k_t + i_t)\Upsilon) \end{aligned} \quad (3.4.3)$$

Define j_t^i as the derivative of adjustment costs with respect to the level of investment.

The first order condition for optimal investment implies:

$$\begin{aligned} 0 &= -1 - j_t^i + \Upsilon \beta e_{t+1}^k(k_{t+1}) \\ 1 + j_t^i &= \Upsilon \beta e_{t+1}^k(k_{t+1}) \end{aligned} \quad (3.4.4)$$

In words: The marginal cost of an additional unit of investment (the LHS) should be equal to the discounted marginal value of the resulting extra capital (the RHS).

The Envelope theorem says

$$\begin{aligned} e_t^k(k_t) &= f^k(k_t) - j_t^k + \beta e_{t+1}^k(k_{t+1}) \overbrace{\left(\frac{\partial k_{t+1}}{\partial k_t} \right)}^{\Upsilon} \\ &= f^k(k_t) - j_t^k + \underbrace{\beta \Upsilon e_{t+1}^k(k_{t+1})}_{=(1+j_t^i)} \end{aligned} \quad (3.4.5)$$

¹²There are some small differences between the formulation of the model here and in the [marginal q model](#) section. Here, investment costs are paid at the time of investment and the depreciation factor applies to $(k_t + i_t)$ rather than just k_t . These changes simplify the computational solution without changing any key results.

where the underbraced term follows from (3.4.4). So the corresponding $t + 1$ equation can be substituted into (3.4.4) to obtain

$$(1 + \underline{j}_t^i) = \left(f^k(k_{t+1}) + (1 + \underline{j}_{t+1}^i - \underline{j}_{t+1}^k) \right) \Upsilon \beta \quad (3.4.6)$$

which is the Euler equation for investment.

3.4.1 Steady State

Now suppose that a steady state exists in which the capital stock is at its optimal level and is not adjusting, so costs of adjustment are zero: $\underline{j}_t = \underline{j}_{t+1} = \underline{j}_t^i = \underline{j}_{t+1}^i = \underline{j}_t^k = \underline{j}_{t+1}^k = 0$.

If $\underline{j}_t^i = \underline{j}_{t+1}^i = \underline{j}_{t+1}^k$ then (3.4.6) reduces to

$$\begin{aligned} 1 &= \overbrace{\beta}^{=R^{-1}} \Upsilon \left[f^k(\check{k}) + 1 \right] \\ R &= \Upsilon (1 + f^k(\check{k})) \end{aligned} \quad (3.4.7)$$

so that the capital stock is equal to the value that causes its marginal product to match the interest factor, after compensating for depreciation.

3.4.2 Phase Diagram Analysis

Another way to analyze this problem is in terms of the marginal value of capital, $\lambda_t \equiv e_t^k(k_t)$.

Rewrite (3.4.5) as

$$\begin{aligned} \lambda_t &= f^k(k_t) - \underline{j}_t^k + \beta \Upsilon (\lambda_t + \lambda_{t+1} - \lambda_t) \\ &= f^k(k_t) - \underline{j}_t^k + \beta \Upsilon (\lambda_t + \Delta \lambda_{t+1}) \\ (1 - \beta \Upsilon) \lambda_t &= f^k(k_t) - \underline{j}_t^k + \Delta \lambda_{t+1} \\ \lambda_t &= \frac{f^k(k_t) - \underline{j}_t^k + \Delta \lambda_{t+1}}{(1 - \beta \Upsilon)} \end{aligned} \quad (3.4.8)$$

and the phase diagram is constructed using the $\Delta \lambda_{t+1} = 0$ locus. In the vicinity of the steady state, we can assume $\underline{j}_t^k \approx 0$ in which case the $\Delta \lambda_{t+1} = 0$ locus becomes

$$\lambda_t = \frac{f^k(k_t)}{(1 - \beta \Upsilon)} \quad (3.4.9)$$

which implies (since $f^k(k_t)$ is downward sloping in k_t) that the $\Delta \lambda_t = 0$ locus (that is, the $\lambda_t(k_t)$ function that corresponds to $\Delta \lambda_t = 0$) is downward sloping.

The phase diagram is depicted in Figure 3.7.

The steady state of the model will be the point at which $k_{t+1} = k_t = \check{k}$, implying from (3.4.2) a steady-state investment rate of

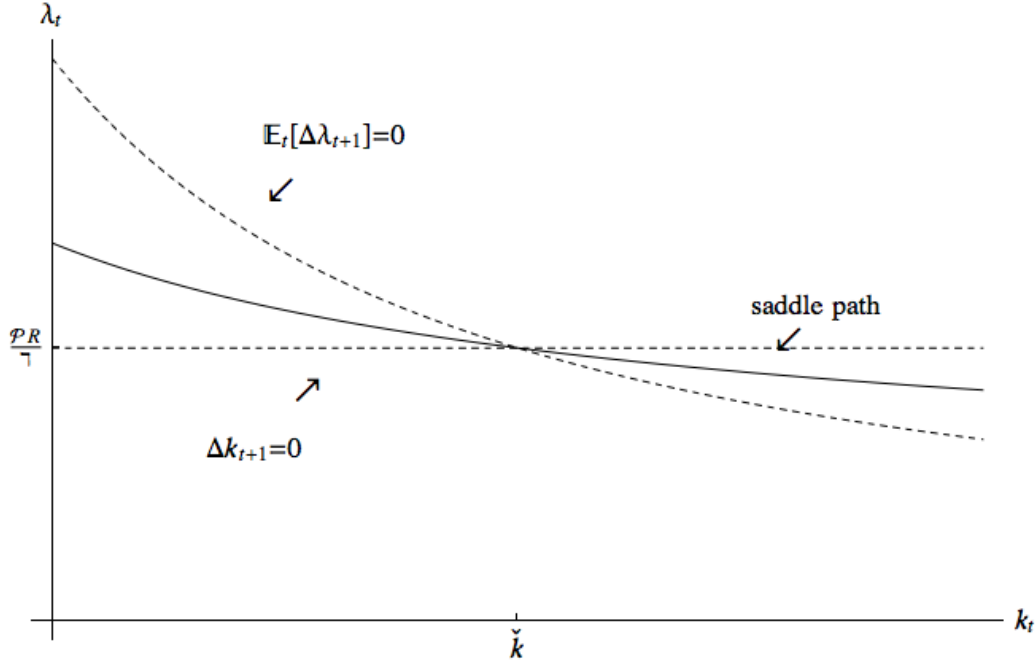
$$\begin{aligned} \check{k} &= (\check{k} + \check{i}) \Upsilon \\ \check{i} &= (1 - \Upsilon) \check{k} / \Upsilon = (\delta / \Upsilon) \check{k} \end{aligned} \quad (3.4.10)$$

and solving (3.4.7) for $f^k(\check{k})$

$$\left(\frac{(1 - \beta \Upsilon)}{\beta \Upsilon} \right) = f^k(\check{k}) \quad (3.4.11)$$

which can be substituted into (3.4.9) to obtain the steady-state value of λ :

$$\check{\lambda} = \left(\frac{R}{\Upsilon} \right). \quad (3.4.12)$$

Figure 3.7: Phase diagram showing the $\Delta\lambda = 0$ and $\Delta k = 0$ loci.

3.4.3 The Entrepreneur's Problem

We now wish to modify the problem in two ways. First, we have been assuming that the firm has only physical capital, and no financial assets. Second, we have been assuming that the manager running the firm only cares about the PDV of profits; suppose instead we want to assume that the firm is a small business run by an entrepreneur who must live off the dividends of the firm, and thus they are maximizing the discounted sum of utility from dividends $u(c_t)$ rather than just the level of discounted profits. (Note that we designate dividends by c_t ; dividends were not explicitly chosen in the q -model version of the problem, because the Modigliani-Miller theorem says that the firm's value is unaffected by its dividend policy).

We call the maximizer running this firm the “entrepreneur.” The entrepreneur's level of monetary assets m_t evolves according to

$$m_{t+1} = f_{t+1} + (m_t - i_t - j_t - c_t)R. \quad (3.4.13)$$

That is, next period the firm's money is next period's profits plus the return factor on the money at the beginning of this period, minus this period's investment and associated adjustment costs, minus dividends paid out (which, having been paid out, are no longer part of the firm's money).

The entrepreneur's Bellman equation can now be written

$$v_t(k_t, m_t) = \max_{\{i_t, c_t\}} u(c_t) + \beta v_{t+1}(k_{t+1}, m_{t+1}) \quad (3.4.14)$$

Derivation: Value is simply the discounted sum of utility from future dividends:

$$\begin{aligned} v_t(k_t, m_t) &= \max_{\{i, c\}_t^\infty} \sum_{n=0}^{\infty} \beta^n u(c_{t+n}) \\ &= \max_{\{i, c\}_t^\infty} \left(u(c_t) + \beta \sum_{n=0}^{\infty} \beta^n u(c_{t+1+n}) \right) \\ &= \max_{\{i_t, c_t\}} u(c_t) + \beta v_{t+1}(k_{t+1}, m_{t+1}). \end{aligned} \quad (3.4.15)$$

Assume that f and j do not depend directly on m_t . That is, their partial derivatives with respect to m_t are zero.

3.4.4 Euler Equation for Dividends

Then we will have

(Here c_t is the firm's dividend payout, so $u'(c_t)$ is the marginal utility of a dividend dollar.)
FOC wrt c_t :

$$u'(c_t) = R\beta v_{t+1}^m \quad (3.4.16)$$

Envelope wrt m_t :

$$v_t^m = R\beta v_{t+1}^m \quad (3.4.17)$$

and combining the FOC with the Envelope theorem we get the usual

$$\begin{aligned} v_t^m &= R\beta v_{t+1}^m \\ &= u'(c_t) \\ &= R\beta u'(c_{t+1}) \\ &= u'(c_{t+1}) \end{aligned} \quad (3.4.18)$$

where the last line follows because we have assumed $R\beta = 1$.

3.4.5 Alternative Formulation

Now note that the value function can be rewritten as

$$v_t(k_t, m_t) = \max_{\{i_t, m_{t+1}\}} u((f_{t+1} - m_{t+1})/R + m_t - i_t - j_t) + \beta v_{t+1}(k_{t+1}, m_{t+1}) \quad (3.4.19)$$

This holds because maximizing with respect to m_{t+1} (subject to the accumulation equation) is equivalent to maximizing with respect to the components of m_{t+1} .

3.4.6 Investment FOC for Entrepreneur

For the version in (3.4.19) the FOC with respect to i_t is

$$u'(c_t)((1 + j_t^i) - f_{t+1}^k \Upsilon/R) = \Upsilon \beta v_{t+1}^k \quad (3.4.20)$$

Derivation: This holds because the derivative of the RHS of (3.4.19) with respect to i_t is

$$u'(c_t) \left(\left(\frac{\partial f_{t+1}}{\partial k_{t+1}} \frac{\partial k_{t+1}}{\partial i_t} \right) / R - \frac{\partial i_t}{\partial i_t} - \frac{\partial j_t}{\partial i_t} \right) + \beta \left(\frac{\partial k_{t+1}}{\partial i_t} \right) v_{t+1}^k(k_{t+1}, m_{t+1}) \quad (3.4.21)$$

(remember that m_{t+1} is a control variable and thus its derivative with respect to investment is zero) so the FOC translates to

$$u'(c_t)(f_{t+1}^k \Upsilon/R - 1 - j_t^i) + \beta \Upsilon v_{t+1}^k = 0 \quad (3.4.22)$$

which reduces to (3.4.20).

3.4.7 Envelope Theorem for Capital

Now we can use the envelope theorem with respect to k_t to show that

$$v_t^k = u'(c_t)(f_{t+1}^k \Upsilon/R - j_t^k) + \beta \Upsilon v_{t+1}^k \quad (3.4.23)$$

This can be seen by directly taking the derivative of the RHS of (3.4.19) with respect to k_t :

$$u'(c_t) \left(\left(\frac{\partial f_{t+1}}{\partial k_{t+1}} \frac{\partial k_{t+1}}{\partial k_t} \right) / R - \frac{\partial j_t}{\partial k_t} \right) + \beta \left(\frac{\partial k_{t+1}}{\partial k_t} \right) v_{t+1}^k \quad (3.4.24)$$

and noting that the Envelope theorem tells us the derivatives with respect to the controls m_{t+1} and i_t are zero while $\partial k_{t+1}/\partial k_t = \Upsilon$.

3.4.8 Euler Equation for Investment (Entrepreneur)

Now we can combine (3.4.20) and (3.4.23) to derive the Euler equation for investment

$$(1 + j_t^i) = \beta \left[f^k(k_{t+1}) + (1 + j_{t+1}^i - j_{t+1}^k) \right]. \quad (3.4.25)$$

Derivation: To see this, start with the Envelope theorem and substitute from (3.4.20),

$$\begin{aligned} v_t^k &= u'(c_t)(f_{t+1}^k \beta/R - j_t^k) + \overbrace{u'(c_t)((1+j_t^i) - f_{t+1}^k \beta/R)}^{\beta v_{t+1}^k} \\ &= u'(c_t)(f_{t+1}^k \beta/R - j_t^k) + u'(c_t)((1+j_t^i) - f_{t+1}^k \beta/R) \\ &= u'(c_t) \left(1 + j_t^i - j_t^k \right) \end{aligned} \quad (3.4.26)$$

which means that we can rewrite (3.4.20) substituting the rolled-forward version:

$$\begin{aligned} u'(c_t)((1+j_t^i) - f_{t+1}^k \beta/R) &= \beta v_{t+1}^k \\ &= \beta u'(c_{t+1}) \left(1 + j_{t+1}^i - j_{t+1}^k \right) \\ (1 + j_t^i) &= \beta \left[f^k(k_{t+1}) + (1 + j_{t+1}^i - j_{t+1}^k) \right] \end{aligned} \quad (3.4.27)$$

where the last line follows because with $R\beta = 1$ we know that $c_{t+1} = c_t$ implying $u'(c_{t+1}) = u'(c_t)$.

3.4.9 Observational Equivalence

Since behavior (for either a firm manager or a consumer) is determined by Euler equations, and the Euler equations for both consumption and investment are identical in this model to the Euler equations for the standard models, there is no observable consequence for investment of the fact that the firm is being run by a utility maximizer, and there is no observable consequence for consumption of the fact that the consumer owns a business enterprise with costly capital adjustment.

3.4.10 Impulse Responses: Monetary Shock

Now consider a firm of this kind that happens to have arrived in period t with positive monetary assets $m_t > 0$ and with capital equal to the steady-state target value $k_t = \check{k}$.

Suppose that a thief steals all the firm's monetary assets.

The consequences for the firm are depicted in Figure 3.8.

Dividends follow a random walk. Thus, there is a one-time downward adjustment to the level of dividends to reflect the stolen money. Thereafter dividends are constant, as are monetary assets (which are constant at zero forever).

The theft of the money has no effect on investment or the capital stock, because the firm's investment decisions are made on the basis of whether they are profitable and the theft of the money has no effect on the profitability of investments.

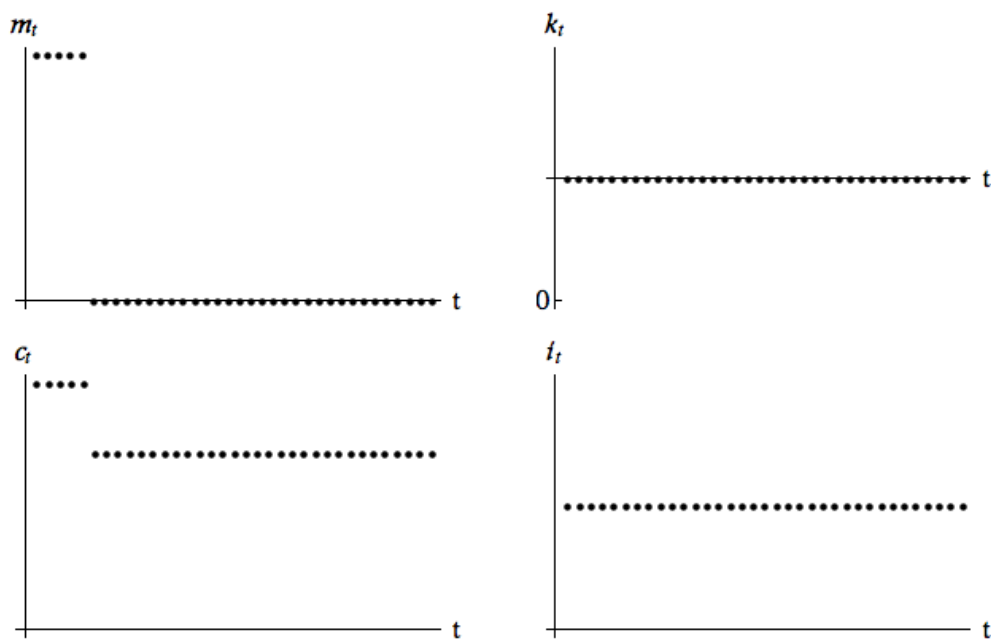
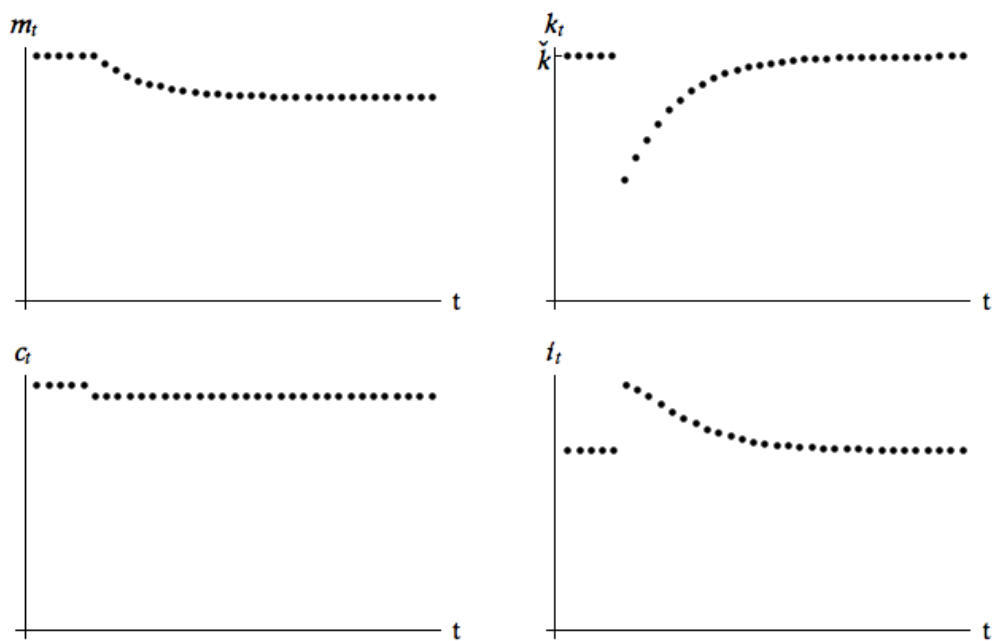
3.4.11 Impulse Responses: Capital Shock

Now consider another kind of shock: The firm's main building gets hit by a meteor, destroying some of the firm's capital stock.

The results are depicted in Figure 3.9.

Again, because dividends follow a random walk, what the firm's managers do is to assess the effect of the meteor shock on the firm's total value and they adjust the level of dividends downward immediately to the sustainable new level of dividends. Thereafter there is no change in the level of dividends.

Investment is more complicated. The firm's capital stock is obviously reduced below its steady-state value by the meteor, so there must be a period of high investment expenditures to bring capital back toward its steady state. However, the firm started out with monetary assets of zero. Therefore

Figure 3.8: Impulse response to a negative shock to m_t (monetary assets stolen).Figure 3.9: Impulse response to a negative shock to k_t (capital destroyed by meteor).

the high initial investment expenditures will be paid for by borrowing, driving the firm's monetary assets to a permanent negative value (the firm goes into debt to pay for its rebuilding). Gradually over time the capital stock is rebuilt back to its target level, and investment expenditures return to zero (or the level consistent with replacing depreciated capital).

3.4.12 Numerical Solution

The solution code uses the following definitions for the production and adjustment cost functions:¹³

$$\begin{aligned} f(k, \ell) &= \Psi k^\alpha \ell^{1-\alpha} \\ j(i, k) &= \frac{\omega k}{2} \left(\frac{i}{k} - \frac{\delta}{\Upsilon} \right)^2 \end{aligned} \quad (3.4.28)$$

where Ψ is the firm's productivity and ℓ is the labor supplied by the entrepreneur (assumed equal to 1).¹⁴

The policy functions are obtained using the method of *reverse shooting*, which is based on recovering for a given k_{t+1} and $e_{t+1}(k_{t+1})$ the values of k_t , i_t and $e_t(k_t)$ consistent with the first order conditions and transition equations.

The reverse-shooting equation for capital comes from a combination of the dynamic budget constraint and the Euler equation. For convenience defining

$$z_t \equiv k_{t+1}/\Upsilon \quad (3.4.29)$$

so that

$$i_t = z_t - k_t, \quad (3.4.30)$$

we can rewrite the investment Euler equation as

$$\begin{aligned} (1 + j_t^i) &= \Upsilon \beta \left(f^k(k_{t+1}) + (1 + j_{t+1}^i - j_{t+1}^k) \right) \\ \left(1 + \omega \left(\frac{\widehat{i_t}}{k_t} - \frac{\delta}{\Upsilon} \right) \right) &= \Upsilon \beta \left(f^k(k_{t+1}) + (1 + j_{t+1}^i - j_{t+1}^k) \right) \end{aligned} \quad (3.4.31)$$

so that

$$\begin{aligned} \frac{z_t}{k_t} - 1 &= \Upsilon \beta \left(f^k(k_{t+1}) + (1 + j_{t+1}^i - j_{t+1}^k) \right) / \omega - 1/\omega + \delta/\Upsilon \\ k_t &= \frac{z_t}{\Upsilon \beta \left(f^k(k_{t+1}) + (1 + j_{t+1}^i - j_{t+1}^k) \right) / \omega - 1/\omega + \delta/\Upsilon + 1}. \end{aligned} \quad (3.4.32)$$

With the values of i_t and j_t obtained from these reverse-shooting equations, the reverse-shooting equation for value is very simple: It is the Bellman equation

$$e_t(k_t) = f_t - i_t - j_t + e_{t+1}(k_{t+1})/R, \quad (3.4.33)$$

where note that f_t and $i_t - j_t$ are direct functions of k_t and i_t which we have already computed.

The steady-state level of capital can be obtained from (3.4.7):

¹³The slight modification to the cost-of-adjustment function (relative to the formulation in the [marginal q model](#) section) reflects the changed timing of depreciation in (3.4.2) compared to the corresponding equation in the [marginal q model](#) section. In the continuous-time limit the two equations become the same, but the formulation here makes the representation of the problem slightly more transparent in the discrete-time computer code.

¹⁴The code also includes the parameters $\mathcal{P}$ and $\mathcal{F}$ (respectively the investment cost after the Investment Tax Credit and the untaxed portion of earnings) in order to study the impact of tax policies. Both parameters are however assumed equal to 1 and thus the equations in the code are equivalent to the simpler ones described in this section.

$$\begin{aligned}
R/\Upsilon - 1 &= f^k(\check{k}) \\
\frac{1}{\alpha} (R/\Upsilon - 1) &= \check{k}^{\alpha-1} \\
\left(\frac{1}{\alpha} (R/\Upsilon - 1) \right)^{1/(\alpha-1)} &= \check{k}
\end{aligned} \tag{3.4.34}$$

while the steady-state value of λ comes from substituting $\check{k}$ into (3.4.9). Steady-state value is straightforward to compute, given that in the steady state the capital stock and amount of investment are constant:

$$\begin{aligned}
\check{e} &= \sum_{n=0}^{\infty} (f(\check{k}) - \check{i})\beta^n \\
&= (R/r)(f(\check{k}) - \check{i}).
\end{aligned} \tag{3.4.35}$$

The reverse shooting routine starts its backwards iterations from a $k_{\hat{t}}$ level very close to the steady state of the model. The accuracy of the solution is improved if we approximate $i_{\hat{t}}$ with a first order Taylor expansion using the derivative of investment at the steady state:

$$i_{\hat{t}} = \check{i} + \check{i}^k \epsilon \tag{3.4.36}$$

where the $\check{\cdot}$ identifies variables at the steady state and ϵ is the deviation from the steady state. $\check{i}^k$ is computed by differentiating the Euler equation with respect to k :

$$(j_t^{ii} i_t^k + j_t^{ik}) = \Upsilon \beta \left(f_t^{kk} \Upsilon (1 + i_t^k) + (j_t^{ik} \Upsilon (1 + i_t^k) + j_t^{ii} i_t^k - j_t^{kk} \Upsilon (1 + i_t^k) - j_t^{ki} i_t^k) \right) \tag{3.4.37}$$

which at the steady-state where $i_{\hat{t}}^k = i_t^k = \check{i}^k$ becomes:

$$(j_t^{ii} \check{i}^k + j_t^{ik}) = \Upsilon \beta \left(f_t^{kk} \Upsilon (1 + \check{i}^k) + (j_t^{ik} \Upsilon (1 + \check{i}^k) + j_t^{ii} \check{i}^k - j_t^{kk} \Upsilon (1 + \check{i}^k) - j_t^{ki} \check{i}^k) \right) \tag{3.4.38}$$

and given any particular set of parameter values $\check{i}^k$ can be found using standard numerical rootfinding methods.

Finally since the problem is solved under perfect foresight the entrepreneur's consumption function is simply:

$$c_t = \left(\frac{R - (R\beta)^{1/\rho}}{R} \right) (m_t + e_t(k_t) - f_t) \tag{3.4.39}$$

because this corresponds to the solution to a perfect foresight consumption problem in which the consumer has monetary resources m_t and net nonmonetary financial resources $e_t(k_t) - f_t$ (see the [perfect foresight CRRA](#) section).

3.5 Investment and Capital Market Imperfections

Every model of investment we have seen so far assumes that capital markets are perfect. A project is valued at its expected payoff and its risk, it is undertaken whenever that value exceeds the cost of acquiring and installing the capital, and the cost is governed by an economy-wide interest rate. The amount of investment that results is efficient.

Capital markets in the real world are not like this, and the largest reason is asymmetric information. A firm knows more about its own projects than the people it borrows from. Two distinct problems follow. Under adverse selection, lenders cannot tell good credit risks from bad ones, so the market rate reflects the average risk; the good risks are unwilling to borrow at that rate, and the market fills with bad ones. Under moral hazard, the money having already been lent, the borrower has an incentive to do things that are not in the lender's interest.

3.5.1 Sources of External Finance

One place these problems show up is in the structure of the financial system itself. Funds flow from savers to borrowers by two routes. They travel directly when a household buys a firm's stocks or bonds, and indirectly when a household deposits at a bank that then lends to the firm. Asymmetric information is worse along the direct route, because a household knows far less about a firm's projects than the firm does, and because no single household has a stake large enough to make paying for monitoring worthwhile.

Financial intermediaries exist largely because of this. A bank becomes a specialist in acquiring information about firms, which is more efficient done at scale, and so gets around some of the problem. It does not get around all of it, and there remains asymmetric information along the indirect route as well (and between depositors and the bank).

The composition of firms' external finance reflects this. Of the external funds raised by U.S. nonfinancial businesses between 1970 and 1985, loans supplied 61.9 percent, bonds 29.8 percent, and stocks 2.1 percent. Most financing is done indirectly, through intermediaries, which is what the information problems predict. The skew is in fact sharper than those numbers suggest, since financial intermediaries rather than households hold much of the stock and most of the bonds. Note also how small a role equity plays. The model we turn to now explains why a debt contract is generally the optimal arrangement between a lender and a borrower.

3.5.2 Costly State Verification

The formal treatment follows Romer [2011]. It is deliberately specific, since it takes up one particular information problem, but it generalizes to others and the qualitative conclusions survive.

An entrepreneur can undertake a project requiring one unit of resources. Project output is distributed uniformly on $[0, 2\mu]$, so expected output is μ , and μ varies across entrepreneurs. The entrepreneur has wealth W and must borrow $1 - W$ from outsiders. Because all of her money is tied up in the project, her payment back to lenders cannot exceed the project's output, so the lenders bear some of the risk. Everyone is risk neutral, the risk-free interest rate is r , and outside investors are competitive, so in equilibrium their expected rate of return must equal r . The entrepreneur undertakes the project when

$$\mu - \mathbb{E}[\text{payment to outsiders}] > (1 + r)W. \quad (3.5.1)$$

Suppose first that outside investors can observe output without cost, so that they know what the firm knows. Every project with $\mu > 1 + r$ is financed, and the loan contract is written so that the expected payment to the investor is $(1 - W)(1 + r)$. The investor is content, earning exactly what lending at the risk-free rate would have paid. So is the entrepreneur, whose expected income is

$$\mu - (1 - W)(1 + r) = W(1 + r) + \mu - (1 + r), \quad (3.5.2)$$

which exceeds $W(1 + r)$ precisely because $\mu > 1 + r$.

Now let the investor pay a cost $c > 0$ to observe the entrepreneur's output. We take up this case, "costly state verification," because it is the most straightforward of the information problems to analyze, and we should be honest that in practice other frictions matter more. We also assume the outsider's wealth exceeds $1 - W$, so that a single investor can fund a project. (Drop that assumption and lenders begin waiting for someone else to pay the verification cost, which is a complication of its own.)

The expected payment to the investor must cover the required return and the verification costs it expects to bear,

$$\mathbb{E}[\text{payment to investor}] = (1 + r)(1 - W) + \mathbb{E}[\text{verification costs}], \quad (3.5.3)$$

and the entrepreneur keeps whatever output remains. Her income is therefore maximized by the contract that minimizes expected verification costs, and that contract is debt. The outsider receives a fixed payment D whenever output is high enough to cover it, verifying nothing; when output falls short, the outsider verifies and takes everything there is. Verifying always would be wasteful, and a payment that varied with output without verification would be worthless, since the entrepreneur would report that things had turned out badly every time.

3.5.3 The Equilibrium Debt Contract

The face value D is endogenous, so we solve for it. Write the lender's expected net receipts as $\mathcal{R}(D)$. If $D \geq 2\mu$ the lender always takes the whole output and always verifies, so $\mathcal{R} = \mu - c$. If $D < 2\mu$, then with probability $(2\mu - D)/2\mu$ output exceeds D and the lender receives D without verifying, while with probability $D/2\mu$ output falls short and the lender verifies and collects $D/2$ on average (the mean of a uniform draw conditional on landing below D). Collecting terms,

$$\mathcal{R}(D) = D - \frac{D^2}{4\mu} - \frac{Dc}{2\mu}, \quad (3.5.4)$$

so that $\mathcal{R}$ is rising as long as $2\mu - c > D$. Expected net receipts therefore increase with D up to $D = 2\mu - c$ and fall thereafter, leveling out at $D = 2\mu$.

Equilibrium sets $\mathcal{R}(D)$ equal to the return the lender requires, $(1 + r)(1 - W)$. Three cases arise as that requirement rises. When it is low the equation has a single relevant root on the rising branch. When it is higher there are two roots, but the larger of them is not a competitive equilibrium, because another investor would undercut anyone who offered it. When the requirement exceeds the maximum of $\mathcal{R}$, no lending takes place at all, since the entrepreneur cannot meet the investor's required return in expected value.

Because the investor verifies whenever output falls below the equilibrium face value, which happens with probability $D^*/2\mu$, expected verification costs are $(D^*/2\mu)c$. Writing that wedge as an agency cost $\mathcal{A}(c, r, W, \mu)$, the project is undertaken only when

$$\mu > 1 + r + \mathcal{A}(c, r, W, \mu), \quad (3.5.5)$$

and differentiation shows that the wedge grows with the verification cost and with the interest rate, and shrinks with the entrepreneur's wealth and with the quality of her project. Models built on moral hazard or adverse selection instead of costly state verification deliver much the same comparative statics.

3.5.4 Implications

Four things follow, and we think each of them marks a real departure from the q theory of the [q model](#).

Agency costs raise the cost of external finance and reduce investment, since (3.5.5) demands a strictly better project than the frictionless condition $\mu > 1 + r$ would. Some worthwhile projects (those with $1 + r < \mu < 1 + r + \mathcal{A}$) simply go unfunded.

Output and interest rates acquire indirect effects on investment alongside the direct ones the q model already delivers. Higher output raises current profitability and with it the internal funds W the

entrepreneur can put into the project. Interest rates move the face value D and therefore the expected verification costs.

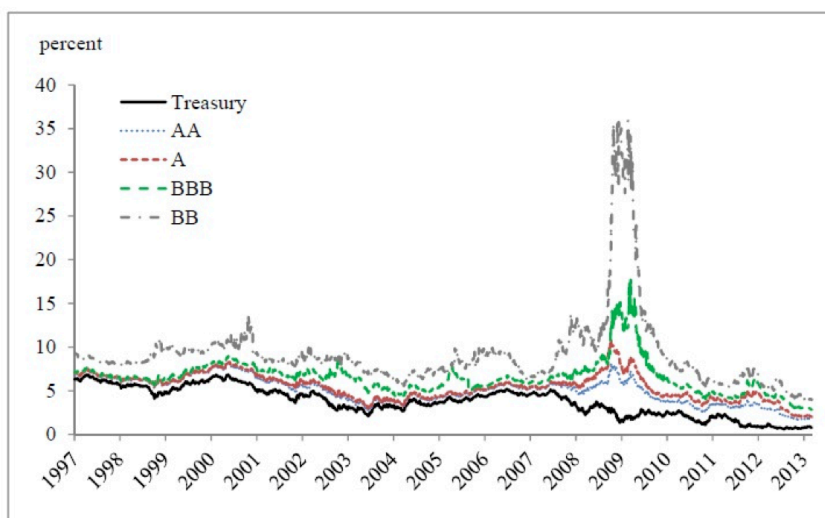
Variables with no place in the q model begin to matter, cash flow most of all. Firms with more of their own resources fund projects more cheaply, and the empirical literature testing that link is large and mostly finds it. Unfortunately, the finding is harder to interpret than it looks, because cash flow moves with profitability, which ought to affect investment even when information is symmetric (a firm having a good year is both richer and, plausibly, facing better projects). We take that identification problem up in [investment and cash flow](#).

Changes in the financial system itself alter investment, by making verification harder or easier. The collapse of the banking system during the Great Depression destroyed precisely the specialists who had been evaluating projects cheaply, which raised the cost of investment and helped drive it down.

3.6 Risk Premia on Financial and Nonfinancial Bonds

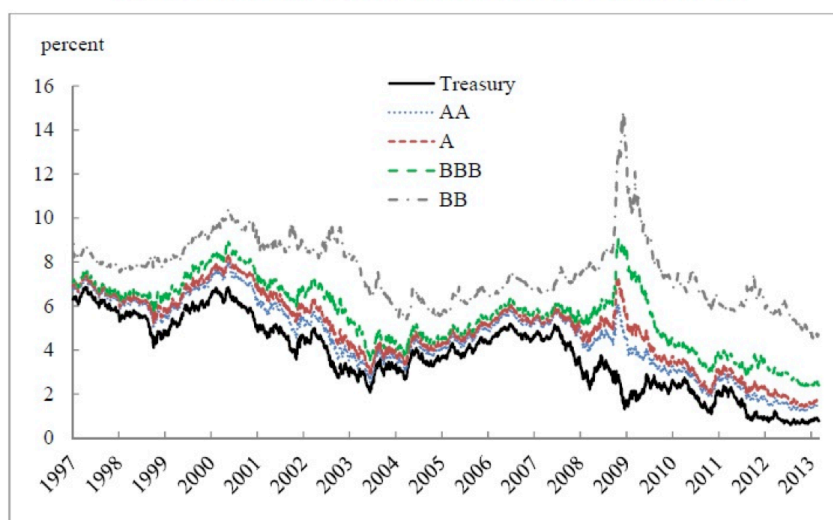
These figures are from Chapter 3 of Matthew Raskin's 2013 JHU dissertation.

Figure 2: 5-Year Treasury Yield and Financial Corporate Bonds



Note: the Treasury yield is the yield on a notional par security with semi-annual coupons estimated from off-the-run Treasury coupon securities, taken from the dataset of Gürkaynak, Sack and Wright (2007). The corporate yields are derived by fitting Nelson-Siegel curves to underlying corporate bond yields provided by Bank of America/Merrill Lynch.

Figure 1: 5-Year Treasury Yield and Nonfinancial Corporate Bonds



Note: the Treasury yield is the yield on a notional par security with semi-annual coupons estimated from off-the-run Treasury coupon securities, taken from the dataset of Gürkaynak, Sack and Wright (2007). The corporate yields are derived by fitting Nelson-Siegel curves to underlying corporate bond yields provided by Bank of America/Merrill Lynch.

3.7 Investment and Cash Flow With Imperfect Capital Markets

Many models of capital market imperfections reach the conclusion that financing an investment project is cheaper with “internal” than with “external” funds. (Internal funds are those that the firm owns, for example, in its bank account; external funds are obtained from outsiders like banks or new investors). For a lucid exposition of an example of models of this kind, see the discussion of financial market imperfections in Romer [2011].

Such models tend to be highly specialized, with entrepreneurs who miraculously receive exogenously specified endowments of “internal” cash and who make a one- (or at most two-) period investment decision. This is a striking contrast with the generality of the canonical q model of investment in which optimizing firms make investment decisions that take into account infinite future paths of adjustment costs, marginal products of capital, taxes, and all other features of the environment.

But models with imperfect capital markets are always very clear on who receives what cash, when, and why. In contrast, the power of the q model comes at a high cost: The model has no implications whatsoever for the firm’s decisions about when and how to make payments to shareholders and lenders.

3.7.1 Modigliani-Miller

To see this, consider the benchmark discrete-time framework in the [marginal \$q\$ model](#) section without taxes or depreciation, and (as in that section) assume that the firm issues or repurchases shares to maintain a number of shares s_t outstanding equal to the size of the physical capital stock (so $s_t = k_t$ in every period). Suppose we assume that the firm has a “dividend policy”: In every period, it pays to each shareholder a dividend equal to the flow of revenues per share.

In this case, the value of all the firm’s shares s_t (the equity value of the firm) will be

$$\hat{e}_t(s_t) = \max_{\{i\}_t^\infty} \mathbb{E}_t \left[\sum_{n=0}^{\infty} \beta^n (\pi_{t+n} - \xi_{t+n}) \right]. \quad (3.7.1)$$

(Notice that because the firm always has $s_t = k_t$, the value of the firm’s shares defined here $\hat{e}_t(s_t)$ is identical to the value of the firm’s capital $e_t(k_t)$ defined in the [marginal \$q\$ model](#) section.)

Now suppose the firm decides to deviate, just a little bit, from this dividend policy. It issues an extra share in period t , which (assuming capital markets are efficient) raises an amount of money equal to the share price λ_t ; it invests the proceeds in the riskless asset, earning return R , then pays out the resulting cash to shareholders in the next period. However, we assume that the firm continues to maximize its value. This modifies the infinite sum to:

$$\max_{\{i\}_t^\infty} \mathbb{E}_t \left[\sum_{n=0}^{\infty} \beta^n (\pi_{t+n} - \xi_{t+n}) \right] + \lambda_t - R\beta\lambda_t. \quad (3.7.2)$$

but since by assumption $\beta = R^{-1}$ this expression reduces to the original expression.

The point can be extended to show that the firm can raise an arbitrary amount of money in one period, paying it back in another, or vice versa, without having any effect on its value or its optimal investment plans.

This is an example of the famous theorem by Modigliani and Miller [1958], who showed that under perfect capital markets the value of the firm is identical whether its investment is financed by equity, debt, or any combination of the two.

It is precisely to get around this result that models of imperfect capital markets were invented, since from the beginning many economists did not believe that the result is a plausible description of reality (including, I believe, Modigliani and Miller). Indeed, the existence (and high salaries) of corporate financial officers constitute a proof that the proposition is not true: if it were true, CFOs would have no effect on the value of the firm, and could all be fired (thus saving the firm their salaries!). What remains true, however, is that models of investment that relax the assumption of perfect capital markets are much more difficult to work with (and harder to extract implications from) than models that assume capital markets are perfect.

A testable result that generally emerges from models of capital market imperfections, however, is that the amount of a firm's investment depends on the amount of cash the firm has on hand, with which it can finance that investment.

3.7.2 Testing for the Failure of Modigliani-Miller

The [marginal q model](#) section shows that the q model can be manipulated (abstracting from depreciation, taxes, and other complications) to yield an implication for investment dynamics of approximately the following kind:

$$\iota \approx (q - 1)/\omega. \quad (3.7.3)$$

The simplest class of imperfect capital markets models is one in which the firm is constrained to pay dividends to shareholders equal to whatever is its flow of cash after subtracting off expenses. But the rate of return that the firm must pay for external funds exceeds the return it can earn on internal funds. Without going into details, the rough implication of this is that an empirical model of the firm's investment choices should augment (3.7.3) with a measure of the funds the firm has in its hands at the time the investment decision is made:

$$\iota \approx \psi_0 + \psi_1 q + \psi_2 \pi. \quad (3.7.4)$$

A large literature spurred by the seminal paper of Fazzari et al. [1988] estimates equations like this, and virtually always finds a statistically significant estimate of ψ_2 , interpreted as indicating the importance of “cash flow” for investment decisions. But there are many criticisms of this literature, and the importance of imperfections in capital markets remains a lively area of debate. My own view is that the most persuasive evidence of large deviations from the benchmark of the MM theorem is the fact that the financial services industry (including corporate CFOs) absorbs a very substantial amount of resources every year. Why would such a large industry exist if the informational problems were easy to solve? It would not, especially in the internet era, where it would be a trivial matter to match up borrowers and lenders in the absence of informational and enforcement difficulties.

Chapter 4

Growth Theory

The Ramsey-Cass-Koopmans model is the workhorse of this chapter. A representative household chooses consumption over an infinite horizon, with capital as the only state variable. Equilibrium traces a saddle path in the consumption-capital plane. The first several handouts stay inside that framework and add pieces to it: government purchases and taxation, adjustment costs on investment, and a decentralization that lets competitive markets reproduce the planner's allocation.

We then turn to the model's central embarrassment. Unless productivity growth arrives from outside, the model predicts no long-run growth in per capita income at all (the thing it was built to explain enters as an assumption). The endogenous growth literature answers by finding some factor that households can accumulate without bound, and the handouts on Romer, Rebelo, and Lucas each pick a different candidate. A final handout swaps the infinitely lived household for Blanchard's perpetual youth structure, which restores a finite planning horizon while keeping the model tractable.

4.1 The Ramsey/Cass-Koopmans (RCK) Model

Ramsey [1928], followed much later by Cass [1965] and Koopmans [1965], formulated the canonical model of optimal growth for an economy with exogenous ‘labor-augmenting’ technological progress.

4.1.1 The Budget Constraint

The economy has a perfectly competitive production sector that uses a Cobb-Douglas aggregate production function

$$Y = F(K, L) = K^\alpha (ZL)^{1-\alpha} \quad (4.1.1)$$

to produce output using capital and labor.¹ Labor hours (the same as population) increases exogenously at a constant rate²

$$\dot{L}/L = \xi \quad (4.1.2)$$

and Z is an index of labor productivity that grows at rate

$$\dot{Z}/Z = \phi. \quad (4.1.3)$$

Thus, technological progress allows each worker to produce perpetually more as time goes by with the same amount of physical capital.³ The quantity ZL is known as the number of ‘efficiency units’ of labor in the economy.

Aggregate capital accumulates according to

$$\dot{K} = Y - C - \delta K. \quad (4.1.4)$$

Lower case variables are the upper case version divided by efficiency units, i.e.

$$\begin{aligned} y &= Y/(ZL) \\ &= K^\alpha (ZL)^{1-\alpha}/(ZL) \\ &= (K/ZL)^\alpha \\ &= k^\alpha. \end{aligned} \quad (4.1.5)$$

Note that

$$\begin{aligned} \dot{k} &\equiv \left(\frac{dk}{dt} \right) = \left(\frac{\dot{K}ZL - K(\dot{Z}L + Z\dot{L})}{(ZL)^2} \right) \\ &= \dot{K}/ZL - k(\dot{Z}/Z + \dot{L}/L) \\ &= \dot{K}/ZL - (\phi + \xi)k \end{aligned} \quad (4.1.6)$$

which means that (4.1.4) can be divided by ZL and becomes

$$\begin{aligned} \dot{K}/ZL &= y - c - \delta k \\ \dot{k} + (\phi + \xi)k &= f(k) - c - \delta k \\ \dot{k} &= f(k) - c - (\phi + \xi + \delta)k. \end{aligned} \quad (4.1.7)$$

A steady-state will be a point where $\dot{k} = 0$.

¹All Roman variables are functions of time, but putting time subscripts on everything would clutter the notation, so we do it only where necessary for clarity.

²The ancient Greek philosophers captured eternal truths; therefore, Greek letters represent constants whose value never changes.

³This is the definition of ‘labor-augmenting’ (Harrod-neutral) productivity growth; with a Cobb-Douglas production function, it turns out to be essentially the same as ‘capital-augmenting’ productivity growth, also known as Hicks-neutral, as well as output-neutral (‘Solow-neutral’) progress.

Equation (4.1.7) yields a first candidate for an optimal steady-state of the growth model: It seems reasonable to argue that the best possible steady-state is the one that maximizes c . This is the “golden rule” optimality condition of Phelps [1961], an article well worth reading; this is one of the chief contributions for which Phelps won the Nobel prize.

4.1.2 The Social Planner’s Problem

Now suppose that there is a social planner whose goal is to maximize the discounted sum of CRRA utility from per-capita consumption:

$$\max \int_0^\infty \left(\frac{(C/L)^{1-\rho}}{1-\rho} \right) e^{-\vartheta t}. \quad (4.1.8)$$

But $C/L = Z(C/ZL) = Zc$. Recall that for a variable growing at rate ϕ ,

$$Z_t = Z_0 e^{\phi t} \quad (4.1.9)$$

so if the economy started off in period 0 with productivity Z_0 , by date t we can rewrite

$$\begin{aligned} C/L &= cZ \\ &= cZ_0 e^{\phi t}. \end{aligned} \quad (4.1.10)$$

Using (4.1.10) and the other results above, we can rewrite the social planner’s objective function as

$$\begin{aligned} \int_0^\infty \left(\frac{(Zc)^{1-\rho}}{1-\rho} \right) e^{-\vartheta t} &= Z_0^{1-\rho} \int_0^\infty \left(\frac{c^{1-\rho}}{1-\rho} \right) e^{-\vartheta t} e^{(1-\rho)\phi t} \\ &= Z_0^{1-\rho} \int_0^\infty u(c) e^{((1-\rho)\phi - \vartheta)t}. \end{aligned} \quad (4.1.11)$$

Thus, defining $\nu = \vartheta - (1-\rho)\phi$ and normalizing the initial level of productivity to $Z_0 = 1$, the complete optimization problem can be formulated as

$$\max \int_0^\infty u(c) e^{-\nu t} \quad (4.1.12)$$

subject to

$$\dot{k} = k^\alpha - c - (\xi + \phi + \delta)k, \quad (4.1.13)$$

which has a discounted Hamiltonian representation

$$\mathcal{H}(k, c, \lambda) = u(c) + (k^\alpha - c - (\xi + \phi + \delta)k)\lambda. \quad (4.1.14)$$

The first discounted Hamiltonian optimization condition requires $\partial \mathcal{H} / \partial c = 0$:

$$\begin{aligned} c^{-\rho} &= \lambda \\ -\rho c^{-\rho-1} \dot{c} &= \dot{\lambda}. \end{aligned} \quad (4.1.15)$$

The second discounted Hamiltonian optimization condition requires:

$$\begin{aligned} \dot{\lambda} &= \nu \lambda - \partial \mathcal{H} / \partial k \\ &= \nu \lambda - \lambda(f'(k) - (\xi + \phi + \delta)) \\ \dot{\lambda} / \lambda &= (\nu + (\xi + \phi + \delta) - f'(k)) \\ \left(\frac{-\rho c^{-\rho-1} \dot{c}}{c^{-\rho}} \right) &= (\nu + (\xi + \phi + \delta) - f'(k)) \\ \rho \dot{c} / c &= (f'(k) - \nu - (\xi + \phi + \delta)) \\ \dot{c} / c &= \rho^{-1} \underbrace{(f'(k) - (\xi + \phi + \delta) - \nu)}_{\equiv \dot{r}} \end{aligned} \quad (4.1.16)$$

where the definition of $\dot{r}$ is motivated by thinking of $f'(k) - (\xi + \phi + \delta)$ as the interest rate net of depreciation and dilution.

This is called the “modified golden rule” (or sometimes the “Keynes-Ramsey rule” because it was originally derived by Ramsey with an explanation attributed to Keynes).

Thus, we end up with an Euler equation for consumption growth that is just like the Euler equation in the perfect foresight partial equilibrium consumption model, except that now the relevant interest rate can vary over time as $f'(k)$ varies.

Substituting in the modified time preference rate gives

$$\begin{aligned}\dot{c}/c &= \rho^{-1}(f'(k) - (\xi + \delta + \phi) - \vartheta + (1 - \rho)\phi) \\ &= \rho^{-1}(f'(k) - (\xi + \delta) - \vartheta - \rho\phi),\end{aligned}\tag{4.1.17}$$

and finally note that defining per capita consumption $\chi = C/L$ so that $c = \chi Z^{-1}$,

$$\begin{aligned}\dot{c} &= \left(\dot{\chi} Z^{-1} \right) = \dot{\chi} Z^{-1} - (\chi Z^{-1}) \dot{Z}/Z \\ \dot{c}/c &= \left(\dot{\chi} Z^{-1} \right) / (\chi Z^{-1}) = \dot{\chi}/\chi - \phi\end{aligned}\tag{4.1.18}$$

and since (4.1.17) can be written

$$\dot{c}/c = \rho^{-1}(f'(k) - (\xi + \delta) - \vartheta) - \phi,\tag{4.1.19}$$

we have

$$\dot{\chi}/\chi = \rho^{-1}(f'(k) - (\xi + \delta) - \vartheta)\tag{4.1.20}$$

so the formula for per capita consumption growth (as a function of k) is identical to the model with no growth (equation (4.1.17) with $\phi = 0$). Any important differences between the no-growth model and the model with growth therefore must come through the channel of differences in k .

4.1.3 The Steady State

The assumption of labor augmenting technological progress was made because it implies that in steady-state, per-capita consumption, income, and capital all grow at rate ϕ .⁴

$\dot{c}/c = 0$ implies that at the steady-state value of $\check{k}$,

$$\begin{aligned}f'(\check{k}) &= \vartheta + \xi + \delta + \rho\phi \\ \alpha \check{k}^{\alpha-1} &= \vartheta + \xi + \delta + \rho\phi \\ \check{k} &= \left(\frac{\vartheta + \xi + \delta + \rho\phi}{\alpha} \right)^{\frac{1}{\alpha-1}} \\ \check{k} &= \left(\frac{\alpha}{\vartheta + \xi + \delta + \rho\phi} \right)^{\frac{1}{1-\alpha}}\end{aligned}\tag{4.1.21}$$

Thus, the steady-state k will be higher if capital is more productive (α is higher), and will be lower if consumers are more impatient, population growth is faster, depreciation is greater, or technological progress occurs more rapidly.

4.1.4 A Phase Diagram

While the RCK model has an analytical solution for its steady-state, it does not have an analytical solution for the transition to the steady-state. The usual method for analyzing models of this kind is a phase diagram in c and k .

⁴See Grossman et al. [2016] for a discussion of the realism of this requirement.

The first step in constructing the phase diagram is to take the differential equations that describe the system and find the points where they are zero. Thus, from (4.1.7) we have that $\dot{k} = 0$ implies

$$c = f(k) - (\phi + \xi + \delta)k \quad (4.1.22)$$

and we have already solved for the (constant) $\check{k}$ that characterizes the $\dot{c}/c = 0$ locus. These can be combined to generate the borders between the phases in the phase diagram, as illustrated in Figure 4.1.

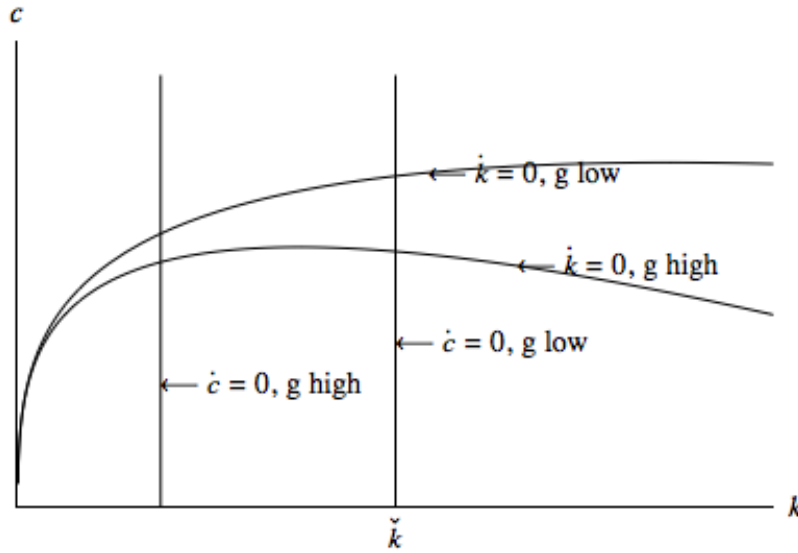


Figure 4.1: $\dot{c}/c = 0$ and $\dot{k} = 0$ Loci

4.1.5 Transition

Actually, as stated so far, the solution to the problem is very simple: The consumer should spend an infinite amount in every period. This solution is not ruled out by anything we have yet assumed (except possibly the fact that once k becomes negative the production function is undefined).

Obviously, this is not the solution we are looking for. What is missing is that we have not imposed anything corresponding to the intertemporal budget constraint. In this context, the IBC takes the form of a “transversality condition,”

$$\lim_{t \rightarrow \infty} \lambda_t e^{(\phi + \xi)t - \int_0^t r_\tau d\tau} k_t = 0. \quad (4.1.23)$$

The intuitive purpose of this unintuitive equation is basically to prevent the capital stock from becoming negative or infinity as time goes by.

Obviously a capital stock that was negative for the entire future could not satisfy the equation. And a capital stock that is too large will have an arbitrarily small interest rate, which will result in the LHS of the TVC being a positive number, again failing to satisfy the TVC.

Figure 4.2 shows three paths for c and k that satisfy (4.1.17) and (4.1.7). The topmost path, however, is clearly on a trajectory toward zero then negative k , while the bottommost path is heading toward an infinite k . Only the middle path, labelled the “saddle path,” satisfies both (4.1.17) and (4.1.7) as well as the TVC (4.1.23).

4.1.6 Interactive Notebooks

An explicit numerical solution to the Ramsey problem, with a description of a solution method and its mathematical/computational underpinnings, is available [here](#).

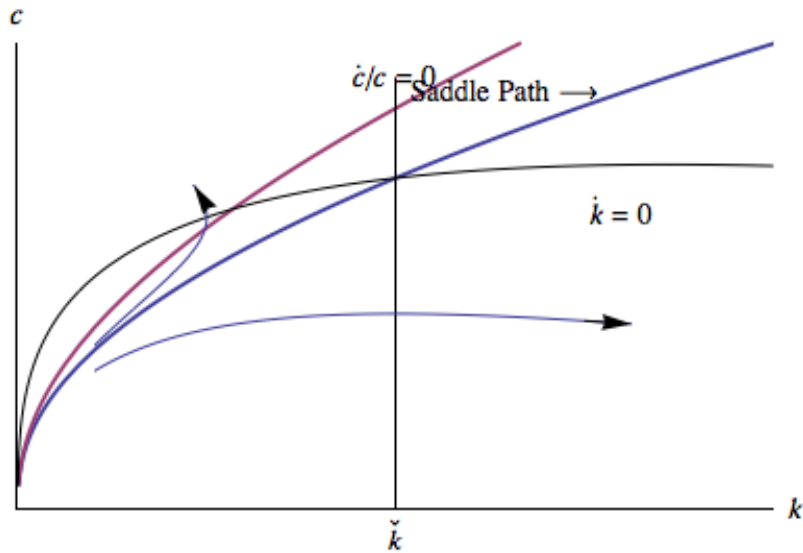


Figure 4.2: Transition to the Steady State

4.1.7 Appendix: Numerical Solution

The RCK model does not have an analytical solution, which means that numerical methods must be used to find out the model's quantitative implications for transition paths.

The method of solution of these kinds of models is not important for the purposes of first year graduate macroeconomics; this appendix has been written as a reference for more advanced students who might be beginning their research on growth models.

The most straightforward method of numerical solution for perfect foresight models of this kind is called the 'time elimination' method. It starts from the fact that

$$\left(\frac{dc/dt}{dk/dt} \right) = dc/dk. \quad (4.1.24)$$

Note from (4.1.17) that we can write

$$\dot{c} = \rho^{-1}(f'(k) - \vartheta - (\xi + \delta) - \rho\phi)c, \quad (4.1.25)$$

so we can obtain

$$dc/dk = \left(\frac{(\alpha k^{\alpha-1} - \vartheta - (\xi + \delta) - \rho\phi)c}{\rho(k^{\alpha} - c - (\phi + \xi + \delta)k)} \right) \quad (4.1.26)$$

which is a differential equation with no analytical solution. Many numerical math packages can solve differential equations numerically, yielding a numerical version of the $c(k)$ function.

There is one problem, however, which is that at the steady-state values of c and k both numerator and denominator of this equation are zero. The alternative is to solve the differential equation twice: Once for a domain extending from $k = 0$ to $\check{k} - \epsilon$, yielding $c_{-}(k)$, and once for a domain from $\check{k} + \epsilon$ to some large value of k , yielding $c_{+}(k)$. The true consumption policy function can then be approximated by interpolating between the upper endpoint of $c_{-}(k)$ and the lower endpoint of $c_{+}(k)$.

For further details of the numerical solution of this model, see this [Jupyter notebook](#), or clone the repo and, in the cloned directory, run the corresponding python program: `ipython RamseyCassKoopmans.py`.

4.2 The Ramsey/Cass-Koopmans (RCK) Model with Government

This section solves the Ramsey/Cass-Koopmans (RCK) model with government.⁵

For simplicity we assume no technological progress, population growth, or depreciation, and a continuum of consumers with mass 1 distributed on the unit interval as per the [Aggregation](#) appendix.

Consider first the case where government is financed by constant lump-sum taxes of amount τ per period, and spending is at rate $\mathbf{x}$ per period, and suppose the government has a balanced budget requirement so that $\mathbf{x} = \tau$ in every period and there is no government debt. We suppose further that government spending yields no utility. The individual's optimization problem (leaving out the i subscripts that we used in the previous section, but understanding that they are implicitly present) is now

$$\int_0^\infty u(c_t) e^{-\vartheta t} \quad (4.2.1)$$

subject to the household DBC

$$\dot{a}_t = r_t a_t + w_t - c_t - \tau, \quad (4.2.2)$$

which has Hamiltonian representation

$$\mathcal{H}(a_t, c_t, \lambda_t) = u(c_t) + (r_t a_t + w_t - c_t - \tau) \lambda_t \quad (4.2.3)$$

The first Hamiltonian optimization condition requires $\partial \mathcal{H}_t / \partial c_t = 0$:

$$\begin{aligned} c_t^{-\rho} &= \lambda_t \\ -\rho c_t^{-\rho-1} \dot{c}_t &= \dot{\lambda}_t \end{aligned} \quad (4.2.4)$$

The second Hamiltonian optimization condition requires:

$$\begin{aligned} \dot{\lambda}_t &= \vartheta \lambda_t - (\partial \mathcal{H}_t / \partial a_t) \lambda_t \\ &= \vartheta \lambda_t - \lambda_t r_t \\ \dot{\lambda}_t / \lambda_t &= (\vartheta - r_t) \\ \dot{c}_t / c_t &= \rho^{-1} (r_t - \vartheta). \end{aligned} \quad (4.2.5)$$

Finally, the household's behavior must satisfy a transversality constraint, which is equivalent to the intertemporal budget constraint:

$$\begin{aligned} \int_0^\infty c_t \mathfrak{R}_t^{-1} &= a_0 + \int_0^\infty \mathbf{y}_t \mathfrak{R}_t^{-1} - \int_0^\infty \tau \mathfrak{R}_t^{-1} \\ \mathbf{C}_0 &= a_0 + \mathbf{Y}_0 - \mathbf{T}_0 \end{aligned} \quad (4.2.6)$$

which says that the present discounted value of consumption must equal the current net physical wealth plus human wealth minus the PDV of taxes.

Now consider the problem from the standpoint of a social planner who has the same utility function as the individual consumers. If the social planner wants to spend a constant amount $\mathbf{x}$ per period, the social planner's budget constraint is

$$\dot{k}_t = f(k_t) - \delta k_t - c_t - \mathbf{x} \quad (4.2.7)$$

which reflects the fact that the social planner divides total net output between consumption and government spending. This leads to Hamiltonian

$$\mathcal{H}(k_t, c_t, \lambda_t) = u(c_t) + \lambda_t (f(k_t) - \delta k_t - c_t - \mathbf{x}), \quad (4.2.8)$$

yielding the first order condition

⁵The treatment is similar to that in Blanchard and Fischer [1989]; see that source for more details.

$$\dot{c}_t/c_t = \rho^{-1}(f'(k_t) - \delta - \vartheta). \quad (4.2.9)$$

Now recall from the section on [decentralizing the RCK model](#) that

$$f(k_t) = \hat{r}_t k_t + w_t \quad (4.2.10)$$

where the gross return on capital $\hat{r}_t$ is equal to the net return r_t plus the depreciation rate.

Thus, the social planner's DBC is:

$$\begin{aligned} \dot{k}_t &= f(k_t) - \delta k_t - c_t - \mathbf{x} \\ &= \hat{r}_t k_t - \delta k_t + w_t - c_t - \mathbf{x} \\ &= r_t k_t + w_t - c_t - \mathbf{x}, \end{aligned} \quad (4.2.11)$$

which is equivalent to the household's budget constraint (4.2.2) when $a_t = k_t$ and $\tau = \mathbf{x}$. As discussed in the [decentralizing the RCK model](#) section, $a_t = k_t$ must hold in equilibrium for identical households, and $\tau = \mathbf{x}$ was the balanced budget assumption that we started off with.

The same equivalence holds for the lifetime constraints. Integrating the planner's DBC gives

$$\begin{aligned} \int_0^\infty c_t \mathfrak{R}_t^{-1} &= k_0 + \mathbf{Y}_0 - \int_0^\infty \mathbf{x} \mathfrak{R}_t^{-1} \\ &= k_0 + \mathbf{Y}_0 - \mathbf{X}_0, \end{aligned} \quad (4.2.12)$$

where $\mathbf{X}_0$ is the PDV of government spending. Under the two conditions just imposed, this is the household's IBC. The initial stocks coincide, $k_0 = a_0$, because households hold no government debt in the balanced budget case; and the two PDVs coincide, $\mathbf{T}_0 = \mathbf{X}_0$, because a government that balances its budget in every period collects taxes worth exactly what it spends. Planner and household therefore face the same lifetime constraint as well as the same flow constraint.

Note that the $\dot{c}_t = 0$ locus in the phase diagram is unchanged by changing τ and $\mathbf{x}$. However, the $\dot{k}_t = 0$ locus is shifted down by amount $\tau = \mathbf{x}$.

Now what happens if the government does not face a balanced budget requirement? Specifically, suppose we continue to have the same constant amount of spending per period but now want to consider the effect of allowing taxes to vary over time, which we denote by a subscript on τ_t . Suppose d is the level of government bonds (debt); the government's Dynamic Budget Constraint is

$$\dot{d}_t = \mathbf{x} + r_t d_t - \tau_t, \quad (4.2.13)$$

which says that debt must rise by the amount by which spending exceeds taxes.

The government's IBC will be the integral of its DBC:

$$\begin{aligned} d_0 + \int_0^\infty \mathbf{x} \mathfrak{R}_t^{-1} &= \int_0^\infty \tau_t \mathfrak{R}_t^{-1} \\ d_0 + \mathbf{X}_0 &= \mathbf{T}_0 \end{aligned} \quad (4.2.14)$$

and we assume $d_0 = 0$ so that the government starts out with no debt (to maintain comparability with the previous example).

The DBC of the idiosyncratic family also changes. They can now own either capital k_t or government debt d_t . If the family is to be indifferent between the two forms of assets, the interest rate must be the same.

$$\begin{aligned} c_t + \dot{a}_t &= w_t + r_t a_t - \tau_t \\ a_t &= k_t + d_t. \end{aligned} \quad (4.2.15)$$

Now the family's IBC becomes

$$\begin{aligned} \int_0^\infty c_t \mathfrak{R}_t^{-1} &= k_0 + d_0 + \mathbf{Y}_0 - \int_0^\infty \tau_t \mathfrak{R}_t^{-1} \\ &= k_0 + d_0 + \mathbf{Y}_0 - \mathbf{T}_0. \end{aligned} \quad (4.2.16)$$

Note: Nowhere in this equation does the time path of taxes matter;

all that matters is the PDV of taxes. And the time path of taxes also does not enter the $\dot{c}/c$ equation. Thus, the path of consumption over time is unaffected by the path of taxes over time!

This is not so surprising when you realize that it is simply the Ricardian equivalence proposition in this perfect foresight framework.

However, now consider the case where there is a tax on capital income at rate τ . Furthermore, for simplicity suppose that the government rebates all of the tax revenue in a lump sum per capita, and suppose depreciation $\delta = 0$. Thus the household budget constraint becomes

$$\dot{a}_t = r_t(1 - \tau)a_t + w_t - c_t + \mathbf{z}_t \quad (4.2.17)$$

where $\mathbf{z}_t$ is the per-capita size of the lump-sum rebates,

$$\mathbf{z}_t = \tau r_t k_t. \quad (4.2.18)$$

The household's Hamiltonian becomes

$$\mathcal{H}_t(c_t, a_t, \lambda_t) = u(c_t) + \lambda_t(r_t(1 - \tau)a_t + w_t - c_t + \mathbf{z}_t). \quad (4.2.19)$$

The crucial difference between this situation and the previous one is that now the effective rate of return on saving has been decreased, so that $\partial \mathcal{H}_t / \partial a_t$ is now $r_t(1 - \tau)$ rather than r_t . Ultimately this produces a consumption Euler equation of

$$\dot{c}_t / c_t = \rho^{-1}((1 - \tau)r_t - \vartheta) \quad (4.2.20)$$

which implies that the economy will be in equilibrium at

$$\begin{aligned} f'(\bar{k})(1 - \tau) &= \vartheta \\ f'(\bar{k}) &= \vartheta / (1 - \tau) \end{aligned} \quad (4.2.21)$$

so that the equilibrium level of the marginal product of capital is higher, and the capital stock must therefore be lower, than before the capital taxation was instituted.

Notice, however, that because the taxes are being rebated, the *aggregate* budget constraint does not change when the tax is imposed:

$$\begin{aligned} \dot{k}_t &= r_t(1 - \tau)k_t + w_t - c_t + \tau k_t r_t \\ &= r_t k_t + w_t - c_t. \end{aligned} \quad (4.2.22)$$

Thus the social planner will choose exactly the same amount of consumption as before the tax was instituted.

The crucial point is that if an individual household saves more and thus causes next year's capital stock to be a bit higher, the *personal* benefit to that household is essentially zero. The higher taxes that the household will pay next year will be distributed to the entire population in a lump sum, so the saver will get nothing. The higher saving of this individual household is basically a positive externality from the point of view of the other consumers in the economy. However, if the social planner forces the economy as a whole to save more, the social planner receives all of the extra tax revenue.

4.3 A q-Ramsey Model

Consider a Ramsey economy in which the capital stock cannot be freely adjusted; instead, as in the q model of investment, capital is subject to quadratic costs of adjustment.⁶

The dynamic budget constraint is

$$\begin{aligned} F(k_t) &= k_t + k_t^\alpha \\ a_t &= F(k_t) - c_t - j_t \\ k_{t+1} &= a_t \end{aligned} \tag{4.3.1}$$

where for analytical simplicity we neglect capital depreciation (though the illustrative figures below show results of a model that properly includes depreciation) and the cost-of-adjustment function takes the form

$$j(i, k) = (k/2)(i/k)^2\omega \tag{4.3.2}$$

for some constant ω , so that the cost of adjustment incurred in period t is given by

$$j_t = j(\overbrace{a_t - k_t}^{\equiv i_t}, k_t). \tag{4.3.3}$$

A social planner is assumed to maximize the discounted sum of utility from consumption c_t , where the utility function is CRRA, $u(c) = c^{1-\rho}/(1-\rho)$.

The social planner's problem can be rewritten in the form of a Bellman equation,

$$\begin{aligned} v(k_t) &= \max_{c_t} u(c_t) + \beta v(k_{t+1}) \\ \text{s.t.} \\ k_{t+1} &= F_t - c_t - j_t. \end{aligned}$$

Because (given k_t) choosing a_t is equivalent to choosing c_t ,⁷ we can rewrite the problem as:

$$v(k_t) = \max_{a_t} u(F(k_t) - a_t - j(a_t - k_t, k_t)) + \beta v(a_t). \tag{4.3.4}$$

The first order condition is found by setting the derivative w.r.t. a_t to zero:

$$(1 + j_t^i)u'(c_t) = \beta v^k(\underbrace{a_t}_{=k_{t+1}}). \tag{4.3.5}$$

The envelope theorem tells us that the marginal value of capital does not depend on its effect on the investment policy function $a(k)$:

$$\begin{aligned} v^k(k_t) &= (F'(k_t) + j^i(a_t - k_t, k_t) - j^k(a_t - k_t, k_t))u'(c_t) \\ &\equiv \lambda_t, \end{aligned} \tag{4.3.6}$$

(where notice that λ_t does not have the simple interpretation of a share price as in the [marginal q model](#) section because here it involves $u'(c_t)$).

Substituting period $t+1$'s version of (4.3.6) into (4.3.5) allows us to rewrite the Euler equation in the form:

$$\begin{aligned} u'(c_t)(1 + j_t^i) &= \beta(F'(k_{t+1}) + j_{t+1}^i - j_{t+1}^k)u'(c_{t+1}) \\ u'(c_t) &= \beta \underbrace{(F'(k_{t+1}) + j_{t+1}^i - j_{t+1}^k)}_{R_{t+1}} u'(c_{t+1}). \end{aligned} \tag{4.3.7}$$

⁶This section descends from a handout Ilhan Ozen wrote up from a 2005 final exam question, revised by Damiano Sandri in 2009 for the growth sequence.

⁷Solving (4.3.1) for c_t and substituting this expression in the place of c_t in the Bellman equation, and then substituting a_t for k_{t+1} .

This economy reduces to a standard Ramsey model when the cost of adjustment parameter is set to $\omega = 0$, because all the j terms disappear so that the interest factor becomes the usual $R_{t+1} = F'(k_{t+1}) = 1 + f'(k_{t+1})$. The presence of adjustment costs does not change the steady state of the model (because in steady state, adjustment costs are zero), but reduces the speed of convergence toward that steady state. This can be seen by considering the policy functions plotted in Figure 4.3, where the solid lines reflect the solution to a model with $\omega = 0$ (the standard Ramsey model) while the dashed lines reflect a model with a high cost of adjustment (the ‘ q -Ramsey’ model).

The differences between the solid and the dashed loci indicate that a faster rate of convergence to the steady state requires a high level of i below the steady state at which $k = \check{k}$ and low level of i when k is above $\check{k}$. Higher adjustment costs work against fast convergence, since, when k is below the steady state (and positive investment is needed to increase k toward $\check{k}$, adjustment costs reduce investment, while they increase investment (making it less negative) when k is above the equilibrium. In both cases, the difference occurs because adjusting capital involves convex costs, and thus it is optimal to proceed slowly in moving the capital stock to minimize those costs. Interestingly, even though the optimal choices of investment and consumption change quite substantially in the model with a larger adjustment cost parameter, the actual size of costs of adjustment borne is quite modest (the dashed line for j_t is barely distinguishable from the horizontal axis except very far from the steady state). This tells us that even if the observed costs paid are not very large, those costs can have a large effect in changing behavior away from the frictionless optimum.

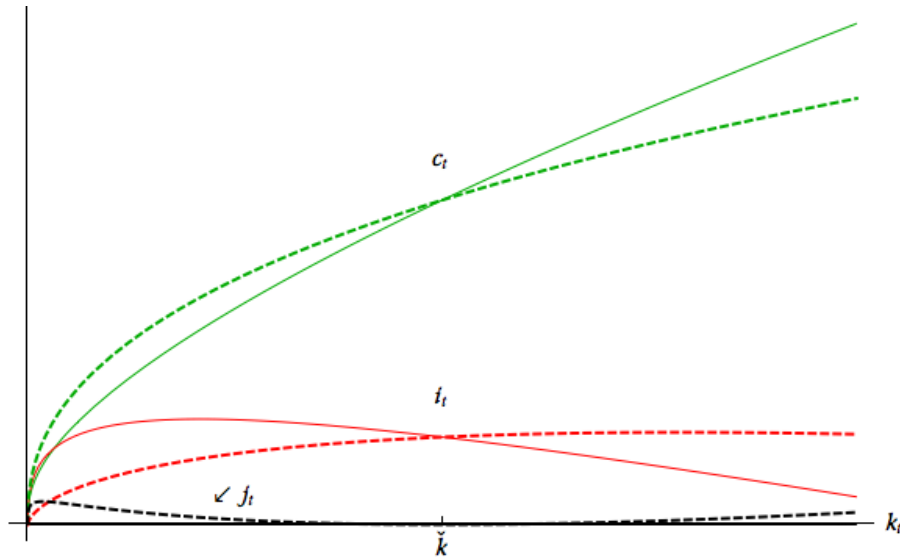
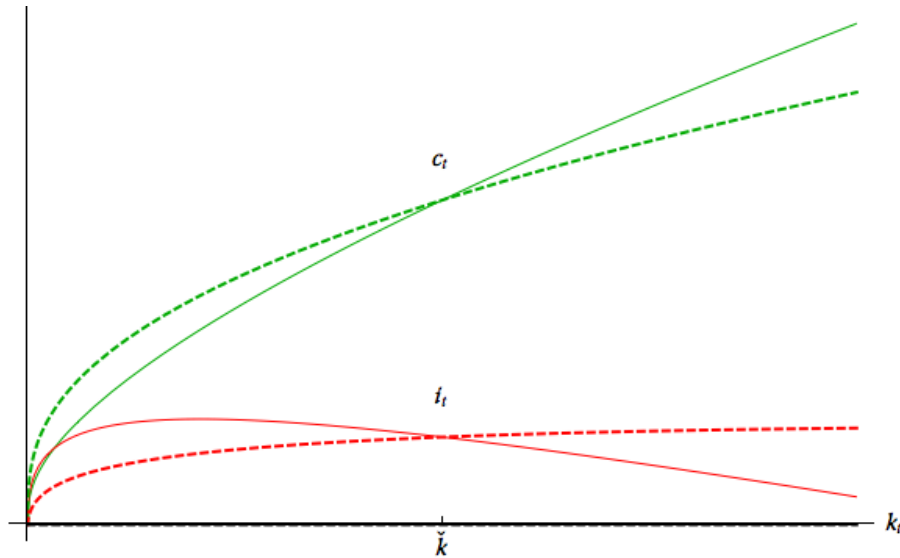


Figure 4.3: Solid Loci: Standard Ramsey Model; Dashed: With Costs of Adjustment

Increasing the desired degree of consumption smoothing, captured by the coefficient of relative risk aversion ρ , leads to similar implications. Figure 4.4 shows that a higher ρ (the dashed loci), implies again lower investment below the steady state and higher above it. This is now caused by a low intertemporal elasticity of substitution: if the economy falls below steady state, a larger ρ implies that the representative agent is less willing to cut consumption in order to boost investment and quickly return to the steady state. Similarly, the increase in consumption above the steady state is more moderate, thus leading to a smaller reduction in investment and a more gradual return to equilibrium.

By comparing the policy functions, we have thus seen that either an intensified consumption smoothing motive (higher ρ) and or a stronger investment smoothing motive (higher ω) have similar implications: they restrain sharp adjustments to consumption and investment, thus slowing down the speed of convergence to the steady state.

We now consider the responses of the model to several shocks, starting from steady state. Figure 4.5 shows the economy's dynamics following the destruction of part of the capital stock. In the standard Ramsey $\omega = 0$ model (black), this leads to an increase in the marginal productivity of capital which boosts investment. In the $\omega > 0$ model with adjustment costs (red), the level of investment actually falls. This is because costs of adjustment are assumed to be relative to the size of the capital stock,


 Figure 4.4: Higher ρ (Dashed Loci) Has a Similar Effect to Adjustment Costs

and with a shrunken capital stock the original level of investment would incur very large costs of adjustment. Investment therefore drops to a level that is large relative to the (shrunken) capital stock but nevertheless smaller than its initial level. Even this lower investment level, though, is large relative to the new lower level of the capital stock, and so the capital stock rises back toward the original equilibrium, just more slowly than in the frictionless model.

Consumption drops due to the negative wealth effect and the need to finance investment. But since investment is lower initially in the model with investment costs, consumption can be higher initially (the first red consumption dot is above the first black one, post-shock).

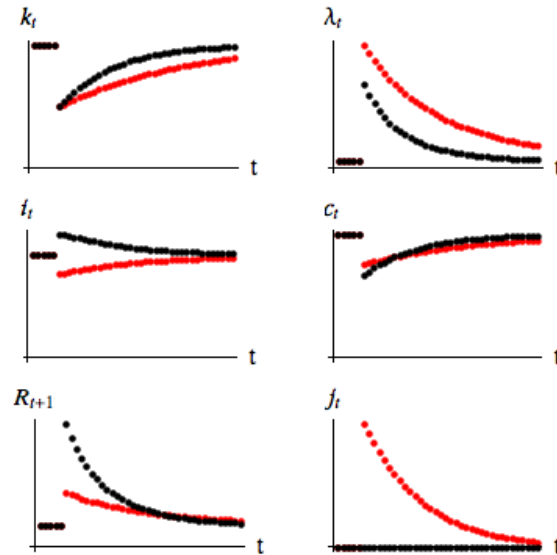

 Figure 4.5: Impulse response functions to 50% destruction of the capital stock $\omega = 0$ (standard Ramsey) in black; $\omega > 0$ (q -Ramsey) in red

Figure 4.6 shows the dynamics triggered by an increase in patience, captured by a permanent rise in β . The most striking difference is in the interest factor R .

In the Ramsey model with no investment costs, the interest rate is simply the marginal product of capital. Here, it must also take account of costs of adjustment. Since costs of adjustment are high when the economy is trying to change the size of the capital stock, the interest rate is lower.

This result is interesting because one problem with using the Ramsey model for studying business cycle dynamics is that the aggregate capital stock barely moves at all over such a short time period

as a business cycle, so the non- q Ramsey model has no hope of matching empirical interest rate fluctuations.

Adding costs of adjustment allows much bigger movements in R and thus gives the model a fighting chance.

Given the lower interest rate (and its implications through the consumption Euler equation), the growth rate of consumption after the increase in patience will be less than in the standard Ramsey model.

Even though consumption drops less, λ_t rises more. Recall that λ_t is a composition of the marginal utility of consumption and the “share price” of ownership of a unit of capital.

The extra rise in λ reflects the fact that the existing capital is more valuable in a period when the rate of investment will be high (going forward), so the market value of a unit of “installed” capital rises to above the purchase price of a unit of capital (which is always 1). This can be interpreted as a boom in asset prices.

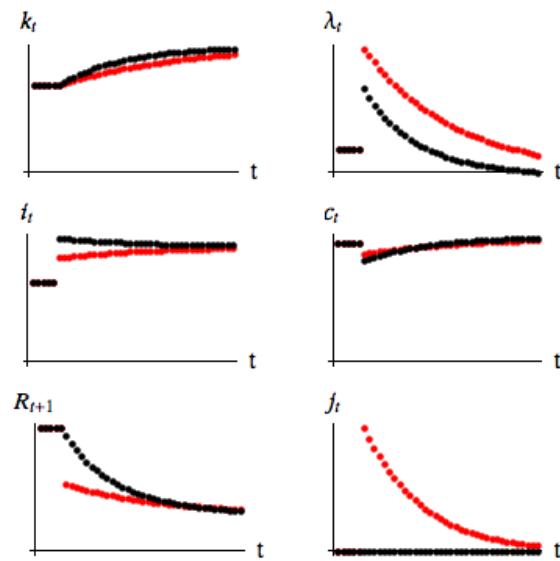


Figure 4.6: Impulse responses to an increase in patience (higher β)

Black: Ramsey; Red: q -Ramsey

4.4 Decentralizing the Ramsey/Cass-Koopmans Model

This section shows that under certain very special conditions the behavior of an economy composed of distinct individual households will replicate the social planner's solution to the Ramsey/Cass-Koopmans model.

4.4.1 The Consumer's Problem

Consider first the problem of an individual infinitely lived consumer indexed by i who has some predetermined set of expectations for how the aggregate net interest rate r_t and wage rate W_t will evolve.

At date t household i owns some capital $k_{t,i}$, and can in principle also borrow; designate the net debt of household i in period t as $d_{t,i}$; since we will be examining a perfect foresight solution with perfect capital markets, the interest rate on debt must match the rate of return on assets. This means that all that really matters is the household's total net asset position,

$$x_{t,i} = k_{t,i} - d_{t,i}. \quad (4.4.1)$$

Each household is endowed with one unit of labor, which it supplies exogenously, earning a wage rate $W_{t,i}$.

Each household solves:

$$\max \int_0^\infty u(c_{t,i}) e^{-\vartheta t} dt \quad (4.4.2)$$

subject to the budget constraint

$$\dot{x}_{t,i} = W_t + r_t x_{t,i} - c_{t,i}. \quad (4.4.3)$$

Integrating the household's dynamic budget constraint and assuming a no-Ponzi-game transversality condition yields the intertemporal budget constraint, which says that the present discounted value of consumption must match the PDV of labor income plus the current stock of net wealth:

$$\mathbb{P}_{0,i}(c) = \mathbb{P}_{0,i}(W) + x_{0,i}. \quad (4.4.4)$$

The formulas for these PDV's are a bit awkward because they must take account of the fact that interest rates are varying over time.

To make the formulas a bit simpler, define the compound interest factor

$$\mathbf{R}_t^{-1} = \exp\left(-\int_0^t r_\tau d\tau\right), \quad (4.4.5)$$

which is simply the compound interest term needed to convert a value at date t to its PDV as of time 0.

With this definition in hand we can write the IBC as

$$\int_0^\infty c_{t,i} \mathbf{R}_t^{-1} dt = h_{0,i} + x_{0,i} \quad (4.4.6)$$

where $h_{0,i}$ is human wealth,

$$h_{0,i} = \int_0^\infty W_t \mathbf{R}_t^{-1} dt. \quad (4.4.7)$$

Each household solves the standard optimization problem taking the future paths of wages and interest rates as *given*. Thus the Hamiltonian⁸ is

$$\mathcal{H}(c_{t,i}, x_{t,i}, \lambda_{t,i}) = u(c_{t,i}) + (r_t x_{t,i} + W_t - c_{t,i}) \lambda_{t,i} \quad (4.4.8)$$

⁸See the [RCK model](#) section for the discounted Hamiltonian optimality conditions and the [discrete and continuous time](#) appendix for the intuition of the logic behind the Hamiltonian.

which implies that the first optimality condition is the usual $u'(c_{t,i}) = \lambda_{t,i}$.
The second optimality condition is

$$\begin{aligned}\dot{\lambda}_{t,i} &= \vartheta \lambda_{t,i} - (\partial \mathcal{H} / \partial x) \\ \dot{\lambda}_{t,i} &= \vartheta \lambda_{t,i} - \lambda_{t,i} r_t \\ \dot{\lambda}_{t,i} / \lambda_{t,i} &= (\vartheta - r_t)\end{aligned}\tag{4.4.9}$$

leading eventually to the usual first order condition for consumption:

$$\dot{c}_{t,i} / c_{t,i} = \rho^{-1}(r_t - \vartheta).\tag{4.4.10}$$

Note (for future use) that the RHS of this equation does not contain any components that are idiosyncratic: The consumption growth rate will be identical for every household. The same is true of the expression for human wealth, equation (4.4.7).

4.4.2 The Firm's Problem

Now we assume that there are many perfectly competitive small firms indexed by j in this economy, each of which has a production function identical to the aggregate Cobb-Douglas production function.

Perfect competition implies that individual firms take the interest rate $\hat{r}_t$ and wage rate $\hat{W}_t$ to be exogenous. Hence firms solve

$$\max_{\{K_{t,j}, L_{t,j}\}} F(K_{t,j}, L_{t,j}) - \hat{W}_t L_{t,j} - \hat{r}_t K_{t,j}\tag{4.4.11}$$

where $\hat{r}_t$ and $\hat{W}_t$ are the rental rates for a unit of capital and a unit of labor for one period. Note that, dividing by $L_{t,j}$, this is equivalent to

$$\max_{\{k_{t,j}\}} \underbrace{k_{t,j}^\alpha}_{f(k_{t,j})} - \hat{W}_t - \hat{r}_t k_{t,j}.\tag{4.4.12}$$

The first order condition for this problem implies that

$$f'(k_{t,j}) = \hat{r}_t.\tag{4.4.13}$$

Under perfect competition firms must make zero profits in equilibrium, which means, by Eulers Theorem, that:

$$f(k_{t,j}) = \hat{W}_t + \hat{r}_t k_{t,j}.\tag{4.4.14}$$

4.4.3 Equilibrium At a Point in Time

Thus far, we have solved the consumer's and the firm's problems from the standpoint of atomistic individuals. It is now time to consider the behavior of an aggregate economy composed of consumers and firms like these.

We assume that the population of households and firms is distributed along the unit interval and the population masses sum to one, as per the [Aggregation](#) appendix. Thus, aggregate assets at time t can be defined as the sum of the assets of all the individuals in the economy at time t ,

$$X_t = \int_0^1 x_{t,i} di\tag{4.4.15}$$

while per capita assets are aggregate assets divided by aggregate population,

$$x_t = X_t / 1.\tag{4.4.16}$$

Similarly, normalizing the population of firms to one yields

$$k_t = K_t/1. \quad (4.4.17)$$

Up to this point, we have allowed for the possibility that different households might have different amounts of net worth. We now impose the assumption that every household is identical to every other household. This assumption rules out the presence of any debt in equilibrium (if all households are identical, they cannot all be in debt: who would they owe the money to?). Indeed, in this case, the aggregate capital stock per capita will equal the aggregate level of net worth, $k_t = x_t$.⁹

Thus, households' expectations about W_t and r_t determine their saving decisions, which in turn determine the aggregate path of k_t .

There is one important subtlety here, however. In writing the consumer's budget constraint, we designated r_t as the net amount of income that would be generated by owning one more unit of net worth (e.g. capital). But if we have depreciation of the capital stock, the net return to capital will be equal to the marginal product *minus* depreciation. The discussion of the firm's optimization problem did not consider depreciation because the firms do not own any capital;

instead, they make a payment $\hat{r}_t$ to the households for the privilege of using the households' capital. Thus the net increment to a household's wealth if the household holds one more unit of capital will be

$$r_t = \hat{r}_t - \delta. \quad (4.4.18)$$

There is no depreciation of labor, so the labor market equilibrium will be

$$\begin{aligned} W_t &= \hat{W}_t \\ &= f(k_t) - \hat{r}_t k_t. \end{aligned} \quad (4.4.19)$$

4.4.4 The Perfect Foresight Equilibrium

We assume that every household knows the aggregate production function, and understands the behavior of all the other households and firms in the economy. Understanding all of this, suppose that households have some set of beliefs about the future path of the aggregate capital stock per capita $\{k_t\}_{t=0}^{\infty}$. This belief about k_t will imply beliefs about wages and interest rates as well $\{W_t, r_t\}_{t=0}^{\infty}$.

The final assumption is that the equilibrium that comes about in this economy is the "perfect foresight equilibrium." That is, consumers have the sets of beliefs such that, if they have those beliefs and act upon them, the actual outcome turns out to match the beliefs.

Note now that using the fact that $x_t = k_t$ in the perfect foresight equilibrium we can rewrite the household's budget constraint as

$$\begin{aligned} \dot{k}_t &= r_t k_t + W_t - c_t \\ &= (\hat{r}_t - \delta)k_t + W_t - c_t. \end{aligned} \quad (4.4.20)$$

Reproducing from (4.4.10),

$$\begin{aligned} \dot{c}_t/c_t &= \rho^{-1}(r_t - \vartheta) \\ &= \rho^{-1}(\hat{r}_t - \delta - \vartheta). \end{aligned} \quad (4.4.21)$$

Now compare these to the equations derived for the social planner's problem (with population growth and productivity growth zero) in a previous section:

$$\begin{aligned} \dot{k}_t &= f(k_t) - \delta k_t - c_t \\ &= \hat{r}_t k_t + W_t - \delta k_t - c_t \\ &= (\hat{r}_t - \delta)k_t + W_t - c_t \end{aligned} \quad (4.4.22)$$

⁹The results do not change if we permit differences in the levels of wealth across households, but this is because we are assuming CRRA utility, perfect certainty, perfect capital markets, and various other things. When any of these assumptions is relaxed, the distribution of assets does matter. For exploration of this more complex and realistic framework, see Carroll [1992], Aiyagari [1994], Krusell and Smith [1998], Carroll [2000b].

and

$$\dot{c}_t/c_t = \rho^{-1}(f'(k_t) - \delta - \vartheta). \quad (4.4.23)$$

Since the equilibrium value of $\hat{r}_t = f'(k_t)$, (4.4.23) = (4.4.21). And (4.4.20) is identical to (4.4.22). Thus, aggregate behavior of this economy is identical to the behavior of the social planner's economy!

This is a very convenient result, because it means that if we are careful about the exact assumptions we make we can often solve a social planner's problem and then assume that the solution also represents the results that would obtain in a decentralized economy.

The social planner's solution and the decentralized solution are the same because they are maximizing the same utility function with respect to the same factor prices (r_t and W_t).

When will the decentralized solution *not* match the social planner's solution? One important case is when there are externalities in the behavior of individual households; another possible case is where there is idiosyncratic risk but no aggregate risk; basically, whenever the household's budget constraint or utility function differs in the right ways from the aggregate budget constraint or the social planner's preferences, there can be a divergence between the two solutions.

4.5 The Romer (1986) Model of Growth

Romer [1986] relaunched the growth literature with a paper that presented a model of increasing returns in which there was a stable positive equilibrium growth rate that resulted from endogenous accumulation of knowledge. This was an important break with the existing literature, in which technological progress had largely been treated as completely exogenous.¹⁰

In Romer's model, firm j 's production function is of the form

$$y_{t,j} = A_t F(k_{t,j}, \ell_{t,j}) \quad (4.5.1)$$

where aggregate output-augmenting technological progress is captured by A_t . Its capital accumulates without depreciation,

$$\dot{k}_{t,j} = i_{t,j}. \quad (4.5.2)$$

Firms and individuals are distributed along the unit interval with a total mass of 1, as in the [Aggregation](#) appendix (and, importantly, there is no population growth). Thus, aggregate investment is, e.g.,

$$I_t = \int_0^1 i_{t,j} dj. \quad (4.5.3)$$

Romer assumes that the aggregate stock of knowledge in the economy is proportional to the cumulative sum of past aggregate investment

$$\Xi_t = \int_{-\infty}^t I_v dv \quad (4.5.4)$$

which, not coincidentally, is identical to the size of the aggregate capital stock,

$$K_t = \int_{-\infty}^t I_v dv. \quad (4.5.5)$$

Romer makes the crucial assumption that the effect of the stock of knowledge determines productivity via

$$A_t = \Xi_t^\eta \quad (4.5.6)$$

where $\eta < 1$. Thus, suppressing the t subscript, the firm-level Cobb-Douglas production function can be written

$$y_j = k_j^\alpha \ell_j^{1-\alpha} \Xi^\eta \quad (4.5.7)$$

which is CRS at the firm level in (k, ℓ) holding aggregate knowledge Ξ fixed.

Aggregate output is

$$Y = K^\alpha L^{1-\alpha} \Xi^\eta \quad (4.5.8)$$

Dividing by the size of the labor force L (or, equivalently, normalizing to $L = 1$), we have

$$y = k^\alpha \Xi^\eta. \quad (4.5.9)$$

Now assume that households maximize a typical CRRA utility function, but each household ignores the trivial effect its own investment decision has on aggregate knowledge. Thus from the individual firm/consumer's perspective, the marginal product of capital is $\alpha k_{t,i}^{\alpha-1} \ell_{t,i}^{1-\alpha} \Xi_t^\eta$. If we normalize the model by assuming that the aggregate quantity of labor adds up to $L_t = 1$, we can set up and solve the Hamiltonian to obtain

$$\dot{c}_{t,i}/c_{t,i} = \rho^{-1}(\alpha k_{t,i}^{\alpha-1} \ell_{t,i}^{1-\alpha} \Xi_t^\eta - \vartheta). \quad (4.5.10)$$

¹⁰See also the prescient paper by Arrow [1962]; Alfred Marshall articulated similar ideas in the late 19th century.

But if all households are identical and $\Xi_t = K_t$, this means that aggregate consumption per capita evolves according to

$$\begin{aligned}\dot{c}_t/c_t &= \rho^{-1}(\alpha k_t^{\alpha-1} \Xi_t^\eta - \vartheta) \\ &= \rho^{-1}(\alpha k_t^{\alpha+\eta-1} - \vartheta).\end{aligned}\tag{4.5.11}$$

A balanced growth path can occur in this economy if $\alpha + \eta = 1$, in which case

$$\dot{c}_t/c_t = \rho^{-1}(\alpha - \vartheta)\tag{4.5.12}$$

so there is constant growth forever at a rate that depends on the degree of impatience and capital's share in output.

Note finally that the steady-state growth rate that would be chosen by the social planner is

$$\dot{c}_t/c_t = \rho^{-1}(\alpha + \eta - \vartheta),\tag{4.5.13}$$

because the social planner would take into account the fact that the externalities imply that there are higher returns to capital accumulation at the social level than at the individual level.

Thus, this model implies that capital accumulation should be subsidized if the social planner wants to induce the private economy to move toward the social optimum.

4.6 Generic Analysis of Endogenous Growth Models

The neoclassical theory of economic growth, as formulated by Solow

Solow [1956], and Cass [1965]-Koopmans [1965], attributes virtually all long-run growth to technological

progress. The level of technology is taken to be an exogenously growing factor outside the model. So the baseline model of growth says that growth is caused mostly by a factor that is not in the model.

It is important to understand why Solow and others made this assumption. The answer is that models of perfect competition are the simplest existing models of firm behavior, with well-understood implications.

But models with perfect competition require constant returns to scale, and say that all factors of production are paid their marginal product; and Eulers Theorem says that the sum of the factor payments exhausts all output:

$$Y = F_K K + F_L L. \quad (4.6.1)$$

Thus, the perfectly competitive firm has no money left over with which to finance basic research, invent patentable technologies, or do anything other than meet the payroll of production workers and pay the cost of capital. Thus, no firm can afford to pay for technological research, and there is therefore no alternative to the assumption that technological progress occurs exogenously.

This unsatisfactory situation persisted for a long time, but a famous paper by Paul Romer [1986] led the way to the formulation of a new generation of models that allow a role for investment in knowledge to affect growth.

The Romer paper spawned a great deal of further theoretical work by a host of others, but a penetrating paper by Sergio Rebelo [1991] provided a succinct summary of the key feature of all of these models. This section summarizes the Rebelo point as interpreted by Barro and Sala-i-Martin [1995].

4.6.1 The Key Point

Rebelo's key observation is as follows. Consider the class of models with a Cobb-Douglas aggregate production function in capital and labor:

$$Y_t = AK_t^\alpha L_t^\nu \quad (4.6.2)$$

where no restriction is made on the ν and α coefficients.

(Recall that Solow, Cass, and Koopmans all assumed $\nu + \alpha = 1$;

Rebelo is relaxing this restriction). Now suppose for simplicity that in this economy saving is a constant proportion of gross income.

In continuous time, the growth of the capital stock is given by

$$\dot{K}_t = sAK_t^\alpha L_t^\nu - \delta K_t. \quad (4.6.3)$$

Suppose further that population growth is constant at

$$\dot{L}_t/L_t = \xi, \quad (4.6.4)$$

and as usual define per-capita variables as the aggregate normalized by population, e.g. $k_t = K_t/L_t$. Then the aggregate per-capita accumulation equation can be rewritten

$$\begin{aligned} \dot{k}_t &= sAk_t^\alpha L_t^{\nu+\alpha-1} - (\delta + \xi)k_t \\ \dot{k}_t/k_t &= sAk_t^{\alpha-1} L_t^{\nu+\alpha-1} - (\delta + \xi). \end{aligned} \quad (4.6.5)$$

4.6.2 Characteristics of the Steady State

If this model has a steady-state growth rate, that rate must satisfy

$$\dot{k}_t/k_t = \gamma \quad (4.6.6)$$

for some constant γ . (The value of A does not affect the conclusions from here on, and so we will assume without loss of generality that $A = 1$). From (4.6.5), this implies that

$$\gamma = s k_t^{\alpha-1} L_t^{\nu+\alpha-1} - (\delta + \xi). \quad (4.6.7)$$

Take the time derivative of this equation to obtain

$$\begin{aligned} 0 &= s \left((\alpha - 1) k_t^{\alpha-2} \dot{k}_t L_t^{\nu+\alpha-1} + k_t^{\alpha-1} (\nu + \alpha - 1) \dot{L}_t L_t^{\nu+\alpha-2} \right) \\ &= k_t^{\alpha-1} L_t^{\nu+\alpha-1} \left((\alpha - 1) \dot{k}_t/k_t + (\nu + \alpha - 1) (\dot{L}_t/L_t) \right) \\ 0 &= (\alpha - 1) \dot{k}_t/k_t + (\dot{L}_t/L_t) (\nu + \alpha - 1) \\ &= (\alpha - 1) \gamma + \xi (\nu + \alpha - 1). \end{aligned} \quad (4.6.8)$$

Using this equation, we can construct a complete catalog of the possible circumstances under which steady-state growth γ can be different from zero endogenously.

Possibilities for Steady State Growth

1. $\nu + \alpha = 1$ (Constant Returns Models)

- (a) $\{\nu, \alpha\} < 1 \rightarrow \gamma = 0$ (Solow case)
- (b) $\nu = 0, \alpha = 1$: Rebelo [1991] AK growth model

2. $\nu + \alpha > 1$ (Increasing Returns Models)

- (a) $\{\nu, \alpha\} < 1$

- i. $\xi > 0 \rightarrow \gamma = \overbrace{\left(\frac{\nu + \alpha - 1}{1 - \alpha} \right)}^{>0} \xi$
- ii. $\xi = 0 \rightarrow \gamma = 0$

- (b) $\nu > 0, \alpha = 1$

- i. $\xi > 0 \rightarrow 0 = \xi(\nu + \alpha - 1)$ which is not satisfied for any γ
- ii. $\xi = 0 \rightarrow 0 = 0$ which can be satisfied for any γ

So whatever the details of endogenous growth models may be, in the end any model that generates perpetual growth without exogenous technological progress must be mathematically reducible to a form like that of either 1.b., 2.a.i., or 2.b.ii.

It is worth delving a bit further into why the Solow case cannot generate perpetual growth. The answer can be understood using the Solow growth accounting framework, which says that there are only three sources of long-run growth: technology, labor, and capital. Thus, there are only two potential sources of growth of output per unit of labor: Technology and an increase in the capital/labor ratio. But if $\alpha < 1$, the gross marginal product of capital approaches zero as the capital/labor ratio approaches infinity; subtracting out depreciation, eventually the net marginal product of capital becomes negative. Thus, capital accumulation can sustain growth only so long.

The bottom line is that there are only two configurations of the model that are capable of generating perpetual growth in a way that makes any sense: 1.b. (the Rebelo AK model) and 2.a.ii.

What this means is that any model that aims to permit perpetual long-run growth must ultimately boil down to a structure in which there are constant returns to scale for some set of factors of production that can jointly be accumulated forever.

4.7 The Rebelo AK Growth Model

Rebelo [1991] examines a model in which a social planner maximizes the discounted sum of utility in an economy with an AK production function:

$$\max_{\{C\}_0^\infty} \int_0^\infty u(C) e^{-\vartheta t} dt \quad (4.7.1)$$

subject to

$$\dot{K} = AK - C, \quad (4.7.2)$$

where we have assumed zero population growth and zero depreciation to make the analysis less cluttered.

This problem can be solved with the same Hamiltonian apparatus we used to solve the Ramsey/Cass-Koopmans model. In particular, with CRRA utility with risk aversion ρ one can show that optimal behavior requires

$$\dot{C}/C = \rho^{-1}(A - \vartheta). \quad (4.7.3)$$

Note that this equation comes about because the marginal product of capital in this model is always $A = r$, because $f'(K) = A$. Note further that according to this equation, the growth rate of consumption is always the same; unlike the Cass-Koopmans model with a normal production function, this model has no transitional dynamics.

We can also use (4.7.2) to obtain an expression for the steady-state growth rate:

$$\dot{K}/K = A - C/K. \quad (4.7.4)$$

If the model has a steady-state growth rate of $\gamma = \dot{K}/K$, this equation implies that

$$C/K = A - \gamma. \quad (4.7.5)$$

This is a model with a constant saving rate, because

$$\begin{aligned} S/AK &= (AK - C)/AK \\ &= 1 - A^{-1}(C/K) \\ &= \gamma/A \\ \gamma &= A(S/AK). \end{aligned} \quad (4.7.6)$$

Thus, the steady-state growth rate in a Rebelo economy is directly proportional to the saving rate. A further requirement for the Rebelo model to have a well-defined solution is that

$$\rho^{-1}(A - \vartheta) < A. \quad (4.7.7)$$

Recalling that A is effectively the real interest rate in this model, this equation can be interpreted as the ‘impatience’ condition that we imposed in the infinite horizon perfect foresight consumption model. In fact, the Rebelo AK model is essentially just a way of reinterpreting the perfect foresight infinite horizon consumption problem as a model for economic growth.

The principal distinction is that we usually use the perfect foresight infinite horizon model to analyze circumstances where the agent has both labor and capital income, whereas the Rebelo model rules out labor income by assumption.

4.8 The Lucas Growth Model

Lucas [1988] presents a growth model in which output is generated via a production function of the form

$$Y = AK^\alpha(\ell hL)^{1-\alpha} \quad (4.8.1)$$

where Y , A , and K are as usually defined and $0 < \alpha < 1$, where ℓ is defined as the proportion of total labor time spent working, and h is what Lucas calls the stock of ‘human capital.’

The production function can be rewritten in per-capita terms as

$$y = Ak^\alpha(\ell h)^{1-\alpha} \quad (4.8.2)$$

which is a constant returns to scale production function in k and ℓh .

Capital accumulation proceeds via the usual differential equation,

$$\dot{k} = y - c - (\xi + \delta)k, \quad (4.8.3)$$

while h accumulates according to

$$\begin{aligned} \dot{h} &= \phi h(1 - \ell) \\ \dot{h}/h &= \phi(1 - \ell). \end{aligned} \quad (4.8.4)$$

4.8.1 Discussion

Before analyzing the model, an aside. Mankiw [1995] has persuasively argued for defining ‘knowledge’ as the sum total of technological and scientific discoveries (what is written in textbooks, scholarly journals, websites, and the like), and defining ‘human capital’ as the stock of knowledge that has been transmitted from those sources into human brains via studying.

Recall Rebelo [1991]’s key insight about endogenous growth models: In order to produce perpetual growth, there must be a factor or a combination of factors that can be accumulated indefinitely without diminishing returns. Mankiw points out that since lifetimes are finite, there is a maximum limit to the amount of human capital that an individual can accumulate.

Thus, while increasing human capital (with more years of schooling, for example) may be able to extend the length of the transition period in a growth model, human capital accumulation cannot be the source of perpetual growth. It is more plausible to think that scientific knowledge can be accumulated indefinitely (though presumably there is some limit even to the accumulation of knowledge). These considerations suggest that models of endogenous growth should focus more on understanding the process of fundamental research and technological development than on human capital accumulation as Mankiw defines it.

With this distinction in mind, there are (at least) two interpretations of the Lucas model. One is at the aggregate level.

Here we can think of ℓ as the fraction of the population engaged in useful work to produce goods and services, while proportion $1 - \ell$ is not working in conventional boring jobs that require asking customers questions like “Would you like fries with that?” but instead is producing ‘knowledge’ by conducting scientific and technological research.

The other interpretation is at the level of an individual agent. Such an agent can be thought of as operating his or her own production function of the form in (4.8.2), where $(1 - \ell)$ is now interpreted as the proportion of the time this individual spends studying and ℓ is the time spent working.

From the point of view of Mankiw’s distinction, it is hard to interpret Lucas’s model as being either about human capital accumulation or about knowledge. It can’t be about human capital because h can be accumulated without bound, and without diminishing returns, neither of which makes sense for an individual. It can’t be about generalized knowledge, because the optimization problem reflects the return for an individual, while only a trivial proportion of total knowledge (in Mankiw’s sense) is contributed by any single individual.

Given these considerations, it probably makes more sense to think of the model as a tool for normative than for positive analysis.

4.8.2 The Solow Version

We analyze first the ‘Solow’ version of the model, in which the saving rate is exogenously fixed at s . Thus the capital accumulation equation becomes

$$\begin{aligned}\dot{k} &= sy - (\xi + \delta)k \\ \dot{k}/k &= s(y/k) - (\xi + \delta) \\ &= sAk^{\alpha-1}(\ell h)^{1-\alpha} - (\xi + \delta) \\ &= sA(k/h)^{\alpha-1}\ell^{1-\alpha} - (\xi + \delta).\end{aligned}\tag{4.8.5}$$

This equation tells us that the steady-state growth rate in this model (if one exists) requires a constant ratio of k to h . Thus, k and h must be growing at the same rate in equilibrium.

Further insight can be obtained by defining $\hat{A} = A\ell^{1-\alpha}$ and rewriting the per-capita production function as

$$y = \hat{A}k^{\alpha}h^{1-\alpha}.\tag{4.8.6}$$

If we define a measure of ‘broad capital’ as the combination of physical capital and human capital,

$$\kappa \equiv k^{\alpha}h^{1-\alpha},\tag{4.8.7}$$

the model becomes

$$y = \hat{A}\kappa.\tag{4.8.8}$$

So if ℓ is constant and if k and h are growing at the same rate, then the exponent on ‘broad capital’ is 1, and we are effectively back at the usual Rebelo AK model.

The key assumption that permits this to work is the accumulation equation for human capital, which is itself like an AK model, in the sense that the exponent on human capital in the accumulation equation for human capital is one. Human capital can be accumulated without bound and without diminishing returns.

4.8.3 The Ramsey/Cass-Koopmans Version

Lucas does not examine the Solow version of the model with a constant saving rate, but instead the version in which a social planner solves for the optimal perfect foresight paths of the two state variables k and h . It’s not worth going through the math here; I’ll just present the conclusion, which is that the steady-state growth rate is

$$\dot{c}/c = \rho^{-1}(\phi - \theta).\tag{4.8.9}$$

Note that this confirms the crucial role of the CRS accumulation equation for human capital: The key parameter that corresponds to the interest rate is ϕ , the parameter that determines the efficiency of human capital accumulation in equation (4.8.4).

Lucas also solves a version of the model in which there is an externality to human capital. The idea here is that each person is more productive if they are surrounded by other people with high levels of human capital. Specifically, in this version of the model the individual’s production function is

$$y_i = Ak_i^{\alpha}(\ell_i h_i)^{1-\alpha} \bar{h}^{\psi}\tag{4.8.10}$$

where $\bar{h}$ is average human wealth in the population (and the other variables reflect the values for the individual).

Working through the decentralized solution, Lucas shows that the steady-state growth rate of human capital for an individual consumer will be

$$\gamma_h = \left(\frac{\rho^{-1}(\phi - \theta)}{1 + \psi(1 - 1/\rho)/(1 - \alpha)} \right). \quad (4.8.11)$$

Since every individual is assumed to be identical, the growth rate of aggregate human capital (and everything else) is the same as the rate for this individual.

It is easy to see that if there is no externality to human capital accumulation (that is, if $\psi = 0$), this solution is identical to (4.8.9). If there is an externality, its effect depends on whether $1/\rho$ is greater than, equal to, or less than 1. This is because the effect depends on whether the externality causes the saving rate to rise or to fall (since in endogenous growth models, saving is the source of all growth). The $\bar{h}$ externality is like an increase in the interest rate, and thus its effect will be determined by the balance between the income and substitution effects. We know that for $\rho = 1$ the income and substitution effects exactly offset each other, leaving consumption unchanged in response to an increase in the interest rate, which is why (4.8.11) collapses to (4.8.9) for $\rho = 1$. If consumers are very willing to cut current consumption in exchange for higher future consumption (that is, if the intertemporal elasticity of substitution ($1/\rho$) is greater than 1), then the externality boosts saving and therefore growth. If consumers have an intertemporal elasticity of less than one, the income effect outweighs the substitution effect, saving falls, and growth is slower.

Lucas also shows that this decentralized solution is suboptimal, because individual consumers do not obtain the full benefits to society of increasing their own stock of knowledge. Devoting more time to h_i accumulation they increase $\bar{h}$, which benefits all others in the economy in addition to themselves. Lucas shows, unsurprisingly, that the socially optimal solution requires greater investment in human capital accumulation than is obtained in the decentralized model. He also derives an optimal subsidy to human capital accumulation that corrects the externality and induces households to invest the socially optimal amount in human capital.

4.9 The Blanchard (1985) Model of Perpetual Youth

This section analyzes a way to relax the standard assumption of infinite lifetimes in the Ramsey/Cass-Koopmans growth model. The trick, introduced by Blanchard [1985], is to assume that the economy is populated by agents who face a constant probability of death.

Thus, an agent who has lived a thousand years is no more likely to die in the next year than an agent who was born yesterday.

Time is measured continuously. If there is an instantaneous probability d of dying, the probability (as viewed by a person alive in period t) of still being alive (not dead) in period τ is

$$\aleph_t^\tau = e^{-d(\tau-t)}. \quad (4.9.1)$$

Assuming the instantaneous utility function is $u(c) = \log c$, and that the pure rate of time preference is ϑ , explain why the objective function at time t of an individual household in this model will be to maximize

$$\int_t^\infty (\log c_\tau) e^{-(\vartheta+d)(\tau-t)} d\tau. \quad (4.9.2)$$

where c_τ is the consumer's consumption at time τ . For convenience, you may wish to define

$$\hat{\vartheta} = \vartheta + d. \quad (4.9.3)$$

Answer:

A consumer with a pure time preference rate of zero will downweight the utility he receives conditional on being alive by the probability that he is still alive, yielding a discounted utility of

$$\begin{aligned} \int_t^\infty (\log c_\tau) \aleph_t^\tau e^{-\vartheta(\tau-t)} d\tau &= \int_t^\infty (\log c_\tau) e^{-\vartheta(\tau-t)} e^{-d(\tau-t)} d\tau \\ &= \int_t^\infty (\log c_\tau) e^{-(\vartheta+d)(\tau-t)} d\tau \\ &= \int_t^\infty (\log c_\tau) e^{-\hat{\vartheta}(\tau-t)} d\tau \end{aligned} \quad (4.9.4)$$

where $\hat{\vartheta} = \vartheta + d$.

A similar point holds in the discrete time model. A sensible thing to assume is that if you have died before $t + 1$, you get zero utility in $t + 1$ and after. Thus, in the two-period context, if the probability of death between $T - 1$ and T was zero we would have

$$V_{T-1} = \max u(C_{T-1}) + \left(\frac{1}{1 + \vartheta} \right) u(C_T) \quad (4.9.5)$$

while if there is a probability d of dying between T and $T + 1$ value would be:

$$\begin{aligned} V_{T-1} &= \max u(C_{T-1}) + (1 - d) \left(\frac{1}{1 + \vartheta} \right) u(C_T) + d\beta \cdot 0 \\ &= \max u(C_{T-1}) + \left(\frac{1 - d}{1 + \vartheta} \right) u(C_T) \\ &\approx \max u(C_{T-1}) + \left(\frac{1}{1 + \vartheta + d} \right) u(C_T) \end{aligned} \quad (4.9.6)$$

But behavior in this case is virtually indistinguishable from the behavior that would be induced if the consumer had a time preference rate of $\vartheta + d$. In continuous time, the approximation in (4.9.6) becomes exact.

If the probability of death is constant, the expected remaining life for an agent of any age is given by

$$\int_0^\infty d\tau e^{-d\tau} d\tau = d^{-1}. \quad (4.9.7)$$

which we will call the agent's 'horizon.' For example, if the chances of dying per year are $d = 1/50$, then the agent's horizon is 50 years.

We will assume that at every instant of time, a large cohort, whose size is normalized to be d , is born.

For an economy that has existed forever, explain why the formula for the aggregate population at time t , P_t , will be

$$P_t = d \int_{-\infty}^t \aleph_s^t ds \quad (4.9.8)$$

and show that this formula implies that the population is $P_t = 1$.

Answer:

The aggregate population will be the sum of the still-alive persons from all past generations.

The proportion of a population born at time s that is still living at time τ is $\aleph_s^\tau$ by equation (4.9.1).

Thus, the absolute size at time τ of a cohort of size d born at time s is $d\aleph_s^\tau = de^{-d(\tau-s)}$. The economy's

total population will be the sum of the populations of all the cohorts that are currently living. Since the economy has existed forever, there will be remaining members of every cohort back to $s = -\infty$.

Thus, indexing each cohort by a time index s , (4.9.8) is simply the sum of the populations of all currently living members of every generation.

Substituting the formula for $\aleph$, the integral becomes

$$\begin{aligned} P_t &= d \int_{-\infty}^t e^{-d(t-s)} ds \\ &= d \int_t^\infty e^{-d(s-t)} ds \\ &= d \int_0^\infty e^{-d\tau} d\tau \\ &= dd^{-1} \\ &= 1 \end{aligned} \quad (4.9.9)$$

where the second line comes from a change of variables $\tau = s - t$ and (4.9.9) follows from the hint.

Now some notation. We will define variable $x(s, t)$ as the value at date t of the variable x for a consumer who was born at date s .

Thus, $c(s, t)$ is consumption at t of a consumer born at s .

Suppose that the consumers in this economy do not have a bequest motive. If they have positive assets at the instant when they die, they are no happier than if they had zero assets. This means that if someone were willing to pay them something while they are still alive for the right to inherit their assets whenever they die, these consumers would happily take that deal.

For a consumer with wealth $w(s, t)$ who has probability of dying d , the flow value of the right to inherit that wealth is $dw(s, t)$. We will therefore assume that insurance companies exist that pay a consumer with wealth $w(s, t)$ an amount $dw(s, t)$ in exchange for the right to receive that consumer's wealth when he dies. (The insurance company will make zero profits).

But notice that from the standpoint of the consumer, this is equivalent to saying that the interest rate received on wealth is higher by amount d .

Now suppose the marginal product of capital in this perfectly-competitive economy is constant at r and suppose an agent born in s receives exogenous labor income in period t of $y(s, t)$.

This plus the insurance scheme implies that the agent's dynamic budget constraint is given by

$$\dot{w}(s, t) = (r + d)w(s, t) + y(s, t) - c(s, t). \quad (4.9.10)$$

Define the 'effective' interest rate as viewed by a consumer as $\hat{r} = r + d$.

Write the current-value Hamiltonian for the individual's maximization problem and use it to show that the growth rate of consumption in period t for a consumer born at time s is given by

$$\begin{aligned} \left(\frac{\dot{c}(s, t)}{c(s, t)} \right) &= \hat{r} - \hat{\vartheta} \\ &= r - \vartheta. \end{aligned} \quad (4.9.11)$$

Answer:

The current-value Hamiltonian is written

$$\mathcal{H}(c, w, \lambda) = \log c(s, t) + \lambda(\hat{r}w(s, t) + y(s, t) - c(s, t)) \quad (4.9.12)$$

so the first optimality condition $\partial \mathcal{H} / \partial c(s, t) = 0$ implies

$$\begin{aligned} 1/c(s, t) &= \lambda \\ -\dot{c}(s, t)/c(s, t)^2 &= \dot{\lambda} \end{aligned} \quad (4.9.13)$$

and the second optimization condition implies

$$\begin{aligned} \dot{\lambda} &= \hat{\vartheta}\lambda - \partial \mathcal{H} / \partial w(s, t) \\ &= \hat{\vartheta}\lambda - \hat{r}\lambda \\ \left(\frac{\dot{\lambda}}{\lambda} \right) &= \hat{\vartheta} - \hat{r} \\ \left(\frac{-\dot{c}(s, t)c(s, t)}{c(s, t)^2} \right) &= \hat{\vartheta} - \hat{r} \\ \left(\frac{\dot{c}(s, t)}{c(s, t)} \right) &= \hat{r} - \hat{\vartheta} \\ &= r - \vartheta. \end{aligned} \quad (4.9.14)$$

Use the first order condition for consumption and the intertemporal budget constraint implied by (4.9.10) to show that the level of consumption in time t for an individual born at time s is

$$c(s, t) = \hat{\vartheta}(w(s, t) + h(s, t)) \quad (4.9.15)$$

where recall that $\hat{\vartheta} = \vartheta + d$ and human wealth is

$$h(s, t) = \int_t^\infty (y(s, \tau) / \hat{\mathcal{R}}_t^\tau) d\tau \quad (4.9.16)$$

where

$$\hat{\mathcal{R}}_t^\tau = e^{\int_t^\tau (r_\mu + d) d\mu} \quad (4.9.17)$$

(this is simply the compound discount factor necessary to take account of time-varying interest rates; if interest rates are constant at r it reduces to the usual term $e^{-r(\tau-t)}$).

Answer:

The IBC says that the PDV of consumption must equal wealth plus the PDV of future labor income:

$$\begin{aligned}\int_t^\infty (c(s, \tau)/\hat{\mathcal{R}}_t^\tau) d\tau &= w(s, t) + \int_t^\infty y(\tau, t)/\hat{\mathcal{R}}_t^\tau d\tau \\ \int_t^\infty (c(s, \tau)/\hat{\mathcal{R}}_t^\tau) d\tau &= w(s, t) + h(s, t).\end{aligned}\tag{4.9.18}$$

But if $\dot{c}(s, t)/c(s, t) = \hat{r}_t - \hat{v}$ then

$$c(s, \tau) = c(s, t) e^{\int_s^\tau \hat{r}_\mu d\mu} e^{-\hat{v}(\tau-t)}\tag{4.9.19}$$

implying

$$\begin{aligned}\int_t^\infty (c(s, \tau)/\hat{\mathcal{R}}_t^\tau) d\tau &= \int_t^\infty c(s, t) e^{\int_s^\tau \hat{r}_\mu d\mu} e^{-\hat{v}(\tau-t)} e^{-\int_s^\tau \hat{r}_\mu d\mu} d\tau \\ &= c(s, t) \int_t^\infty e^{-\hat{v}(\tau-t)} d\tau \\ &= c(s, t)/\hat{v}\end{aligned}\tag{4.9.20}$$

so that the IBC becomes:

$$\begin{aligned}c(s, t)/\hat{v} &= w(s, t) + h(s, t) \\ c(s, t) &= \hat{v}(w(s, t) + h(s, t)).\end{aligned}\tag{4.9.21}$$

Suppose we define upper-case variables as the aggregate value across all generations currently living of the corresponding lower-case value, e.g.

aggregate consumption is

$$C(t) = \int_{-\infty}^t dc(\tau, t) \aleph_\tau^t d\tau.\tag{4.9.22}$$

(Easy) Show that the aggregate level of consumption in this economy is

$$C(t) = \hat{v}(W(t) + H(t)).\tag{4.9.23}$$

Answer:

$$\begin{aligned}C(t) &= \int_{-\infty}^t dc(\tau, t) \aleph_\tau^t d\tau \\ &= \int_{-\infty}^t d\hat{v}(w(\tau, t) + h(\tau, t)) \aleph_\tau^t d\tau \\ &= \hat{v}(W(t) + H(t))\end{aligned}\tag{4.9.24}$$

from the definition of $W(t)$ and $H(t)$.

Suppose all living agents in this economy receive the same noncapital income, $y(s, t) = Y(t)$. Since every member of the population has the same income, and the size of the population is one, aggregate income will also be $Y(t)$. Aggregate human wealth at t is therefore

$$H(t) = \int_t^\infty (Y(\tau)/\hat{\mathcal{R}}_t^\tau) d\tau\tag{4.9.25}$$

where

$$\dot{H}(t) = \hat{r}_t H(t) - Y(t).\tag{4.9.26}$$

The differential equation for aggregate wealth can be shown to be

$$\dot{W}(t) = w(t, t) - dW(t) + \int_{-\infty}^t \dot{w}(\tau, t) d e^{-d(t-\tau)} d\tau, \quad (4.9.27)$$

where $w(t, t) = 0$ is the wealth of newly born generations, $dW(t)$ is the wealth of those who are dying at the moment, and the last term is the change in wealth for those who neither die nor are born in this period. But (4.9.10) implies that

$$\int_{-\infty}^t \dot{w}(\tau, t) d e^{-d(t-\tau)} d\tau = (r + d)W(t) + Y(t) - C(t) \quad (4.9.28)$$

so (4.9.27) becomes

$$\dot{W}(t) = rW(t) + Y(t) - C(t). \quad (4.9.29)$$

Collecting, writing out $\hat{r} = r + d$ and $\hat{\vartheta} = \vartheta + d$, and dropping the (t) arguments gives us the following equations for aggregate variables:

$$\begin{aligned} C &= (d + \vartheta)(H + W) \\ \dot{H} &= (r + d)H - Y \\ \dot{W} &= rW + Y - C \end{aligned} \quad (4.9.30)$$

Use these equations to show that in this economy

$$\dot{C} = (r - \vartheta)C - d(d + \vartheta)W \quad (4.9.31)$$

Hint: Differentiate (4.9.30) and substitute out for H by solving (4.9.30) for H .

Answer:

Time differentiate (4.9.30) and substitute for $\dot{H}$, $\dot{W}$, and H to get

$$\begin{aligned} \dot{C} &= (d + \vartheta) [(r + d)H - Y + rW + Y - C] \\ &= (d + \vartheta) [(r + d)H + rW] - C(d + \vartheta) \\ &= (d + \vartheta) \left[(r + d) \left(\frac{C}{d + \vartheta} - W \right) + rW \right] - C(d + \vartheta) \\ &= (r + d)C + (d + \vartheta) [rW - (r + d)W] - C(d + \vartheta) \\ &= (r - \vartheta)C - (d + \vartheta)dW. \end{aligned} \quad (4.9.32)$$

Now assume there is a standard production function $F(K) = K^\alpha - \delta K$ and assume perfect competition so that the net interest rate r is equal to the net marginal product of capital,

$$r = F'(K) = \alpha K^{\alpha-1} - \delta \quad (4.9.33)$$

and the aggregate capital stock at time t is the same as aggregate nonhuman wealth, $K(t) = W(t)$. The aggregate accumulation equation is just the usual

$$\dot{K} = K^\alpha - \delta K - C. \quad (4.9.34)$$

Show that the following equation

$$d(d + \vartheta)K = C(\alpha K^{\alpha-1} - \delta - \vartheta) \quad (4.9.35)$$

describes the $\dot{C} = 0$ locus. Use this equation to show that

$$\begin{aligned} \lim_{C \rightarrow 0} K &= 0 \\ \lim_{C \rightarrow \infty} K &= ((\vartheta + \delta)/\alpha)^{1/\alpha-1}. \end{aligned} \quad (4.9.36)$$

Answer:

Rewriting the $\dot{C}$ equation as a function of K yields

$$\dot{C} = (\alpha K^{\alpha-1} - \vartheta - \delta)C - (d + \vartheta)dK. \quad (4.9.37)$$

The $\dot{C} = 0$ locus is therefore given by

$$\begin{aligned} 0 &= (\alpha K^{\alpha-1} - \vartheta - \delta)C - d(d + \vartheta)K/C \\ d(d + \vartheta)K/C &= (\alpha K^{\alpha-1} - \delta - \vartheta) \\ d(d + \vartheta)K &= C(\alpha K^{\alpha-1} - \delta - \vartheta) \end{aligned} \quad (4.9.38)$$

In the limit as $C \rightarrow 0$ this expression approaches

$$d(d + \vartheta)K = 0 \quad (4.9.39)$$

which can be true only if $\lim_{\{C \rightarrow 0\}} K = 0$.

On the other hand, as $C \rightarrow \infty$ we know that K must remain finite (because the DBC does not allow infinite accumulation of K ; infinite K would imply infinite depreciation which could never be paid for by a production function with diminishing marginal returns), which means that

$$\begin{aligned} \lim_{\{C \rightarrow \infty\}} \alpha K^{\alpha-1} &= (\vartheta + \delta) \\ K^* &= ((\vartheta + \delta)/\alpha)^{1/\alpha-1}. \end{aligned} \quad (4.9.40)$$

Using your results from the previous question and the aggregate dynamic budget constraint, draw the phase diagram for this model and discuss how and why it differs from the phase diagram for the standard Ramsey-Cass/Koopmans model. (please use K^* to designate the RCK steady-state capital stock and $\bar{K}$ to designate this model's steady-state K .)

Answer:

Using (4.9.34), the $\dot{K} = 0$ locus is

$$C = K^\alpha - \delta K \quad (4.9.41)$$

which yields the usual hump-shaped $\dot{K} = 0$ locus.

Rewriting the $\dot{C}$ equation as a function of K yields

$$\dot{C} = (\alpha K^{\alpha-1} - \vartheta - \delta)C - (d + \vartheta)dK. \quad (4.9.42)$$

The $\dot{C} = 0$ locus is therefore given by

$$\begin{aligned} 0 &= (\alpha K^{\alpha-1} - \vartheta - \delta)C - d(d + \vartheta)K/C \\ d(d + \vartheta)K/C &= (\alpha K^{\alpha-1} - \delta - \vartheta) \\ d(d + \vartheta)K &= C(\alpha K^{\alpha-1} - \delta - \vartheta) \end{aligned} \quad (4.9.43)$$

In the limit as $C \rightarrow 0$ this expression approaches

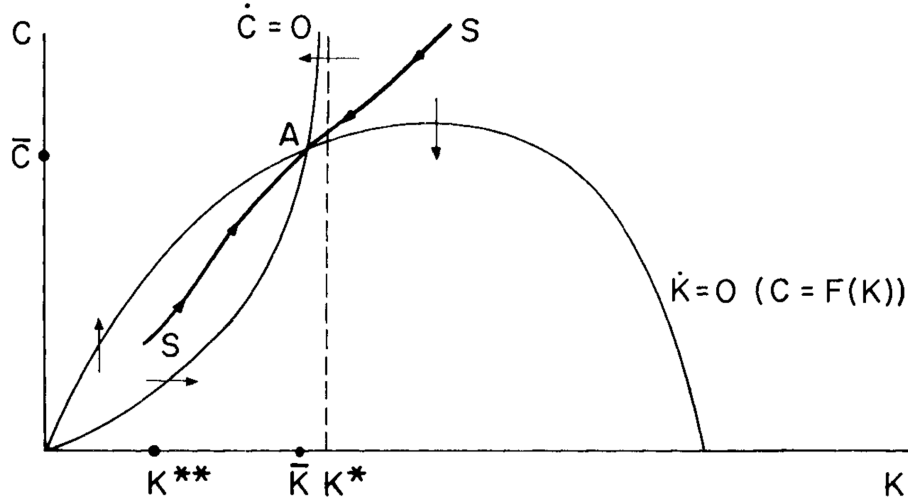
$$d(d + \vartheta)K = 0 \quad (4.9.44)$$

which can be true only if $\lim_{\{C \rightarrow 0\}} K = 0$.

On the other hand, as $C \rightarrow \infty$ we know that K must remain finite (because the DBC does not allow infinite

accumulation of K), which means that

$$\begin{aligned} \lim_{\{C \rightarrow \infty\}} \alpha K^{\alpha-1} &= (\vartheta + \delta) \\ K^* &= ((\vartheta + \delta)/\alpha)^{1/(\alpha-1)}. \end{aligned} \quad (4.9.45)$$



In the infinite horizon economy we have $d = 0$ and so the steady-state interest rate would be the K^* where $\alpha K^{\alpha-1} - \delta = \vartheta$. But since K/C and $d(d + \vartheta)$ are

strictly positive, in this finite-horizon economy we would have $\dot{C}/C < 0$ at $K = K^*$. It is clear therefore that in order for (4.9.38) to hold we will need $\alpha K^{\alpha-1}$

to be larger than it is at K^* , which is to say we need a higher steady-state interest rate, and thus we need a lower steady-state capital stock, which is depicted in the figure as $\bar{K}$.

This makes sense because the finite-horizon consumers in this economy discount the future more than the representative agent does, because they die but a representative agent does not.

These results are combined in the figure, which shows that the intersection of the $\dot{C} = 0$ locus intersects the $\dot{K} = 0$ locus at point A which corresponds to a lower level of the capital stock than in the infinite horizon model.

Now consider the introduction of a government that finances spending either by lump-sum taxes or by debt. Its dynamic budget constraint is

$$\dot{D} = rD + G - T \quad (4.9.46)$$

where D is government debt, G is government spending, and T is a lump-sum per capita tax. Defining

$$\mathcal{R}_t^s = e^{\int_t^s r_v dv} \quad (4.9.47)$$

as the compound interest factor between time t and time s , the government is also required to satisfy the transversality condition

$$\lim_{t \rightarrow \infty} D_t / \mathcal{R}_t^s = 0. \quad (4.9.48)$$

Consider the following fiscal policy experiment. Until time t there has been no government ($G_s = D_s = T_s = 0 \forall s < t$). At date t the government issues a quantity D of debt and announces that future lump sum taxes will be imposed in amounts exactly large enough to pay the interest on this debt (so subsequently, $\dot{D} = 0$ forever). The government rebates the proceeds of its sale of debt to the public as a per-capita lump sum of D per person. The government will never engage in any spending (aside from paying interest on the debt). Define the new variables

$$\begin{aligned}\mathcal{W} &= K + D \\ \mathcal{Y} &= Y - T \\ \mathcal{H} &= \int_t^\infty \mathcal{Y} / \hat{R}_t^s ds\end{aligned}\tag{4.9.49}$$

Explain why the effect of this policy is to modify the aggregate specification of the economy to

$$\begin{aligned}C &= (d + \vartheta)(\mathcal{H} + \mathcal{W}) \\ \dot{\mathcal{W}} &= r\mathcal{W} + \mathcal{Y} - C \\ \dot{\mathcal{H}} &= (r + d)\mathcal{H} - \mathcal{Y}\end{aligned}\tag{4.9.50}$$

Answer:

The effect of the government policy is twofold. On the one hand, the distribution of government bonds increases the consumers' wealth W by an amount equal to the value of the bonds received, resulting in a new definition of wealth $\mathcal{W}$ which includes the bonds. On the other hand, the higher value of taxes off to infinity reduces the consumers' human wealth by an amount equal to the present discounted value of the taxes. The change in (redefined) wealth is now net income $Y - \vartheta$ minus consumption.

Use these equations to show that the new dynamic equations for the economy are

$$\begin{aligned}\dot{C} &= (r - \vartheta)C - d(d + \vartheta)(K + D) \\ \dot{K} &= K^\alpha - \delta K - C\end{aligned}\tag{4.9.51}$$

Show how the policy change affects the economy over time, using a phase diagram and a diagram showing the dynamics of aggregate consumption after the policy is introduced. Explain the impact of the policy on different generations in the economy.

Answer:

Time differentiating (4.9.50) yields

$$\begin{aligned}\dot{C} &= (d + \vartheta)(\dot{\mathcal{H}} + \dot{\mathcal{W}}) \\ &= (d + \vartheta)(r\mathcal{W} + (r + d)\mathcal{H} - C)\end{aligned}\tag{4.9.52}$$

Now solve (4.9.50) for $\mathcal{H}$,

$$\mathcal{H} = \left(\frac{C}{d + \vartheta} \right) - \mathcal{W}\tag{4.9.53}$$

and substitute into (4.9.52) to obtain

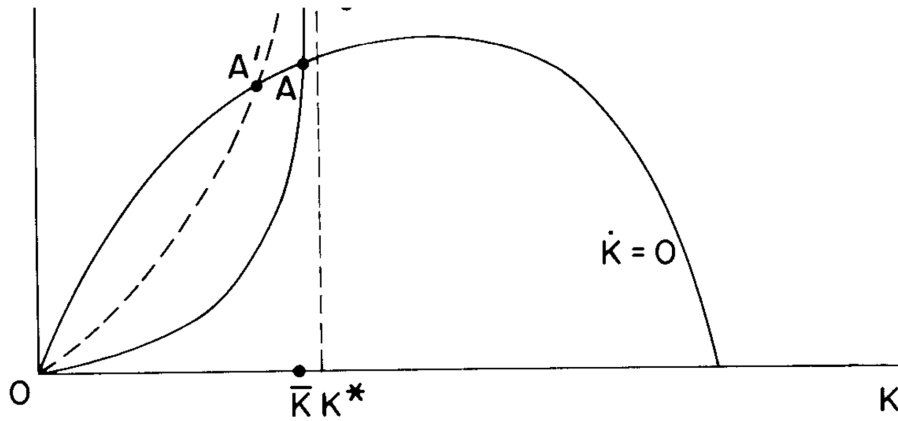
$$\begin{aligned}\dot{C} &= (d + \vartheta)(r\mathcal{W} + (r + d)(C(d + \vartheta)^{-1} - \mathcal{W}) - C) \\ &= (r + d)C - (d + \vartheta)C - d(d + \vartheta)\mathcal{W} \\ &= (r - \vartheta)C - d(d + \vartheta)(K + D).\end{aligned}\tag{4.9.54}$$

Since we are assuming that D is a constant (after the fiscal

experiment), any combination of C and W that would have been on the $\dot{C} = 0$ locus before the policy shift now has a value $\dot{C} = -d(d + \vartheta)D$. This means that the C that would restore

$\dot{C} = 0$ must be a larger C , which says that the $\dot{C} = 0$ locus shifts up (or, equivalently, to the left). Thus, the new equilibrium will be at a lower value of K and a higher interest rate.

The phase diagram shows that the new equilibrium point A' is to the left of the original equilibrium. This is because at a given level of the aggregate capital stock, consumers spend more because $\mathcal{W} > W$. Thus, Ricardian equivalence does not hold in this model, because a tax cut today financed by a future perpetual tax is a transfer of resources from future consumers to today's consumers, and there are no altruistic links that make current consumers offset this by saving more on behalf of future generations.



The next figure shows the path of consumption per capita in this economy. Prior to time 0, the economy was in its steady-state equilibrium at the level of consumption C_0 corresponding to the equilibrium labeled A in the phase diagram. At time 0, the fiscal policy is carried out. The fiscal policy immediately increases consumption because it amounts to a transfer of resources from future to current generations. However, the higher level of consumption runs down the capital stock per capita, and so over time consumption asymptotically approaches a new, lower equilibrium level of consumption C' .

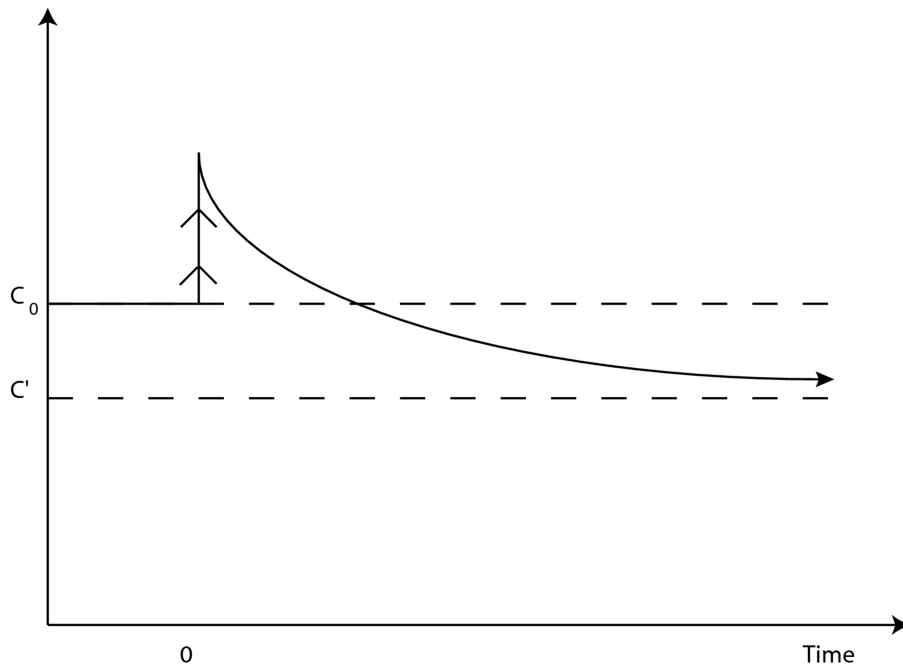
Blanchard shows that if the model is changed so that each consumer's income declines exponentially at rate γ after birth, the result is equivalent to assuming that future labor income is discounted at an interest rate that is higher by γ . He further shows that the equations of motion of the model change to

$$\begin{aligned}\dot{C} &= (\alpha K^{\alpha-1} - \delta + \gamma - \vartheta)C - (d + \gamma)(d + \vartheta)K \\ \dot{K} &= K^\alpha - \delta K - C.\end{aligned}\tag{4.9.55}$$

Note that it is possible to rewrite the $\dot{C} = 0$ locus as

$$(\alpha K^{\alpha-1} - \delta + \gamma - \vartheta)C = (d + \gamma)(d + \vartheta)K\tag{4.9.56}$$

Use this equation to show that as $C \rightarrow \infty$, the $\dot{C} = 0$ locus asymptotes to



$$r = \alpha K^{\alpha-1} - \delta = \vartheta - \gamma \quad (4.9.57)$$

Thus, for a large enough value of γ it is possible that the net interest rate in this economy could be negative. Draw a phase diagram corresponding to an equilibrium with a negative net interest rate, and comment on why this is an interesting case to think about. In particular, what new light does it shed on the fiscal policy experiment examined above?

Answer:

As C goes to infinity on the LHS of (4.9.56), the only way the equation can continue to hold is if

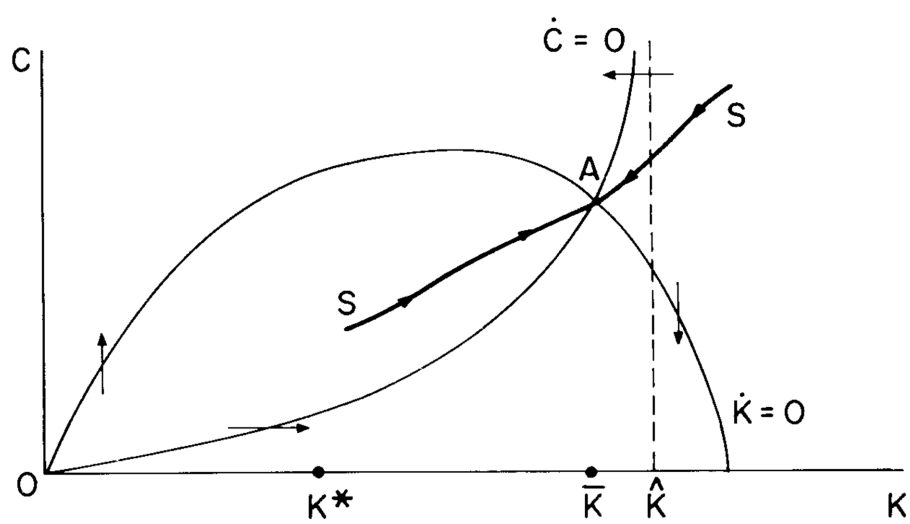
$$\lim_{C \rightarrow \infty} (\alpha K^{\alpha-1} - \delta + \gamma - \vartheta) = 0 \quad (4.9.58)$$

which implies (4.9.57).

The new phase diagram shows the $\dot{C} = 0$ locus intersecting the $\dot{K} = 0$ locus to the right of the maximum of the $\dot{K} = 0$ locus.

This is implied by the fact that the net interest rate is negative, which means that the net interest rate could be increased by reducing the capital stock.

The reason this is an interesting case is that in this case it is possible for the economy to be in a condition of dynamic inefficiency, just as in the 2-period OLG models discussed early in the class. The idea is to think of declining labor income as a way to generate a ‘life cycle’ saving motive. In such a case the fiscal experiment examined above is interesting because it could rescue an economy with too much capital from a state of dynamic inefficiency.



4.10 Gross Saving and Growth in the RCK Model

In the neoclassical growth model with labor-augmenting technological progress at rate γ , utility function $u(c) = c^{1-\rho}/(1-\rho)$, time preference rate ϑ and depreciation rate δ the steady-state will be at the point where the growth rate of consumption is equal to the growth rate of labor-augmenting technological progress, γ ,

$$\begin{aligned}\frac{\dot{C}}{C} &= \rho^{-1}(f'(k) - \delta - \vartheta) \\ &= \gamma\end{aligned}\tag{4.10.1}$$

which implies that

$$\begin{aligned}f'(k) &= \rho\gamma + \vartheta + \delta \\ \alpha k^{\alpha-1} &= \rho\gamma + \vartheta + \delta \\ k &= \left[\frac{\rho\gamma + \vartheta + \delta}{\alpha} \right]^{\frac{1}{\alpha-1}}.\end{aligned}\tag{4.10.2}$$

The aggregate gross saving rate is defined as

$$\begin{aligned}s &= \frac{y - c}{y} \\ &= \frac{k^\alpha - c}{k^\alpha} \\ &= 1 - c/k^\alpha.\end{aligned}\tag{4.10.3}$$

In steady-state by definition

$$\dot{k} = 0\tag{4.10.4}$$

but from the capital accumulation equation we know that

$$\dot{k} = k^\alpha - (\delta + \gamma)k - c\tag{4.10.5}$$

so in steady-state

$$c = k^\alpha - (\delta + \gamma)k.\tag{4.10.6}$$

This can be substituted into (4.10.3) to obtain

$$\begin{aligned}s &= 1 - \frac{k^\alpha - (\delta + \gamma)k}{k^\alpha} \\ &= (\delta + \gamma)k^{1-\alpha}\end{aligned}\tag{4.10.7}$$

and the expression for the steady-state level of capital per capita can be substituted in to yield

$$\begin{aligned}s &= (\delta + \gamma) \left(\frac{\rho\gamma + \vartheta + \delta}{\alpha} \right)^{-1} \\ &= \left(\frac{\alpha(\gamma + \delta)}{\delta + \vartheta + \rho\gamma} \right)\end{aligned}\tag{4.10.8}$$

The derivative of this expression with respect to γ is

$$\begin{aligned}\frac{ds}{d\gamma} &= \left(\frac{\alpha(\delta + \vartheta + \rho\gamma) - \alpha(\gamma + \delta)\rho}{(\delta + \vartheta + \rho\gamma)^2} \right) \\ &= \left(\frac{\alpha(\delta(1 - \rho) + \vartheta)}{(\delta + \vartheta + \rho\gamma)^2} \right).\end{aligned}\tag{4.10.9}$$

This will be positive if its numerator is positive, i.e. if

$$\begin{aligned}\rho\delta &< \vartheta + \delta \\ \rho &< 1 + \vartheta/\delta.\end{aligned}\tag{4.10.10}$$

A typical assumption is $\vartheta = .04$ and $\delta = .08$, implying that the steady-state relationship between saving and growth in the neoclassical model is positive only if the coefficient of relative risk aversion ρ is less than 1.5. Typically we assume values of ρ in the range from 2 to 5, so the model leads us to expect a negative relationship between saving and growth.

Chapter 5

DSGE Models

Adding productivity shocks to the growth model turns a theory of accumulation into a theory of business cycles. Almost nothing in that literature has a closed form solution.

The Brock-Mirman case is the exception. Logarithmic utility, Cobb-Douglas production, and complete depreciation within the period together yield an exact policy rule, which is why we study this model closely. It is the benchmark against which numerical solutions to harder problems can be checked (a solver that cannot reproduce Brock-Mirman to machine precision has a bug, not an approximation error). The Prescott real business cycle model gives up that luxury in exchange for realism, and asks how much of observed output volatility technology shocks alone can account for.

5.1 The Brock-Mirman Stochastic Growth Model

Brock and Mirman [1972] provided the first optimizing growth model with unpredictable (stochastic) shocks.

The social planner's goal is to solve the problem:

$$\max \mathbb{E} \left[\sum_{n=0}^{\infty} \beta^n \log C_{t+n} \right] \quad (5.1.1)$$

s.t.

$$\begin{aligned} K_{t+1} &= Y_t - C_t \\ Y_{t+1} &= A_{t+1} K_{t+1}^\alpha \end{aligned} \quad (5.1.2)$$

where A_t is the level of productivity in period t , which is now allowed to be stochastic (alternative assumptions about the nature of productivity shocks are explored below). Note the key assumption that the depreciation rate on capital is 100 percent.

In this model the capital stock is not useful as a state variable: Because capital has a 100 percent depreciation rate, all that matters to the consumer when choosing how much to consume is how much income they have now, and not how that income breaks down into a part due to K and a part due to A .

The first step is to rewrite the problem in Bellman equation form

$$V_t(Y_t) = \max_{C_t} \log C_t + \beta \mathbb{E}_t[V_{t+1}(Y_{t+1})] \quad (5.1.3)$$

and take the first order condition:

$$\begin{aligned} u'(C_t) &= \beta \mathbb{E}_t \left[A_{t+1} \alpha K_{t+1}^{\alpha-1} u'(C_{t+1}) \right] \\ \frac{1}{C_t} &= \beta \mathbb{E}_t \left[\frac{A_{t+1} \alpha K_{t+1}^{\alpha-1}}{C_{t+1}} \right] \\ 1 &= \beta \mathbb{E}_t \left[\underbrace{\alpha A_{t+1} K_{t+1}^{\alpha-1}}_{\equiv \mathbf{R}_{t+1}} \frac{C_t}{C_{t+1}} \right] \end{aligned} \quad (5.1.4)$$

where our definition of $\mathbf{R}_{t+1}$ helps clarify the relationship of this equation to the usual consumption Euler equation (and you should think about why this is the right definition of the interest factor in this model).

Now we show that this FOC is satisfied by the consumption function $C_t = \kappa Y_t$, where $\kappa = 1 - \alpha\beta$. To see this, note first that the proposed consumption rule implies that $K_{t+1} = (1 - \kappa)Y_t$.

The first order condition says

$$\begin{aligned} 1 &= \beta \mathbb{E}_t \left[\alpha \frac{A_{t+1} K_{t+1}^\alpha}{K_{t+1}} \frac{\kappa Y_t}{\kappa Y_{t+1}} \right] \\ &= \beta \mathbb{E}_t \left[\alpha \frac{Y_{t+1}}{K_{t+1}} \frac{\kappa Y_t}{\kappa Y_{t+1}} \right] \\ &= \beta \left[\alpha \frac{Y_t}{K_{t+1}} \right] \\ &= \beta \left[\alpha \frac{Y_t}{Y_t - C_t} \right] \\ &= \beta \left[\alpha \frac{Y_t}{Y_t(1 - \kappa)} \right] \\ &= \beta \left[\alpha \frac{1}{(1 - \kappa)} \right] \\ (1 - \kappa) &= \alpha\beta \\ \kappa &= 1 - \alpha\beta. \end{aligned} \quad (5.1.5)$$

An important way of judging a macroeconomic model and deciding whether it makes sense is to examine the model's implications for the dynamics of aggregate variables. Defining lower case variables as the log of the corresponding upper case variable, this model says that the dynamics of the capital stock are given by

$$\begin{aligned} K_{t+1} &= (1 - \kappa)Y_t \\ &= \alpha\beta A_t K_t^\alpha \\ k_{t+1} &= \log \alpha\beta + a_t + \alpha k_t \end{aligned} \tag{5.1.6}$$

which tells us that the dynamics of the (log) capital stock have two components: One component (a_t) mirrors whatever happens to the aggregate production technology; the other is serially correlated with coefficient α equal to capital's share in output.

Similarly, since log output is simply $y = a + \alpha k$, the dynamics of output can be obtained from

$$\begin{aligned} y_{t+1} &= a_{t+1} + \alpha k_{t+1} \\ &= \alpha(\log K_{t+1}) + a_{t+1} \\ &= \alpha(\log \alpha\beta Y_t) + a_{t+1} \\ &= \alpha(y_t + \log \alpha\beta) + a_{t+1} \end{aligned} \tag{5.1.7}$$

so the dynamics of aggregate output, like aggregate capital, reflect a component that mirrors a and a serially correlated component with serial correlation coefficient α .

The simplest assumption to make about the level of technology is that its log follows a random walk:

$$a_{t+1} = a_t + \epsilon_{t+1}. \tag{5.1.8}$$

Under this assumption, consider the dynamic effects on the level of output from a unit positive shock to the log of technology in period t (that is, $\epsilon_{t+1} = 1$ where $\epsilon_s = 0 \forall s \neq t+1$). Suppose that the economy had been at its original steady-state level of output $\check{y}$ in the prior period. Then the expected dynamics of output would be given by

$$\begin{aligned} y_t &= \check{y} + a_t \\ \mathbb{E}_t[y_{t+1}] &= \check{y} + a_t + \alpha a_t \\ \mathbb{E}_t[y_{t+2}] &= \check{y} + a_t + \alpha a_t + \alpha^2 a_t \end{aligned} \tag{5.1.9}$$

and so on, as depicted in Figure 5.1.

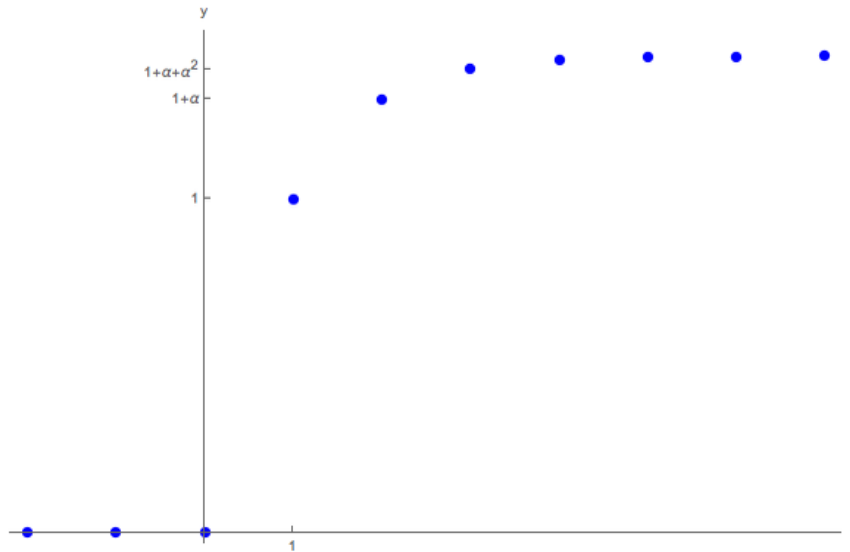


Figure 5.1: Dynamics of Output With a Random Walk Shock

Also interesting is the case where the level of technology follows a white noise process,

$$a_{t+1} = \check{a} + \epsilon_{t+1}. \quad (5.1.10)$$

The dynamics of income in this case are depicted in Figure 5.2.

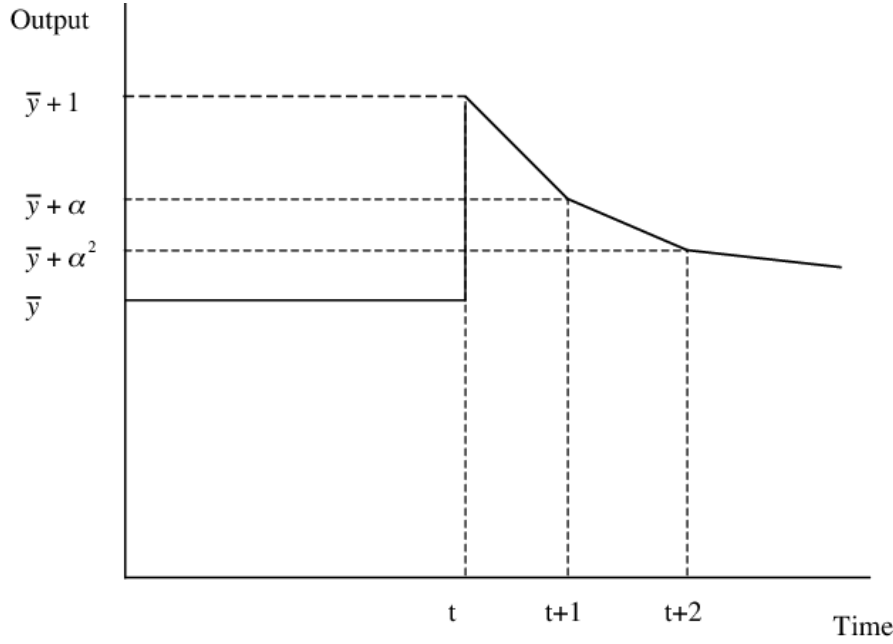


Figure 5.2: Dynamics of Output With A White Noise Shock

The key point of this analysis, again, is that the dynamics of the model are governed by two components: The dynamics of the technology shock, and the assumption about the saving/accumulation process.

For further analysis, consider a nonstochastic version of this model, with $A_t = 1 \forall t$. The consumption Euler equation is

$$\frac{C_{t+1}}{C_t} = (\beta R_{t+1})^{1/\rho} \quad (5.1.11)$$

But this is an economy with no technological progress, so the steady-state interest rate must take on the value such that $C_{t+1}/C_t = 1$. Thus we must have $\beta R = 1$ or $R = 1/\beta$.

We can further derive the steady state level of capital of a nonstochastic version of the model in which $a_t = a \forall t$ from (5.1.6):

$$\begin{aligned} k &= \log \alpha \beta + a + \alpha k \\ (1 - \alpha)k &= \log \alpha \beta + a \\ k &= \frac{\log \alpha \beta + a}{1 - \alpha} \end{aligned} \quad (5.1.12)$$

The nonstochastic version of the model is of course not very interesting, except as a point of comparison to the stochastic version of the model. But what could be meant by the ‘steady state’ of a stochastic model that never settles down? We can define a ‘stochastic steady state’ for such models in a number of (potentially) different ways:

- The location (if one exists) to which the model will converge after an arbitrarily long period in which no shocks occurred $a_{t+1} = a_t \forall t$ (even if in every period agents *expect* that shocks will occur)
- The mean value of some variable in the model (say, K)
- The value of some state variable, say $\check{K}$, such that $\mathbb{E}_t[K_{t+1}] = K_t$ if $K_t = \check{K}$.

We consider here the last of these, which we will show reduces (in this special case) to the same equation as for the nonstochastic version of the model, $K_{t+1} = K_t$. To see this, rewrite the Euler equation as:

$$\begin{aligned}
1 &= \beta \mathbb{E}_t \left[\underbrace{\alpha A_{t+1} K_{t+1}^{\alpha-1}}_{\equiv R_{t+1}} \frac{C_t}{C_{t+1}} \right] \\
1 &= \beta \mathbb{E}_t \left[\alpha A_{t+1} K_{t+1}^{\alpha-1} \frac{Y_t}{Y_{t+1}} \right] \\
1 &= \beta \mathbb{E}_t \left[\alpha A_{t+1} K_{t+1}^{\alpha-1} \frac{A_t K_t^\alpha}{A_{t+1} K_{t+1}^\alpha} \right] \\
1 &= \beta \mathbb{E}_t \left[\alpha K_{t+1}^{\alpha-1} \frac{A_t K_t^\alpha}{K_{t+1}^\alpha} \right] \\
1 &= \beta \left[\alpha K_{t+1}^{\alpha-1} \frac{A_t K_t^\alpha}{K_{t+1}^\alpha} \right]
\end{aligned} \tag{5.1.13}$$

where the expectations operator disappears because no variables are stochastic (the A_{t+1} terms in the numerator and denominator cancel, and K_{t+1} is directly chosen in t so is known). For any given A_t , the steady state where $K_{t+1} = K_t = \check{K}$ is then where

$$\begin{aligned}
1 &= \beta \alpha A_t \check{K}_t^{\alpha-1} \\
\check{k} &= \frac{\log \beta \alpha + a_t}{1 - \alpha}
\end{aligned} \tag{5.1.14}$$

which is the generalization of the nonstochastic solution derived in (5.1.12).

The result that the nonstochastic and stochastic steady states are the same is special to the Brock-Mirman model; it is NOT true of many other models of growth; it occurs here because the linearity of the consumption function, among other special assumptions. Furthermore, the third of our possible definitions of a steady state will generally differ at least a little bit from either of the first two.

5.2 The Prescott Real Business Cycle Model

This section presents the elements of the original Real Business Cycle model of aggregate fluctuations, as laid out by Prescott [1986], along with a few critiques articulated by Summers [1986] and others.

Consider a representative household whose goal is to maximize

$$\mathbb{E}_t \left[\sum_{t=0}^{\infty} \beta^t u(c_t, z_t) \right] \quad (5.2.1)$$

where z_t is the fraction of time the representative agent spends at leisure (not working); the alternative to leisure is the number of hours you work, which will be designated ℓ_t , and the time endowment is normalized to 1, so that

$$\ell_t + z_t = 1. \quad (5.2.2)$$

Assume that the structure of the utility function is

$$\begin{aligned} u(c, z) &= \left(\frac{(c^{1-\zeta} z^\zeta)^{1-\rho}}{1-\rho} \right) \\ u^c &= (c^{1-\zeta} z^\zeta)^{-\rho} c^{-\zeta} z^\zeta (1-\zeta) \\ u^z &= (c^{1-\zeta} z^\zeta)^{-\rho} c^{1-\zeta} z^{\zeta-1} \zeta. \end{aligned} \quad (5.2.3)$$

Think about the maximum amount of income that could be gained if the representative agent worked every waking hour:

$$y_t = W_t. \quad (5.2.4)$$

The representative agent can then think of deciding to ‘purchase’ two things with this endowment of income: leisure z_t whose price is W_t , or consumption, whose price is normalized to one.

Over the past century in the U.S., wages have risen very substantially, but hours worked have not declined much if at all. (Ramey and Francis [2009]). What kind of utility function implies that the budget share of a good (leisure) remains constant even as the price of the good changes sharply? A Cobb-Douglas utility function. Hence the assumption that utility is obtained from a Cobb-Douglas aggregate of consumption and leisure is consistent with the lack of a strong trend in hours worked per worker.

Since workers are choosing how many hours to work as well how much to consume, a first order condition will characterize the optimal choice between consumption and leisure within a period. In particular, the price of leisure is W_t and the price of consumption is 1, so the ratio of the marginal utility of leisure to the marginal utility of consumption should be

$$\left(\frac{W_t}{1} \right) = \left(\frac{u_t^z}{u_t^c} \right). \quad (5.2.5)$$

To see this, note that the consumer’s goal is to

$$\max_{\{c_t, z_t\}} u(c_t, z_t). \quad (5.2.6)$$

Suppose the consumer has decided to spend a given amount χ_t in period t on a combination of consumption and leisure,

$$c_t + W_t z_t = \chi_t \quad (5.2.7)$$

Then (5.2.6) becomes

$$\max_{\{z_t\}} u(\chi_t - W_t z_t, z_t) \quad (5.2.8)$$

for which the FOC is

$$\begin{aligned} -u_t^c W_t + u_t^z &= 0 \\ W_t &= (u_t^z / u_t^c). \end{aligned} \quad (5.2.9)$$

Returning to (5.2.5)

$$\begin{aligned} \left(\frac{W_t}{1} \right) &= \left(\frac{u_t^z}{u_t^c} \right) \\ W_t &= \left(\frac{c_t^{1-\zeta} z_t^{\zeta-1}}{c_t^{-\zeta} z_t^{\zeta}} \right) \left(\frac{\zeta}{1-\zeta} \right) \\ &= \left(\frac{c_t}{z_t} \right) \left(\frac{\zeta}{1-\zeta} \right) \end{aligned} \quad (5.2.10)$$

or

$$W_t z_t / c_t = \left(\frac{\zeta}{1-\zeta} \right). \quad (5.2.11)$$

Since we know that z has been roughly constant over long periods of time, this implies that as wages rise, consumption rises by roughly the same amount.

One of the original promises of the DSGE literature was to calibrate its business-cycle models based on either long-run facts (like the lack of a trend in z) or on micro data (like intertemporal elasticities estimated using household data). So how is ζ calibrated?

If wages are defined as per unit of labor, then if on average consumption roughly equals labor income $c = (1 - z)w$ we have

$$\begin{aligned} \left(\frac{wz}{w(1-z)} \right) &= \left(\frac{\zeta}{1-\zeta} \right) \\ \left(\frac{1-z}{z} \right) &= \left(\frac{1-\zeta}{\zeta} \right) \\ 1/z - 1 &= 1/\zeta - 1 \\ z &= \zeta. \end{aligned} \quad (5.2.12)$$

So ζ should be calibrated to be equal to the proportion of their available (i.e. non-sleep) time people spend not working. A 40-hour work week (along with 8 hours of sleep a day) would yield $\zeta = 2/3$. Among other taste parameters, Prescott chooses log utility ($\lim_{\rho \rightarrow 1} u(c)$) and $\beta = 0.96$.

The aggregate production function is assumed to be Cobb-Douglas,

$$\begin{aligned} K_{t+1} &= \overbrace{K_t + Y_t - C_t}^{1-\delta} \\ Y_t &= A_t K_t^{1-\nu} L_t^\nu \end{aligned} \quad (5.2.13)$$

where $\ell_t = 1 - z_t$ is labor supply per member of the working population and $L_t = \ell_t \mathfrak{H}_t$ is aggregate labor input, and where $\mathfrak{H}_t = \sum_i \mathfrak{h}_i$ is the aggregate amount of hours available to members of the working population. Constant income shares and perfect competition imply

$$\begin{aligned} F_L L / Y &= \nu A_t K_t^{1-\nu} L_t^{\nu-1} L_t / Y_t \\ &= \nu \end{aligned} \quad (5.2.14)$$

so that labor's share of GDP is roughly constant. Prescott sets labor's share to a constant 64 percent, and chooses a depreciation rate of $\delta = 0.10$.

The crucial assumption, however, is about the productivity process, since 'technology shocks' are assumed to drive business cycles.

Prescott defines the 'hat' operator $\hat{\cdot}$ as:

$$\hat{X}_t = \left(\frac{X_t - X_{t-1}}{X_{t-1}} \right) \quad (5.2.15)$$

Table 5.1: Standard deviations of key variables, data versus model

	σ_z	σ_y	σ_n
US Data	0.76	1.76	1.67
Model	0.76	1.48	0.76

which implies from the production function that

$$\hat{A}_t \approx \hat{Y}_t - \nu \hat{L}_t - (1 - \nu) \hat{K}_t. \quad (5.2.16)$$

(this is just the Solow residual).

Prescott ‘estimates’ a productivity process that takes the form

$$\hat{A}_t = \phi_t + \epsilon_t \quad (5.2.17)$$

with a standard deviation of $\sigma_\epsilon = 0.76$ per quarter.

Prescott makes sufficient assumptions (perfect competition, etc.) so that the social planner’s problem is the same as the decentralized solution. With log utility, the social planner’s problem is

$$\max \mathbb{E}_t \left[\sum_{t=0}^{\infty} \beta^t ((1 - \zeta) \log c_t + \zeta \log(1 - \ell_t)) \right] \quad (5.2.18)$$

subject to

$$\begin{aligned} k_{t+1} &= \mathfrak{T}k_t + A_t k_t^{1-\nu} \ell_t^\nu - c_t \\ \hat{A}_t &= \phi + \epsilon_t. \end{aligned} \quad (5.2.19)$$

Prescott argues that the way to judge the model is by whether it produces plausible statistics for standard deviations of the key variables. He produces a table that argues it does:

Since the first column is calibrated, it isn’t a test of the model.

The second column comes out of the model, and isn’t too bad a fit.

However, the third column is a terrible fit. What it says is that labor input is much more variable over the course of the business cycle than this model would suggest.

What’s going on? To understand the answer, we need to understand why hours fluctuate in this model at all. Recall that we deliberately constructed the model (by choosing a utility function that was Cobb-Douglas in consumption and leisure) in a way designed to prohibit any long-run response of hours worked to wages. Since hours worked are being chosen freely on a day-by-day basis by workers in this model, there must be some incentive that causes them to be willing to put up with short-term variation in hours (over the business cycle).

The answer is that transitory productivity shocks provide an incentive to work harder some times than others. In particular, if there is a temporary positive productivity shock you will be willing to work longer hours than usual, while if there is a negative productivity shock everybody wants to take a vacation.

To see this formally, consider again the first order conditions from the maximization problem. We showed in (5.2.11) that

$$\begin{aligned} W_t z_t / c_t &= \left(\frac{\zeta}{1 - \zeta} \right) \\ &= W_{t+1} z_{t+1} / c_{t+1}. \end{aligned} \quad (5.2.20)$$

Now note that since (5.2.18) is separable in consumption and leisure the intertemporal FOC will imply that

$$\begin{aligned} 1/c_t &= R_{t+1} \beta / c_{t+1} \\ c_{t+1} &= R_{t+1} \beta c_t. \end{aligned} \quad (5.2.21)$$

Combining this with (5.2.20) gives

$$\begin{aligned} W_t z_t &= W_{t+1} z_{t+1} / R_{t+1} \beta \\ z_{t+1} / z_t &= \left(\frac{R_{t+1} \beta}{W_{t+1} / W_t} \right) \\ \hat{z}_{t+1} &\approx -\hat{w}_{t+1} + (r_{t+1} - \vartheta) \end{aligned} \tag{5.2.22}$$

What this equation tells us is that there are two ways to make z_t (and therefore ℓ_t) fluctuate over the business cycle:

1. Make transitory movements in real wages induce a strong labor supply response

- Problem: A large number of microeconomic studies have estimated the elasticity of labor supply with respect to wages to be much too small to explain the observed fluctuations in hours over the business cycle. Indeed, there is even controversy about whether real wages rise or fall over the business cycle. But even if we accept that wages fall during recessions, the evidence at the micro level does not seem to suggest that people are willing to dramatically change their work hours in response to transitory fluctuations in wages

2. Make movements in interest rates induce a strong labor

supply response (the idea is that you will work hard during the period of high interest rates in order to earn more cash which can be invested to take advantage of the high interest rate; in thinking about this, consider that a high value of R_{t+1} will induce high leisure growth between t and $t + 1$ by resulting in low leisure in period t)

- Problem: If this is the mechanism, there should at the same time be a strong consumption response:

$$\begin{aligned} W_t z_t / c_t &= W_{t+1} z_{t+1} / c_{t+1} \\ c_{t+1} / c_t &= W_{t+1} z_{t+1} / W_t z_t \end{aligned} \tag{5.2.23}$$

so if the labor supply response is not being driven by wage differences, there should be a one-for-one comovement of consumption with leisure; that is, recessions should be periods of high consumption and booms should be periods of low consumption!

This latter is a quite general problem with the DSGE framework in which fluctuations in employment over the business cycle are driven by voluntary changes in hours worked.

Appendix A

Mathematical Background

We use the results collected here repeatedly in the main chapters, but deriving them where they are first needed would interrupt the argument.

Two of them concern approximation, showing how to replace a continuous lognormal distribution with a small number of equiprobable points, in the univariate and then the bivariate case, so that a computer can evaluate expectations by simple averaging. The rest is reference material. It covers the conditions under which the behavior of many heterogeneous households collapses into a single aggregate relationship, a side-by-side comparison of the discrete and continuous time formulations of the Ramsey problem, and a list of the mathematical facts this literature leans on most often (geometric sums and logarithmic approximations, mostly).

A.1 Aggregation For Dummies (Macroeconomists)

Consider an economy populated by a set of agents distributed uniformly along the unit interval with a total population mass of 1. That is, for $i \in [0, 1]$ the probability distribution function is $f(i) = 1$ and $f(i) = 0$ elsewhere; the CDF on the $[0, 1]$ interval is therefore $F(i) = i$, implying an aggregate population mass of $F(1) = 1$.

Agent i 's value of variable $\bullet$ at date t is $\bullet_{t,i}$.

Thus aggregate consumption is

$$\begin{aligned} C_t &= \int_0^1 c_{t,i} f(i) di, \\ &= \int_0^1 c_{t,i} di, \end{aligned} \tag{A.1.1}$$

and a similar notation applies to other variables.

Since the aggregate population is normalized to 1, capital letters refer not only to aggregate variables but also to per capita variables, since per-capita consumption is aggregate consumption divided by aggregate population:

$$\left(\frac{\int_0^1 c_{t,i} f(i) di}{\int_0^1 f(i) di} \right) = \left(\frac{C_t}{1} \right) = C_t. \tag{A.1.2}$$

Each individual agent is infinitesimally small, and can therefore neglect the effects of its own actions on aggregates.

A.1.1 Blanchard Lives

For many purposes the assumption that economic agents live forever is useful;

but for other purposes it is necessary to be able to analyze agents with finite horizons. Blanchard [1985] introduced a tractable framework that permits analysis of many of the key issues posed by finite lifetimes.

The key assumption is that the probability of death is independent of the agent's age. (This is similar to the Calvo [1983] assumption that the probability that a firm will change its prices is independent of the time elapsed since the last price change).

The most convenient formulation of the model is one in which the number of dying individuals is always equal to the number of newborn individuals, so that the population remains constant.

Discrete Time

As above, suppose that the population alive at time t is arranged on the unit interval.

The probability of death is d (and the probability of not dying is $D = 1 - d$).

Then for a person living at any location $i \in [0, 1]$, expected remaining lifetime including the current period will be

$$\begin{aligned} 1 + \overbrace{(1 - D)}^D + (1 - D)^2 + \dots &= \left(\frac{1}{1 - (1 - D)} \right) \\ &= 1/D. \end{aligned} \tag{A.1.3}$$

If a new cohort of size D has been born each period since the beginning of time, the total population will be given by the size of a new cohort D multiplied by the expected lifetime D^{-1} :

$$DD^{-1} = 1, \tag{A.1.4}$$

so that the mass of the aggregate population is constant at 1, as above.

Continuous Time

Blanchard's original treatment was in continuous time, with a constant rate of death d , so that the probability of remaining alive (not dead) after t periods for a consumer born in period 0 is¹

$$\aleph_t = e^{-dt} \quad (\text{A.1.5})$$

so that the expected life span is

$$\int_0^\infty e^{-d\tau} d\tau = 1/d \quad (\text{A.1.6})$$

and if the flow arrival rate of new population is d (that is, at each instant a flow of new population arrives at rate d) then again the population mass is constant at

$$d/d = 1. \quad (\text{A.1.7})$$

Population Growth

Now suppose that the population in the discrete-time model is growing by a factor $\Xi = (1 + \xi)$ from period to period; if the number of newborns in period 0 was 1, then the number of newborns in period t is given by

$$\Xi^t. \quad (\text{A.1.8})$$

In this framework we want to keep track of the relative population of each cohort compared to the size of the newborn cohort. At age z , the cohort that was born in period 0 will be of relative size

$$(\mathcal{D}/\Xi)^z \quad (\text{A.1.9})$$

The total relative populations will be

$$1 + (\mathcal{D}/\Xi) + (\mathcal{D}/\Xi)^2 + \dots = \left(\frac{1}{1 - \mathcal{D}/\Xi} \right) \quad (\text{A.1.10})$$

so that if in period 0 the population was of size $(1 - \mathcal{D}/\Xi)$ then the sizes of the relative populations will add up to one even as the absolute population grows by the factor Ξ .

¹The number of people who will die during the first period in the continuous-time model is measured by the difference between an initial population of size 1 and the size of the population remaining alive after one period, $1 - \aleph_1 = 1 - e^{-d}$. ExpEps in the [Math Facts](#) list implies that $e^{-d} \approx 1 - d$; therefore the proportion who have died will be approximately $1 - (1 - d) = d$. Hence for small death rates, in order for the same population to survive for one period in a continuous-time model with death rate d and the discrete-time model with death rate D , one would need $d \approx D$.

A.2 An Equiprobable Approximation to the Bivariate Lognormal

Economic agents face risks of many kinds, which may mutually covary. A stock broker, for example, is likely to earn a salary bonus that is positively related to the performance of the stock market; if that broker also has personal stock investments, his financial wealth and labor income will be positively correlated.

The first part of this section presents a convenient (and empirically realistic) formulation in which a consumer faces two shocks (which can be interpreted as a shock to noncapital income and a shock to the rate of return) that are distributed according to a multivariate lognormal that allows for correlation between them. The second part describes a computationally simple and convenient method for approximating that joint distribution.

A.2.1 Theory

Consider a consumer who faces both a risk to transitory noncapital income²

$$\theta_{1,t+1} \equiv \log \Theta_{1,t+1} \sim \mathcal{N}(-0.5\sigma_1^2, \sigma_1^2) \quad (\text{A.2.1})$$

and a risky log rate-of-return that is affected by following factors: the riskless rate r ;

a risk premium φ ; an additional constant ζ (whose purpose will become clear below); a component that is linearly related to $\theta_{1,t+1}$; and an independent shock $\theta_2 \sim \mathcal{N}(-0.5\sigma_2^2, \sigma_2^2)$:

$$\mathbf{r}_{t+1} \equiv \log \mathbf{R}_{t+1} = r + \varphi + \zeta + \omega\theta_{1,t+1}(\sigma_2/\sigma_1) + \theta_{2,t+1} \quad (\text{A.2.2})$$

for some constant ω . Since $(\sigma_2/\sigma_1)\omega\theta_{1,t+1}$ is the only component of $\mathbf{r}_{t+1}$ that covaries with $\theta_{1,t+1}$,

$$\begin{aligned} \text{cov}(\theta_{1,t+1}, \mathbf{r}_{t+1}) &= \text{cov}(\theta_{1,t+1}, (\sigma_2/\sigma_1)\omega\theta_{1,t+1}) \\ &= \omega(\sigma_2/\sigma_1) \underbrace{\text{cov}(\theta_{1,t+1}, \theta_{1,t+1})}_{=\sigma_1^2} \\ &= \omega\sigma_2\sigma_1. \end{aligned} \quad (\text{A.2.3})$$

Equation (A.2.2) yields a description of the return process in which the parameter ω controls the correlation between the risky log return shock and the risky log labor income shock. If $\omega = 0$ the processes are independent.

Now we want to find the value of ζ such that the mean risky return is unaffected by σ_1^2 (so that we will be able to understand clearly the distinct effects of labor income risk, the independent component of rate-of-return risk σ_2^2 , and the correlation between labor income risk and rate-of-return risk, ω). Thus, we want to find the ζ such that

$$\mathbb{E}_t[\mathbf{R}_{t+1}] = e^{r+\varphi} \quad (\text{A.2.4})$$

regardless of the values of σ_1^2 and σ_2^2 . We therefore need:

$$\begin{aligned} \mathbb{E}[e^{\zeta + (\sigma_2/\sigma_1)\omega\theta_{1,t+1} + \theta_{2,t+1}}] &= 1. \\ \log \mathbb{E}[e^{\zeta + (\sigma_2/\sigma_1)\omega\theta_{1,t+1} + \theta_{2,t+1}}] &= 0. \end{aligned} \quad (\text{A.2.5})$$

Using standard facts about lognormals (cf. [Math Facts](#)), and for convenience defining $\hat{\omega} = (\sigma_2/\sigma_1)\omega$, we have

$$\begin{aligned} 0 &= \zeta - 0.5\hat{\omega}\sigma_1^2 - 0.5\sigma_2^2 + 0.5\hat{\omega}^2\sigma_1^2 + 0.5\sigma_2^2 \\ &= \zeta - 0.5\sigma_1^2\hat{\omega}(1 - \hat{\omega}) \\ \zeta &= 0.5(\hat{\omega} - \hat{\omega}^2)\sigma_1^2 = 0.5(\omega\sigma_2\sigma_1 - \omega^2\sigma_2^2). \end{aligned} \quad (\text{A.2.6})$$

Because the two shocks are independent by construction, the joint distribution takes a convenient form. Stacking $\theta_{1,t+1}$ and $\theta_{2,t+1}$ into a vector $\vec{\theta}$, we can write

²The assumed distribution has the property $\mathbb{E}[\Theta_{1,t+1}] = 1$, cf. [Math Facts](#).

$$\vec{\theta} \sim \mathcal{N}(\vec{\mu}, \Sigma) \quad (\text{A.2.7})$$

where $\vec{\mu} = \{-0.5\sigma_1^2, -0.5\sigma_2^2\}'$ and the covariance matrix Σ is diagonal, carrying σ_1^2 and σ_2^2 .³

A.2.2 Computation

A key step in the computational solution of any model with uncertainty is the calculation of expectations. Writing $\tilde{\Theta}_1 \equiv \tilde{\Theta}_{1,t+1}$ and $\tilde{\mathbf{R}} \equiv \mathbf{R}_{t+1}$ and $\mathbb{E}[\bullet] = \mathbb{E}_t[\bullet_{t+1}]$, the expectation of some function h that depends on the realization of the risky return $\mathbf{R}$ and the labor income shock is:

$$\mathbb{E}[h(\tilde{\Theta}_1, \tilde{\mathbf{R}})] = \int_{\underline{\Theta}_1}^{\bar{\Theta}_1} \int_{\underline{\mathbf{R}}}^{\bar{\mathbf{R}}} h(\tilde{\Theta}_1, \tilde{\mathbf{R}}) dF(\tilde{\Theta}_1, \tilde{\mathbf{R}}) \quad (\text{A.2.8})$$

where $F(\tilde{\Theta}_1, \tilde{\mathbf{R}})$ is the joint cumulative distribution function. Standard numerical computation software can compute this double integral, but at such a slow speed as to be almost unusable.

Computation of the expectation can be massively speeded up by advance construction of a numerical approximation to $F(\tilde{\Theta}_1, \tilde{\mathbf{R}})$.

Such approximations generally take the approach of replacing the distribution function with a discretized approximation to it; appropriate weights $w_{i,j}$ are attached to each of a finite set of points indexed by i and j , and the approximation to the integral is given by:

$$\mathbb{E}[h(\tilde{\Theta}_1, \tilde{\mathbf{R}})] \approx \sum_{i=1}^n \sum_{j=1}^m h(\hat{\Theta}_1[i, j], \hat{\mathbf{R}}[i, j]) w[i, j] \quad (\text{A.2.9})$$

where the $\hat{\Theta}_1$ and $\hat{\mathbf{R}}$ matrices contain the conditional means of the two variables in each of the $\{i, j\}$ regions. Various methods are used for constructing the weights $w[i, j]$ and the nodes (the i and j points for Θ_1 and $\mathbf{R}$).

Perhaps the most popular such method is Gauss-Hermite interpolation (see Judd [1998] for an exposition, or Kopecky and Suen [2010] for some alternatives). Here, we will pursue a particularly intuitive alternative: Equiprobable discretization. In this method, $m = n$ and boundaries on the joint CDF are determined in such a way as to divide up the total probability mass into submasses of equal size (each of which therefore has a mass of n^{-2}). This is conceptually easier if we represent the underlying shocks as statistically independent, as with $\theta_{1,t+1}$ and $\theta_{2,t+1}$ above; in that case, each submass is a square region in the Θ_1 and Θ_2 grid. We then compute the average value of Θ_1 and $\mathbf{R}$ conditional on their being located in each of the subdivisions of the range of the CDF. Since, in this specification, $\mathbf{R}$ is a function of Θ_1 , the $\mathbf{R}$ values are indexed by both i and j , but since we have written Θ_1 as IID, the representation of the approximating summation is even simpler than (A.2.9):

$$\mathbb{E}[h(\tilde{\Theta}_1, \tilde{\mathbf{R}})] \approx n^{-2} \sum_{i=1}^n \sum_{j=1}^n h(\hat{\Theta}_1[i], \mathbf{R}(\hat{\Theta}_1[i], \hat{\Theta}_2[j])) \quad (\text{A.2.10})$$

where the function $\mathbf{R}(\Theta_1, \Theta_2)$ is implicitly defined by (A.2.2).

Details can be found in the Mathematica notebook associated with this section. A particular example, in which $\sigma_2^2 = \sigma_1^2$ and $\omega = 0.5$, is illustrated in Figure A.1; the red dots reflect the height of the approximation to the CDF above the conditional mean values for Θ_1 and $\mathbf{R}$ within each of the equiprobable regions.

³An alternative would be to work with $\theta_{1,t+1}$ and $\mathbf{r}_{t+1}$ directly, which would require a multivariate normal with nonzero off-diagonal elements. The two approaches are mathematically indistinguishable, and we take the independent one because it is the more convenient to integrate against below.

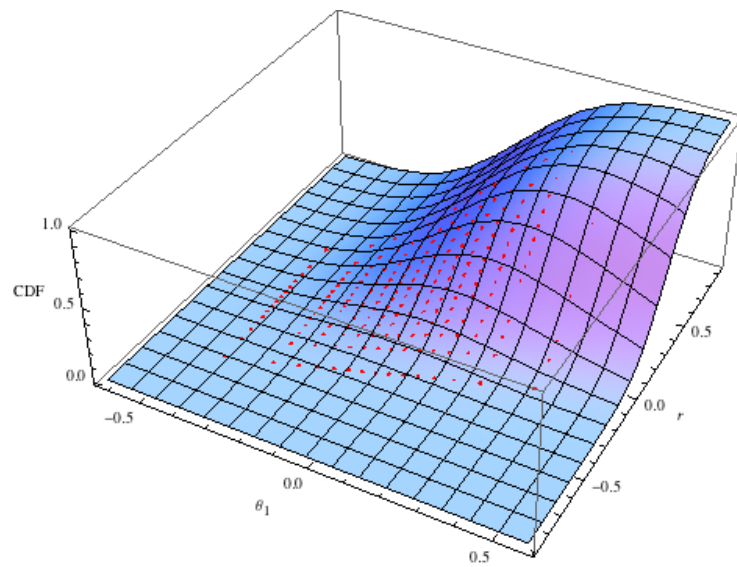


Figure A.1: ‘True’ CDF With Approximation Points in Red for $\omega = 0.5$

A.3 Equiprobable Approximation to Bivariate Lognormal Returns

This section starts by defining a convenient notation to represent two rate-of-return shocks that are distributed according to a multivariate lognormal that allows for nonzero covariances. It next presents a computationally simple method for constructing a numerical approximation to that joint distribution. The final section uses both the numerical approximation, and more accurate (but enormously slower) standard numerical integration tools to assess the accuracy of the Campbell-Vicera analytical approximation to the solution to the optimal portfolio choice problem described in Portfolio-Multi-CRRA.

A.3.1 Statistical Theory

Consider a set of two normally distributed risks

$$\begin{aligned}\theta_{2,t+1} &\sim \mathcal{N}(-0.5\Sigma^2, \Sigma^2) \\ \theta_{1,t+1} &\sim \mathcal{N}(-0.5(x\Sigma)^2, (x\Sigma)^2)\end{aligned}\tag{A.3.1}$$

which are statistically independent ($\theta_{1,t+1} \perp \theta_{2,t+1}$) even though the scale of shock 1's standard deviation is determined by the proportionality factor x multiplied by the value of shock 2's standard deviation, Σ . (This permits us to increase or decrease the size of both risks by changing Σ and to adjust the relative size of risk 1 vs risk 2 by adjusting x .)

The θ are interpreted as the logarithms of level variables Θ , and the means of the log variables have been chosen such that the expectations of the levels are independent of the size of the risk (cf. [Math Facts](#)). That is, defining $\Theta_i = \mathbb{E}_t[e^{\theta_{i,t+1}}]$ for $i \in \{1, 2\}$:

$$\begin{aligned}\Theta_i &\equiv \mathbb{E}_t[\Theta_{i,t+1}] = 1 \quad \forall i \\ \log \mathbb{E}_t[\Theta_{i,t+1}] &= 0 \quad \forall i.\end{aligned}\tag{A.3.2}$$

From the first shock we can construct a log rate-of-return variable that can be represented equivalently in either of two ways:

$$\begin{aligned}\mathbf{r}_{1,t+1} &= \overbrace{\mathbf{r}_1 + 0.5(x\Sigma)^2}^{\equiv \hat{\mathbf{r}}_1} + \theta_{1,t+1} \\ &= \underbrace{\hat{\mathbf{r}}_1 - 0.5(x\Sigma)^2}_{\equiv \mathbf{r}_1} + \underbrace{0.5(x\Sigma)^2 + \theta_{1,t+1}}_{\equiv \theta_{1,t+1}}\end{aligned}\tag{A.3.3}$$

where $\theta_{1,t+1} \sim \mathcal{N}(0, (x\Sigma)^2)$; the notation for θ is motivated by the fact that the addition of the extra term cancels the nonzero mean of the original θ . Then

$$\begin{aligned}\mathbf{R}_{1,t+1} &\equiv e^{\mathbf{r}_{1,t+1}} \\ \mathbf{R}_1 &\equiv \mathbb{E}_t[\mathbf{R}_{1,t+1}] = e^{\mathbf{r}_1 + (x\Sigma)^2/2} \\ \log \mathbb{E}_t[\mathbf{R}_{1,t+1}] &= \mathbf{r}_1 + (x\Sigma)^2/2 = \hat{\mathbf{r}}_1\end{aligned}\tag{A.3.4}$$

where note to avoid confusion that $\hat{\mathbf{r}}_1 \neq \mathbb{E}_t[\mathbf{r}_{1,t+1}]$ while $\log \mathbb{E}_t[\mathbf{R}_{1,t+1}] \neq \mathbf{r}_1 = \mathbb{E}_t[\mathbf{r}_{1,t+1}]$.

Using $\mathbb{E}_t[\theta_{2,t+1}] = -0.5\Sigma^2$ and $\mathbb{E}_t[(\omega/x)\theta_{1,t+1}] = -0.5((\omega/x)\Sigma)^2$, we can by analogy define a second return

$$\begin{aligned}\mathbf{r}_{2,t+1} &\equiv \hat{\mathbf{r}}_2 + \zeta + (\omega/x)\theta_{1,t+1} + \theta_{2,t+1} \\ &= \hat{\mathbf{r}}_2 + \zeta + (\omega/x)(\theta_{1,t+1} - 0.5(x\Sigma)^2) + (\theta_{2,t+1} - 0.5\Sigma^2) \\ &= \underbrace{\hat{\mathbf{r}}_2 + \zeta - 0.5(\omega/x)(x\Sigma)^2 - 0.5\Sigma^2}_{\mathbf{r}_2} + (\omega/x)\theta_{1,t+1} + \theta_{2,t+1}\end{aligned}\tag{A.3.5}$$

for some constants ω and ζ .

Since $(\omega/x)\theta_{1,t+1}$ is the only component of $\mathbf{r}_{2,t+1}$ that is correlated with $\mathbf{r}_{1,t+1}$,

$$\begin{aligned}
\text{cov}(\mathbf{r}_{1,t+1}, \mathbf{r}_{2,t+1}) &= \text{cov}(\mathbf{r}_{1,t+1}, (\omega/x)\mathbf{r}_{1,t+1}) \\
&= (\omega/x) \underbrace{\text{cov}(\mathbf{r}_{1,t+1}, \mathbf{r}_{1,t+1})}_{=x^2\Sigma^2} \\
&= \omega x \Sigma^2.
\end{aligned} \tag{A.3.6}$$

Thus, the parameter ω controls the covariance between the risky returns. If we set $\omega = 0$ then $\mathbf{r}_{1,t+1} \perp \mathbf{r}_{2,t+1}$ (the returns are independent).

Next we want to find the value of ζ such that the expected level of the return is unaffected by Σ (so that we will be able to explore independently the distinct effects of the components of each shock and their covariance):

$$\mathbf{R}_2 \equiv \mathbb{E}_t[\mathbf{R}_{2,t+1}] = e^{\dot{\mathbf{r}}_2} \tag{A.3.7}$$

regardless of the values of x and Σ . Using (A.3.5), we therefore need:

$$\begin{aligned}
\mathbb{E}_t[e^{\zeta + (\omega/x)\theta_{1,t+1} + \theta_{2,t+1}}] &= 1 \\
\log \mathbb{E}_t[e^{\zeta + (\omega/x)\theta_{1,t+1} + \theta_{2,t+1}}] &= 0.
\end{aligned} \tag{A.3.8}$$

Using standard facts about lognormals (cf. [Math Facts](#)), and for convenience defining $\hat{\omega} = (\Sigma/x)\omega$, we have

$$\begin{aligned}
0 &= \zeta - 0.5\hat{\omega}x^2 - 0.5\Sigma^2 + 0.5\hat{\omega}^2x^2 + 0.5\Sigma^2 \\
&= \zeta - 0.5x^2\hat{\omega}(1 - \hat{\omega}) \\
\zeta &= 0.5(\hat{\omega} - \hat{\omega}^2)x^2 = 0.5(\omega x - \omega^2)\Sigma^2
\end{aligned} \tag{A.3.9}$$

which means that we can rewrite (A.3.5) directly as

$$\begin{aligned}
\mathbf{r}_{2,t+1} &= \dot{\mathbf{r}}_2 + 0.5(\omega x - \omega^2)\Sigma^2 + (\omega/x)\theta_{1,t+1} + \theta_{2,t+1} \\
&= \dot{\mathbf{r}}_2 + 0.5(\omega x - \omega^2)\Sigma^2 - 0.5(\omega/x)(x\Sigma)^2 - 0.5\Sigma^2 + (\omega/x)\boldsymbol{\theta}_{t+1,1} + \boldsymbol{\theta}_{2,t+1} \\
&= \dot{\mathbf{r}}_2 + 0.5\omega x \Sigma^2 - 0.5\omega x \Sigma^2 - 0.5\omega^2 \Sigma^2 - 0.5\Sigma^2 + (\omega/x)\boldsymbol{\theta}_{t+1,1} + \boldsymbol{\theta}_{2,t+1} \\
&= \underbrace{\dot{\mathbf{r}}_2 - (1 + \omega^2)0.5\Sigma^2}_{\mathbf{r}_2} + (\omega/x)\boldsymbol{\theta}_{t+1,1} + \boldsymbol{\theta}_{2,t+1}.
\end{aligned} \tag{A.3.10}$$

Hence, from the independent mean-one lognormally distributed shocks Θ_1 and Θ_2 , we have constructed a pair of jointly lognormally distributed shocks whose covariance is controlled by the parameter ω , whose relative and absolute variances are controlled by the parameters x and Σ , and whose means are $\mathbf{r}_1$ and $\mathbf{r}_2$.

To sum up, the process can be described in either of two ways:

$$\begin{pmatrix} \mathbf{r}_{1,t+1} \\ \mathbf{r}_{2,t+1} \end{pmatrix} \sim \begin{pmatrix} \mathcal{N}(\mathbf{r}_1, x^2\Sigma^2) \\ \mathcal{N}(\mathbf{r}_2, \Sigma^2) \end{pmatrix} = \begin{pmatrix} \mathcal{N}(\dot{\mathbf{r}}_1 - 0.5(x\Sigma)^2, x^2\Sigma^2) \\ \mathcal{N}(\dot{\mathbf{r}}_2 - (1 + \omega^2)0.5\Sigma^2, \Sigma^2) \end{pmatrix} \tag{A.3.11}$$

with covariance matrix

$$\begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix} \equiv \begin{pmatrix} (x\Sigma)^2 & x\omega\Sigma^2 \\ x\omega\Sigma^2 & (\omega^2 + 1)\Sigma^2 \end{pmatrix} \tag{A.3.12}$$

where a final useful result that follows from (A.3.12) and (A.3.6) is

$$\begin{aligned}
\text{corr}(\mathbf{r}_{1,t+1}, \mathbf{r}_{2,t+1}) &\equiv \text{cov}(\mathbf{r}_{1,t+1}, \mathbf{r}_{2,t+1})/\sigma_1\sigma_2 \\
&= \omega/(1 + \omega^2)^{0.5}.
\end{aligned} \tag{A.3.13}$$

A.3.2 Computational Theory

To reduce clutter, define $\tilde{\mathbf{R}}_i = \mathbf{R}_{i,t+1}$ and interpret $\mathbb{E}$ as $\mathbb{E}_t[\cdot]$, so that we can write the expectation of some function h that depends on the realization of the return shocks as:

$$\mathbb{E}[h(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2)] = \int_{\mathbf{R}_1}^{\tilde{\mathbf{R}}_1} \int_{\mathbf{R}_2}^{\tilde{\mathbf{R}}_2} h(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2) dF(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2) \quad (\text{A.3.14})$$

where $F(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2)$ is the cumulative distribution function for a multivariate lognormal with the covariance matrix defined by (A.3.12).⁴

Standard numerical computation software can compute this double integral, but at such a slow speed as to be unusable for many purposes.

Computation of the expectation can be massively speeded up by advance construction of a numerical approximation to $F(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2)$.

Such approximations often take the approach of replacing the distribution function with a discretized approximation to it; appropriate weights $w[i, j]$ are attached to each of a finite set of points indexed by i and j , and the approximation to the integral is given by:

$$\mathbb{E}[h(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2)] \approx \sum_{i=1}^n \sum_{j=1}^m h(\hat{\mathbf{R}}_1[i, j], \hat{\mathbf{R}}_2[i, j]) w[i, j] \quad (\text{A.3.15})$$

where various methods are used for constructing the weights $w[i, j]$ and the nodes (corresponding to the $\{i, j\}$ pairs). The matrices $\hat{\mathbf{R}}_1$ and $\hat{\mathbf{R}}_2$ contain the conditional mean values of $\tilde{\mathbf{R}}_1$ and $\tilde{\mathbf{R}}_2$ associated with each of the regions.

Perhaps the most popular such method is Gauss-Hermite interpolation (see Judd [1998] for an exposition, or Kopecky and Suen [2010] for a recent candidate for a better choice). Here, we will pursue a particularly simple and intuitive alternative: Equiprobable discretization. In this method, $m = n$ and boundaries on the joint CDF are determined in such a way as to divide up the total probability mass into submasses of equal size (each of which therefore has a mass of n^{-2}). This is conceptually easier if we represent the underlying shocks as statistically independent, as with Θ_1 and Θ_2 above; in that case, each submass is a square region in the Θ_1 and Θ_2 grid. We then compute the average value of Θ_1 and Θ_2 conditional on their being located in each of the subdivisions of the range of the CDF. Since Θ_1 and Θ_2 are IID, the representation of the approximating summation is even simpler than (A.3.15):

$$\mathbb{E} \left[h \left(\tilde{\mathbf{R}}_1, \tilde{\mathbf{R}}_2 \right) \right] \approx n^{-2} \sum_{i=1}^n \sum_{j=1}^n h \left(\mathbf{R}_1(\hat{\Theta}_1[i], \hat{\Theta}_2[j]), \mathbf{R}_2(\hat{\Theta}_1[i], \hat{\Theta}_2[j]) \right) \quad (\text{A.3.16})$$

where $\mathbf{R}_i(\Theta_1, \Theta_2)$ are the (linear) functions relating the return shocks to the IID shocks.

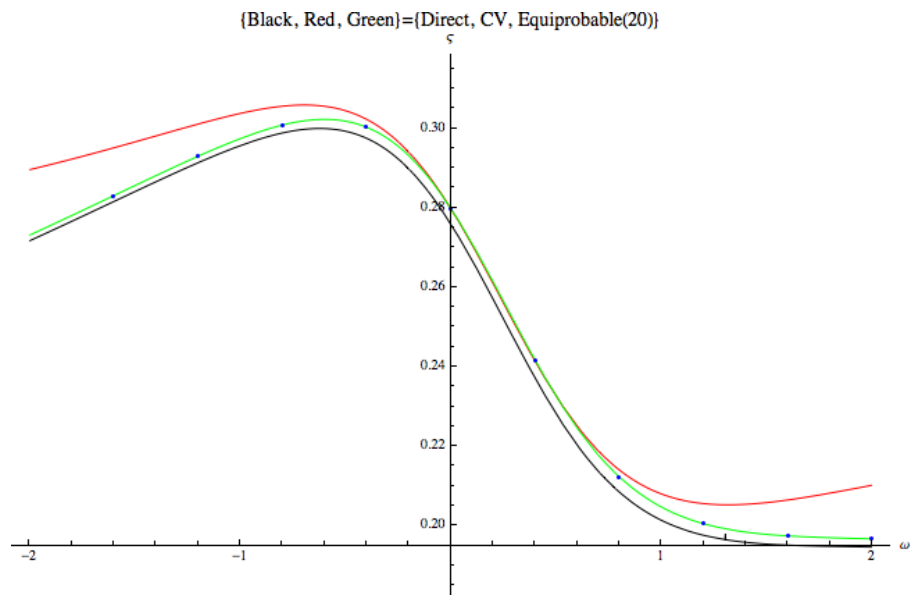
A.3.3 Computational Results

Figure A.2 compares the computed optimal portfolio share for a numerical solution using the built-in numerical optimizer and maximization functions (the lowest, black, locus), the Campbell-Viceira solution (the highest, red locus) and an equiprobable approximation using 20 approximation points (green, middle) as well as the solution using the equiprobable approximation at an evenly-spaced grid of points (blue dots).

Careful examination indicates that the numerical approximation is quite close to the full numerical solution, while the CV approximation diverges substantially from the numerical answer. The tradeoff is that the equiprobable solution is about 2000 times slower than the CV approximation, while the direct solution is more than 100 times slower than the equiprobable solution. Depending on the requirements of the problem being examined, these differences in efficiency can make a tremendous difference in the feasibility of a research project.⁵

⁴Under the lognormal assumption, $\tilde{\mathbf{R}}_i = 0$ and $\tilde{\mathbf{R}}_i = \infty$ for both shocks $i \in \{1, 2\}$.

⁵Details can be found in the Mathematica notebook associated with this section.

Figure A.2: Portfolio Share By Method as a Function of ω

A.4 Ramsey Growth in Discrete and Continuous Time

This section solves a continuous-time version of the Ramsey/Cass-Koopmans (RCK) model using the Hamiltonian method, and shows the relationship between that method and the discrete-time approach.

The problem is to choose a path of consumption per capita c from the present moment (arbitrarily called time 0) into the infinite future, $\{c\}_0^\infty$, that solves the problem

$$\max_{\{c\}_0^\infty} \int_0^\infty u(c) e^{-\vartheta t} \quad (\text{A.4.1})$$

subject to

$$\begin{aligned} \dot{k} &= f(k) - c - (\xi + \delta)k \\ k &> 0 \quad \forall t \end{aligned} \quad (\text{A.4.2})$$

where ϑ is the time preference rate, ξ is the population growth rate, and δ is the depreciation rate. (In continuous time, we think of all variables as implicitly being a function of time, but it is cumbersome to write, e.g., $c(t)$ everywhere, so the time argument is omitted; we are also thinking of the initial value of capital at date 0 as being a ‘given’ in the problem, so that $k(0) = \bullet$ for some specific value of $\bullet$).

To emphasize the similarity between the continuous-time and the discrete-time solutions where we have typically used the roman V to denote value, for the continuous-time problem we define ‘curly’ value as a function of the initial level of capital as $\mathcal{V}(k)$.

The current-value (discounted) Hamiltonian is

$$\mathcal{H}(k, c, \lambda) = u(c) + (f(k) - c - (\xi + \delta)k)\lambda \quad (\text{A.4.3})$$

where k is the state variable, c is the control variable, and λ is the costate variable.

λ is the continuous-time equivalent of a Lagrange multiplier, so its value should be equivalent to the value of relaxing the corresponding constraint by an infinitesimal amount. But the constraint in question is the capital-accumulation constraint. Thus λ should be equal to the value of having a tiny bit more capital, $\frac{\mathcal{V}(k+\Delta k) - \mathcal{V}(k)}{\Delta k}$. In other words, you can think of $\lambda = \mathcal{V}'(k)$.

The first necessary Hamiltonian condition for optimality is

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} &= 0 \\ u'(c) &= \lambda \\ u'(c) &= \mathcal{V}'(k). \end{aligned} \quad (\text{A.4.4})$$

Note the similarity between (A.4.4) and the result we usually obtain by using the Envelope theorem in the discrete-time problem,

$$u'(c_t) = V'(k_t). \quad (\text{A.4.5})$$

Thus, you can use the intuition you (should have) developed by now about why the marginal utility of consumption should be equal to the marginal value of extra resources to understand this Hamiltonian optimality condition.

The second necessary condition is

$$\begin{aligned} \dot{\lambda} &= \vartheta \lambda - \lambda(f'(k) - (\xi + \delta)) \\ \left(\frac{\dot{\lambda}}{\lambda}\right) &= \vartheta - (f'(k) - (\xi + \delta)) \end{aligned} \quad (\text{A.4.6})$$

which expresses the growth rate of λ at an annual rate (because the interest rate r and time preference rate ϑ are measured at an annual rate).

To interpret this in terms of our discrete-time model, begin with the condition

$$V'(k_t) = R\beta V'(k_{t+1}). \quad (\text{A.4.7})$$

The final necessary condition is just that the accumulation equation for capital is satisfied,

$$\dot{k} = f(k) - c - (\xi + \delta)k. \quad (\text{A.4.8})$$

This is the continuous-time equivalent of what we have previously called the Dynamic Budget Constraint.

Up to now in this course we haven't thought very much about what the time period is. Generally, we have expressed things in terms of yearly rates, so that for example we might choose $R = 1.04$ and $\beta = 1/(1 + \vartheta) = 1/(1.04)$ to represent an interest rate of 4 percent and a discount rate of 4 percent.

One of the attractive features of the time-consistent model we have been using is that it generates self-similar behavior as the time interval is changed. Thus if we wanted to solve a quarterly version of the model we would choose $R = 1.01$ and $\beta = 1/1.01$ and it would imply consumption of almost exactly $1/4$ of the amount implied by the annual model, so that four quarters of such behavior would aggregate to the prediction of the annual model.

To put this in the most general form, suppose R and β correspond to 'annual rate' values and we want to divide the year into m periods. Then the appropriate interest rate and discount factor on a per-period basis would be $R^{1/m}$ and $\beta^{1/m}$. Thus the discrete-time equation could be rewritten

$$V'(k_t) = R^{1/m} \beta^{1/m} V'(k_{t+1}) \quad (\text{A.4.9})$$

where the time interval is now $1/m$ th of a year (e.g. if $m=52$, we're talking weekly, so that period $t+1$ is one week after period t). Now we can use our old friend, the fact that $e^z \approx 1 + z$, to note that this is approximately

$$\begin{aligned} V'(k_t) &\approx [e^r]^{1/m} [e^{-\vartheta}]^{1/m} V'(k_{t+1}) \\ &= e^{(1/m)r} e^{-(1/m)\vartheta} V'(k_{t+1}) \\ &= e^{(1/m)(r-\vartheta)} V'(k_{t+1}) \\ &= e^{(1/m)(r-\vartheta)} (V'(k_t) + \Delta V'(k_{t+1})) \\ 1 &= e^{(1/m)(r-\vartheta)} \left(\frac{V'(k_t) + \Delta V'(k_{t+1})}{V'(k_t)} \right) \\ e^{(1/m)(\vartheta-r)} &= \left(\frac{V'(k_t) + \Delta V'(k_{t+1})}{V'(k_t)} \right) \\ V'(k_t) \left(\underbrace{e^{(1/m)(\vartheta-r)} - 1}_{\approx 1 + (1/m)(\vartheta-r) - 1} \right) &= \Delta V'(k_{t+1}) \\ \frac{\Delta V'(k_{t+1})}{V'(k_t)} &\approx (1/m)(\vartheta - r) \\ \frac{m \Delta V'(k_{t+1})}{V'(k_t)} &\approx (\vartheta - r). \end{aligned} \quad (\text{A.4.10})$$

We defined the interest rate and time preference rate on an annual basis, but the time interval between t and $t+1$ is only $(1/m)$ th of a year. Thus $m \Delta V'(k_{t+1})$ expresses the speed of change in $V'(k_t)$ at an annual rate.

Now, note that since the effective interest rate in this model is $f'(k) - (\xi + \delta)$, equation (A.4.10) is basically the same as (A.4.6) since $\lambda = \mathcal{V}'(k)$ and $m \Delta V'(k_{t+1}) = (d/dt) \mathcal{V}'(k) = \dot{\lambda}$. Hence, the second optimality condition in the Hamiltonian optimization method is basically equivalent to the condition $V'(k_t) = R \beta V'(k_{t+1})$ from the discrete-time optimization method!

The final required condition (the transversality constraint) is

$$\lim_{t \rightarrow \infty} \lambda k e^{-\vartheta t} = 0 \quad (\text{A.4.11})$$

The translation of this into the discrete-time model is

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t) k_t = 0. \quad (\text{A.4.12})$$

Consider the simple model with a constant gross interest rate R and CRRA utility. In that model, recall that $c_{t+1} = (R\beta)^{1/\rho} c_t$. Thus considered from time zero (A.4.12) becomes

$$\begin{aligned}
 \lim_{t \rightarrow \infty} \beta^t (c_0 ((R\beta)^{1/\rho})^t)^{-\rho} k_t &= 0 \\
 &= c_0^{-\rho} \beta^t [(R\beta)^{t/\rho}]^{-\rho} k_t \\
 &= c_0^{-\rho} \beta^t \beta^{-t} R^{-t} k_t \\
 &= c_0^{-\rho} R^{-t} k_t \\
 &\rightarrow \lim_{t \rightarrow \infty} R^{-t} k_t = 0.
 \end{aligned} \tag{A.4.13}$$

What this says is that you cannot behave in such a way that you expect k_t to grow faster than the interest rate forever.⁶

This is the infinite-horizon version of the intertemporal budget constraint.

Among the infinite number of time paths of c and k that will satisfy the first order conditions above, only one will also satisfy this transversality constraint, because all the others imply a violation of the intertemporal budget constraint.

Now differentiate (A.4.4) with respect to time

$$\dot{c} u''(c) = \dot{\lambda} \tag{A.4.14}$$

and substitute this into equation (A.4.6) to get

$$\begin{aligned}
 \frac{\dot{c} u''(c)}{u'(c)} &= (\vartheta - (f'(k) - (\xi + \delta))) \\
 \dot{c} &= -\frac{u'(c)}{u''(c)} (f'(k) - (\xi + \delta) - \vartheta)
 \end{aligned} \tag{A.4.15}$$

using the fact derived earlier that for a CRRA utility function $u(c) = c^{1-\rho}/(1-\rho)$, $-u''(c)c/u'(c) = \rho$, this becomes

$$\begin{aligned}
 &= (c/\rho)(f'(k) - (\xi + \delta) - \vartheta) \\
 \dot{c}/c &= \rho^{-1}(f'(k) - (\xi + \delta) - \vartheta)
 \end{aligned} \tag{A.4.16}$$

⁶Note that this also rules out negative k_t values that grow faster than the interest rate.

A.5 Math Facts Useful for Graduate Macroeconomics

The following collection of facts is useful in many macroeconomic models. No proof is offered in most cases because the derivations are standard elements of prerequisite mathematics or microeconomics classes; this section is offered as an aide memoire and for reference purposes.

Throughout this document, typographical distinctions should be interpreted as meaningful; for example, the variables $\mathbf{r}$ and $\tilde{r}$ are different from each other, like x and y .

Furthermore, a version of a variable without a subscript should be interpreted as the population mean of that variable. Thus, if $\mathbf{R}_{t+1}$ is a stochastic variable, then $\mathbf{R}$ denotes its mean value.

A.5.1 Utility Functions

Fact: CRRALim

$$\lim_{\rho \rightarrow 1} \left(\frac{c^{1-\rho} - 1}{1 - \rho} \right) = \log c \quad (\text{A.5.1})$$

This follows from L'Hopital's rule⁷ because for any $\rho \neq 1$ the derivative exists,

$$u'(c) = c^{-\rho}, \quad (\text{A.5.2})$$

and $\lim_{\rho \rightarrow 1} c^{-\rho} = 1/c$ but $\int (1/c) = \log c$.

Thus, we can conclude that as $\rho \rightarrow 1$, the behavior of the consumer with $u(c) = c^{1-\rho}/(1-\rho)$ becomes identical to the behavior of a consumer with $u(c) = \log c$.⁸

Fact: FinSum

$$\sum_{i=0}^T \gamma^i = \left(\frac{1 - \gamma^{T+1}}{1 - \gamma} \right) \quad (\text{A.5.3})$$

Fact: InfSum

If $0 < \gamma < 1$, then

$$\sum_{i=0}^{\infty} \gamma^i = \left(\frac{1}{1 - \gamma} \right) \quad (\text{A.5.4})$$

Fact: FinSumMult

$$\sum_{i=0}^T i\gamma^i = \left(\frac{\gamma + \gamma^{T+1}(T(\gamma - 1) - 1)}{(1 - \gamma)^2} \right) \quad (\text{A.5.5})$$

⁷Given recent decrees of the relevant authorities, the circumflex may need to be eliminated from future versions of these notes. Which is OK, because the gentleman in question paid someone smarter for the result anyway.

⁸Recall that *behavior* is not affected by adding a constant to the utility function ...

Fact: InfSumMult

If $0 < \gamma < 1$, then

$$\sum_{i=0}^{\infty} i\gamma^i = \left(\frac{\gamma}{(1-\gamma)^2} \right) \quad (\text{A.5.6})$$

A.5.2 ‘Small’ Number Approximations

Sometimes economic models are written in continuous time and sometimes in discrete time. Generically, there is a close correspondence between the two approaches, which is captured (for example) by the future value of a series that is growing at rate g .

$$e^{gt} \text{ corresponds to } (1+g)^t \equiv G^t. \quad (\text{A.5.7})$$

The words ‘corresponds to’ are not meant to imply that these objects are mathematically identical, but rather that these are the corresponding ways in which constant growth is treated in continuous and in discrete time; while for small values of g they will be numerically very close, continuous-time compounding does yield slightly different values after any given time interval than does discrete growth (for example, continuous growth at a 10 percent rate after 1 year yields $e^{0.1} \approx 1.10517$ while in discrete time we would write it as $G = 1.1$.)

Many of the following facts can be interpreted as manifestations of the limiting relationships between continuous and discrete time approaches to economic problems. (The continuous time formulations often yield simpler expressions, while the discrete formulations are useful for computational solutions; one of the purposes of the approximations is to show how the discrete-time solution becomes close to the corresponding continuous-time problem as the time interval shrinks).

Fact: TaylorOne

For ϵ near zero (‘small’), a first order Taylor expansion of $(1+\epsilon)^\zeta$ around 1 yields

$$(1+\epsilon)^\zeta \approx 1 + \epsilon\zeta \quad (\text{A.5.8})$$

Fact: TaylorTwo

For ϵ near zero (‘small’), a second order Taylor expansion of $(1+\epsilon)^\zeta$ around 1 yields

$$\begin{aligned} (1+\epsilon)^\zeta &\approx 1 + \zeta\epsilon + \epsilon^2\zeta(\zeta-1)/2 \\ &= 1 + \left(1 + \left(\frac{\zeta-1}{2} \right) \epsilon \right) \zeta\epsilon \end{aligned} \quad (\text{A.5.9})$$

Fact: LogEps

For ϵ near zero (‘small’)

$$\log(1+\epsilon) \approx \epsilon \quad (\text{A.5.10})$$

Fact: ExpEps

For ϵ near zero (‘small’)

$$(1 + \epsilon) \approx e^\epsilon \quad (\text{A.5.11})$$

Fact: OverPlus

For ϵ near zero ('small')

$$1/(1 + \epsilon) \approx 1 - \epsilon \quad (\text{A.5.12})$$

Fact: MultPlus

For ϵ and ζ near zero ('small')

$$(1 + \epsilon)(1 + \zeta) \approx 1 + \epsilon + \zeta \quad (\text{A.5.13})$$

Fact: ExpPlus

For real numbers ϵ and ζ

$$\exp(\zeta) \exp(\epsilon) = \exp(\zeta + \epsilon) \quad (\text{A.5.14})$$

Fact: SmallSmallZero

If ϵ is small and ζ is small then $\epsilon\zeta$ can be approximated by zero.

A.5.3 Statistics/Probability Facts**Fact: SumNormsIsNorm**

If $\tilde{r}_{t+1} \sim \mathcal{N}(\tilde{r}, \sigma_{\tilde{r}}^2)$ and $\mathbf{r}_{t+1} \sim \mathcal{N}(\mathbf{r}, \sigma_{\mathbf{r}}^2)$ and $\tilde{r}_{t+1}$ and $\mathbf{r}_{t+1}$ are independent (written $\tilde{r}_{t+1} \perp \mathbf{r}_{t+1}$), then

$$\tilde{r}_{t+1} + \mathbf{r}_{t+1} = \mathcal{N}(\tilde{r} + \mathbf{r}, \sigma_{\tilde{r}}^2 + \sigma_{\mathbf{r}}^2) \quad (\text{A.5.15})$$

Fact: ELogNorm

If from the viewpoint of period t the stochastic variable $\mathbf{R}_{t+1}$ is lognormally distributed with mean $\mathbf{r}$ and variance $\sigma_{\mathbf{r}}^2$, $\mathbf{r}_{t+1} \sim \mathcal{N}(\mathbf{r}, \sigma_{\mathbf{r}}^2)$, then

$$\mathbb{E}_t[e^{\mathbf{r}_{t+1}}] = e^{\mathbf{r} + \sigma_{\mathbf{r}}^2/2} \quad (\text{A.5.16})$$

Fact: ELogNormMeanOne

If from the viewpoint of period t the stochastic variable $\mathbf{R}_{t+1}$ is lognormally distributed with mean $-\sigma^2/2$ and variance $\sigma_{\mathbf{r}}^2$, $\log \mathbf{R}_{t+1} \sim \mathcal{N}(-\sigma^2/2, \sigma^2)$, then

$$\mathbb{E}_t[e^{\mathbf{r}_{t+1}}] = e^{-\sigma_{\mathbf{r}}^2/2 + \sigma_{\mathbf{r}}^2/2} = e^0 = 1 \quad (\text{A.5.17})$$

Fact: LogELogNorm

If $\mathbf{R}_{t+1}$ is lognormally distributed as in ELogNorm, then

$$\begin{aligned}\log \mathbb{E}_t[\mathbf{R}_{t+1}] &= \mathbb{E}_t[\log \mathbf{R}_{t+1}] + \sigma_{\mathbf{r}}^2/2 \\ &= \mathbf{r} + \sigma_{\mathbf{r}}^2/2\end{aligned}\tag{A.5.18}$$

This follows from taking the log of both sides of the equation in ELogNorm.

Fact: NormTimes

If $\mathbf{r}_{t+1} \sim \mathcal{N}(\mathbf{r}, \sigma_{\mathbf{r}}^2)$, then

$$\gamma \mathbf{r}_{t+1} \sim \mathcal{N}(\gamma \mathbf{r}, \gamma^2 \sigma_{\mathbf{r}}^2)\tag{A.5.19}$$

Fact: MeanOne

If $\log \mathbf{R}_{t+1} \sim \mathcal{N}(-\sigma_{\mathbf{r}}^2/2, \sigma_{\mathbf{r}}^2)$, then

$$\mathbb{E}_t[\mathbf{R}_{t+1}] = 1\tag{A.5.20}$$

for any value of $\sigma_{\mathbf{r}}^2 \geq 0$.

This follows from substituting $-\sigma_{\mathbf{r}}^2/2$ for $\mathbf{r}$ in ELogNorm.

Fact: LogMeanMPS

If $\log \mathbf{R}_{t+1} \sim \mathcal{N}(\mathbf{r} - \sigma_{\mathbf{r}}^2/2, \sigma_{\mathbf{r}}^2)$, then

$$\log \mathbb{E}_t[\mathbf{R}_{t+1}] = \mathbf{r}\tag{A.5.21}$$

for any value of $\sigma_{\mathbf{r}}^2 \geq 0$.

This follows from substituting $\mathbf{r} - \sigma_{\mathbf{r}}^2/2$ for $\mathbf{r}$ in ELogNorm and taking the log.

Fact: ELogNormTimes

If $\log \hat{\mathbf{R}}_{t+1} = \gamma \log \mathbf{R}_{t+1}$ where $\log \mathbf{R}_{t+1} \sim \mathcal{N}(\mathbf{r}, \sigma_{\mathbf{r}}^2)$, then

$$\mathbb{E}_t[\hat{\mathbf{R}}_{t+1}] = e^{\gamma \mathbf{r} + \gamma^2 \sigma_{\mathbf{r}}^2/2}\tag{A.5.22}$$

Fact: LogELogNormTimes

If $\log \hat{\mathbf{R}}_{t+1} = \gamma \log \mathbf{R}_{t+1}$ where $\log \mathbf{R}_{t+1} \sim \mathcal{N}(\mathbf{r}, \sigma_{\mathbf{r}}^2)$, then

$$\log \mathbb{E}_t[\hat{\mathbf{R}}_{t+1}] = \gamma \mathbf{r} + \gamma^2 \sigma_{\mathbf{r}}^2/2\tag{A.5.23}$$

This follows from taking the log of ELogNormTimes.

A.5.4 Other Facts

Fact: EulersTheorem

If $Y = F(K, L)$ is a constant returns to scale production function, then

$$Y = F_K K + F_L L, \tag{A.5.24}$$

and if this production function characterizes output in a perfectly competitive economy then F_K is the interest factor and F_L is the wage rate.

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